Polynomial Presentations of the Integers over the Rationals: Conditional Constructions, Lower Bounds, and Number-Field Obstructions

JUVa

6 November 2026

Abstract

We study explicit polynomial presentations of the ordinary integers in the rationals, starting from a pinned revision of the repository *diophantine-z-q*. Under its named arithmetic inputs and Sun's simultaneous prime-pattern theorem, we construct an indexed polynomial in 41 total variables, including the input, with degree at most $2^{9566673}$. A 43-variable alternative has degree at most 2^{236}. The direct substitution of the Lucas polynomial into approximate logarithmic ratios removes two retained coordinates and one finite algebraic coordinate. It also gives quartics with 1315 variables. An affine change reduces their coefficient height to $64 \cdot 3^{2^{28}}$; an alternative with 1335 variables has coefficient height less than 2^{1020}. Their complete sparse coefficient tables are independently checked. The previously implemented 558-variable construction and its source-preserving alternatives remain available for comparison.

No construction in two through ten total variables is established. Unconditional lower bounds exclude two total variables at every degree. We prove that two quadratic equations, in any number of coordinates, cannot have an infinite input image contained in a number field's ring of integers. A quadratic system defining that ring consequently needs at least five total coordinates and at least three independent quadratic equations. Further results explain precise limits of finite algebraic encoding and fixed denominator tests without mistaking them for universal variable lower bounds.

Over every specified number field, a finite system can be combined into one equation without adding variables. An explicit division-norm construction gives degree at most $D\lceil\sqrt{s}\rceil$ for s equations of degree at most D; the preferred quadratic systems yield degree 72. These are faithful encodings, not general integrality theorems. Exact counterexamples show that the literal newer quartics accept all inputs over $\mathbb{Q}(i)$, the unshifted indexed equations reject certain algebraic integers, and the fixed cyclic elliptic argument fails in an explicit totally real quartic extension. A translated square repairs the finite input contacts but not that arithmetic obstruction. All hypotheses, proofs, computational scopes, and unresolved dependencies are stated separately.

Keywords. Diophantine definability; rational points; existential formulas; polynomial witness compression; quartic equations; finite-algebra presentations; number fields.

Contents

1 Definitions, objectives, and the scope of the results **6**

2 From integer equations to rational equations **7**
2.1 The precise contribution of MRDP 8

3 Several equations versus one equation **9**
3.1 Finite alternatives and nonzero conditions 10
3.2 The degree of a universal combining polynomial 11

4 The arithmetic structure retained from the source **12**

5 An economical semilocal condition **12**
5.1 An explicit three-variable semilocal definition 12

6 Rational relation combining and finite algebraic choices **14**
6.1 Combining square conditions without assuming integrality 14
6.2 One variable for finite choices and nonzero conditions 15

7 Three rational coordinates for an elliptic point over a quadratic field **17**
7.1 Application to elliptic doubling 18
7.2 Guards and finite unions without selector variables 19

8 One witness for a constructible finite-fiber projection **19**
8.1 Conjunctions and the literature on existential rank 21
8.2 Finite-fiber compression over other fields 22

9 An explicit polynomial for a guarded finite algebra **24**
9.1 Explicit degree and coefficient bounds for the specification 26

10 A finite-fiber presentation with 557 retained coordinates **28**
10.1 Point occurrences 28
10.2 The pole blocks and their zero fibers 29
10.3 The smaller anchor 29
10.4 The remaining parameters and algebraic choices 30
10.5 The emitted 558-variable source specification 31

11 Removing a fixed size condition without adding variables **33**

12 Coefficient bounds for a fixed elliptic exclusion product **34**
12.1 A bound in a finite algebra 36

13 A shared-point construction **37**
13.1 Fixed data and the complete rational system 38
13.2 The finite algebra and the variable count 40
13.3 Local tests and the fixed preparation-zero predicate 41
13.4 Height transfer with the original numerical constant 42
13.5 Soundness of the definition 43
13.6 Completeness and compatible simultaneous orders 43
13.7 Completeness of the nine pole occurrences 44
13.8 Preserving the scalar certificate when its point is shared 46
13.9 Removing the scalar preparation coordinate 47

13.10Fewer finite choices and smaller algebra rank 51

14 The compiled shared-point polynomials with Lucas proxies 53
14.1 Retained coordinates and rational parametrization 53
14.2 Integral triangular monic equations 53
14.3 Quadratic exceptions and geometric simplicity 54
14.4 The fixed elliptic exclusion product 55

15 Testing the integer relation without proxy coordinates 56
15.1 An elementary substitution principle 56
15.2 The replacement in the shared construction 57
15.3 Exact coordinate and finite-algebra counts 58

16 Changing the input shift without adding coordinates 59

17 The compiled polynomials with forty-one and forty-three coordinates 60

18 The quadratic compilation with Lucas proxy coordinates 61
18.1 Comparison order fifty-two with an explicit larger anchor 61
18.2 The domains of the division equations 64
18.3 The exact arithmetic conventions 64
18.4 Reusing polynomial evaluations 65
18.5 Finite choices at degree four 66
18.6 Removing redundant quadratic coordinates 66

19 Quartics obtained from the direct Lucas test 67

20 Reducing the anchor coefficients and shortening its constant chain 69
20.1 An affine change in the three private anchor coordinates 69
20.2 Eliminating the last coordinate in the constant chain 70
20.3 Inlining the three initial constants 71

21 Exact quartics and their computational verification 72
21.1 Removing constants without expanding their decimal digits 73
21.2 Verification boundaries 74

22 A fully specified polynomial with 854 variables 74
22.1 The finite list of rational equations 74
22.2 Eliminating the denominator witness 76

23 An explicit reduction to 853 variables 78
23.1 The pole denominators are automatic over the rationals 78
23.2 Combining two square roots and the nonzero conditions 78

24 Unconditional lower bounds 79
24.1 One witness and Hilbert irreducibility 80
24.2 A topological obstruction to rational parametrizations 81
24.3 Quadratic polynomials 81
24.4 Cubic cones and the sharp degree obstruction 82

24.5 A stronger one-witness obstruction 83

25 Stronger obstructions for small polynomial presentations 84
25.1 Curves and rational parametrizations 84
25.2 Two quadrics in projective four-space 86
25.3 Nonnegative polynomials in three variables 88
25.4 Further necessary conditions 88
25.5 Why the unrestricted two-witness case remains difficult 89

26 A restriction on the degree in two witnesses 90
26.1 Quaternion obstructions at integral specializations 90
26.2 Degenerate conics and the simultaneous theorem 91

27 Lower bounds for systems of polynomial equations 93
27.1 Integral-valued functions on simple varieties 93
27.2 Two quadrics in any dimension 95
27.3 Restrictions on the number of coordinates 96

28 What generic projection can and cannot compress 97

29 Geometric limitations of compressing the existing witnesses 99
29.1 Finite algebra requires geometric finiteness 99
29.2 The rank-one point group is dense in a surface 100

30 Limits of finite tests on denominator powers 101
30.1 Multiplying local obstructions loses information 101
30.2 Fixed rational pullbacks cannot characterize integrality 102

31 What rational pairing and existential rank do, and do not, supply 104
31.1 Integer pairing is not rational pairing 104
31.2 A recent explicit result has a restricted domain 105
31.3 Existential rank gives the appropriate distinction 105
31.4 What remains necessary for a count of at most ten 105

32 Quantitative consequences for rational Diophantine problems 106

33 Number fields: what transfers, and what does not 107
33.1 An exact counterexample to extending the released polynomial 108
33.2 A general obstruction to uniform scalar extension 108
33.3 Combining equations over an arbitrary number field 109
33.4 Precise transfer principles 110
33.5 The remaining arithmetic question 111

34 What the newer constructions do over number fields 112
34.1 The four literal quartics accept every input over $\mathbb{Q}(i)$ 112
34.2 An exact integral input rejected by the unshifted sources 113
34.3 A totally real quartic obstruction to the finite elliptic anchor 114
34.4 A domain assertion that does extend to fields with a real place 115

34.5 No one fixed existential formula works over every number field 116
34.6 What remains open after these checks 116

35 Combining equations over number fields by division norms 117
35.1 A rational norm form adapted to a specified number field 117
35.2 The degree-36 form and the untransformed direct systems 119
35.3 The preceding degree-37 form and its comparison bounds 120
35.4 The finite-algebra compiler over every number field 121

36 Field behavior of the final quartic transformations 121
36.1 The affine translation works over every field of characteristic zero 122
36.2 Eliminating a private squaring coordinate preserves the system 122
36.3 Inlining three fixed constants 123
36.4 The preferred degree-72 norm specifications 123

37 The remaining optimization problem 124
37.1 What a reduction to ten coordinates would have to accomplish 124
37.2 Degree and coefficients at a fixed coordinate count 125
37.3 The arithmetic hypotheses still requiring independent review 125
37.4 Number fields and the limits of a uniform formula 125

38 A conditional geometric obstruction 126

A Permissible precision reductions 126

B Audit of the arithmetic source and the remaining dependencies 128
B.1 Scope and conclusion 128
B.2 The local obstruction and norm argument 128
B.3 The reciprocal logarithm, common point, and final estimates 129
B.4 The two substantial arithmetic dependencies 130
B.5 Consequences for the optimization claims 131

C Reproducibility and the scope of the computations 131
C.1 Exact objects and independent checks 132
C.2 Identifying the principal polynomial files 132

D Reproducing the constructions with Lucas proxies 133

E Reproducing the direct constructions and the field counterexamples 134
E.1 Indexed formulas and quartic tables 134
E.2 Independent checks for the field assertions 135
E.3 Rebuild and integrity records 136

F Verification of the new source specification 136

1 Definitions, objectives, and the scope of the results

Definition 1.1. Let k be a field. A set $S \subseteq k$ is *Diophantine over* k if there are $r \geq 0$ and $P \in k[T, Y_1, \ldots, Y_r]$ such that

$$t \in S \quad \Longleftrightarrow \quad \exists (y_1, \ldots, y_r) \in k^r : \ P(t, y_1, \ldots, y_r) = 0.$$

The variable T is the input. We call r the number of witnesses and $r + 1$ the total number of variables. Degrees in this article are total degrees, including the input. For a nonzero polynomial over $\mathbb{Q}$, clear numerical denominators and divide the coefficients by their greatest common divisor. Its *coefficient height* is the maximum absolute value of the resulting primitive integer coefficients.

We compare the constructions lexicographically: first the total number of variables, then degree, then coefficient height. Under this order, a smaller-variable polynomial of very high degree is preferable to a quartic with more variables. A useful practical comparison should nevertheless retain the alternatives with moderate coefficients and complete coefficient tables. A circuit node used to *write* a polynomial is not a witness variable. Conversely, a newly quantified coordinate constrained to equal a circuit value does count as a witness.

We use the upstream source and a fork at the pinned revision

`28e87b1b62af0effd954605af7f70b2699dd5672.`

The released package is `quartics/quartic-12504`; its manuscript is `quartic_12504.tex`. The repository's own documentation distinguishes finite algebraic verification from verification of the arithmetic theorem; its Lean material is incomplete [1]. We retain that distinction throughout.

Write $\mathcal{S}$ for the rational existential system specified by the manuscript's reciprocal comparison source, with precisions 64 and 16, and its scalar order 48. Write $S_{\mathcal{S}}$ for its projected rational solution set in the ordinary input X.

Assumption 1.2 (The source arithmetic assertion)**.** *The set $S_{\mathcal{S}}$ is the ordinary subset $\mathbb{Z} \subset \mathbb{Q}$.*

This is the assertion of the source manuscript's theorem on the arithmetic interpretation of the revised source. It is not an additional conjecture introduced by the optimizations. The source-preserving transformations establish equality with $S_{\mathcal{S}}$; calling their outputs definitions of $\mathbb{Z}$ requires Assumption 1.2. Replaying a certificate for polynomial substitution does not prove this assumption. In particular, the present article does not claim a new independent proof of the full arithmetic manuscript.

The new shared-point construction uses more information than the bare equality in Assumption 1.2: it uses the manuscript's stated cyclic-group, pole, formal-logarithm, height, and Lucas results, together with Sun's simultaneous prime-pattern theorem [8, Theorem 4.1]. Its soundness and completeness are proved from these explicitly named inputs. This additional theorem permits comparison points to be shared across different selected primes. We distinguish this arithmetic modification from the unconditional algebraic transformations of the released source. The lower-bound theorems require neither arithmetic assumption.

The source release has 12,504 total variables, 11,939 quadratic residuals, 135,721 monomials in their sum of squares, and maximum coefficient

$$685783949153203534233601 < 2^{80}.$$

These are exact algebraic statistics, independently replayed here. The constructions differ in their form of explicitness and coefficient representation.

Total variables	Degree	Coefficients	Output supplied
41	$\le 2^{9566673}$	$\le 2^{2^{86098382}}$	Indexed polynomial with fixed elliptic exclusion product
43	$\le 2^{236}$	$\le 2^{2^{240}}$	Indexed semilocal alternative with coefficient DAG
558	$\le 2^{3122}$	$\le 2^{2^{3127}}$	Implemented source-preserving indexed polynomial
1315	4	$64\kappa < 2^{429496736}$	Complete table over $\mathbb{Z}[s]$, with $s^2 = \kappa$
1335	4	$< 2^{1020}$	Complete integer coefficient table
12,504	4	$< 2^{80}$	Unchanged released package

All counts include the input. The first two and the two smaller quartic rows use the shared-point arithmetic argument and the direct Lucas test. The 558-variable row is a source-preserving algebraic construction. Large displayed degree and height inequalities are proved upper bounds, not exact expansion statistics. Here $\kappa = 3^{2^{28}}$ and $s = 3^{2^{27}}$ are fixed integers; neither counts as a coordinate. The indexed polynomials specify every matrix and finite product but do not expand their enormous tables. Both 41- and 43-variable specifications also have separately checked versions with $\beta = X^2 + 2402$, with the same displayed bounds.

For comparison, earlier 43- and 45-variable indexed formulas, the untransformed 1315-variable table, the 1338-, 1339-, 1367- and 1391-variable quartics, source-preserving 853-variable circuit, and 12,449–12,493-variable quartics are retained as reviewable comparison constructions. Their proofs supply several ingredients used below. None of these counts is claimed to be globally minimal. Under the stated arithmetic inputs, the unrestricted gap is between three and 41 total variables. For a sum of squares of quadratic residuals, the proved gap is between five and 1315. Degree four is the least possible degree for one defining equation if the conditional construction is valid. A definition using at most ten total variables remains unproved.

Allowing several equations helps with degree and presentation but does not change the minimum number of variables when degree is unrestricted. Over $\mathbb{Q}$, their sum of squares combines any finite system without a new variable. Over any specified number field, the division-norm construction gives the same conclusion with a controlled degree. The general and field-specific assertions are proved separately in this article.

2 From integer equations to rational equations

Suppose first that an integer-coefficient polynomial $P(T, \mathbf{Y})$ of degree d defines $\mathbb{Z}$ in $\mathbb{Q}$, using r witnesses. For $f \in \mathbb{Z}[X_1, \dots, X_m]$, form

$$R_f(\mathbf{X}, \mathbf{Y}_1, \dots, \mathbf{Y}_m) = f(\mathbf{X})^2 + \sum_{i=1}^{m} P(X_i, \mathbf{Y}_i)^2. \tag{1}$$

Proposition 2.1. *The equation $f = 0$ has a solution in $\mathbb{Z}^m$ if and only if $R_f = 0$ has a solution over $\mathbb{Q}$. The construction is effective, uses $m(r+1)$ rational variables, and has degree at most $2\max\{\deg f, d\}$.*

Proof. Given an integer zero of f, the defining property of P supplies a witness tuple for each coordinate, so every summand in (1) vanishes. Conversely, a rational sum of squares vanishes only when every summand vanishes. Thus any rational zero of R_f satisfies $f = 0$ and $P(X_i, \mathbf{Y}_i) = 0$ for every i. Each X_i is therefore an integer. The construction consists only of copying, squaring, and adding finite polynomials; the bounds follow by inspection. □

For example, an equation $f(a, b, c) = 0$ is changed into a rational equation with three separate integrality tests, one for each of a, b, c. Its witnesses are allowed to have denominators. The tests force the three original coordinates to be integers; they need not force the added witnesses to be integers.

There is a better degree bound for the quartics actually supplied here. They have the form $P = \sum_j q_j^2$, with $\deg q_j \leq 2$. In that case P is nonnegative on rational inputs, so

$$f(\mathbf{X})^2 + \sum_{i=1}^{m} P(X_i, \mathbf{Y}_i) = 0 \tag{2}$$

already gives the desired reduction. Its degree is at most $\max\{2 \deg f, 4\}$, with the same variable count. Squaring each copy of P a second time is unnecessary.

Moreover every polynomial f can be evaluated by a finite sequence of additions and multiplications. Introduce one rational coordinate for each intermediate value, and impose equations such as $z - u - v = 0$ or $z - uv = 0$, together with the equation that the final value is zero. These equations have degree at most two, and for fixed $\mathbf{X}$ their intermediate coordinates are uniquely determined. Combining their squares with the integrality quartics produces a quartic rational-solvability problem for *every* f. If g intermediate coordinates are used, the variable count is $m(r + 1) + g$. These intermediate coordinates need no separate integrality predicates: they already equal the prescribed polynomial values. This trade preserves degree at the cost of variables, so it is secondary under this ordering.

2.1 The precise contribution of MRDP

The Matiyasevich–Robinson–Davis–Putnam theorem identifies the computably enumerable subsets of $\mathbb{Z}^n$ with the integer Diophantine sets, with effective and uniform polynomial representations [2, 4]. A set is computably enumerable if an algorithm can list its members, without necessarily deciding nonmembership. The theorem over nonnegative integers gives the version over all integers by the usual signed encodings and four-square representations of nonnegative variables.

Corollary 2.2. *If $\mathbb{Z}$ is Diophantine in $\mathbb{Q}$, then the computably enumerable subsets of $\mathbb{Q}^n$ are exactly its Diophantine subsets. In particular, rational polynomial solvability is computably enumerable complete under many-one reductions, and hence undecidable.*

Proof. Use the unique coprime numerator–positive denominator encoding of each rational coordinate. If $S \subseteq \mathbb{Q}^n$ is computably enumerable, its set of such integer encodings is computably enumerable. MRDP supplies an integer polynomial for that set. Replace each integer quantifier by a rational quantifier and an integrality test $P = 0$; add the equations $b_i x_i - a_i = 0$. The conjunction is one polynomial by a sum of squares, and has exactly the desired rational projection. Conversely, enumerate all rational witness tuples and evaluate a given polynomial exactly; this enumerates every Diophantine set. Uniformity gives many-one hardness from any computably enumerable set, and direct enumeration gives membership in that class. □

Applying this argument to a universal computably enumerable relation also gives a universal rational Diophantine polynomial, with its index as an input parameter. This does not make the small integer universal polynomials into definitions over rational witnesses without modification. Nor is an abstract Diophantine model of arithmetic automatically a definition of the ordinary embedded subset $\mathbb{Z} \subset \mathbb{Q}$. A model, an undecidability reduction, and a definition of that particular subset are different assertions. All the conclusions in this section become conditional applications to the source by Assumption 1.2.

3 Several equations versus one equation

Multiple equations can make the degree smaller and the construction easier to inspect. They do not lower the minimum possible number of variables when degree is unrestricted: over the rationals, and over every number field, a finite conjunction can be combined into one equation without adding a variable. The degree and field of the combining polynomial must nevertheless be recorded.

Proposition 3.1. *Let K be a number field with a real embedding, and let $F_1, \dots, F_s \in K[T, Y_1, \dots, Y_r]$ have degree at most D. The system $F_1 = \cdots = F_s = 0$ has exactly the same K-solutions as the single equation*

$$P = \sum_{i=1}^{s} F_i^2 = 0.$$

No new variable is introduced, and $\deg P \leq 2D$. For a single residual one may simply take $P = F_1$. If $K = \mathbb{Q}$, $P \neq 0$, all residuals have integer coefficients of absolute value at most H, and there are $n = r + 1$ total coordinates, then

$$H(P) \leq s\binom{n+D}{D} H^2.$$

Dividing the coefficients by their common integer divisor can only improve this bound.

Proof. Apply a real embedding of K to a zero of the displayed sum. Every summand is a nonnegative real number, so all of them vanish; injectivity of the embedding gives $F_i = 0$ in K. The converse and the degree bound are immediate. There are at most $M = \binom{n+D}{D}$ monomials in each residual. For a fixed monomial in F_i^2, choosing one exponent determines the other, so its coefficient is a sum of at most M products, each of absolute value at most H^2. Summing over i proves the bound. □

The real-embedding condition is weaker than total reality. It is essential to this particular combining formula: over $\mathbb{Q}(i)$, the two nonzero values $1, i$ have squares summing to zero. A replacement over other number fields is supplied by division algebras.

Corollary 3.2. *For any number field K, a system of $s \geq 1$ polynomial equations of degrees at most D can be combined into one equation in the same coordinates of degree at most*

$$D\lceil\sqrt{s}\rceil.$$

For a specified K, a combining polynomial with rational coefficients suffices. This is a statement about equivalence of solution sets over that field; it does not identify a changed field's solution set with its ring of integers.

Proof. Put $d = \lceil \sqrt{s} \rceil$. The division-algebra construction of Theorem 35.1 supplies a homogeneous form N of degree d in d^2 variables such that $N(\mathbf{z}) = 0$ for $\mathbf{z} \in K^{d^2}$ exactly when $\mathbf{z} = 0$. Substitute

$$(F_1, \dots, F_s, 0, \dots, 0)$$

into this form. Its zero condition is precisely the original conjunction, and its degree is at most dD. □

For example, the preferred quartic presentations in 1315 and 1335 total coordinates come from 1272 and 1292 quadratic residuals, respectively. In both cases $\lceil \sqrt{s} \rceil = 36$. The number-field compiler therefore combines each residual system into an equation of degree at most 72 without changing its coordinate count. The supplied determinant specifications have degree exactly 72: the input occurs only as $+X_0^2$ in the first residual, which multiplies the identity matrix, so the coefficient of X_0^{72} is one. Over $\mathbb{Q}$ the sum-of-squares quartic is better in degree. Over a specified nonreal number field a field-adapted reduced-norm construction preserves the conjunction, whereas the sum of squares need not do so. Neither claim says that these base-changed residuals define the new ring of integers.

Corollary 3.3. *For a fixed subset S of a number field K, the least total number of coordinates in a definition of S by one existential polynomial equation equals the least total number in a definition by a finite system of existential polynomial equations. The values are understood to be infinite if there is no such definition.*

Proof. A single equation is a finite system, giving one inequality. The preceding combination preserves every coordinate, giving the reverse inequality. □

Thus a system defining $\mathbb{Z}$ in at most ten total variables would immediately give a single polynomial in at most ten total variables. If its equations are quadratic over $\mathbb{Q}$, the combined polynomial would be quartic. The system may be much easier to discover or display, but it does not bypass a lower bound on the variable count.

3.1 Finite alternatives and nonzero conditions

Suppose $P_j(T, \mathbf{Y}_j) = 0$ are finitely many alternative presentations, with r_j witness coordinates. Pad the lists to $r = \max_j r_j$ using unused coordinates, and set

$$P = \prod_j P_j.$$

Over a field its existential projection is precisely the union of the original projections. Its total degree is at most $\sum_j \deg P_j$, and no selector coordinate is needed. This also combines a finite union of systems after combining each conjunction. The assertion is about finite products with fixed index sets, not a quantified integer choice.

Nonzero conditions require separate attention. The conjunction $G_1 \neq 0, \dots, G_t \neq 0$ is equivalent over a field to

$$u \prod_{j=1}^{t} G_j - 1 = 0$$

with one additional witness u. The new residual has degree at most $1 + \sum_j \deg G_j$. This generic method therefore costs one coordinate, even though conjunction and finite disjunction of existing

equations do not. The cost cannot simply be ignored: the one-coordinate set $K \setminus \{0\}$ is defined by the nonzero condition $T \neq 0$, but it is not the common zero set of any finite family of one-variable polynomials. Such a zero set is finite unless every polynomial is zero, in which case it is all of K. Absorbing a guard into an existing witness requires a separate equivalence, such as the reciprocal constructions proved below.

3.2 The degree of a universal combining polynomial

Here a universal combining polynomial in s inputs means a polynomial $A \in K[Z_1, \dots, Z_s]$ with

$$A(\mathbf{z}) = 0 \quad \Longleftrightarrow \quad \mathbf{z} = 0 \qquad (\mathbf{z} \in K^s).$$

Its defining property concerns all residual-value tuples. It is stronger than the property needed for one particular list of residuals, whose value tuples may satisfy additional relations.

Proposition 3.4. *For $s \geq 2$, a universal combining polynomial has degree at least two. If its degree is at most two, it is a homogeneous anisotropic quadratic form. Consequently:*

1. *if K has a real embedding, the minimum degree is two for every $s \geq 2$;*
2. *if K is totally imaginary, the minimum degree is two for $2 \leq s \leq 4$, and three for $5 \leq s \leq 9$.*

For totally imaginary K and $s \geq 10$, the present argument only gives

$$3 \leq \textit{minimum universal-combiner degree} \leq \lceil \sqrt{s} \rceil.$$

No optimality of the last upper bound is asserted.

Proof. The constant term is zero. A nonzero linear form in at least two variables has a nonzero vector in its kernel, so degree one cannot work. Write a polynomial of degree at most two as $A = q + \ell$, with q quadratic homogeneous and ℓ linear. If both parts are nonzero, choose $v \in K^s$ with $q(v)\ell(v) \neq 0$. The field is infinite, so two proper hypersurfaces cannot cover its affine space. With $t = -\ell(v)/q(v) \neq 0$ one has

$$A(tv) = t^2 q(v) + t\ell(v) = 0,$$

contradicting universality. If $q = 0$, the linear argument already applies. Therefore $\ell = 0$, and q must be anisotropic.

A real embedding permits the anisotropic form $Z_1^2 + \cdots + Z_s^2$ in every dimension. For a totally imaginary number field, the u-invariant is four: every quadratic form in at least five variables is isotropic, while a quaternion division algebra supplies an anisotropic norm form in four variables. The upper restriction follows from Hasse–Minkowski and the local u-invariant four; quaternion existence follows by ramifying at two finite places. See [14, Theorems 14.6.1 and 14.6.9] and the explicit summary in [29, Section 5.1]. Restricting an anisotropic four-variable form to a coordinate subspace treats $s = 2, 3, 4$. For $5 \leq s \leq 9$, a degree-three division norm supplies the upper bound, and the quadratic obstruction makes it sharp. The general upper bound is Corollary 3.2. □

For $s = 1$ the minimum is one. These combining-polynomial restrictions do not impose the same degree bound on a particular definition: dependencies among its residuals can allow cheaper combinations. Conversely, the general degree lower bound for a system defining $\mathcal{O}_K$ is only two, since linear systems are affine spaces and have empty, constant, or surjective input projection. The much stronger restrictions for quadratic systems are proved in Section 27.

4 The arithmetic structure retained from the source

The polynomial is assembled from a considerably smaller list of arithmetic requirements. It uses

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \qquad E_D : y^2 = x^3 - Dx,$$

and the point

$$P_* = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2} \right).$$

The source's generator assertion identifies $2E_D(K)$ with the infinite cyclic group generated by P_*. Its points can therefore encode integer indices. This assertion is a substantive arithmetic input; the elementary chart substitutions proved below do not require us to infer it from a computer search.

The input is converted to $\beta = X^2 + 2$. Four points encode indices n, s, t, z, and three rational proxy coordinates satisfy a Lucas-type equation. Formal-group comparisons at selected primes make the proxies congruent to the indices. A common denominator and forty-one affine contact expressions convert these congruences into height bounds. A separate scalar batch bounds the height of β in terms of $|s|$. The anchor forces $|s|$ beyond the fixed cutoff. Finally Lucas growth contradicts the height bounds unless the Lucas residual is zero and $\beta = n \in \mathbb{Z}$. A rational number whose square is an integer is an integer, so $X \in \mathbb{Z}$ follows.

In the other direction, the manuscript chooses sufficiently divisible elliptic indices, uses a simultaneous ordinary-prime pattern, constructs private comparison points, and obtains pole witnesses. The difficult inputs include its quantitative five-contact estimate, its restricted converse and pole-completeness argument, the prime-pattern theorem, and the cyclic elliptic-group assertion. We preserve these inputs and their precise witness scopes. Our changes operate on the resulting existential presentation.

Two forms of verification should be separated. The released certificate is a finite algebraic object: it checks lowering, elimination, invertible row operations, variable maps, and expansion into a sum of squares. The arithmetic manuscript must additionally prove both soundness and completeness of its source. The former can be checked in seconds by exact integer arithmetic. It cannot establish the latter by itself. The proofs and evidence supplied below explicitly identify which task each calculation performs.

5 An economical semilocal condition

Throughout this section a witness variable ranges over $\mathbb{Q}$ unless another field is specified. An equivalence of polynomial presentations means an equivalence of their projected rational solution sets in every displayed input. Such an equivalence is independent of whether the arithmetic theorem that identifies the original projected set with $\mathbb{Z}$ has been verified. The next section matches these statements to the actual residual system.

5.1 An explicit three-variable semilocal definition

Write

$$R = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)} = \{t \in \mathbb{Q} : v_2(t) \geq 0,\ v_3(t) \geq 0\},$$

with $v_p(0) = +\infty$. The following specialization of classical quaternion methods is particularly economical. Daans proves three-quantifier existential definability for every finite intersection of

valuation rings in a global field; he traces the corresponding rational cases back to Robinson [5, Proposition 4.2 and Remark 4.3]. Here all constants and all exceptional cases are made explicit.

Proposition 5.1. *For every $t \in \mathbb{Q}$,*

$$t \in R \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \quad u^2 + 6v^2 - 2w^2 - 3t^2 + 2 = 0.$$

Proof. Consider the quaternion algebra $B = (2, 3)_{\mathbb{Q}}$ with generators $i^2 = 2$, $j^2 = 3$, $ij = -ji$. Its ramification set consists exactly of 2 and 3. Indeed its Hilbert symbol at 3 equals the Legendre symbol $(2/3) = -1$, and at 2 the formula $(2, u)_2 = (-1)^{(u^2-1)/8}$ for odd u gives $(2, 3)_2 = -1$. It splits at every other finite place because both parameters are units at those odd primes, and it splits at infinity because both parameters are positive. In particular B is a division algebra over $\mathbb{Q}$.

We use the standard quadratic embedding criterion for quaternion algebras: a quadratic field $L/\mathbb{Q}$ embeds in B if and only if every ramified place of B remains a field in L. This is the quadratic case of the Albert–Brauer–Hasse–Noether theorem and the central simple algebra embedding criterion; equivalently, extending scalars to L kills the local invariant $1/2$ precisely when the local extension has degree two. See [14, Proposition 14.6.7].

Suppose first that $t \neq 0$, and put

$$\delta = 1 - \frac{2}{3t^2}.$$

This number is nonzero: $t^2 = 2/3$ has no rational solution. At 2, put $m = v_2(t)$. If $m > 0$, then $v_2(\delta) = 1 - 2m$ is odd, so δ is not a square in $\mathbb{Q}_2$. If $m = 0$, then $t^2 \equiv 1 \pmod 8$ and $\delta \equiv 3 \pmod 8$, again a nonsquare. If $m < 0$, then $\delta \in 1 + 8\mathbb{Z}_2$, hence it is a square in $\mathbb{Q}_2$. At 3, if $m = v_3(t) \geq 0$ then $v_3(\delta) = -1 - 2m$ is odd; if $m < 0$, then $\delta \in 1 + 3\mathbb{Z}_3$ and Hensel's lemma makes it a square. Consequently

$$\delta \notin \mathbb{Q}_2^{\times 2}, \quad \delta \notin \mathbb{Q}_3^{\times 2} \quad \Longleftrightarrow \quad t \in R.$$

The embedding criterion shows that this is equivalent to the existence of a pure quaternion $xi + yj + zij$ with square δ, that is,

$$2x^2 + 3y^2 - 6z^2 = \delta.$$

This equivalence also excludes the case in which δ is a rational square: a division quaternion algebra cannot contain a noncentral element whose square is a nonzero rational square, since $(\alpha - s)(\alpha + s) = 0$ would force $\alpha = \pm s$. A nonzero pure quaternion has reduced trace zero, so it cannot be such a nonzero rational scalar. Multiplying the displayed equation by $3t^2$ and setting $u = 3ty$, $v = tx$, $w = 3tz$ gives the equation in the statement. These substitutions are invertible for $t \neq 0$. Finally $t = 0$ belongs to R and is represented by $(u, v, w) = (0, 0, 1)$. □

Lemma 5.2. *Let $F(A, B) = A^2 + AB - B^2$. For every $a, b \in \mathbb{Q}$ and $p = 2, 3$,*

$$v_p(F(a, b)) = 2 \min\{v_p(a), v_p(b)\}.$$

In particular $(a, b) \in R^2$ if and only if $F(a, b) \in R$.

Proof. The assertion is immediate for $a = b = 0$. Otherwise scale a, b by a common power of p until both are integral and at least one is a unit. The reduction of $A^2 + AB - B^2$ has no nontrivial zero over either $\mathbb{F}_2$ or $\mathbb{F}_3$: over $\mathbb{F}_2$ one checks the three nonzero pairs, and over $\mathbb{F}_3$ the discriminant 5 is a nonsquare. Thus the scaled value is a unit, proving the equality. □

Corollary 5.3 (Five-variable replacement for the anchor)**.** *Fix rational inputs f_0, f_1, C with $C \neq 0$. The condition $f_0/C, f_1/C \in R$ is equivalent to the existence of five rational variables r, D, u, v, w satisfying the three quadratic equations*

$$\begin{aligned} r &= f_0^2 + f_0 f_1 - f_1^2, \\ D &= C^2, \\ u^2 + 6v^2 - 2w^2 - 3r^2 + 2D^2 &= 0. \end{aligned}$$

Proof. By Lemma 5.2, the desired condition is $r/C^2 \in R$, with $r = F(f_0, f_1)$. Apply Proposition 5.1 to $t = r/C^2$ and multiply its equation by C^4. Multiplication of each of its three witnesses by C^2 is a bijection of $\mathbb{Q}$, giving the stated equation. □

The released anchor uses two scaling variables and two copies of a seven-witness trace-sum condition, hence sixteen witnesses in this module. Corollary 5.3 replaces that module by five witnesses. It therefore saves eleven variables before any other change and retains quadratic residuals, so their sum of squares remains quartic. The exact global counts and independent incidence checks are given in the next section. The three displayed residuals have integer coefficients of absolute value at most six. This is an exact replacement, with no change to the arithmetic meaning or strength of the anchor.

The same method handles arbitrary finite tuples. For a finite nonempty set S of rational primes and a positive integer m, choose a monic polynomial $g \in \mathbb{Z}[T]$ of degree m whose reductions are irreducible modulo every $p \in S$; finite-field irreducible polynomials and the Chinese remainder theorem supply one. If α is a root, let

$$N_m(X_1, \ldots, X_m) = N_{\mathbb{Q}(\alpha)/\mathbb{Q}}(X_1 + X_2\alpha + \cdots + X_m\alpha^{m-1}).$$

For $p \in S$ the extension is unramified and inert at p, and its power basis is an integral basis locally. Therefore

$$v_p(N_m(\mathbf{x})) = m \min_i v_p(x_i).$$

This proves, with no new existential variables, that all coordinates of $\mathbf{x}$ belong to $\bigcap_{p\in S} \mathbb{Z}_{(p)}$ if and only if $N_m(\mathbf{x})$ does. It is the norm-form version of [5, Proposition 4.8]. In the present two-prime, two-input case the polynomial $T^2 - T - 1$ gives precisely the form F above.

6 Rational relation combining and finite algebraic choices

6.1 Combining square conditions without assuming integrality

Matiyasevich and Robinson's relation-combining theorem is an essential source of small integer Diophantine representations [3, 4]. Its integer-domain hypotheses must not be silently replaced by rational-domain hypotheses. The appropriate rational version is Zhang–Sun [7, Theorem 1.2]. Their separation parameter is $(k + \sum A_i^2)(1 + \sum A_i^{-2})$, and their theorem assumes that every A_i is nonzero. The following slightly simpler separation bound also works and has a direct proof.

Lemma 6.1. *Let $k \geq 1$. For nonzero $A_1, \ldots, A_k \in \mathbb{Q}$, put*

$$W = \sum_{i=1}^{k} (A_i^2 + A_i^{-2}), \qquad J(T) = \prod_{\boldsymbol{\epsilon} \in \{\pm 1\}^k} \left(T - \sum_{i=1}^{k} \epsilon_i W^{i-1} \sqrt{A_i} \right).$$

Then $J \in \mathbb{Q}[T]$, and J has a rational root if and only if all A_i are rational squares. Every complex root of J is nonzero.

Proof. Independently changing the formal sign of each square root fixes the product, so its coefficients are rational polynomials in the A_i and W. Let $m = \min_i \sqrt{|A_i|}$, $M = \max_i \sqrt{|A_i|}$, and $\rho = M/m \geq 1$. The two summands corresponding to maximal and minimal absolute value show that $W \geq 2\rho^2 \geq 1 + \rho$; the same inequality is immediate when both extrema occur at one index. Therefore for every j,

$$\sum_{i<j} MW^{i-1} < \frac{MW^{j-1}}{W-1} \leq mW^{j-1}.$$

No nonempty signed subsum of the radicals can vanish, since its last term has larger absolute value than the sum of the preceding terms. In particular every root is nonzero. If a displayed root is rational, every automorphism of the multiquadratic splitting field fixes it. Subtracting its conjugate would give a nonempty signed subsum if any radical changed sign. Thus every radical is fixed by the full Galois group and is rational. The converse is immediate. □

There is also an explicit one-variable polynomial that enforces the nonzero condition itself. Put

$$S = \prod_{i=1}^{k} A_i, \qquad D = S^2, \qquad N = D\sum_i A_i^2 + \sum_i \prod_{j\neq i} A_j^2$$

and define

$$K_k(\mathbf{A}, U) = \prod_{\boldsymbol{\epsilon}\in\{\pm 1\}^k} \left(1 - U\sum_{i=1}^{k} \epsilon_i S^{2k-2i+1} N^{i-1}\sqrt{A_i}\right).$$

Independent sign changes show that this is an integer-coefficient polynomial in $\mathbf{A}, U$. If $S = 0$ it is identically 1. If $S \neq 0$, the signed sum in each factor is $S^{2k-1}\sum_i \epsilon_i W^{i-1}\sqrt{A_i}$, so Lemma 6.1 proves

$$\exists U \in \mathbb{Q} : K_k(\mathbf{A}, U) = 0 \quad \Longleftrightarrow \quad A_i \in \mathbb{Q}^{\times 2} \text{ for all } i.$$

This polynomial has degree 2^k in U; total degree grows exponentially with k. To allow zero square inputs, take the finite product, over $I \subseteq \{1, \ldots, k\}$, of

$$\sum_{i\notin I} A_i^2 + K_{|I|}((A_i)_{i\in I}, U)^2,$$

using zero in place of K_0 for the empty subset. One shared rational U suffices for all branches. This treats zero inputs exactly, but its expanded degree is correspondingly large.

Private square roots can be replaced in this way only when no other condition needs their particular values or signs. If roots occur elsewhere, the complete finite algebraic system must be compressed, as below. For example, replacing $y^2 = A,\ y = 1$ by the bare condition that A is a square would be invalid.

6.2 One variable for finite choices and nonzero conditions

The useful general principle is that the number of finite algebraic choices need not equal the number of rational witnesses. Sun's recent preprint [8, Sections 10.3–10.6] uses this principle for a bounded-variable undecidability construction. The elementary algebra below is independently verifiable

and does not depend on that preprint's height or prime-pattern estimates. Its undecidability result is not itself a definition of $\mathbb{Z}$ inside $\mathbb{Q}$.

Lemma 6.2. *Let A be a reduced finite-dimensional $\mathbb{Q}$-algebra of dimension $d \geq 1$, generated by $N \geq 1$ elements $a_1, \ldots, a_N$. Some integer*

$$1 \leq b \leq 1 + (N-1)\binom{d}{2}$$

makes $z_b = a_1 + ba_2 + \cdots + b^{N-1}a_N$ have distinct values under all d embeddings $A \to \overline{\mathbb{Q}}$. For such a b, the characteristic polynomial of multiplication by z_b has a rational root if and only if A has a $\mathbb{Q}$-algebra homomorphism to $\mathbb{Q}$.

Proof. Two different embeddings differ on a generator, so equality of their values on z_b is a nonzero polynomial equation in b of degree at most $N-1$. At most $(N-1)\binom{d}{2}$ values are bad. For a good value the characteristic polynomial has precisely the distinct embedding values as its roots. If one value is rational, the absolute Galois group fixes that value and hence its unique corresponding embedding; the values of all generators are therefore rational. Conversely a rational homomorphism supplies a rational root. □

Lemma 6.3. *Suppose*

$$P_{\mathbf{x}}(T) = T^d + \sum_{j=1}^{d} \frac{N_j(\mathbf{x})}{B(\mathbf{x})} T^{d-j}$$

is monic on $B \neq 0$, and let $H(\mathbf{x})$ be the product of the numerators of all further required inequations. Then

$$\Psi(\mathbf{x}, U) = 1 + \sum_{j=1}^{d} N_j(\mathbf{x}) B(\mathbf{x})^{2j-1} H(\mathbf{x})^j U^j$$

has a rational root in U exactly when $BH \neq 0$ and $P_{\mathbf{x}}$ has a nonzero rational root.

Proof. If $BH = 0$, all nonconstant terms vanish and $\Psi = 1$. If $BH \neq 0$, set $L = B^2H$. For $U \neq 0$ the identity

$$\Psi(\mathbf{x}, U) = (LU)^d P_{\mathbf{x}}(1/(LU))$$

proves the assertion, and $\Psi(\mathbf{x}, 0) = 1$ excludes $U = 0$. □

Zero roots are covered by taking the two finite alternatives $P(T)$ and $P(T-1)$: a rational root of one of them is nonzero. Thus nonzero conditions need no additional inverse variable once one finite-choice variable is already present. Equations depending only on $\mathbf{x}$ can be imposed by adding their squares to Ψ^2, and finite alternatives can be combined by multiplying the resulting polynomials, using the same U in every factor.

For completeness, this also handles nonreduced finite algebras and additional equations involving their generators. Quotient by the ideal of those equations, and discard factors violating an inequation by localizing at its value; these are finite-dimensional algebra operations. Pass to the reduced quotient. In characteristic zero its nilradical is the kernel of the trace pairing $(a, b) \mapsto \mathrm{Tr}(ab)$: a product decomposition into Artinian local factors reduces the assertion to the nondegeneracy of the trace pairing of a separable field extension. Gaussian elimination then computes this quotient. When coefficients depend on free parameters, branch on the finitely many pivots being zero or nonzero and record all denominators. On every branch apply Lemma 6.2,

enforce the nonzero discriminant of the selected characteristic polynomial, and apply Lemma 6.3. This is an effective finite construction when the original algebra has a fixed finite basis, for example for a tower of monic equations. It preserves exceptional specializations.

Dropping the separation condition is unsound: $x^2 = y^2 = 2$ has no rational pair, but the characteristic polynomial for $x+y$ is $T^2(T^2-8)$ and has the rational root zero. Its repeated roots signal precisely the collision. Using $x+2y$ instead gives $T^4 - 20T^2 + 36$, with no rational root.

7 Three rational coordinates for an elliptic point over a quadratic field

The following elementary elimination reduces the ambient coordinate count of a point over a quadratic field. It trades degree for variables, which is appropriate when the number of variables is the first optimization criterion. The square-root expressions below are reconstructed rational functions, not new quantified variables.

Proposition 7.1 (Eliminating one coordinate of a quadratic square root)**.** *Let F be a field of characteristic different from two, let $\delta \in F^\times$ be a nonsquare, and write $s^2 = \delta$. Fix $j \in F$ and put $\lambda = j^2 - \delta \neq 0$. For $A, B, C \in F$ with $C \neq 0$, define*

$$U = C^2 - A - jB, \qquad V = \frac{U}{\lambda}, \qquad b_1 = \frac{B/2 + jV}{C}, \qquad b_0 = C - jb_1.$$

Then the following are equivalent:

1. $(b_0 + b_1 s)^2 = A + Bs$;
2. $C^2V - (B/2 + jV)^2 = 0$;
3. $\Gamma_j(A, B, C) = 0$, *where*

$$\Gamma_j(A,B,C) = 4\delta C^4 - 4\big((j^2+\delta)A + 2j\delta B\big)C^2 + \big(2jA + (j^2+\delta)B\big)^2.$$

The correspondence identifies all nonzero square roots $b = b_0 + b_1 s$ satisfying $b_0 + jb_1 \neq 0$ with the nonzero solutions C of the last equation, by $C = b_0 + jb_1$.

Proof. First suppose $b^2 = A + Bs$ and $C = b_0 + jb_1 \neq 0$. Comparing coefficients gives

$$A = b_0^2 + \delta b_1^2, \qquad B = 2b_0 b_1, \qquad C^2 = A + jB + \lambda b_1^2.$$

Hence $V = b_1^2$ and $Cb_1 = B/2 + jV$, proving the reconstruction formulas and the second assertion.

Conversely, suppose the second assertion holds and use the displayed reconstruction formulas. Because $C \neq 0$, it follows that $b_1^2 = V$. Consequently

$$b_0 b_1 = Cb_1 - jb_1^2 = B/2$$

and

$$\begin{aligned} b_0^2 + \delta b_1^2 &= C^2 - 2jCb_1 + (j^2+\delta)V \\ &= C^2 - jB - (j^2 - \delta)V = A. \end{aligned}$$

Thus $b^2 = A + Bs$, and $b \neq 0$ follows from $C = b_0 + jb_1 \neq 0$. Finally, multiplication of the second assertion by $-4\lambda^2$ gives exactly $\Gamma_j = 0$. This also proves that the two constructions are inverse to one another on the specified charts. □

For $F = \mathbb{Q}$ and $\delta = 2$, this becomes

$$\Gamma_j(A, B, C) = 8C^4 - 4\big((j^2+2)A + 4jB\big)C^2 + \big(2jA + (j^2+2)B\big)^2.$$

The root coordinates can equivalently be reconstructed as

$$b_0 = \frac{-4C^2 + 2j^2A + j(j^2+2)B}{2(j^2-2)C},$$
$$b_1 = \frac{2jC^2 - 2jA - (j^2+2)B}{2(j^2-2)C}.$$

Thus neither V, b_0, nor b_1 consumes an additional variable.

Lemma 7.2 (A common chart for finitely many points)**.** *Let $b^{(1)}, \dots, b^{(N)}$ be nonzero elements of $\mathbb{Q}(\sqrt{\delta})$, where δ is a rational nonsquare. There exists $j \in \{0, 1, \dots, N\}$ such that*

$$b_0^{(r)} + j b_1^{(r)} \neq 0 \qquad (1 \le r \le N).$$

Proof. For a fixed nonzero pair (b_0, b_1), the expression $b_0 + jb_1$ vanishes for at most one rational j. If $b_1 = 0$, it never vanishes. At most N values are therefore excluded, and the displayed set has $N + 1$ distinct elements. □

7.1 Application to elliptic doubling

Let $K = \mathbb{Q}(\sqrt{2})$ and let

$$E_D : \; b^2 = a^3 - Da, \qquad D \in K^\times.$$

The assumption $D \neq 0$ makes this Weierstrass cubic nonsingular. Write $a = a_0 + a_1\sqrt{2}$ and $a^3 - Da = A + B\sqrt{2}$ with $A, B \in \mathbb{Q}$. An affine point with nonzero ordinate is therefore represented, on the j-chart, by only three rational coordinates (a_0, a_1, C), subject to $C \neq 0$ and $\Gamma_j(A, B, C) = 0$. For fixed D, this polynomial has total degree at most six in these three coordinates: A, B have degree at most three and the leading terms come from a square of their linear combination.

Reconstruct b as above and put

$$h = \frac{3a^2 - D}{2b}, \qquad \xi = h^2 - 2a, \qquad \eta = h(a - \xi) - b.$$

These rational functions give the usual doubling formula $[2](a, b) = (\xi, \eta)$. Since $b \neq 0$, the doubled point is affine. Conversely, every nonidentity point of $2E_D(K)$ has an affine preimage with nonzero ordinate: a preimage equal to the identity or to a point with ordinate zero would double to the identity. For a list of N such doubled points, choose a halfpoint of each and then apply Lemma 7.2. Thus the union of the $N + 1$ common charts represents all such lists.

There is no change to the group-theoretic assertions required by the original construction. In particular, the identity point must be handled separately if a surrounding formula permits it. In the proposed application, the 211 point occurrences are constrained to be nonidentity by their original nonvanishing conditions; this condition is part of the application, not a conclusion of the elimination alone.

7.2 Guards and finite unions without selector variables

Every rational function used in a fixed chart can be written as a quotient of rational polynomials. Suppose the required denominator factors in that branch are $d_1, \dots, d_r$, and set $\Delta = \prod_{k=1}^{r} d_k$. The single equation

$$Z\Delta - 1 = 0$$

with one rational witness Z enforces all the nonvanishing conditions at once. After imposing this equation, clearing those denominators from the remaining branch equations preserves both directions of the existential assertion. In particular, the factors C_i must be guarded; merely imposing $\Gamma_j = 0$ at $C_i = 0$ would introduce points outside the displayed chart.

If a branch is given by rational polynomial residuals $f_{j,1}, \dots, f_{j,r_j}$, including its own equation $Z\Delta_j - 1 = 0$, set

$$H_j = \sum_{k=1}^{r_j} f_{j,k}^2.$$

Over $\mathbb{Q}$, the union of the branches is then represented by

$$\prod_{j=0}^{N} H_j = 0.$$

All branches may reuse the same list of witness-variable names: a zero of the product belongs to at least one factor, and that factor alone must satisfy its denominator guard. Requiring the guards of inactive branches would generally destroy completeness and is not part of this construction. Additional finite choices, such as the 256 fixed height-class choices, can be included in the same product over pairs of branch indices without selector variables.

8 One witness for a constructible finite-fiber projection

A constructible set is a finite union of locally closed algebraic sets. A projection has finite geometric fibers if every fiber over every point with coordinates in an algebraic closure is a finite set. Finite rational fibers alone are not the hypothesis used here.

Theorem 8.1. *Let $X \subseteq \mathbb{A}^{d+m}_{\mathbb{Q}}$ be constructible over $\mathbb{Q}$, and suppose the coordinate projection $\pi : X \to \mathbb{A}^{d}_{\mathbb{Q}}$ has finite geometric fibers. There is an effectively constructible polynomial $P \in \mathbb{Z}[\mathbf{x}, U]$ such that, for every $\mathbf{a} \in \mathbb{Q}^d$,*

$$\mathbf{a} \in \pi(X(\mathbb{Q})) \quad \Longleftrightarrow \quad \exists U \in \mathbb{Q} : \ P(\mathbf{a}, U) = 0.$$

Proof. We give the construction including its exceptional loci. Decompose X into finitely many locally closed affine pieces; represent an inequation $h \neq 0$ by the coordinate relation $hv = 1$ when forming a coordinate ring. The added coordinate is uniquely determined, so geometric fibers stay finite. Decompose the image closure into irreducible pieces and restrict accordingly. On one piece let R be the coordinate ring of the irreducible base closure B, let $L = \mathrm{Frac}(R)$, and let A be the reduced coordinate ring of the affine source. Its generic fiber algebra A_L is finite-dimensional, reduced, and separable over L; it is a finite product of finite separable field extensions.

The primitive element theorem for a finite separable algebra gives $A_L \simeq L[T]/(h)$ with h monic. Clearing the denominators of a primitive element, of expressions for the source generators in that primitive element, and of their finitely many relations gives a nonzero $g \in R$ for which

$$A_g \simeq R_g[T]/(h), \qquad h \in R_g[T] \text{ monic.}$$

For precision, the map is surjective after clearing the generator expressions; its kernel vanishes over L and is therefore zero because $R_g[T]/(h)$ is free over the domain R_g. Any intermediate relation which vanishes only over L is made zero by a further localization. Thus this is an actual isomorphism on an open base set, not just a bijection on generic points. Consequently a rational point of $B \cap D(g)$ has a rational lift exactly when the specialized monic h has a rational root. The residual part lies over a proper closed subset of B, so induction on the dimension of the image closure gives finitely many such charts. Each chart comes with polynomial equations $E_i(\mathbf{x}) = 0$ for its base closure and one or more polynomial inequations; multiply the latter together to obtain $G(\mathbf{x}) \neq 0$. Choose polynomial representatives in the ambient base space for all elements of R.

It remains to express each chart with one quantified variable. Write its monic equation, after choosing a common coefficient denominator, as

$$h(T) = T^n + \sum_{j=1}^{n} \frac{N_j(\mathbf{x})}{D(\mathbf{x})} T^{n-j}.$$

Add $D \neq 0$ to the recorded chart conditions. A rational root of h may be zero; to cover this exactly, use the two fixed alternatives $h(T)$ and $h(T-1)$. If r is a rational root of h, then the two polynomials have roots r and $r+1$, at least one of which is nonzero. Conversely a rational root of either polynomial gives a rational root of h. The shifted polynomial is still monic and has the same permissible common denominator D.

For either alternative, with its coefficient numerators again denoted N_j, set

$$\Psi(\mathbf{x}, U) = 1 + \sum_{j=1}^{n} N_j(\mathbf{x}) D(\mathbf{x})^{2j-1} G(\mathbf{x})^j U^j.$$

At $DG = 0$ this polynomial is identically 1. At $DG \neq 0$, put $L_0 = D^2 G$. Then, for $U \neq 0$,

$$\Psi(\mathbf{x}, U) = (L_0 U)^n h(1/(L_0 U)),$$

where h now denotes the selected alternative. Also $\Psi(\mathbf{x}, 0) = 1$. Hence a rational zero of Ψ exists exactly when all inequations hold and that alternative has a nonzero rational root. This construction cannot acquire spurious solutions at a vanished denominator, because its value there is exactly one.

The polynomial

$$Q(\mathbf{x}, U) = \Psi(\mathbf{x}, U)^2 + \sum_i E_i(\mathbf{x})^2$$

therefore describes that chart and alternative. Multiply Q over the finite list of charts and the two alternatives, always reusing the same variable U, and clear numerical coefficient denominators. This gives P. A product is zero if and only if some factor is zero, and over $\mathbb{Q}$ each displayed sum of squares is zero if and only if all of its summands vanish. The empty image is represented by $P = 1$. All the steps are effective operations in polynomial rings over $\mathbb{Q}$, including ideal decomposition

and localization, followed by finite-dimensional algebra over the resulting function fields. The induction terminates because the dimension of every exceptional base strictly decreases. This effectivity assertion does not mean that these computations have been executed for the source system. □

Corollary 8.2. *A finite-fiber presentation retaining d rational coordinates, one of which is the designated input, yields a polynomial in at most $d+1$ total variables, with at most d existential witnesses and the same rational input projection. In particular, the presentation with $d = 557$ proved below gives at most 558 total variables. Its identification with $\mathbb{Z}$ retains the arithmetic hypothesis of the source.*

The elementary reciprocal transformation in this proof agrees with [8, Section 10.6, equation (10.8)]; the chart argument above extends it from the finite algebras explicitly treated there to all constructible sets with finite geometric fibers. This citation supports the algebraic device only. The bounded-variable undecidability theorem in that recent preprint and its analytic arithmetic inputs are not used. Its convention is that a monic degree-n polynomial is written $T^n + \sum_{j=1}^{n}(N_j/D)T^{n-j}$: the reciprocal polynomial has exponent U^j and denominator-clearing factor D^{2j-1}, not D^{j-1}. The two shifts are essential if zero could be the only rational root. All mathematical claims used here are proved above; no result about the arithmetic of that preprint is needed.

8.1 Conjunctions and the literature on existential rank

The general relation between witness number and finite fibers is part of the theory of existential rank and essential fiber dimension developed by Daans, Dittmann, and Fehm [6, Corollary 4.21]. Here *existential rank* means the minimum number of existential quantifiers in a definition of a specified set. Our finite-fiber proof is a direct constructive treatment of the special case needed here, including exceptional strata and nonzero conditions. We make no claim that the general quantifier-saving phenomenon is new.

For clarity, it gives the following useful known intersection bound [6, Theorem 1.4], also stated in [5, Theorem 2.5], with an explicit constructive proof over the rationals.

Proposition 8.3. *Let $D_i \subseteq \mathbb{Q}^n$, for $1 \le i \le r$ and $r \ge 1$, be defined by one polynomial equation with $m_i \ge 1$ existential witnesses. Their intersection has a one-polynomial definition with at most*

$$1 + \sum_{i=1}^{r}(m_i - 1)$$

witnesses. Such a definition can be constructed effectively from the given polynomials.

Proof. Write the ith polynomial as

$$F_i(\mathbf{x}, \mathbf{u}_i, Z_i) = \sum_{j=0}^{d_i} a_{ij}(\mathbf{x}, \mathbf{u}_i) Z_i^j,$$

where $\mathbf{u}_i$ has $m_i - 1$ coordinates and the witness lists are private to their respective predicates. Let C_i be the closed condition that all coefficients a_{ij} vanish. Replace the zero set of F_i by the constructible set

$$(\{F_i = 0\} \setminus C_i) \ \cup \ (C_i \cap \{Z_i = 0\}).$$

This replacement preserves the projection to $(\mathbf{x}, \mathbf{u}_i)$ over any field: a nonzero polynomial has its usual roots, and an identically zero polynomial permits the choice $Z_i = 0$. Each geometric fiber in Z_i is now finite. Nonzero constant specializations give empty fibers and cause no exception.

Take the fiber product of these presentations over $\mathbf{x}$. It has finite geometric fibers over $(\mathbf{x}, \mathbf{u}_1, \ldots, \mathbf{u}_r)$. Theorem 8.1 replaces all the Z_i by one additional variable. Quantifying the remaining $\sum_i (m_i - 1)$ private coordinates proves the formula and its effectiveness. Genuinely shared coordinates remain in $\mathbf{x}$; they are never silently duplicated or identified. □

For example, each of the forty-two pole predicates has a four-witness presentation after retaining its three free pole parameters and encoding its remaining finite choices by one variable. The proposition combines them with $42 \cdot 4 - 41 = 127$ witnesses. This is exactly the 126 retained pole coordinates and one common finite-choice variable already used in the global finite-fiber construction. It does not provide another forty-one-variable saving on top of that construction.

8.2 Finite-fiber compression over other fields

The one-variable compression theorem does not require an ordering. The place where rational squares were used was the conjunction of finitely many equations. Over any field that is not algebraically closed, a homogenized irreducible polynomial supplies the needed replacement.

Lemma 8.4. *Let K be a field that is not algebraically closed. There is a homogeneous polynomial $H(A, B) \in K[A, B]$ of degree $e \geq 2$ such that*

$$H(a, b) = 0 \quad \Longleftrightarrow \quad a = b = 0 \qquad (a, b \in K).$$

Consequently any finite system of polynomial equations over K can be combined into one polynomial equation without introducing variables. For r equations of degrees at most D, a balanced combination has degree at most $e^{\lceil \log_2 r \rceil} D$ when $r \geq 1$.

Proof. Choose a monic irreducible polynomial $h(T) \in K[T]$ of degree $e \geq 2$, and homogenize it:

$$H(A, B) = B^e h(A/B).$$

This notation denotes the polynomial obtained by homogenization, also at $B = 0$. If $b \neq 0$, then $H(a, b) = 0$ would give a root a/b of h in K, contradicting irreducibility. If $b = 0$, monicity gives $H(a, 0) = a^e$, so $a = 0$. Conversely $H(0, 0) = 0$.

To combine residuals $f_1, \ldots, f_r$, pad the list with zero polynomials to a power of two, and repeatedly apply H to pairs. At each stage a combined residual vanishes exactly when both its inputs vanish. After $\lceil \log_2 r \rceil$ stages, the one remaining equation is equivalent to the whole system. Each stage multiplies the degree bound by at most e. □

Theorem 8.5. *Let K be a field of characteristic zero that is not algebraically closed. Let $X \subseteq \mathbb{A}_K^{d+m}$ be constructible over K, and suppose every geometric fiber of the coordinate projection $\pi : X \to \mathbb{A}_K^d$ is finite. There is a polynomial $P(\mathbf{x}, U) \in K[\mathbf{x}, U]$ such that*

$$\mathbf{a} \in \pi(X(K)) \quad \Longleftrightarrow \quad \exists U \in K : \ P(\mathbf{a}, U) = 0 \qquad (\mathbf{a} \in K^d).$$

For a number field K, one may take $P \in \mathcal{O}_K[\mathbf{x}, U]$, and the construction is effective from equations and inequations for X. More generally it is effective whenever the coefficient field is presented with the algebraic operations needed for factorization, elimination, and primitive-element computation, and an irreducible polynomial of degree at least two is available effectively.

Proof. The algebraic part of Theorem 8.1 works over K. Indeed, K is infinite and perfect. After passing to reduced affine pieces and localizing on each irreducible image closure, the generic finite algebra is a product of finite separable field extensions of the image's function field. The primitive element theorem gives a monic polynomial presentation. Clearing the finitely many denominators extends the presentation over a nonempty open subset. The complement has smaller image dimension, so induction produces finitely many charts, including every exceptional finite fiber.

Thus, on each chart, a point lies in the image precisely when

$$f_1(\mathbf{x}) = \cdots = f_r(\mathbf{x}) = 0, \qquad g(\mathbf{x}) \neq 0,$$

and a certain monic polynomial

$$h(T) = T^n + \sum_{j=1}^{n} \frac{N_j(\mathbf{x})}{B(\mathbf{x})} T^{n-j}$$

has a root in K. All coefficient denominators are nonzero on the chart; enlarge its guard to include them. For $\epsilon \in \{0, 1\}$, let $h_\epsilon(T) = h(T - \epsilon)$ and write its coefficients with the same common denominator B. Define

$$\Psi_\epsilon(\mathbf{x}, U) = 1 + \sum_{j=1}^{n} N_{\epsilon,j}(\mathbf{x}) B(\mathbf{x})^{2j-1} g(\mathbf{x})^j U^j.$$

When $Bg = 0$ or $U = 0$, this expression is 1. Otherwise, with $L = B^2 g$, the exact identity

$$\Psi_\epsilon(\mathbf{x}, U) = (LU)^n h_\epsilon(1/(LU))$$

shows that it has a zero precisely when h_ϵ has a nonzero root. A root of h gives a nonzero root of at least one of $h(T)$ and $h(T - 1)$; in particular a zero root becomes the root one after the translation. Hence these two choices cover all roots.

For each choice of chart and ϵ, apply Lemma 8.4 to the residuals $f_1, \ldots, f_r, \Psi_\epsilon$. This gives one equation in $\mathbf{x}, U$ equivalent over K to their conjunction. Take the product of these equations over the finitely many charts and the two translations. A finite product vanishes if and only if one factor vanishes, which proves the assertion without adding another witness. For an empty constructible set take $P = 1$.

If K is a number field, clearing a common numerical denominator places the finitely many coefficients in $\mathcal{O}_K$ without changing the zero set. All algebraic operations in the construction are effective over a number field. The more general effective assertion follows under the stated algorithmic hypotheses. □

For number fields, Lemma 33.6 supplies a quadratic form in four arguments that vanishes only at the zero tuple. A balanced tree of that form combines r residuals of degree at most D with degree at most $2^{\lceil \log_4 r \rceil} D$, again without adding variables. Thus the same witness bound is available even for totally imaginary number fields; the failure of sums of squares there is not an obstruction to this compression argument.

Why the field hypothesis cannot simply be omitted. Over an algebraically closed field K, take X to be the origin in $\mathbb{A}^2_K$, with the identity projection to $\mathbb{A}^2_K$. All geometric fibers are finite. Its rational image nevertheless cannot be written as

$$\{(a, b) \in K^2 : \exists U \in K,\ P(a, b, U) = 0\}$$

for a single polynomial P. If P has positive degree in U, then every point in the nonempty open subset where its leading U-coefficient is nonzero has a lift, since every nonconstant polynomial over K has a root. The image therefore cannot be a single point. If P is independent of U, a nonconstant polynomial in two variables has a zero set of dimension at least one whenever it has a zero; a constant gives either the empty set or the whole plane. Neither possibility gives exactly the origin. Thus non-algebraic-closedness is essential for this general one-equation, one-additional-variable statement.

9 An explicit polynomial for a guarded finite algebra

Let $R = \mathbb{Z}[x_1, \dots, x_b]$. Consider a triangular monic presentation

$$A = R[z_1, \dots, z_m]/(f_1, \dots, f_m), \qquad f_i \in R[z_1, \dots, z_i],$$

where f_i is monic of degree $e_i \geq 1$ in z_i. Its standard monomials $\prod_i z_i^{a_i}$, $0 \leq a_i < e_i$, form an R-basis. In particular A is free of rank $n = \prod_i e_i$. For $a \in A$, write M_a for its multiplication matrix in this ordered monomial basis. The matrices M_a commute, and their entries are integer polynomials in the base coordinates. They are completely specified by repeated reduction using the monic equations; they need not be materialized to specify a determinant involving them.

Fix elements $G, E_1, \dots, E_r \in A$. We assume that in every geometric fiber, every point at which G is nonzero has local length one. This condition holds, for example, if G contains the product of the diagonal derivatives $\partial f_i/\partial z_i$ as a factor. We seek the rational base points admitting a rational point of A with

$$G \neq 0, \qquad E_1 = \cdots = E_r = 0.$$

For $0 \leq \ell \leq mn(n+1)/2$, define

$$\lambda_\ell = 1 + \sum_{i=1}^{m} \ell^i z_i, \qquad M = M_{G\lambda_\ell}, \qquad H(V) = \det(I - VM) = \sum_{k=0}^{n} a_k V^k.$$

Here and below $a_k \in R$ depend on the fixed integer ℓ; always $a_0 = 1$. Also put

$$R_i(V) = \operatorname{Tr}\big(M_{E_i} \operatorname{adj}(I - VM)\big).$$

This is a polynomial in $R[V]$ of degree at most $n-1$.

For each $1 \leq d \leq n$, set

$$\begin{aligned}
H_d(V) &= \sum_{k=0}^{d} a_k V^k, \\
g &= a_d \operatorname{Res}_V(H_d, H_d'), \qquad L = a_d^2 g, \\
\Psi(U) &= 1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j, \\
\widetilde{R}_i(U) &= (LU)^{n-1} R_i\big(1/(LU)\big), \\
Q_{\ell,d}(\mathbf{x}, U) &= \Psi(U)^2 + \sum_{k=d+1}^{n} a_k^2 + \sum_{i=1}^{r} \widetilde{R}_i(U)^2.
\end{aligned}$$

The resultant is the Sylvester determinant with the specified degree bounds d and $d-1$. The expression for $\widetilde{R}_i$ denotes polynomial coefficient reversal, hence defines a polynomial also at $LU = 0$. Every displayed expression has integer coefficients.

Theorem 9.1. *The explicitly specified polynomial*

$$P(\mathbf{x}, U) = \prod_{\ell=0}^{mn(n+1)/2} \prod_{d=1}^{n} Q_{\ell,d}(\mathbf{x}, U)$$

has the following property. For every $\mathbf{a} \in \mathbb{Q}^b$, there exists $U \in \mathbb{Q}$ with $P(\mathbf{a}, U) = 0$ if and only if there is a rational point of the fiber algebra $A_{\mathbf{a}}$ satisfying $G \neq 0$ and all $E_i = 0$. The formula uses only one coordinate in addition to the b base coordinates; its formal algebra generators are eliminated by the determinants and are not quantified coordinates.

Proof. Fix a rational base point. First consider one factor $Q_{\ell,d}$. At $a_d g = 0$ or $U = 0$, its reciprocal residual Ψ equals one. Otherwise, with $V = 1/(LU)$,

$$\Psi(U) = (LU)^d \frac{H_d(V)}{a_d}.$$

Thus a rational zero of $Q_{\ell,d}$ implies $a_d g U \neq 0$, all coefficients a_k with $k > d$ vanish, and $H(V) = H_d(V) = 0$. The resultant condition makes this a simple root of H. Since $H(0) = 1$, the root V is nonzero.

The rational matrix $I - VM$ has a one-dimensional kernel. Every multiplication matrix M_{z_i} preserves it and therefore acts on it by a rational scalar ζ_i. Applying the algebra relations to this common eigenline shows that $(\zeta_1, \ldots, \zeta_m)$ is a rational point of the fiber. Its $G\lambda_\ell$-value is $1/V$, so its G-value is nonzero.

To verify the closed conditions, write $J = \operatorname{Tr}(\operatorname{adj}(I - VM))$. At a root of H, the adjugate identity gives $M \operatorname{adj}(I - VM) = V^{-1} \operatorname{adj}(I - VM)$. Differentiating the determinant then yields

$$J = -VH'(V) \neq 0.$$

The columns of the adjugate lie in the common eigenline, so

$$R_i(V) = E_i(\boldsymbol{\zeta}) J.$$

Since $LU \neq 0$, the equation $\widetilde{R}_i(U) = 0$ is equivalent to $R_i(V) = 0$, and hence $E_i(\boldsymbol{\zeta}) = 0$. This proves the reverse implication without assuming that the rest of the fiber algebra is reduced.

Conversely, suppose a suitable rational point exists. Over an algebraic closure, multiplication by $G\lambda_\ell$ has the value $G(p)\lambda_\ell(p)$ at every geometric point p, with multiplicity equal to its local length. If $G(p) = 0$, its factor in $\det(I - VM)$ is one, even if the local algebra is nonreduced. If $G(p) \neq 0$, its local length is one by hypothesis.

At any such point p, the polynomial $G(p)(1 + \sum_{i=1}^{m} \ell^i z_i(p))$ is nonzero as a polynomial in ℓ, because its constant coefficient is $G(p) \neq 0$. For distinct such points p, q, their difference is also nonzero: if all coefficients vanished, first $G(p) = G(q) \neq 0$, and then all their coordinates would agree. Each of these polynomials has degree at most m. With at most n geometric points, there are at most $mn(n+1)/2$ bad values of ℓ in total, counting zero values and collisions. Thus one of the specified integers makes all nonzero guarded eigenvalues distinct and nonzero.

For that ℓ, the polynomial H is squarefree of some degree $1 \leq d \leq n$. The given rational point provides a nonzero rational root V of H. Its closed residuals vanish, so the trace identities give $R_i(V) = 0$. The leading coefficient and resultant are nonzero. Taking $U = 1/(LV)$ therefore gives a zero of the corresponding factor $Q_{\ell,d}$, and hence of P. □

The degree cases are part of the literal finite formula. They include every exceptional drop in degree of H. The discriminant condition excludes bad separating choices, and the finite bound on ℓ proves that this exclusion loses no valid rational point. In particular the construction does not leave a generic primitive-element argument or an unperformed exceptional-stratum decomposition implicit.

9.1 Explicit degree and coefficient bounds for the specification

The following bounds concern the expanded polynomial defined by the indexed determinant formula. They do not require expanding its matrices or its final finite product. For a polynomial f, let $\|f\|_1$ denote the sum of the absolute values of its coefficients. Then $\|fg\|_1 \le \|f\|_1\|g\|_1$ and $\|f+g\|_1 \le \|f\|_1 + \|g\|_1$.

Give every base coordinate weight one, and assign a positive integer weight w_i to each algebra generator z_i. Choose the weights so that every nonleading term of its monic relation has weight at most $e_i w_i$. Reduction by these relations never increases weighted degree. Writing $W = \sum_i (e_i - 1)w_i$, the entries of a multiplication matrix M_f therefore have base degree at most $\operatorname{wdeg}(f) + W$: the input basis monomial has weight at most W, and the output basis monomial has nonnegative weight.

For a matrix over the polynomial ring, use the maximum column sum of the coefficient norms of its entries. This norm is submultiplicative. If

$$z_i^{e_i} + \sum_{k<e_i} c_{ik} z_i^k = 0,$$

the block companion description gives

$$\|M_{z_i}\| \le \max\left\{1, \sum_{k<e_i} \|M_{c_{ik}}\|\right\}.$$

Extending to later monic generators preserves this estimate, since multiplication by an earlier element is repeated on diagonal blocks, up to a simultaneous permutation of rows and columns. Thus weights and bounds $\|M_{z_i}\| \le 2^{b_i}$ can be computed recursively from the coefficient DAG. The source compiler records the entire table, and the independent checker verifies the corresponding inequalities.

For the point quartics the weights can be taken to be two. A coefficient bound for the companion matrix is the sum of the bounds for its two nonzero lower coefficients. For each of the subsequent quadratic generators, its nonzero branch has only the constant coefficient LM, so one may take

$$w_i \ge \tfrac{1}{2}\operatorname{wdeg}(LM), \qquad b_i \ge \log_2 \|M_{LM}\|.$$

The zero branch has relation $z_i = 0$ and satisfies the same bounds. Consequently one uniform table applies to all 2^{86} fixed branches.

Put $n = 2^{508}$ for a uniform rank bound, $m = 297$, and $T = mn(n+1)/2$ for the largest separating parameter. Suppose the recorded bounds give

$$\operatorname{wdeg}(G) \le w_G, \quad \|M_G\| \le 2^{b_G}, \qquad \operatorname{wdeg}(E_i) \le w_E, \quad \|M_{E_i}\| \le 2^{b_E}.$$

Set

$$b_\lambda = \lceil \log_2(m+1) \rceil + \max_i \{ i \lceil \log_2 T \rceil + b_i \},$$
$$\alpha = w_G + \max_i w_i + W, \qquad \beta = b_G + b_\lambda,$$
$$\alpha_E = w_E + W.$$

These bound the entry degrees and operator norm of $M_{G\lambda_\ell}$, uniformly for all prescribed ℓ and all source branches.

The coefficients of $H(V) = \det(I - VM)$ have base degree at most $A_H = n\alpha$, and its full coefficient norm is at most 2^{B_H} with $B_H = n(\beta+1)$. Indeed the coefficient norm of a determinant is bounded by the product of its column sums. This bound already includes the sum over determinant permutations; no additional factorial is needed. By multilinearity in columns,

$$R_i(V) = \frac{\partial}{\partial t} \det(I - VM + tM_{E_i}) \bigg|_{t=0}$$

has base coefficient degree at most $A_R = (n-1)\alpha + \alpha_E$ and coefficient norm at most 2^{B_R}, where

$$B_R = \lceil \log_2 n \rceil + b_E + (n-1)(\beta+1).$$

This accounts for the n possible columns containing M_{E_i}.

For each degree case $d \le n$, the Sylvester matrix for $\mathrm{Res}(H_d, H_d')$ has $2d-1$ rows. Every row has coefficient norm at most $d\|H\|_1$. Hence uniform bounds for the reciprocal scale $L = a_d^3 \mathrm{Res}(H_d, H_d')$ are

$$A_L = (2n+2)A_H,$$
$$B_L = (2n+2)B_H + (2n-1)\lceil \log_2 n \rceil.$$

The reciprocal residual has degree at most $n(A_L+1)$ and coefficient norm at most $2^{nB_L + \lceil \log_2(n+1) \rceil}$. The reversed trace residuals have degree at most $A_R + (n-1)(A_L+1)$ and coefficient norm at most $2^{B_R + (n-1)B_L}$. Together with the high-coefficient conditions $a_k = 0$, these give a bound for the sum of squares by doubling the largest degree, doubling the largest logarithmic norm bound, and adding $\lceil \log_2(n+87) \rceil$ to the latter. For the reciprocal estimate, no division of coefficient norms is used: its jth coefficient is the polynomial $a_{d-j} a_d^{3j-1} \mathrm{Res}(H_d, H_d')^j$. Its degree is at most jA_L and its logarithmic norm bound is at most $j(3B_H + B_{\mathrm{Res}}) = jB_L$.

The number of final factors is also explicit. Write $n_0 = 4^{211}$. For a mask with k nonzero quadratic branches, its rank is $n_k = n_0 2^k$. There are $\binom{86}{k}$ such masks, and its number of separating-parameter and degree choices is $n_k(1 + mn_k(n_k+1)/2)$. Including the 212 point charts and 256 height choices gives exactly

$$B_{\mathrm{tot}} = 54272 \left[n_0 3^{86} + \frac{297}{2} \left(n_0^3 9^{86} + n_0^2 5^{86} \right) \right]$$

factors. Multiplying the uniform branch degree and logarithmic coefficient-norm bounds by B_{tot} bounds the complete polynomial. All quantities here are exact integers, apart from the displayed ceiling notation, which is evaluated by integer bit lengths.

For the emitted coefficient specification the resulting convenient rounded bounds are

$$\deg P \le 2^{3122}, \qquad H(P) \le 2^{2^{3127}}.$$

The machine-readable specification records the stronger unrounded integer degree bound and the stronger integer bound on $\log_2 H(P)$. These numbers are rigorous upper bounds for the literal indexed polynomial, not measurements of an expanded coefficient table.

10 A finite-fiber presentation with 557 retained coordinates

We now give a source-specific application of Theorem 8.1. The count is a theorem about an existential polynomial obtainable by elimination. It is not the degree or variable count of an already expanded coefficient table. Its identification with the ordinary integers remains conditional on the arithmetic correctness of the source definition, exactly as for any projection-preserving optimization of that definition.

All geometric fibers in this section are taken after extension to an algebraic closure of $\mathbb{Q}$. This qualification matters: $\mathbb{Q}(\sqrt{2}) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$ is the product of two copies of $\overline{\mathbb{Q}}$, not a field. Whenever a quadratic-field inverse occurs, we retain the inequation that its rational norm is nonzero. Its value is then uniquely determined on both factors, including on geometric fibers.

10.1 Point occurrences

The root module has six nonidentity point occurrences, with labels A, S, T, Z, height J, and common comparison J. Each of the 41 comparison calls has five more, with labels H, X_0, X_1, X_2, X_3. Thus the total is

$$6 + 41 \cdot 5 = 211.$$

Replace each such point by the double of a halfpoint (a, b) on the fixed curve $E_D : b^2 = a^3 - Da$ over $K = \mathbb{Q}(\sqrt{2})$. This uses the exact nonidentity $2E_D(K)$ membership statement of the source. Retain a_0, a_1, where $a = a_0 + a_1\sqrt{2}$, and leave b_0, b_1 as coordinates to be eliminated. Keep the original nonzero guards on the doubled point as well as the norm guard on b. All downstream point coordinates are rational functions of these halfpoint coordinates.

This point substitution can be checked by explicit rational maps, without a group-membership assertion as a black box. Write $D = d^2$ for the source curve. From a halfpoint (a, b) set

$$u = \frac{2b}{a^2 + D}, \qquad v = \frac{a^2 - 2da - D}{a^2 + D}, \qquad w = \frac{a^2 + 2da - D}{a^2 + D}, \qquad h = vw.$$

Then put $f = u^2$, $x = 1/u^2$, $y = h/u^3$, and $z = -u/h$. On the guarded domain these satisfy every old full-chart equation:

$$v^2 + df = 1, \quad w^2 + v^2 = 2, \quad h = vw, \quad zh + u = 0, \quad xf = 1, \quad yz + x = 0.$$

In addition,

$$x = \frac{(a^2 + D)^2}{4b^2}, \qquad y = \frac{(a^2 + D)(a^4 - 6Da^2 + D^2)}{8b^3},$$

so $(x, y) = [2](a, b)$ by the duplication formulas.

Conversely, the old full chart forces u, v, w to be nonzero. With $A = (1 + v)(1 + w)$ and $S = v + w$, define

$$a = \frac{A}{u^2}, \qquad b = \frac{SA}{u^3}.$$

These formulas always give a valid halfpoint. Indeed $v^2 + w^2 = 2$ and $du^2 = 1 - v^2$ imply

$$A(S^2 - A) = -(1 - v^2)^2, \qquad A^2 + (1 - v^2)^2 = 2SA.$$

The first identity gives $b^2 = a^3 - d^2a$; the second, together with

$$A^2 - 2(1 - v^2)A - (1 - v^2)^2 = 2vSA,$$
$$A^2 + 2(1 - v^2)A - (1 - v^2)^2 = 2wSA,$$

shows that the first set of formulas recovers the same u, v, w. None of the new denominators vanishes. In fact $1+v=0$ or $1+w=0$ would give $du^2=0$; and $v+w=0$ would give $v^2=w^2=1$, with the same contradiction. Thus A, S, b, a^2+D are nonzero. This proves both directions of the full-chart substitution directly. The original guards on the output exclude any remaining exceptional halfpoints in the forward map.

For every fixed geometric value of a_0, a_1, the equation $b^2 = a^3 - Da$ has at most four geometric solutions for (b_0, b_1). Indeed, after the splitting $K \otimes \overline{\mathbb{Q}} \simeq \overline{\mathbb{Q}} \times \overline{\mathbb{Q}}$, it becomes two independent equations $b_+^2 = c_+$ and $b_-^2 = c_-$, each with at most two roots. Restricting to the guarded open subset cannot increase this number. The retained point coordinates therefore number

$$211 \cdot 2 = 422.$$

10.2 The pole blocks and their zero fibers

There is one pole block from the height call and one from each of the 41 comparison calls, hence 42 in total. The source block has input b and private variables t, U, e, p, q, s, h, with $B = 17b$, satisfying

$$\begin{aligned} U(1+bt^2) &= b(1-bt^2), \\ p^2 &= B(1-e)(1+3e), \\ h &= e(B-U^2), \\ Bq^2 &= (s^2-Bh)(s^2+3Bh). \end{aligned}$$

Retain t, e, s from each block. The factor $1+bt^2$ never vanishes on this system, even on geometric fibers. Otherwise $bt^2 = -1$ and the first equation would give $0 = 2b$, a contradiction. Thus U is uniquely determined; h is then uniquely determined, p has at most two possibilities, and q has at most two possibilities when $b \neq 0$.

The exceptional case $b = 0$ must be treated. The source equations then force

$$U = p = h = s = 0,$$

but they do not constrain the private variable q. Replace this fiber by $q = 0$. This does not change any rational existential projection: q occurs nowhere outside this block, and replacing it by zero preserves all its equations when $b = 0$. The same normalization makes the geometric fiber finite. It is a constructible condition, given by the union of the branch $b \neq 0$ and the branch $b = 0, q = 0$.

Consequently these blocks contribute

$$42 \cdot 3 = 126$$

retained coordinates and at most four eliminated choices per block.

10.3 The smaller anchor

Let $C \neq 0$ be the fixed source anchor constant and let f_0, f_1 be the two reciprocal coordinates supplied to the anchor. Put

$$F = f_0^2 + f_0 f_1 - f_1^2.$$

By Proposition 5.1 and Lemma 5.2, the exact anchor condition is equivalent to

$$u^2 + 6v^2 - 2w^2 - 3F^2 + 2C^4 = 0 \tag{3}$$

with three rational witnesses. This follows by applying the semilocal formula to F/C^2 and multiplying by C^4; multiplication of each of its three witnesses by C^2 is a bijection of $\mathbb{Q}$. Retain v, w. Once the point coordinates and their finite choices are fixed, F is fixed and (3) is monic quadratic in u. Thus two retained coordinates suffice for this block, with at most two eliminated geometric choices. The old two sets of five retained anchor coordinates have been replaced by these two.

10.4 The remaining parameters and algebraic choices

Retain the seven root coordinates

$$X, \quad s, \quad z, \quad \theta, \quad e, \quad u_c, \quad v_c.$$

Here X is the ordinary input, s, z are two of the three index proxies, θ is the common approximation to $\sqrt{2}$, e is the common coordinate denominator, and u_c, v_c are the two common contact parameters. The third index proxy, denoted τ to avoid confusing it with a retained pole variable, satisfies the monic equation

$$\tau^2 - s(s-2)\tau z + (s-2)^2 z^2 - 1 = 0.$$

It therefore contributes at most two geometric choices and no retained coordinate. The eight height-class bits satisfy $\epsilon^2 = \epsilon$ and contribute only finitely many choices. Every remaining arithmetic gate is determined by its inputs; every inverse is unique on its retained guarded open subset. Fixed constant constructions contribute no free coordinates.

One nonvanishing point in this claim deserves explicit attention. The compressed reciprocal-logarithm equations are $\Lambda p_0 = G_0$, $A_i = C_i p_i$, $A_i \Lambda = G_i$. Their coefficients $G_i = G_J(Df_i^2)$ do not vanish for $f_i \in K$. Here

$$G_J(W) = 1 + \sum_{j=1}^{J} a_j W^j, \quad a_1 = -2/5, \quad a_j = a_{j-1}\frac{4j-5}{4j+1} \ (j \geq 2),$$

with $J \geq 2$ in every application. The extension of the two-adic valuation to K is

$$v(a + b\sqrt{2}) = \min\{v_2(a), v_2(b) + \tfrac{1}{2}\}.$$

The fixed D has valuation zero and all a_j have valuation one. If $v(f) \geq 0$, the constant term of $G_J(Df^2)$ is uniquely least in valuation; if $v(f) < 0$, then $v(f) \leq -1/2$ and the last nonconstant term is uniquely least. Thus $G_J(Df^2) \neq 0$. The displayed equations force $\Lambda, p_i \neq 0$ on rational solutions. Imposing their rational norm guards before eliminating them therefore loses no rational solutions. These guards are still retained on geometric fibers: the valuation argument is not being asserted over an algebraic closure.

For clarity, the passage from each compressed point chart back to the full one is also explicit. Its equations are

$$f_i = u_i^2, \quad v_i^2 + df_i = 1, \quad p_i^2 = x_i - Df_i, \quad x_i f_i = 1.$$

The nonzero p_i just proved and the identity $f_i p_i^2 = v_i^2(1 + df_i)$ force $v_i \neq 0$. Set $z_i = -1/p_i$, $y_i = x_i p_i$, $h_i = u_i p_i$, and $w_i = u_i p_i / v_i$. Substitution gives all full-chart equations. Conversely the

full chart gives $p_i = y_i f_i$. The ratio variables are then determined by $\Lambda = G_0/p_0$, $A_i = G_i/\Lambda$, and $C_i = A_i/p_i$. All other difference and reciprocal-coordinate guards of the original source are retained.

Proposition 10.1. *After the exact point-membership substitution, the exact anchor replacement, and the harmless zero-fiber normalization of the private pole variable q, the source has a constructible presentation whose projection to*

$$422 + 126 + 2 + 7 = 557$$

retained rational coordinates has finite geometric fibers. Its rational existential projection onto X is unchanged.

Proof. The exact equivalences of the substitutions have been established above. In a fiber of the retained coordinates, first choose the ordinate coordinates of the 211 halfpoints, in finitely many ways. The doubled points and all their downstream rational functions are then determined wherever their guards hold. Next choose the index proxy τ and the eight bits. This determines the rational input of each pole block. The normalized pole equations have finitely many solutions for their remaining private variables by the case analysis above. Finally the anchor ordinate u has at most two choices. Additional equations or inequations only discard choices. All remaining auxiliary coordinates are unique functions of these choices. Thus every geometric fiber is finite, including exceptional and empty fibers.

For an explicit uniform upper bound on the number of choices, the argument gives

$$4^{211} \cdot 4^{42} \cdot 2 \cdot 2 \cdot 2^8 = 2^{516}.$$

This is only a bound on the size of a geometric fiber of this presentation, not on the degree of the final polynomial after stratification and branch multiplication. □

Corollary 10.2. *There is an effectively constructible integer-coefficient polynomial in at most 558 total variables, one of them the input X, whose projected rational zero set equals that of the source system. Conditional on that source system defining $\mathbb{Z}$ in $\mathbb{Q}$, the polynomial is a Diophantine definition of $\mathbb{Z}$ using at most 557 witnesses.*

Proof. Apply Theorem 8.1 with $d = 557$ to Proposition 10.1. Its polynomial uses the 557 retained coordinates and one additional coordinate. Existentially quantifying all except X gives the stated projection. □

The corollary is an existence and effective-construction bound, with explicitly proved projection equivalence. An actual output polynomial of this size requires completing the finite-algebra elimination and the exceptional-stratum computation. No expanded coefficient table, degree, coefficient height, or claim of minimum variable count is asserted here.

10.5 The emitted 558-variable source specification

The coefficient compiler starts with the pinned 855-variable rational system, whose 297 residuals consist of 211 point equations, 84 pole equations, one anchor equation, and one Lucas equation. Its independent source audit is retained as a separate layer. Keep the 557 base coordinates of the finite-fiber presentation: 422 point abscissa coordinates, 126 pole parameters, two anchor parameters, and the seven remaining root parameters including the input.

For every point chart put $Y = 2C$, write $a^3 - Da = A + B\sqrt{2}$, and set

$$K = (j^2 + 2)A + 4jB, \qquad J = 2jA + (j^2 + 2)B.$$

The point equation becomes the monic integer polynomial

$$Y^4 - 2KY^2 + 2J^2 = 0.$$

Its derivative is nonzero on every valid point chart. Indeed $Y \neq 0$ there, and

$$K^2 - 2J^2 = (j^2 - 2)^2(A^2 - 2B^2) \neq 0.$$

If the derivative vanished at a nonzero root, then $Y^2 = K$ and the point equation would contradict this identity. The last norm is nonzero because the reconstructed halfpoint ordinate has nonzero norm. Thus including all 211 point derivatives among the guards does not discard valid rational solutions.

Complete the Lucas equation by putting

$$V = 2\tau - a, \qquad a = s(s-2)z, \qquad \Delta = a^2 - 4((s-2)^2z^2 - 1).$$

Then $V^2 - \Delta = 0$ and $\tau = (V + a)/2$. In the remaining residual numerators and guards, substitute $C_i = Y_i/2$ and this expression for τ. At each addition gate clear to the largest power of two occurring in its children; at products add the exponents, and at power gates multiply them. This gives exact integer polynomial numerators, with recorded certificates

$$\widetilde{H} = 2^k H\big(C_i = Y_i/2, \tau = (V + a)/2\big).$$

No vanishing denominator is introduced by these numerical factors.

Each of the 85 other finite witnesses occurs in just its own quadratic residual and in no denominator guard. Its cleared equation is $Ly^2 + M = 0$, where L, M depend only on the base coordinates, the point roots, and the completed Lucas root. It has the exact finite-case replacement

$$\begin{aligned} \epsilon = 0: &\quad Z = 0, \quad M = 0, \\ \epsilon = 1: &\quad Z^2 + LM = 0, \quad LZ \neq 0. \end{aligned}$$

For the second case recover $y = Z/L$. For the first recover $y = 0$. Conversely, if $Ly^2 + M = 0$ and $Ly \neq 0$, use the second case with $Z = Ly$. Otherwise $M = 0$, and the first case is possible. In particular a fiber with $L = M = 0$ is handled by choosing the private witness zero, without changing any other equation. This includes the exceptional zero pole input. Apply the same replacement to the completed Lucas equation with $L = 1$, $M = -\Delta$.

There are consequently 297 monic generators: 211 quartics and 86 linear-or-quadratic choices. For each fixed mask, their monic triangular algebra has rank

$$n = 4^{211}2^{\mathrm{popcount}(\epsilon)} \leq 2^{508}.$$

The guard is the original guard product, multiplied by the point derivatives and by LZ in every nonzero quadratic case. It makes all the diagonal derivatives of the monic presentation nonzero. Every guarded geometric point is therefore simple. The only added closed conditions are the 86 equations $(1 - \epsilon_i)M_i = 0$; in a nonzero case the corresponding equation is identically zero. This verifies all hypotheses of Theorem 9.1, including all zero and repeated-root cases.

Apply that theorem for each of the 212 fixed point charts, each of the 256 fixed height corrections, and every one of the 2^{86} masks. Their finite product is the emitted polynomial in 558 external variables. The 297 formal algebra generators are eliminated by the specified matrix determinants. They are not extra witnesses. The input projection agrees with the pinned rational system; its identification with $\mathbb{Z}$ retains the arithmetic hypothesis of that system.

The artifact contains 101637 integer coefficient-DAG nodes and the complete finite determinant/coefficient/product recipe. The independent checker imports no producer. It verifies all 211 point identities, every private quadratic extraction, the Lucas completion, the power-of-two substitution identities, monicity, triangularity, coordinate inventories, and the full table of degree and matrix-norm majorants. A separate exact small-instance interpreter executes 177 selector/degree factors across eight examples, including nilpotent components, excluded double roots, bad separators, nonrational residue fields, and zero coefficients. It also produces a fully expanded degree-16 example. The enormous source multiplication matrices and final branch product are not materialized, and no such computation is claimed.

11 Removing a fixed size condition without adding variables

The following elementary observation is useful when the number of variables has priority over both the degree and the size of the coefficients. It replaces a condition on the index of a point in a cyclic group by a finite product of polynomials. The length of that product can be extremely large.

Proposition 11.1 (Excluding finitely many indices)**.** *Let $K = \mathbb{Q}(\sqrt{2})$, let E/K be an elliptic curve in a Weierstrass equation, and let $P \in E(K)$ have infinite order. Fix an integer $M \geq 1$. There is an explicitly computable polynomial $A_M \in \mathbb{Z}[X_0, X_1]$ such that, for every nonzero integer s,*

$$A_M(x_0([s]P), x_1([s]P)) \neq 0 \quad \Longleftrightarrow \quad |s| > M.$$

Here $x(Q) = x_0(Q) + \sqrt{2}\, x_1(Q)$. The polynomial has degree $2M$ and requires no additional existential variables.

Proof. For $1 \leq n \leq M$, compute integers a_n, b_n, D_n with $D_n > 0$ and

$$x([n]P) = (a_n + b_n\sqrt{2})/D_n.$$

Such a representation is obtained by the usual exact group law and clearing the two rational denominators. Define

$$A_M(X_0, X_1) = \prod_{n=1}^{M} \big((D_n X_0 - a_n)^2 - 2(D_n X_1 - b_n)^2\big).$$

Each factor is D_n^2 times the field norm of $X_0 + \sqrt{2}X_1 - x([n]P)$. Since the norm of an element of a field vanishes only at zero, a factor vanishes precisely when the two abscissas agree. On a Weierstrass elliptic curve, two points with the same abscissa are negatives of each other. Thus $x([s]P) = x([n]P)$ holds precisely when $[s]P = [n]P$ or $[s]P = [-n]P$. The infinite order of P gives $s = n$ or $s = -n$. The leading homogeneous part of the displayed product is nonzero, so its degree is $2M$. □

An inequality $A_M \neq 0$ is a guard, rather than an extra equation with its own inverse variable. In a construction which already has a common denominator guard, one inserts A_M into that product. In the finite algebra construction below, one inserts it into the element whose nonzero components are selected. In either case it costs no new coordinate.

Corollary 11.2 (Conditional replacement in the repository construction)**.** *Assume the arithmetic soundness argument of the repository, including its height estimate with a constant $H < 2^{4783227}$, and its completeness with arbitrarily large admissible indices s. Replace the packed semilocal anchor by the guard $A_M(x(S)) \neq 0$, where*

$$M = 2^{9566463}.$$

The resulting construction still defines $\mathbb{Z}$. Applying the finite algebra construction to it uses 556 *total variables, including the input.*

Proof. The numerical choice satisfies

$$(20H)^2 < 400\, 2^{9566454} < 2^{9566463} = M.$$

Proposition 11.1 therefore supplies the strict size condition $|s| > (20H)^2$ used in the soundness argument. Completeness is preserved by choosing the admissible index larger than this fixed M. The former packed anchor used three coordinates: two retained coordinates and one coordinate finite over them. Removing them lowers the number of retained coordinates from 557 to 555. The finite algebra construction adds one coordinate to the retained base, giving 556 total coordinates. □

This corollary is an arithmetic modification of the defining system. Unlike elimination of finite fibres, it is not asserted to preserve the projection of an arbitrary system independently of the source arithmetic theorem. It also does not make the enormous finite product practical to expand. Its integer coefficients are specified exactly by a terminating indexed construction. If the contacts or height theorem are changed, the constant M must be chosen again from the new height estimate; the numerical value above must not be reused without that check.

12 Coefficient bounds for a fixed elliptic exclusion product

The finite size condition in the shared-point construction uses fixed multiples of a fixed elliptic point. The number of factors is extremely large, but their coefficients admit elementary uniform bounds. This section gives an exact algorithm defining those coefficients and proves bounds without computing the full product.

Put $k = \mathbb{Q}(\sqrt{2})$. Every $x \in k$ has a unique expression

$$x = \frac{a + b\sqrt{2}}{c}, \qquad a, b, c \in \mathbb{Z}, \quad c > 0, \quad \gcd(a, b, c) = 1.$$

We call (a, b, c) its primitive common-denominator triple and set

$$h_0(x) = \lceil \log_2 \max(1, |a|, |b|, c) \rceil .$$

Thus the triple of zero is $(0, 0, 1)$ and $h_0(0) = 0$. This elementary coefficient height is used only in this section; it is not the canonical elliptic height or the absolute logarithmic Weil height.

Lemma 12.1. *For $x, y \in k$ one has*

$$h_0(x+y) \le h_0(x) + h_0(y) + 1,$$
$$h_0(xy) \le h_0(x) + h_0(y) + 2,$$

and, for $x \neq 0$,

$$h_0(x^{-1}) \le 2h_0(x) + 2.$$

Proof. Write the primitive triples of x, y as (a, b, c) and (a', b', c'). Before cancellation their sum has triple

$$(ac' + a'c,\ bc' + b'c,\ cc')$$

and their product has triple

$$(aa' + 2bb',\ ab' + a'b,\ cc').$$

The coordinates are bounded respectively by $2^{h_0(x)+h_0(y)+1}$ and $3 \cdot 2^{h_0(x)+h_0(y)}$. Passing to a primitive triple can only decrease their maximum. For the inverse use

$$x^{-1} = \frac{ca - cb\sqrt{2}}{a^2 - 2b^2}.$$

The denominator is nonzero because $x \neq 0$. Its absolute value is at most $3 \cdot 2^{2h_0(x)}$, and the numerator coefficients are at most $2^{2h_0(x)}$. Changing all signs if necessary and cancelling the common divisor proves the last bound. □

Consider the fixed elliptic curve and point

$$E: \quad y^2 = x^3 - D_E x, \qquad D_E = 11 - 6\sqrt{2},$$
$$P = \left(\frac{57 - 28\sqrt{2}}{8}, \frac{-660 + 417\sqrt{2}}{32} \right).$$

Direct substitution verifies $P \in E(k)$. For a finite point $R = (x, y)$ put $H_0(R) = \max(h_0(x), h_0(y))$, and set $H_0(\mathcal{O}) = 0$ for the identity. Here $h_0(D_E) = 4$, $h_0(x(P)) = 6$, $h_0(y(P)) = 10$, and $H_0(P) = 10$.

Lemma 12.2. *For every integer $n \ge 1$, the point nP is defined by exact arithmetic in k and*

$$H_0(nP) \le 17n^8.$$

Consequently, if $nP \neq \mathcal{O}$ and $x(nP) = (a_n + b_n\sqrt{2})/c_n$ is written in its primitive common-denominator triple, then

$$\max(|a_n|, |b_n|, c_n) \le 2^{17n^8}.$$

No nontorsion assumption is needed for these inequalities.

Proof. We first establish bounds for doubling and addition of the fixed point. Let $R = (x, y)$ be finite and $H = H_0(R)$. If $y \neq 0$, the usual doubling formulas are

$$m = \frac{3x^2 - D_E}{2y}, \qquad x_2 = m^2 - 2x, \qquad y_2 = m(x - x_2) - y.$$

Lemma 12.1, together with $h_0(2) = 1$ and $h_0(3) = 2$, gives successively

quantity	bound for its coefficient height
$3x^2 - D_E$	$2H + 11$
$(2y)^{-1}$	$2H + 8$
m	$4H + 21$
x_2	$9H + 48$
$x - x_2$	$10H + 49$
y_2	$15H + 73.$

Thus $H_0(2R) \le 15H + 80$. When $R = \mathcal{O}$ or $y = 0$, the result is the identity, so the same bound holds using its specified height zero.

For $R \neq \mathcal{O}, P, -P$, write $P = (x_P, y_P)$ and use

$$m = \frac{y - y_P}{x - x_P}, \qquad x_+ = m^2 - x - x_P, \qquad y_+ = m(x - x_+) - y.$$

The denominator is nonzero: points of a short Weierstrass curve with the same abscissa are opposite. Using the uniform bounds $h_0(x_P), h_0(y_P) \le 10$ gives

quantity	bound for its coefficient height
$y - y_P,\ x - x_P$	$H + 11$
$(x - x_P)^{-1}$	$2H + 24$
m	$3H + 37$
x_+	$7H + 88$
$x - x_+$	$8H + 89$
y_+	$12H + 129.$

It follows that $H_0(R+P) \le 12H + 129$. The exceptional cases obey this bound as well. If $R = \mathcal{O}$ the output has height 10, and if $R = -P$ it has height zero. If $R = P$, the doubling estimate above with $H_0(P) = 10$ bounds $H_0(2P)$ by 223, which is smaller than $12 \cdot 10 + 129$.

To compute nP, begin with the leading binary digit of n, so the initial point is P. Read each subsequent digit from left to right, doubling and then, for a digit one, adding P. Every step is a well-defined group operation; use the identity and the exceptional cases just specified instead of a formula with zero denominator. One binary step replaces a height bound H by at most

$$12(15H + 80) + 129 = 180H + 1089.$$

The same upper bound covers a digit zero. Therefore the next height H' satisfies $H' + 7 \le 180(H + 7)$. There are $r = \lfloor \log_2 n \rfloor$ digits after the leading digit, and hence

$$H_0(nP) + 7 \le 17 \cdot 180^r \le 17 \cdot (2^r)^8 \le 17n^8,$$

since $180 < 2^8$. This proves the asserted weaker bound $H_0(nP) \le 17n^8$ and the bound for the primitive triple. □

12.1 A bound in a finite algebra

Let $R = \mathbb{Z}[X_1, \ldots, X_d]$ and let A be a commutative R-algebra free of positive finite rank, with a specified basis. For $a \in A$ let M_a be its multiplication matrix. If $\|f\|_1$ denotes the sum of the absolute values of the coefficients of a polynomial $f \in R$, define

$$\|M\| = \max_j \sum_i \|M_{ij}\|_1.$$

This norm satisfies the triangle inequality, is submultiplicative, and gives $\|M_c\| = |c|$ for an integer scalar c. The same arguments apply over rational coefficients with the corresponding coefficient norm.

Suppose $N_0, N_1, D_0 \in A$ and, for a real number $b \geq 0$,

$$\|M_{N_0}\|,\ \|M_{N_1}\|,\ \|M_{D_0}\| \leq 2^b.$$

For each $n \geq 1$ with $nP \neq \mathcal{O}$, define

$$\mathcal{E}_n = (c_n N_0 - a_n D_0)^2 - 2(c_n N_1 - b_n D_0)^2 \in A;$$

when $nP = \mathcal{O}$, define $\mathcal{E}_n = 1$. For an integer $M \geq 1$ put $\mathcal{E}_{\leq M} = \prod_{n=1}^{M} \mathcal{E}_n$.

Proposition 12.3. *The multiplication matrices of these factors satisfy*

$$\|M_{\mathcal{E}_n}\| \leq 2^{34n^8+2b+4},$$
$$\|M_{\mathcal{E}_{\leq M}}\| \leq 2^{(2b+38)M^9}.$$

The fixed coefficients in the product are specified by the binary algorithm of Lemma 12.2; they are not additional quantified variables.

Proof. For a finite nP, let $h_n = h_0(x(nP))$. The triangle inequality gives

$$\|M_{c_n N_0 - a_n D_0}\|,\ \|M_{c_n N_1 - b_n D_0}\| \leq 2^{h_n+b+1}.$$

Submultiplicativity and the factor's two terms now give

$$\|M_{\mathcal{E}_n}\| \leq 3 \cdot 2^{2h_n+2b+2} \leq 2^{34n^8+2b+4}.$$

The identity factor when $nP = \mathcal{O}$ obeys the same bound. Multiplying and using $\sum_{n=1}^{M} n^8 \leq M^9$ and $M \leq M^9$ gives

$$\|M_{\mathcal{E}_{\leq M}}\| \leq 2^{34\sum_{n=1}^{M} n^8+(2b+4)M} \leq 2^{(2b+38)M^9},$$

as claimed. This reasoning does not require expansion of the product. □

At a rational specialization for which $D_0 \neq 0$, the value $\mathcal{E}_n$ for a finite nP is zero precisely when

$$\frac{N_0 + N_1\sqrt{2}}{D_0} = x(nP).$$

Indeed $u^2 - 2v^2 = 0$ with $u, v \in \mathbb{Q}$ forces $u = v = 0$. If the left side is the abscissa of a finite point $Q \in E(k)$, the equality is equivalent to $Q = \pm nP$. Thus the product, together with the existing denominator guard, excludes exactly these finitely many abscissas. The convention for $nP = \mathcal{O}$ makes the indexed algorithm total and introduces no unwanted exclusion.

13 A shared-point construction

Scope. The construction below assumes the source manuscript's cyclic-group, five-contact height, scalar-height, and pole-completeness results, and also Theorem 4.1 of Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2. The last theorem is an additional arithmetic input. The number of variables is not inferred from Sun's undecidability theorem. All algebraic operations below have rational coefficients; field operations in $K = \mathbb{Q}(\sqrt{2})$ mean their two rational components.

13.1 Fixed data and the complete rational system

Use

$$d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$
$$P = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32), \qquad 2E(K) = \mathbb{Z}P.$$

For a nonzero point put $f(Q) = 1/x(Q)$ and $z(Q) = -x(Q)/y(Q)$. Write σ for conjugation. Every displayed quotient is accompanied by the inequation that its denominator, or its field norm, is nonzero. The old point charts are replaced by doubles of affine halfpoints.

The ten point occurrences are

$$A, S, T, Z, J_0, H, U_A, U_S, U_T, U_Z.$$

Every occurrence is nonidentity and lies in $2E(K)$. The points A, S, T, Z have integer labels (n, a, t, b), respectively; these labels are used only in the proof and are not additional rational coordinates.

Set $\beta = X^2 + 2$. Use rational proxies ξ, τ, ζ with

$$\tau^2 - \xi(\xi - 2)\tau\zeta + (\xi - 2)^2\zeta^2 - 1 = 0.$$

Only ξ, ζ are retained; τ is a finite algebraic coordinate. Let $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z)$ and $(U_0, U_1, U_2, U_3, U_4) = (H, U_A, U_S, U_T, U_Z)$. Form $V_i = U_i - [4]Q_i$ for $0 \le i \le 4$. These five differences are required to be nonidentity. Affine subtraction also retains the guard $x(U_i) \ne x([4]Q_i)$; completeness will choose a common index different from $0, 4, -4$.

For $e \in \{16, 64\}$ put

$$L_e(Q) = z(Q)G_{e/4-1}(Df(Q)^2), \quad G_m(W) = 1 + \sum_{j=1}^{m} c_j W^j,$$

where $c_1 = -2/5$ and $c_j = c_{j-1}(4j - 5)/(4j + 1)$. The source proves these logarithms nonzero at every required nonidentity point. For $k = (16, 64, 64, 64)$ and $b = (\beta, \xi, \tau, \zeta)$ define

$$L_i = \frac{L_{k_i}(U_i)}{L_{64}(H)}, \qquad L_i' = \sigma\left(\frac{L_{k_i}(V_i)}{L_{64}(V_0)}\right).$$

The eighteen field tests D_j, with orders m_j, are

$f(U_0), \ldots, f(U_4), \sigma f(V_0), \ldots, \sigma f(V_4)$	1 (10 entries)
$L_i - L_i' \quad (1 \le i \le 4)$	k_i
$L_i - b_i \quad (1 \le i \le 4)$	k_i.

There is no approximate square root of two and no common comparison point J in this construction.

Enumerate the eight rational components of $x(A), x(S), x(T), x(Z)$ as $x_1, \ldots, x_8$. On finite branches set $A_0 = 1$ and, for $1 \le r \le 8$, set $A_r = x_r^2$ if $x_r \ne 0$, and $A_r = 1$ if $x_r = 0$. These are expressions, not new variables. The common slope is one rational coordinate λ. Put

$$\Theta_0 = \{0, -1/2, -1, 1/2399, -32/407\}, \qquad F_5(T) = \prod_{\theta \in \Theta_0} (T - \theta), \quad t_r = A_r\lambda.$$

Require $t_r \notin \Theta_0 \cup \{1\}$, and define, for $0 \le r \le 8$,

$$R_r = A_r^{-5}\frac{F_5(t_r)}{(t_r-1)^6}, \qquad \widetilde{D}_{r,j} = \frac{D_j}{(t_r-1)^{m_j+1}},$$

$$W_r = \prod_{j=1}^{18}\left(1 + R_r\left(\frac{\widetilde{D}_{r,j}}{R_r^{m_j}}\right)^2\right), \quad T_r = \mathrm{Tr}_{K/\mathbb{Q}}(W_r^{-1}), \quad B_r = R_r/T_r^2.$$

Require $W_r, T_r \neq 0$ and the following pole condition for each B_r.

Retain two rational coordinates e_r, s_r, set

$$C_r = 17B_r, \qquad h_r = e_r(C_r - B_r^2),$$

and impose the two monic equations

$$p_r^2 = C_r(1-e_r)(1+3e_r), \qquad q_r^2 = \frac{(s_r^2 - C_r h_r)(s_r^2 + 3C_r h_r)}{C_r}.$$

The roots p_r, q_r are finite algebraic coordinates. This is precisely the old pole predicate with preparation variable zero. No new finite multiplier choice is needed. The prime-pattern theorem will be invoked with its odd squarefree avoidance modulus divisible by $3 \cdot 17$. It then makes each selected prime square in $\mathbb{Q}_2, \mathbb{Q}_3$ and $\mathbb{Q}_{17}$. Thus the fixed multiplier 17 works throughout.

For use in the scalar-height block alone, retain the finite set

$$\mathcal{C} = \left\{ \pm \prod_{p \in I} p : I \subseteq \{2,3,5,7,11,13,17\} \right\}.$$

The unchanged scalar-height block. Put

$$r_\beta = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1).$$

Take the six rational coefficients of $f(J_0), f(S), z(S)/z(J_0)$, with orders $(1,1,1,1,48,48)$. Use the source's fixed binary norm construction $N_r(a,b) = a^2 + rb^2$ on the constant leaf 1 and the six normalized tests, and denote its root by W. On a finite branch choose $c \in \mathcal{C}$ and set $b_* = cr_\beta W^2$, $B_* = 17b_*$. Retain t_*, e_*, s_* and define

$$U_* = b_*\frac{1 - b_* t_*^2}{1 + b_* t_*^2}, \qquad h_* = e_*(B_* - U_*^2).$$

Require $b_* \neq 0$, $1 + b_* t_*^2 \neq 0$, and the two monic equations

$$p_*^2 = B_*(1-e_*)(1+3e_*), \qquad q_*^2 = \frac{(s_*^2 - B_* h_*)(s_*^2 + 3B_* h_*)}{B_*}.$$

The extra guard $b_* \neq 0$ removes no required integral-input witnesses: the source constructs W as a unit at a prime dividing the nonzero integer r_β, which has absolute value greater than one.

A fixed finite size condition. Let H be the original five-contact constant, with $1 \le H < 2^{4783227}$, and put $M = 2^{9566463}$. Require

$$\mathcal{A}_M(S) := \prod_{j=1}^{M} N_{K/\mathbb{Q}}\big(x(S) - x([j]P)\big) \neq 0.$$

This is an indexed finite expression with fixed rational coefficients. It forces $|a| > M > (20H)^2$. It has not been expanded. An alternative source replaces this guard by the proved three-witness semilocal anchor; that alternative adds two retained coordinates and one quadratic root.

13.2 The finite algebra and the variable count

For a point occurrence retain a_0, a_1 and set $a = a_0 + a_1\sqrt{2}$, $a^3 - Da = A + B\sqrt{2}$. Introduce a root C of the monic quartic

$$C^4 - AC^2 + B^2/2 = 0,$$

require $C \neq 0$ and $A^2 - 2B^2 \neq 0$, and put

$$b = C + \frac{B}{2C}\sqrt{2}.$$

The quartic gives $b^2 = a^3 - Da$. Use $[2](a, b)$ for the point, retaining all necessary norm guards. This single chart suffices for the specific curve used here; there is no finite chart parameter.

Lemma 13.1 (A single ordinate chart over a quadratic field)**.** *Let F have characteristic different from two, let $K = F(\sqrt{\delta})$ be a quadratic field, and let $d \in K^\times$ satisfy $d^3 \notin F \cup \sqrt{\delta}F$. On $E_d : y^2 = x^3 - d^2x$, every nonidentity $Q \in 2E_d(K)$ has a halfpoint whose ordinate has nonzero F-component. In particular this applies to our curve. More generally it applies over a real quadratic field whenever $d = u + v\sqrt{\delta}$ with $uv \neq 0$.*

Proof. Start with any affine halfpoint, so $b \neq 0$ and $a \neq 0$. Translation by $T_0 = (0, 0)$ gives

$$(a, b) + T_0 = (-d^2/a, d^2b/a^2)$$

and does not change its double. If both ordinates had F-component zero, then b and d^2b/a^2 would be nonzero F-multiples of $\sqrt{\delta}$, so $d^2/a^2 \in F$. Thus $w = a/d$ has square in F and belongs to F or $\sqrt{\delta}F$. But $b^2 = d^3w(w^2 - 1) \in F^\times$. In the first case this forces $d^3 \in F$, and in the second it forces $d^3 \in \sqrt{\delta}F$, both contradictions. The nonzero factor $w(w^2 - 1)$ follows from $b \neq 0$.

Over a real quadratic field, the two components of d^3 are $u(u^2 + 3\delta v^2)$ and $v(3u^2 + \delta v^2)$, both nonzero when $uv \neq 0$. For the present curve, explicitly,

$$d^3 = 45 - 29\sqrt{2}, \qquad d^3\sqrt{2} = -58 + 45\sqrt{2}.$$

The addition formula has the displayed positive sign: its slope is b/a, its new abscissa is $-d^2/a$, and its new ordinate is $(b/a)(a + d^2/a) - b = d^2b/a^2$. □

The quartic derivative is nonzero at the permitted roots. Indeed, with $Y = C^2$, the equation is $Y^2 - AY + B^2/2 = 0$, of discriminant $A^2 - 2B^2 \neq 0$, and $C \neq 0$. Thus its derivative may be recorded explicitly as a guard without excluding any permitted root.

The retained coordinates number

$$1 + 20 + 2 + 1 + 18 + 3 = 45.$$

The remaining coordinates are ten quartic roots, eighteen pole roots, two scalar-height roots and the Lucas root. Their monic triangular degree product is $4^{10}2^{21} = 2^{41}$. Vanishing square radicands are treated by the linear relation that their root is zero; on each nonzero branch its root is guarded nonzero. The finite-fibre theorem therefore supplies one additional rational coordinate, for a total of 46. Finite choices and their strata are combined by products, not by new existential selectors.

13.3 Local tests and the fixed preparation-zero predicate

Lemma 13.2 (A restricted completeness domain with preparation zero)**.** *The preparation-zero pole just specified excludes negative odd valuations outside* $\{2, 3, 17\}$*. It has rational witnesses whenever* b *has square reduced denominator, is positive and square in* $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$*, and satisfies*

$$b < 17, \qquad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2.$$

This assertion uses only the pole-completeness arithmetic input already used by the source.

Proof. The equations are the old equations with preparation variable zero and $U = b$, so the old one-sided valuation obstruction applies. For completeness put $B = 17b$ and $V = B(B - b^2) = 17^2b^2(1 - b/17)$. The stated inequalities make V positive and square at the three specified local fields. They are exactly the sufficient preparation conditions in the source's Lemma `original-seven-parity-completeness`. Its remaining argument constructs rational points in square class B on the two twists and then rescales them to the displayed equations. No further preparation variable is used. Equivalently one may repeat that argument with $U = b$: at a support prime of the square class of B outside the three exceptions, $v_p(b)$ is positive odd, so $1 - b/17$ is a square unit; at a support prime of V the valuation of B is even and cancellation forces B/b^2 to be a square unit. Thus the same disjoint-support and mutual-squareness hypotheses for the simultaneous two-twist construction hold. Its conclusions supply the required witnesses. □

Lemma 13.3 (One place for all local tests)**.** *Let* $R \neq 0$ *be rational, let* $D_j \in K$*, and let* $m_j \geq 1$*. Set*

$$W = \prod_j (1 + R(D_j/R^{m_j})^2), \quad T = \mathrm{Tr}_{K/\mathbb{Q}}(W^{-1}), \quad b = R/T^2,$$

with the displayed nonzero guards and a valid preparation-zero pole certificate for b*. If* $p > 76176$ *and* $v_p(R)$ *is odd, then it is positive and some place* $\mathfrak{p} \mid p$ *satisfies* $v_{\mathfrak{p}}(D_j) \geq m_j$ *for every* j*.*

Proof. Put $\alpha = v_p(R)$. All valuations in $K_{\mathfrak{p}}$ are integral, because p is unramified. Each nonzero nonconstant term of a factor has odd valuation $b_{j,\mathfrak{p}} = \alpha + 2(v_{\mathfrak{p}}(D_j) - m_j\alpha)$, so its factor has valuation $\min(0, b_{j,\mathfrak{p}})$. Put $\lambda_{\mathfrak{p}} = v_{\mathfrak{p}}(W) \leq 0$ and $\lambda = \max_{\mathfrak{p}|p} \lambda_{\mathfrak{p}} \leq 0$. Integrality of trace gives $v_p(T) \geq -\lambda$. Therefore

$$v_p(b) = \alpha - 2v_p(T) \leq \alpha + 2\lambda.$$

The left side is odd and cannot be negative by the pole obstruction. It is consequently positive, so $\alpha > 0$ and $\lambda > -\alpha/2$. At a place attaining λ, each nonpositive summand of $v(W)$ is at least λ. For each nonzero D_j this gives

$$\alpha + 2(v(D_j) - m_j\alpha) > -\alpha/2, \qquad v(D_j) > (m_j - 3/4)\alpha \geq m_j - 3/4.$$

Integral valuations yield $v(D_j) \geq m_j$. Zero terms meet the same conclusion automatically. This is the reciprocal-trace argument of Sun's Lemma 3.1, written here for one shared group. □

Lemma 13.4 (Two anchors compare the four integer labels)**.** *At a prime $p > 76176$, suppose the eighteen field tests have their stated orders at a place $\mathfrak{p} \mid p$. Then every proxy b_i is p-integral and*

$$b_i \equiv n_i \pmod{p^{k_i}}, \qquad (n_1, n_2, n_3, n_4) = (n, a, t, b).$$

This statement does not assume that p splits or that the label of H is a p-unit.

Proof. Write $H = [h]P$, $U_i = [u_i]P$. The first ten tests place H, U_i in the formal group at $\mathfrak{p}$, and V_0, V_i in the formal group at its conjugate place. The guarded denominators ensure $h \neq 0, 4$. The source's formal logarithm identities give the exact ratios

$$r_i = u_i/h, \qquad r_i' = (u_i - 4n_i)/(h - 4).$$

The reciprocal truncations have the same valuations as these ratios and agree with them to relative precision k_i. At least one of h and $h - 4$ is a p-unit; its corresponding exact ratio is integral. The test $v(L_i - L_i') \geq k_i$ then makes the other ratio integral too. Consequently the relative precision becomes absolute precision for both ratios. The remaining test gives $b_i \equiv r_i \equiv r_i' \pmod{p^{k_i}}$. Multiplying by the integer denominators and subtracting yields $4(b_i - n_i) \equiv 0 \pmod{p^{k_i}}$. Since $p \neq 2$, the assertion follows. If p is inert, conjugation acts within the same local field and the same proof applies. No splitting assertion is needed for soundness. □

13.4 Height transfer with the original numerical constant

Let $M_0(t)$ denote the odd-contact radical for the five new contacts, with primes at most 76176 omitted. The map $\phi(t) = 2 + 1/t$ carries these contacts to the original five contacts. Its matrix and inverse are integral with determinant -1, and the coordinate forms differ by fixed rational factors. The prime divisors of the new nonzero coefficients and contact differences, including differences from 1, are

$$2, 3, 5, 7, 11, 37, 109, 439, 2399.$$

Thus no new exceptional prime is introduced. For t avoiding the five contacts,

$$M_0(t) = M_5(\phi(t)), \qquad h(t) \leq h(\phi(t)) + \log 3 \leq (H + \log 3) M_0(t)^{24}.$$

At a prime in $M_0(t_r)$, $t_r - 1$ is a unit and exactly one factor of $F_5(t_r)$ has positive odd valuation. Because A_r^{-5} is a rational square, $v_p(R_r)$ is odd even when A_r is not a unit. The preceding trace lemma applied to $\widetilde{D}_{r,j}$ therefore gives the original, unscaled tests to their stated precision. Hence the two-anchor lemma applies at every prime in every $M_0(t_r)$.

Suppose the Kth powers of all nine radicals divide a nonzero integer N. Then

$$h(\lambda), h(A_r\lambda) \leq (H + \log 3)|N|^{24/K}.$$

For $x_r \neq 0$, the identity $A_r = x_r^2$ gives

$$2h(x_r) = h(A_r) \leq h(\lambda) + h(A_r\lambda).$$

The same inequality for $h(x_r)$ is immediate at $x_r = 0$. Finally,

$$h(x(Q_i)) \leq h(x_{2i-1}) + h(x_{2i}) + \tfrac{3}{2}\log 2 < 7H|N|^{24/K},$$

because $H \geq 1$, $|N| \geq 1$, and $2\log 3 + \frac{3}{2}\log 2 < 5$. The original constant $7H$ has therefore been preserved, rather than replaced by an uncontrolled constant.

13.5 Soundness of the definition

Take a rational solution. Write $a_0 = |a|$ and $B = \max(|n|, |a|, |t|, |b|)$. The finite exclusion gives $a_0 > M > (20H)^2$. The unchanged scalar-height argument gives

$$h(\beta) \le H\sqrt{a_0}.$$

For completeness about its use, its constant leaf is 1 on the all-left path of the binary norm tree, and the other six leaves are the stated tests divided by r_β to their assigned powers. At every odd five-contact of β, the old norm and pole argument gives order 48 for the parameter ratio, hence divisibility of the integer label a by the 48th power of that radical. The old five-contact height theorem gives exactly the displayed estimate.

Set

$$\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1 \in \mathbb{Z}.$$

If $\Delta \neq 0$, the proxy equation and the precision-64 congruences show that each of the nine radicals has 64th power dividing Δ. The source index-height inequality $h(x([m]P)) \ge m^2 - 6$ and the previous subsection give

$$B^2 - 6 < 7H|\Delta|^{3/8} < 14HB^{3/2},$$

where $|\Delta| < 5B^4$ since $B \ge a_0 > 7$. Hence $B < (14H+6)^2 \le (20H)^2$, contradicting $B \ge a_0$. Thus $\Delta = 0$. The source's elementary Lucas-growth lemma, applied to the nonzero label b, now gives

$$\log B > a_0 - 3.$$

If $\beta \neq n$, let N be the absolute numerator of $\beta - n$ in lowest terms. The precision-16 comparisons show that every radical has 16th power dividing N. As $N \le 2e^{h(\beta)}B$, one obtains

$$B^2 < 7HN^{3/2} + 6, \qquad \log B < 2\log(32H) + 3h(\beta).$$

The bounds $a_0 > (20H)^2$ and $H \ge 1$ imply both $h(\beta) < a_0/20$ and $\log(32H) < a_0/20$. Consequently $\log B < a_0/4 < a_0 - 3$, a contradiction. Therefore $\beta = n \in \mathbb{Z}$. Since $\beta = X^2 + 2$ and $X \in \mathbb{Q}$, reduction of the denominator of X gives $X \in \mathbb{Z}$.

13.6 Completeness and compatible simultaneous orders

Fix an integer X and put $n = X^2 + 2$. Use the source's scalar-height completeness construction: choose J_0 in sufficiently deep formal subgroups at the finitely many relevant primes, and then choose a positive label a sufficiently divisible relative to J_0. There are arbitrarily large such labels, so require $a > M$. The source's Lucas lemma supplies positive integers t, b satisfying $\Delta = 0$. Set the proxies to (n, a, t, b). This fixes the four root points, their eight abscissa components and all positive square scales A_r. The scalar norm root is a unit at every prime dividing r_β; as $|r_\beta| > 1$, it is nonzero. Thus the additional scalar guard is satisfied and the original scalar pole witnesses exist.

Temporarily prescribe $H = [h]P$ and $U_i = [hn_i]P$. Each comparison test is now a rational function of the point $[h]P$, obtained using fixed multiplications and translations. At $h = 64$ all its denominators are nonzero: the relevant integer multiples are nonzero, $64 \neq \pm 4$, and the source reciprocal logarithms never vanish there. Continuity in the finitely many groups $E(K_\mathfrak{p})$, for $\mathfrak{p} \mid 2 \cdot 3 \cdot 17$, gives a positive integer L, divisible by 64, and integers $c_p \ge 0$ such that

$$h \equiv 64 \pmod{L} \quad \Longrightarrow \quad v_\mathfrak{p}(D_j) \ge -c_p \quad (\mathfrak{p} \mid p,\ p = 2, 3, 17).$$

One obtains L by taking a common multiple of the indices of small open formal subgroups contained in the required neighborhoods.

There is also a real bound C, independent of all future progression moduli, attainable by arbitrarily large positive h in every nonconstant progression. Here is the compact-group argument. The closure G of $\{([h]P, [h]P^{\sigma}) : h \in \mathbb{Z}\}$ in the product of the real elliptic groups is a compact group with finitely many components. Each component projects onto the identity circle, up to translation, in each real elliptic factor: an infinite cyclic projection is infinite, and its connected closure is the circle. The finitely many poles of the required rational functions therefore remove a set of Haar measure zero from each component. Choose a neighborhood avoiding them in each component and bound all tests there by one C. Multiplication by any positive integer maps G^0 onto itself, so the closure of any nonconstant progression contains one component. Its positive tail is dense in that closure, proving the assertion. This is the argument in Sun's Lemma 8.1; in particular the bound is chosen before the prime pattern and its order moduli.

Choose a positive integer D_0 for which $D_0^2\theta/A_r$ is integral for every contact θ and every r, and put

$$v = D_0^2 w^2, \qquad \lambda = u/v.$$

The contact positions are $u - (\theta/A_r)v$, a finite collection of forms $u + c_j w^2$ with integer c_j. Delete repeated positions; within each fixed r its five positions are distinct. The zero contact gives the common position u. Enlarge the prescribed lower bound for the primes beyond 76176 and beyond all prime factors of the fixed coefficients and scales.

Invoke Sun's Theorem 4.1 with arbitrary prescribed divisibility of w, arbitrarily large u/w^2, and the odd squarefree avoidance modulus containing every odd prime factor of $3 \cdot 17L$ other than seven. It supplies positive prime values at all positions, Gaussian generators $\pi \equiv 9 \pmod{16}$, $\pi \equiv 1 + 3i \pmod{7}$, and $\pi \equiv i$ modulo that avoidance modulus. Its two reduction orders are $d_+ = N(\pi + 1)$ and $d_- = N(\pi - 1)$, with

$$v_2(d_+) = 2, \qquad v_2(d_-) = 6, \qquad \gcd(d_-(p), d_+(q)) = 4$$

for all selected primes p, q. Both orders are prime to the odd part of L: the prime seven follows separately from $d_+ \equiv 6$, $d_- \equiv 2 \pmod{7}$.

Put $U = \operatorname{lcm} d_-$ and $V = \operatorname{lcm} d_+$. The three congruences

$$h \equiv 0 \pmod{U}, \qquad h \equiv 4 \pmod{V}, \qquad h \equiv 64 \pmod{L}$$

are compatible. Their pairwise common odd factors have been removed, and the two-parts are compatible because $v_2(U) = 6$, $v_2(V) = 2$, $64 \mid L$, and $4 \mid 60$. The generalized Chinese remainder theorem supplies a nonconstant progression. Choose a sufficiently large positive h in it attaining the real bound C. This avoids $0, 4, -4$ and every affine-chart exception. At a selected prime choose the place with reduction order d_-. Then H, U_i are formal there and V_0, V_i are formal at the conjugate place of order d_+. Their exact two logarithm ratios equal n_i, so all eighteen tests have their required local orders.

13.7 Completeness of the nine pole occurrences

The determinant of the contact coordinate change is a unit outside the fixed exceptions. More explicitly, for each r,

$$R_r = \frac{\prod_{\theta \in \Theta_0}(u - (\theta/A_r)v)}{v^5(t_r - 1)^6}.$$

The denominator is a rational square, since v is a square. At its five distinct contact primes R_r has valuation one, and at every other prime it has even valuation. The contact primes do not divide v; otherwise two distinct prime positions would share that prime. They do not divide the denominator of a fixed scale or a fixed difference, by the chosen lower bound. Thus $t_r - 1$ is a unit at these primes.

At either place above a contact prime, each factor of W_r has valuation zero or negative. Its inverse has residue one or zero, respectively. At the successful place all factors have residue one. Therefore T_r has residue one or two and is a unit there. It follows that $B_r = R_r/T_r^2$ has positive odd valuation at its contact primes and even valuation at all other primes. In particular its reduced denominator is a square.

The imposed Gaussian congruences give each contact prime $p \equiv 1 \pmod 8$ and $p \equiv 1 \pmod{3 \cdot 17}$. Such a prime is square in each of $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The displayed product formula for R_r, and then division by T_r^2, shows that B_r has the same local square properties. This is the reason a single fixed multiplier 17 suffices for all nine blocks.

It remains to arrange their quantitative preparation-zero conditions. The real bound C was uniform in the later congruence moduli, so for $\lambda \to +\infty$ the fixed scales give, uniformly in the chosen h,

$$R_r \sim A_r^{-6}\lambda^{-1}, \quad W_r = 1 + O(\lambda^{-3}), \quad T_r = 2 + O(\lambda^{-3}), \quad B_r \sim (4A_r^6\lambda)^{-1}.$$

For sufficiently large λ all factors at both real embeddings are positive and $0 < B_r < 17$. At a small prime put $\nu_p = v_p(v) = -v_p(\lambda)$ and $\eta_{p,r} = v_p(A_r)$. For sufficiently large ν_p, direct valuation calculation gives

$$v_p(R_r) = \nu_p - 6\eta_{p,r}, \qquad v_{\mathfrak{p}}(\widetilde{D}_{r,j}) \geq -c_p + (m_j + 1)(\nu_p - \eta_{p,r}).$$

The nonconstant term of its factor consequently has valuation at least

$$3\nu_p + (10m_j - 8)\eta_{p,r} - 2c_p.$$

These bounds tend to infinity with ν_p. Thus W_r and its inverse are arbitrarily close to one at each place above p. Integrality of trace, also at the ramified prime two, gives $v_p(T_r) = v_p(2)$ once the error is sufficiently small. Hence

$$v_p(B_r) = \nu_p - 6\eta_{p,r} - 2v_p(2).$$

Arbitrarily high prescribed divisibility of w makes these valuations meet $v_2(B_r) \geq 3$, $v_3(B_r) \geq 1$, $v_{17}(B_r) \geq 2$ for every r. All such denominator requirements and the real lower bound were chosen before the application of the prime-pattern theorem. The restricted pole-completeness lemma now supplies all nine structural pairs and their square roots. This completes the rational solution for every integral input.

Theorem 13.5 (Conditional forty-six-variable construction)**.** *Assume the arithmetic inputs explicitly stated at the beginning of this section, including Sun's simultaneous prime-pattern theorem. The rational system specified above has input projection precisely $\mathbb{Z}$. It gives, by the finite-algebra construction, an effectively specified polynomial over $\mathbb{Z}$ with at most 46 total variables, including its ordinary rational input. With the three-witness semilocal anchor in place of the indexed finite exclusion, the same conclusion holds with at most 48 total variables.*

Proof. Soundness and completeness were proved in the preceding subsections. The point substitution is exact by the single-chart lemma. On each finite branch all denominators are guarded, all non-base choices lie in the stated finite triangular algebra, and its zero-radicand and repeated-root specializations have the prescribed finite treatment. There are 45 retained coordinates and one final coordinate, or two additional retained anchor coordinates in the semilocal version. Clearing fixed rational coefficients gives integer coefficients. The finite anchor is a specified indexed polynomial, not a claim that its astronomically many factors have been expanded. □

13.8 Preserving the scalar certificate when its point is shared

We verify that the scalar-height point J_0 can be identified with the common comparison point H. This is an arithmetic modification of the system, with the same hypotheses as the shared-point construction. The scalar pole witnesses must be chosen after the final common index has been selected.

Proposition 13.6. *Under the stated source arithmetic inputs and Sun's simultaneous prime-pattern theorem, imposing $J_0 = H$ preserves the input projection of the shared-point construction. Consequently the indexed-exclusion version uses at most 44 total coordinates after finite-algebra compression, and the semilocal-anchor version uses at most 46.*

Proof. Soundness follows from the construction with separate points by assigning the same point to both occurrences. For completeness fix an integer input, write $r = r_\beta$, and choose the signed squarefree correction c so that $cr > 0$ and cr is square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The number cr is a nonzero integer. Let $\mathcal{S}$ be its finite prime support.

Choose a positive integer h_0 divisible by 64 such that $J_* = [h_0]P$ is sufficiently deep in the formal subgroups at all places above $\mathcal{S}$. Next choose the positive label a of S sufficiently divisible relative to h_0, and sufficiently large for the size anchor. The original scalar-completeness argument permits these successive choices. It gives the following explicit conditions on the six scalar tests, normalized by their assigned powers of r: every normalized test is p-integral for $p \mid r$, and every nonconstant normalized test is divisible by p for $p \mid c$ with $p \nmid r$. The denominators used in those tests are nonzero. The source Lucas construction now fixes the two other root labels and all root points, and all four proxies are their integer labels.

These scalar conditions persist when J_* is replaced by $[h]P$ in a sufficiently small neighborhood at every place above $\mathcal{S}$. Indeed each test is a rational function defined at J_*; the coefficient maps from K to its two rational components are continuous in the product of the completions above p. An integral lower valuation bound defines an open condition at a point satisfying it. Simultaneously impose the uniform lower bounds for the eighteen comparison tests at the places above $2, 3, 17$, using continuity at the nonzero indices h_0 and $h_0 - 4$. Their denominators are nonzero by the reciprocal-logarithm lemma and $64 \mid h_0$, $h_0 > 0$.

The resulting finitely many local neighborhoods are preserved by one integer progression

$$h \equiv h_0 \pmod{L}, \qquad 64 \mid L.$$

To see this, put a sufficiently small open formal subgroup inside each translated neighborhood, take a common multiple of its finite group indices, and further multiply by 64.

Use the compact real-group argument to obtain bounds for the comparison tests uniform in every future progression modulus. Choose the fixed denominator divisibilities and real lower bound for the common slope as in the separate-point construction. In the simultaneous prime

theorem, include all odd prime factors of $3 \cdot 17L$, except seven, in its avoidance modulus. Its Gaussian class modulo seven already excludes seven from both reduction orders. With $U = \operatorname{lcm} d_-$ and $V = \operatorname{lcm} d_+$, the hypotheses therefore give

$$\gcd(U, V) = 4, \qquad \gcd(U, L) = 64, \qquad \gcd(V, L) = 4.$$

The three congruences

$$h \equiv 0 \pmod{U}, \qquad h \equiv 4 \pmod{V}, \qquad h \equiv h_0 \pmod{L}$$

are compatible: $64 \mid h_0$ and $4 \mid h_0 - 4$ supply the two remaining conditions. Choose a sufficiently large positive h in this progression attaining the uniform real bounds, and put $J_0 = H = [h]P$. The comparison witnesses and the nine preparation-zero pole blocks are then constructed exactly as before.

It remains to choose the scalar pole witnesses for this final h. Let W be its scalar norm root. At a prime dividing r, the normalized leaves are integral and the constant leaf 1 on the all-left path makes W a unit. At a prime dividing c but not r, the nonconstant leaves are divisible by that prime, so the same tree again has unit root. These conditions hold because the progression preserved them. Since $|r| > 1$, at least one such prime exists and $W \neq 0$.

Thus $b_* = crW^2$ is positive. At primes in $\mathcal{S}$ its valuation is the nonnegative valuation of the integer cr; at all other primes its valuation is even. Its reduced denominator is therefore a square. Its square classes at $2, 3, 17$ equal those of cr, so it is locally square at all three places. The original scalar pole-completeness theorem now supplies its preparation and curve witnesses. No real bound for W, and no new rational coordinate, was required.

Sharing removes one point occurrence, hence two retained abscissa coordinates and one quartic algebra generator. The retained count becomes 43 (or 45 with the semilocal anchor); the finite-algebra construction adds one coordinate. This proves the stated counts. □

13.9 Removing the scalar preparation coordinate

The scalar-height certificate can also use preparation zero, after a different normalization of its contact parameter and a larger fixed set of squareclass corrections. This is a change of arithmetic construction; it is not an identity on the old full witness space. The following argument uses the same explicitly stated arithmetic inputs as the shared-point definition.

Put

$$\delta(T) = T^2 + T + 23, \qquad \epsilon = \frac{204}{23^{12}}, \qquad R_\beta = \epsilon^2 \frac{r_\beta}{\delta(\beta)^4},$$

where $\beta = X^2 + 2$ and r_β is the previous scalar contact polynomial. Its nonvanishing and $\delta(\beta) > 0$ hold for every rational input. Define the two fixed prime sets

$$\mathcal{B} = \{5, 19, 23, 349, 661, 5209\}, \qquad \mathcal{L} = \{2, 3, 11, 17, 37, 67, 71\},$$

and the finite correction set

$$\mathcal{C}' = \left\{ \pm \prod_{p \in I} p : I \subseteq \mathcal{B} \cup \mathcal{L} \right\}.$$

It has 2^{14} elements. All these primes belong to the original exceptional set S_*; the change introduces no new exceptional prime in the height theorem.

Use the same six rational scalar tests and their same orders $(1,1,1,1,48,48)$, with $J_0 = H$. In their fixed binary norm tree replace r_β by R_β, including in every leaf denominator. Thus the six leaves are divided by the assigned powers of R_β; the constant leaf remains 1 on the all-left path. Denote the root by W, and require $W \neq 0$. Choose $c \in \mathcal{C}'$ on a finite branch, and put

$$b = cR_\beta W^2, \qquad B = 17b, \qquad h = e(B - b^2).$$

Keep just the two structural coordinates e, s, together with the two finite square roots

$$p^2 = B(1-e)(1+3e), \qquad q^2 = \frac{(s^2 - Bh)(s^2 + 3Bh)}{B}.$$

The old preparation coordinate t_* and its rational expression U_* are absent. The guard $b \neq 0$ is retained.

Lemma 13.7 (The fixed quadratic and the local correction basis)**.** *The polynomial δ has no root modulo any prime of $\mathcal{L}$. The squareclasses of the seven primes in $\mathcal{L}$ form a basis of*

$$\mathbb{Q}_2^*/\mathbb{Q}_2^{*2} \;\times\; \mathbb{Q}_3^*/\mathbb{Q}_3^{*2} \;\times\; \mathbb{Q}_{17}^*/\mathbb{Q}_{17}^{*2}.$$

If an integer β makes a prime divide both r_β and $\delta(\beta)$, that prime belongs to $\mathcal{B}$. At any finite scalar contact of a rational β outside S_*, $\delta(\beta)$ is a unit.

Proof. Modulo two, $\delta = T^2 + T + 1$. At the odd primes of $\mathcal{L}$ its discriminant -91 is a quadratic nonresidue. These are finite residue computations. For clarity the local squareclass vectors are displayed below. The columns are the valuation at two, the two unit coordinates for the basis $(-1, 5)$ in $\mathbb{Z}_2^*$, the valuation and nonsquare-unit coordinate at three, and the valuation and nonsquare-unit coordinate at seventeen:

2	1	0	0	0	1	0	0
3	0	1	1	1	0	0	1
11	0	1	1	0	1	0	1
17	0	0	0	0	1	1	0
37	0	0	1	0	0	0	1
67	0	1	1	0	0	0	0
71	0	1	0	0	1	0	1

Their matrix has rank seven over $\mathbb{F}_2$, proving the basis claim by the usual local squareclass descriptions.

The relevant evaluations of the quadratic are

$$\begin{aligned} \delta(0) &= 23, \qquad \delta(1) = 5^2, \\ \delta(2401) &= 5^2 \cdot 349 \cdot 661, \\ 32^2\delta(-343/32) &= 5^2 \cdot 5209, \\ 4\delta(1/2) &= 5 \cdot 19. \end{aligned}$$

For an odd prime dividing the scalar contact polynomial, reduction therefore gives exactly the asserted list of possible common primes. The prime two cannot divide $\delta(\beta)$ for integer β. The same evaluation argument shows that at a rational finite contact outside S_* the quadratic is a unit: the contact value and its fixed denominators are integral there, and its displayed evaluation is a unit. This proves both remaining statements. □

Lemma 13.8 (Preservation of scalar soundness)**.** *The modified scalar certificate implies the same estimate*

$$h(\beta) \le H\sqrt{|a|}$$

as the previous scalar-height block.

Proof. First, $\beta = X^2+2$ has no odd contact with infinity at any finite prime. Indeed, if $v_p(\beta) < 0$, then $v_p(X) < 0$ and $v_p(\beta) = 2v_p(X)$ is even. It therefore suffices to consider the finite contacts counted by the source's radical $M_5(\beta)$.

At each such odd contact outside S_*, the preceding lemma makes $\delta(\beta)$ a unit. The fixed factor ϵ and every $c \in \mathcal{C}'$ are also units there. Consequently $v_p(R_\beta) = v_p(r_\beta)$ is positive and odd. The source's no-cancellation argument for $N_R(u,v) = u^2 + Rv^2$ now applies unchanged, with $R = R_\beta$. The pole condition on cRW^2 forces the six original rational tests to their assigned orders. In particular it gives the order-48 divisibility of the scalar index a at every prime in $M_5(\beta)$. Thus $M_5(\beta)^{48} \mid a$, and the original five-contact height theorem gives the stated estimate, with its original constant H. No test has been divided by an extra, possibly nonunit, normalizer: only the contact parameter of the norm construction has changed. □

Lemma 13.9 (Completeness with preparation zero)**.** *For every integral input the shared-point construction can satisfy the modified scalar block together with all of its comparison, height, and size conditions.*

Proof. Fix an integer input, so $\beta \ge 2$ is an integer. First choose the subset of $\mathcal{B}$ in c so that

$$v_p(cr_\beta) \equiv 0 \pmod 2 \quad (p \in \mathcal{B}).$$

The factors ϵ^2 and $\delta(\beta)^4$ do not change these parities. Choose the sign of c so that $cr_\beta > 0$. Finally the independent basis $\mathcal{L}$ supplies a subset making cr_β, and hence cR_β, square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$. The two prime sets are disjoint, so this last choice preserves all prescribed parities at $\mathcal{B}$.

The reduced denominator of cR_β is a square. At the primes of $\mathcal{B}$ its valuation is even. Away from $\mathcal{B}$, if $p \mid \delta(\beta)$, then p divides neither r_β nor c: this follows from the resultant calculation and the absence of roots modulo primes of $\mathcal{L}$. Its valuation is then a multiple of four. At the remaining primes the valuation is nonnegative, since the only fixed denominator of ϵ is supported at $23 \in \mathcal{B}$.

At two, three, and seventeen the quadratic is a unit and the factor $204^2 = (2^2 \cdot 3 \cdot 17)^2$ gives

$$v_2(cR_\beta) \ge 4, \qquad v_3(cR_\beta) \ge 2, \qquad v_{17}(cR_\beta) \ge 2.$$

The largest absolute value of a correction is

$$C_{\max} = \prod_{p \in \mathcal{B}\cup\mathcal{L}} p = 518513917206770054130.$$

For real $\beta \ge 2$ one has

$$\frac{|r_\beta|}{\delta(\beta)^4} < \frac{1202 \cdot 204 \cdot 2}{\beta^3} \le 61302 < 2^{16}.$$

The exact integer inequality

$$2^{16} C_{\max} 204^2 < 23^{24}$$

therefore gives $0 < cR_\beta < 1$. All numerical and finite-field computations in this paragraph are reproduced by the companion exact checker.

It remains to choose the common point and the anchor point so that their norm root W is a unit at the finite support of cR_β, and $|W-1|<1/2$ in the real absolute value. The local condition is the same formal-group construction as before: take $H_*=[h_0]P$ sufficiently deep at those finitely many places, then take S sufficiently deeper relative to H_*. Every nonconstant normalized leaf can consequently have an arbitrarily large positive valuation. Even when $v_p(R_\beta)<0$, this makes each nonconstant summand of every norm node divisible by p; the constant leaf on the all-left path then gives $W\equiv 1 \pmod p$.

We give the simultaneous real choice explicitly, since it is needed to remove the preparation coordinate. Let G be the closure of the cyclic subgroup generated by the pair of real embeddings of P, let G^0 be its identity component, and let m be the order of G/G^0. The closure of any subgroup $k\mathbb{Z}P$ contains G^0; if $m\mid k$, it is exactly G^0. The projection of G^0 to either real elliptic curve is nontrivial, since that projection of P has infinite order. Thus there are small nonempty open sets in G^0 near its identity which avoid the identity in both projections.

Choose such a relatively compact open set Ω, so close to the identity that the two rational coefficients of $f(H)$ are as small as needed for all $H\in\Omega$. Shrink it to have $z(H)$ bounded away from zero at both real embeddings. Choose h_0 positive, divisible by $64m$, satisfying the local formal conditions and with $[h_0]P\in\Omega$. Such a choice follows from the density of the required finite-index subgroup in G^0. Choose the positive index of S in the required finite-index subgroup, satisfying its anchor and local conditions, and with S sufficiently close to the real identity. Then all six scalar tests, including the two coefficients of $z(S)/z(H)$, are as small as prescribed, uniformly for $H\in\Omega$. At zero values of all six tests the fixed norm root is exactly one. Its continuity therefore gives $|W-1|<1/2$, and in particular $0<W<2$, throughout this real choice. The real conditions on S are compatible with any required size lower bound because a positive orbit in the finite-index subgroup returns infinitely often to the chosen open set.

After fixing S and the Lucas root indices, prescribe the common index modulo L, where $64m\mid L$, so that all scalar local conditions persist near h_0 and all small-prime comparison conditions hold. Inside Ω remove the pole loci of the finitely many comparison functions and then choose a nonempty open set with compact closure avoiding those loci. Each locus has empty interior: its projection is a finite preimage of a point under a nonzero multiplication map on a real elliptic curve; all the root labels are nonzero by the completeness choice. The comparison functions are uniformly bounded on the chosen compact set.

The simultaneous prime-pattern theorem is now applied as in the previous sharing proof, including every odd factor of L except seven in its avoidance modulus. Its two order moduli U,V satisfy

$$\gcd(U,V)=4,\quad \gcd(U,L)=64,\quad \gcd(V,L)=4.$$

Thus $h\equiv 0 \pmod U$, $h\equiv 4 \pmod V$, and $h\equiv h_0 \pmod L$ are compatible. The final progression consists of multiples of m and has closure G^0 in the real group. It therefore contains arbitrarily large positive indices whose real points lie in the chosen compact-open neighborhood. The complete comparison and detector construction goes through, and the scalar inequalities and local units are preserved.

For this final common point,

$$b=cR_\beta W^2 \quad \text{satisfies} \quad 0<b<4<17.$$

It has square reduced denominator: at every prime in the support of cR_β the norm root is a unit, and at all other primes its additional valuation is even. It is square at two, three, and seventeen and has valuations there at least $4,2,2$, respectively. The preparation-zero pole-completeness

lemma therefore supplies e, s, p, q. All added guards hold, and no preparation coordinate has been used. □

Theorem 13.10 (Conditional forty-three-variable construction)**.** *Under the same arithmetic inputs as the shared-point construction, the scalar modification above preserves the integer input projection. After finite-algebra compression it uses at most* 43 *total rational coordinates with the indexed finite exclusion, or* 45 *with the semilocal anchor. The corresponding witness counts are* 42 *and* 44*. These are exact construction bounds, not claims that a coefficient table of the final polynomial has been expanded.*

Proof. Soundness changes only the scalar-height input, which was proved unchanged. The preceding lemma proves simultaneous completeness. The scalar preparation coordinate is deleted, while its two square roots are retained as finite algebra generators. Thus the retained base dimension decreases from 43 to 42, or from 45 to 44 with the semilocal anchor. The triangular algebra still has rank at most 2^{39}, or 2^{40}, respectively. The enlarged fixed correction set is combined by a product over its branches and adds no rational coordinate. The finite-algebra theorem supplies one final coordinate, giving the asserted totals. □

13.10 Fewer finite choices and smaller algebra rank

The following two refinements preserve the retained-coordinate counts. They improve the subsequent degree bound by removing unnecessary finite choices and finite algebra generators. They apply in particular to the forty-three-coordinate construction.

Lemma 13.11 (Fixed nonzero height scales)**.** *Write $x(Q) = x_0 + x_1\sqrt{2}$ for any nonidentity root point of the shared construction. One may replace the two piecewise height scales attached to these components by*

$$A_0 = x_0^2, \qquad A_1 = (1 + x_1^2)^2.$$

Both are positive nonzero rational squares. This removes all eight zero-component case parameters and their closed equations, without changing the arithmetic soundness and completeness proofs. The original point-height bound with constant $7H$ still holds.

Proof. The point Q is the double of a halfpoint (a, b) on $y^2 = x^3 - Dx$, with $D = (3 - \sqrt{2})^2$ and $b \neq 0$. The duplication formula gives

$$x(Q) = \left(\frac{a^2 + D}{2b}\right)^2.$$

At each real embedding the numerator $a^2 + D$ is strictly positive and the denominator is nonzero. Thus both real embeddings of $x(Q)$ are positive, and their average x_0 is positive. The assertions about the two scales follow.

Put $T = (H + \log 3)|N|^\alpha$, where α is the radical exponent used in the applicable height comparison. The common slope detector and each scaled detector give

$$h(\lambda), h(A_i\lambda) \leq T, \qquad h(A_i) \leq 2T.$$

For the first component $h(A_0) = 2h(x_0)$, so $h(x_0) \leq T$. For a rational number $x_1 = a/b$ in lowest terms with $b > 0$,

$$h(1 + x_1^2) = \log(a^2 + b^2) \geq 2h(x_1), \qquad h(A_1) \geq 4h(x_1).$$

The displayed equality uses $\gcd(a^2+b^2, b^2) = 1$. Therefore $h(x_1) \le T/2$. The standard height inequality for the fixed quadratic basis gives

$$h(x(Q)) \le h(x_0) + h(x_1) + \tfrac{3}{2}\log 2 \le \tfrac{3}{2}(H + \log 3)|N|^\alpha + \tfrac{3}{2}\log 2 < 7H|N|^\alpha,$$

because $H \ge 1$ and $|N| \ge 1$. Thus the soundness estimates are preserved, with room in the same constant. Every completeness argument uses only that the scales are fixed positive rational squares once the root points are chosen; the new choices have exactly those properties. All nine detector constructions and their simultaneous prime patterns remain available. □

Lemma 13.12 (Parametrizing the first pole conic)**.** *Let $B \in \mathbb{Q}^*$ and consider*

$$p^2 = B(1-e)(1+3e).$$

Its rational points are covered by two rational formulas with one parameter. For a finite choice $\sigma \in \{0,1\}$ retain one rational coordinate t, put $D_t = t^2 + 3B$, require $D_t \neq 0$, and set

$$e = \frac{t^2 + (4\sigma - 1)B}{D_t}, \qquad p = -\frac{4Bt(1-\sigma)}{D_t}.$$

Every permitted assignment satisfies the conic equation, and every old rational point occurs in one of the two choices. In particular the square root coordinate p can be eliminated exactly.

Proof. For $\sigma = 0$,

$$1 - e = \frac{4B}{D_t}, \qquad 1 + 3e = \frac{4t^2}{D_t},$$

which gives the equation. For $\sigma = 1$ the formulas give $e = 1, p = 0$ directly. Conversely, if an old point has $e \neq 1$, put $t = p/(e-1)$. Substitution of its conic equation gives

$$t^2 + 3B = -\frac{4B}{e-1} \neq 0.$$

The formulas with $\sigma = 0$ recover both e and p. If $e = 1$, then $p = 0$; the choice $\sigma = 1, t = 0$ gives this point with denominator $3B \neq 0$. The other point with $p = 0$, namely $e = -1/3$, is already covered by $\sigma = 0, t = 0$. Accordingly no nonzero guard on p or on t is imposed. □

Apply this lemma separately to the first conic in each of the nine coordinate detectors and in the scalar block. Their B values are nonzero by the existing guards. Each block now retains its conic parameter t and its coordinate s, and keeps only the second square root q as a finite algebra generator. The parameter t here parametrizes a conic; it is distinct from the scalar preparation coordinate that was removed in the preceding subsection. The exceptional conic choices are fixed finite parameters, not quantified selectors.

The preferred finite-anchor source therefore has 42 retained coordinates and twenty finite generators: nine quartic point generators, ten quadratic pole generators, and one quadratic Lucas generator. Its algebra rank is at most

$$4^9 2^{11} = 2^{29}.$$

The semilocal source has 44 retained coordinates and one additional quadratic anchor generator, hence rank at most 2^{30}. Finite-algebra compression still gives totals 43 and 45, respectively. The ten conic choices replace the ten now-absent first-root zero/nonzero strata, whereas all eight zero-component choices have disappeared. The generic determinant compiler must use these updated ranks and finite branch lists for its final degree bounds.

14 The compiled shared-point polynomials with Lucas proxies

We apply the explicit finite-algebra construction to the preferred sources `source-43-conic.json.gz` and `source-45-conic-semilocal.json.gz`. This section proves the algebraic conversion performed by `compile_shared_source.py`. The arithmetic argument identifying the source projection with $\mathbb{Z}$ is a separate input; it uses the pinned manuscript, Sun's stated theorem, the scalar normalization of Section 13.9, and the degree refinements of Section 13.10. The compiler preserves the rational projection of each supplied source, including every exceptional quadratic case.

14.1 Retained coordinates and rational parametrization

The finite-anchor source retains 42 rational coordinates: the input, eighteen abscissa coordinates for nine points over $\mathbb{Q}(\sqrt{2})$, two Lucas parameters, one common slope, eighteen parameters for the nine coordinate poles, and two parameters for the scalar pole. Its remaining finite choices consist of nine point roots, one Lucas root, and ten second pole roots. There are twenty formal algebra generators. The semilocal source retains two additional anchor coordinates and has one additional finite anchor root.

The first pole roots have already been eliminated by Lemma 13.12. More explicitly, for each nonzero B the conic

$$p^2 = B(1-e)(1+3e)$$

is parametrized using a retained coordinate t and a fixed choice $\sigma \in \{0,1\}$:

$$e = \frac{t^2 + (4\sigma - 1)B}{t^2 + 3B}, \qquad p = -\frac{4Bt(1-\sigma)}{t^2+3B}, \qquad t^2 + 3B \neq 0.$$

The choice zero covers all points with $e \neq 1$: its inverse is $t = p/(e-1)$, and the conic equation gives $t^2 + 3B = -4B/(e-1) \neq 0$. The choice one covers $e = 1, p = 0$; $t = 0$ is permitted because $B \neq 0$. The point $e = -1/3, p = 0$ is covered by the choice zero with $t = 0$. No nonzero guard on p or t is imposed. Thus this operation preserves the rational input projection while removing one finite quadratic root in each of the ten pole blocks.

The fixed nonzero square height scales of Lemma 13.11 are

$$x_0^2, \qquad (1 + x_1^2)^2.$$

They eliminate the eight component-zero finite choices and their closed conditions. Consequently each preferred source has $16384 \cdot 2^{10} = 2^{24}$ original finite parameter choices: the scalar correction and the ten conic charts. These are fixed product indices, not quantified coordinates.

14.2 Integral triangular monic equations

Write $a = a_0 + a_1\sqrt{2}$ and $a^3 - (11 - 6\sqrt{2})a = A + B\sqrt{2}$. In the fixed point chart, a halfpoint ordinate is reconstructed as

$$b = C + \frac{B}{2C}\sqrt{2}, \qquad C \neq 0, \qquad C^4 - AC^2 + \frac{B^2}{2} = 0.$$

The source variable $Y = 2C$ satisfies the integral monic equation

$$Y^4 - 4AY^2 + 8B^2 = 0.$$

Its derivative $4Y^3 - 8AY$ is an explicit source guard. Hence all retained point roots are simple, including after base change to an algebraic closure. Equivalently, the discriminant of the quadratic in Y^2 is $16(A^2 - 2B^2)$, and the source guards imply both $Y \neq 0$ and $A^2 - 2B^2 \neq 0$.

The Lucas equation has the form

$$\tau^2 - a\tau + b = 0,$$

where a, b are integer polynomials in retained parameters. Set $Z = 2\tau - a$ and $\Delta = a^2 - 4b$. The equation becomes $Z^2 - \Delta = 0$, with inverse $\tau = (Z + a)/2$. This coordinate change is invertible over $\mathbb{Q}$, including when $\Delta = 0$. A repeated Lucas root is therefore not discarded.

Every other finite equation is already monic and quadratic with no linear term. Earlier substitutions can introduce powers of two in its coefficient. Suppose its current form is

$$y^2 + 2^{-k}N = 0,$$

where N is an integer polynomial in retained coordinates and earlier formal generators. With $s = \lceil k/2 \rceil$ and $Z = 2^s y$, it becomes

$$Z^2 + 2^{2s-k}N = 0.$$

The exponent $2s - k$ is zero or one. These changes preserve triangularity and are invertible over $\mathbb{Q}$. The producer records each exact numerator and power of two. The independent checker verifies the identities by induction through the coefficient DAG, without expanding large polynomials.

14.3 Quadratic exceptions and geometric simplicity

For every resulting equation $Z_i^2 + c_i = 0$, take a fixed binary choice between

choice	monic algebra relation	additional condition
0	$Z_i = 0$	$c_i = 0$,
1	$Z_i^2 + c_i = 0$	$Z_i \neq 0$.

The condition $c_i = 0$ is imposed as a closed residual, not as a quotient relation, so it does not destroy freeness of the presented algebra. The second case contributes Z_i to the guard. These cases cover precisely all rational roots. In the zero case the diagonal derivative is one; in the nonzero case it is $2Z_i$, nonzero on the guarded locus.

The compiler forms the original guard from its distinct structural factors, discarding repeated factors and nonzero integer constants. The source rational-square clearings $Z = Dq$, $Z^2 = ND$ for $q^2 = N/D$ are checked exactly, including the guard on D. The closed-condition and indexed-anchor argument denominators are also checked against the recorded guards. After the substitutions, the coefficient numerators are integer polynomials and the discarded numerical denominators are powers of two.

The full guard is the product of the original factors, the nonzero-case quadratic generators, and the indexed anchor when present. Each point derivative is among the original factors. Thus every diagonal derivative of the triangular monic presentation is nonzero on the guarded open subset. At every geometric point of that subset the fiber has local length one. This is the exact hypothesis of Theorem 9.1; generic separability alone is not being used.

There are eleven quadratic choices for the finite-anchor source and twelve for the semilocal source. A mask with k nonzero choices has free rank $4^9 2^k$, with respective maxima 2^{29} and 2^{30}. All source parameters, quadratic masks, separating integers, and degree cases in the theorem are literal finite product indices. Its formal algebra generators are eliminated by the determinant and coefficient operations.

14.4 The fixed elliptic exclusion product

For the finite-anchor source, the guard also contains a fixed product excluding small elliptic indices. The integer

$$M = 2^{9566463}$$

and the fixed point P are part of the specification. Write the variable abscissa as $(N_0 + N_1\sqrt{2})/D$ after exact denominator conversion. For each finite nP, compute its primitive triple

$$x(nP) = \frac{a_n + b_n\sqrt{2}}{c_n}.$$

The exact integer factor is

$$(c_nN_0 - a_nD)^2 - 2(c_nN_1 - b_nD)^2.$$

For an identity multiple the factor is one. The product runs from $n = 1$ to M, and D is already guarded. Since $\sqrt{2} \notin \mathbb{Q}$, a factor vanishes at a rational specialization exactly when the abscissa equals $x(nP)$. The triples are defined by the exact binary multiplication algorithm of Section 12. Thus all integer coefficients and index bounds are specified, with no additional quantified variables. The enormous collection of individual factors is not materialized.

The semilocal source uses its quadratic anchor instead and has no indexed elliptic exclusion product. Its complete coefficient DAG is materialized. Both variants use the outer indexed determinants and finite products of the finite-algebra theorem.

Proposition 14.1. *The preferred emitted specifications define integer polynomials P_{43} and P_{45} in respectively 43 and 45 coordinates, including the input. Their rational input projections equal those of their corresponding supplied sources. They have 42 and 44 existential witnesses, respectively. Subject to the arithmetic characterization of those sources, each is a Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$.*

With $H(P)$ denoting the largest absolute coefficient, the verified bounds are

	maximum algebra rank	$\deg P$	$H(P)$
P_{43}	2^{29}	$2^{9566680}$	$2^{2^{86098389}}$
P_{45}	2^{30}	2^{243}	$2^{2^{2^{246}}}$.

The degree and height entries are upper bounds, not asserted exact values.

Proof. The conic parametrization covers every rational source solution, and all subsequent rational coordinate changes are invertible. The two quadratic cases cover all roots and impose exactly the original equations. Denominator coverage makes the cleared source equations and guards equivalent to the original ones. The fixed anchor clearing is equivalent by the norm identity just proved. Hence every finite branch has its intended rational projection. The diagonal derivative check permits application of Theorem 9.1. Its one added coordinate gives the stated counts, and a finite product realizes the union of all source choices and masks.

For the bounds, apply the weighted reduction and multiplication matrix norm estimates of Section 9.1 with nine quartics and respectively eleven or twelve quadratic choices. Proposition 12.3 bounds the indexed fixed exclusion product. Every generator weight, operator norm estimate, determinant and resultant bound, and finite factor count is recorded in the specification and checked independently. They give the displayed rounded estimates. For the indexed exclusion product the estimates are stored as exact powers of two, so the bound computation itself does not require materializing integers with millions of digits. □

The specifications contain 6224 and 6245 coefficient-DAG nodes. They are exact compact descriptions using integer polynomial operations, coefficient extraction, determinants, and explicitly bounded finite products. They are not expanded coefficient tables or circuits listing every gate of rank-sized matrices. An independent exact interpreter executes the complete theorem on small algebras, including nilpotents, root collisions, degree drops, and zero quadratic cases. The theorem and source conversion proof justify the unexpanded large instances.

The other emitted source variants are preserved in separate packages for independent review of the arithmetic replacements. All use the same compiler and the same proof of algebraic conversion. The package index records their individual counts, bounds, hashes, and passing verification receipts.

15 Testing the integer relation without proxy coordinates

The first new improvement removes the three rational Lucas proxies from the shared-point construction. Two were retained coordinates; the third was a finite quadratic root. Their role was to impose one integer polynomial relation on three elliptic-group labels. The relation can instead be tested directly on the approximate logarithmic ratios. This gives a conditional definition with 41 total rational coordinates, including the input, or 43 with the semilocal anchor.

15.1 An elementary substitution principle

Lemma 15.1. *Let L be a number field, let $\mathfrak{p}$ lie over a rational prime p, and normalize its valuation by $v_{\mathfrak{p}}(p) = 1$. Suppose $z_i \in L$ and $n_i \in \mathbb{Z}$ satisfy*

$$v_{\mathfrak{p}}(z_i) \geq 0, \qquad v_{\mathfrak{p}}(z_i - n_i) \geq k \quad (1 \leq i \leq r).$$

For every $F \in \mathbb{Z}[T_1, \ldots, T_r]$,

$$v_{\mathfrak{p}}\big(F(\mathbf{z}) - F(\mathbf{n})\big) \geq k.$$

Consequently a requirement $v_{\mathfrak{p}}(F(\mathbf{z})) \geq k$ is equivalent to $p^k \mid F(\mathbf{n})$ when k is a positive integer. The same congruence assertion holds for rational-coefficient polynomials whose coefficient denominators are units at p; in that case the final condition is $v_p(F(\mathbf{n})) \geq k$.

Proof. For each monomial, replace its factors one at a time. Every term of the resulting telescoping difference contains some $z_i - n_i$; all its other factors, including the coefficient, are integral at $\mathfrak{p}$. Thus every term has valuation at least k. The nonarchimedean triangle inequality proves the assertion. The final equivalence follows because $F(\mathbf{n})$ is a rational integer and the normalized valuation restricts to v_p on it. □

This principle does not make the ratios equal to integer labels. It transfers a polynomial relation modulo the selected prime powers. The global height comparison remains the step that forces the integer residual to vanish exactly.

For a quantitative general version, write $\|F\|_1$ for the sum of the absolute values of its coefficients. If $\deg F = d$, $|n_i| \leq B$, and $B \geq 1$, then

$$|F(\mathbf{n})| \leq \|F\|_1 B^d.$$

Thus a height argument of the form

$$B^2 - 6 < 7H|F(\mathbf{n})|^{24/k}$$

forces $F(\mathbf{n}) = 0$ whenever

$$k > 12d, \qquad B > \Big((7\|F\|_1^{24/k} + 6)H\Big)^{1/(2-24d/k)}.$$

Indeed, if the integer residual were nonzero, the first inequality and $B, H \geq 1$ would imply the reverse strict inequality on B. For several integer relations one can either test them separately, or use the single relation $\sum_j F_j^2 = 0$. The latter doubles the largest polynomial degree and its precision requirement must be adjusted accordingly.

15.2 The replacement in the shared construction

Retain the nine nonidentity points

$$A, S, T, Z, H, U_A, U_S, U_T, U_Z \in 2E(K),$$

where $K = \mathbb{Q}(\sqrt{2})$, and all the point, subtraction, logarithm and denominator guards of the preferred shared source. Their root labels are (n, a, t, b) for (A, S, T, Z), respectively. These are integers used in the proof; they are not rational coordinates of the polynomial. Put $\beta = X^2 + 2$.

Let L_i, L_i', $1 \leq i \leq 4$, be the primary and conjugate alternate logarithmic ratios from the previous construction, with orders $(16, 64, 64, 64)$. Delete the rational coordinates ξ, ζ and the monic quadratic root τ, together with the equation involving them. Define

$$F(U, V, W) = V^2 - U(U-2)VW + (U-2)^2W^2 - 1.$$

Use the following sixteen field tests in every coordinate detector:

$f(U_0), \ldots, f(U_4), \sigma f(V_0), \ldots, \sigma f(V_4)$	$1 \quad (10 \text{ tests})$
$L_i - L_i' \quad (1 \leq i \leq 4)$	$16, 64, 64, 64$
$L_1 - \beta$	16
$F(L_2, L_3, L_4)$	$64.$

Here $U_0 = H$, $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T, Z)$, and $V_i = U_i - [4]Q_i$, exactly as before. There are ten tests of order one, two of order sixteen, and four of order sixty-four. Every other part of the preferred source is retained: its fixed positive square scales, nine reciprocal-trace detectors, normalized scalar-height block sharing H, conic parametrizations, and the chosen size anchor.

Proposition 15.2 (Preservation of the integer input projection)**.** *Under the explicitly stated arithmetic inputs of the shared construction, the modified system has input projection exactly $\mathbb{Z}$.*

Proof. Consider first a rational solution and a prime counted by one of the nine height radicals. The detector selects a place above that prime at which all sixteen tests have their assigned orders. The two-anchor argument uses only the first ten tests and the four differences $L_i - L_i'$. Hence it still proves

$$L_i \text{ is integral}, \qquad L_i \equiv n_i \pmod{p^{k_i}}, \qquad (n_1, n_2, n_3, n_4) = (n, a, t, b).$$

In particular this conclusion does not require any proxy equation.

Lemma 15.1 now gives

$$F(L_2, L_3, L_4) \equiv \Delta := t^2 - a(a-2)tb + (a-2)^2b^2 - 1 \pmod{p^{64}}.$$

The new order-64 test therefore says exactly that the integer Δ is divisible by p^{64}. The radical divisibilities used in the original height argument are unchanged. If $B = \max(|n|, |a|, |t|, |b|)$ and $\Delta \neq 0$, that argument gives

$$B^2 - 6 < 7H|\Delta|^{3/8}, \qquad |\Delta| < 5B^4.$$

The unchanged anchor contradicts these estimates. Therefore $\Delta = 0$, and the nonzero root label b yields the same Lucas growth inequality $\log B > |a| - 3$.

The normalized scalar block is unchanged and still gives $h(\beta) \leq H\sqrt{|a|}$. The order-16 test $L_1 - \beta$ and the corresponding cross comparison give the same divisibility of the numerator of $\beta - n$. If $\beta \neq n$, the resulting bound

$$\log B < 2\log(32H) + 3h(\beta)$$

contradicts the anchor and the Lucas growth inequality as before. Thus $\beta = n \in \mathbb{Z}$, and $X \in \mathbb{Z}$.

For completeness, start with an integer input. Choose the same integer root labels as in the previous completeness proof, so $F(a, t, b) = 0$. At every selected place, choose the common index and points so that $L_i \equiv n_i \pmod{p^{k_i}}$ and the same holds for the alternate ratios. The lemma then gives $v_{\mathfrak{p}}(F(L_2, L_3, L_4)) \geq 64$, which is the only requirement on this new test. Exact vanishing of $F(L_2, L_3, L_4)$ is not required.

The new test is a fixed polynomial in the old ratios. Its real values are bounded on the same compact sets avoiding the old denominators, and its valuations have finite lower bounds in the same small-prime neighborhoods. Consequently the normalization, prime-pattern, local-square and preparation-zero completeness arguments apply without alteration. No new denominator or exceptional prime is introduced by F itself. This constructs all the remaining witnesses for every integral input. □

15.3 Exact coordinate and finite-algebra counts

With the finite size exclusion, the retained coordinates are

$$\underbrace{1}_{X} + \underbrace{18}_{\text{nine halfpoint abscissa pairs}} + \underbrace{1}_{\lambda} + \underbrace{18}_{\text{nine pole pairs}} + \underbrace{2}_{\text{scalar pole pair}} = 40.$$

The finite algebra has nine quartic point generators and ten quadratic pole generators. Its rank is at most

$$4^9 2^{10} = 2^{28}.$$

The finite-algebra construction adds one final rational coordinate, giving 41 total coordinates and 40 existential witnesses. The semilocal anchor adds two retained coordinates and one quadratic generator, giving rank at most 2^{29} and totals 43 and 42, respectively. The finite correction and conic choices are unchanged. The removed Lucas root has no remaining equation, repeated-root stratum, or denominator guard.

The files `source-41-direct.json.gz` and `source-43-direct-semilocal.json.gz` record the complete coefficient DAGs and finite parameters. The copied producer is `build_direct_source.py`. The frozen earlier source files are unchanged. Final degree and coefficient-height bounds come from the separately checked finite-algebra compilation of these new inputs.

16 Changing the input shift without adding coordinates

The substitution $\beta = X^2 + 2$ is convenient over $\mathbb{Q}$, but its interaction with the five scalar contacts is not stable under extension of the ground field. We first describe a uniform rational modification and then a field-dependent way to avoid every member of a prescribed finite exceptional set. Neither statement changes the elliptic cyclic-group hypothesis.

Proposition 16.1. *Let C be an integer with $C \geq 2$ such that $2401 - C$ is not a nonnegative integer square. Replace $\beta = X^2 + 2$ by $\beta = X^2 + C$ throughout the shared-point construction and the direct Lucas-polynomial refinement. Under their stated arithmetic inputs, the resulting rational input projection is still $\mathbb{Z}$. The retained-coordinate counts and finite-algebra ranks are unchanged. In particular every integer $C \geq 2402$ is permitted.*

Proof. Write $X = a/b$ in lowest terms, with $b > 0$. Then

$$\beta = \frac{a^2 + Cb^2}{b^2}, \qquad \gcd(a^2 + Cb^2, b^2) = 1.$$

Thus β is an integer if and only if X is an integer. Moreover, at any finite prime where β has negative valuation, $v_p(X) < 0$ and $v_p(\beta) = 2v_p(X)$ is even. These are precisely the input-specific facts used in scalar soundness: the finite five-contact radical accounts for every odd contact, without a new contact at infinity. The scalar height argument and the comparisons which force β to equal an integer label therefore remain valid.

For integral X, the new value of β is an integer at least two. It cannot equal $0, 1, -343/32$, or $1/2$. It cannot equal 2401, by the hypothesis on C. Hence the scalar contact product

$$r_\beta = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1)$$

is nonzero. The scalar normalization, local squareclass choices, prime-pattern construction, and simultaneous point choices apply to this integer value of β just as before. Their estimates were uniform for all integral $\beta \geq 2$ away from these contacts; they did not require the special numerical value of the constant two. In particular the same height constant and precision choices are valid. This proves completeness.

Changing a fixed coefficient introduces no coordinate and no new algebra generator. It also does not change the formal degrees of the source expressions. The independently computed upper bounds for the emitted coefficient formulas must nevertheless be recorded separately, since their numerical coefficient sizes can change. □

If a number field K has a real embedding and $C \geq 2402$, then $X^2 + C$ avoids every scalar contact for every $X \in K$. Indeed at that embedding its value is at least 2402, whereas every contact is real and at most 2401. The normalizing denominator $\beta^2 + \beta + 23$ is also nonzero: its real discriminant is -91. Thus $C = 2402$ removes these two specific domain obstructions over every number field with a real embedding. This observation does not imply that the resulting formula defines $\mathcal{O}_K$: the cyclic point group and the arithmetic comparison arguments require their own field-dependent replacements.

Lemma 16.2 (Avoiding a finite set by a translated square)**.** *Let K be a number field and let $\Theta \subset K$ be finite. There is an effectively searchable arithmetic progression of positive integers C such that*

$$\theta - C \notin K^2 \qquad (\theta \in \Theta).$$

In particular $X^2 + C$ avoids Θ for every $X \in K$. The progression contains integers greater than any prescribed bound.

Proof. For each $\theta \in \Theta$, choose a different odd rational prime ℓ_θ which splits completely in the normal closure of $K/\mathbb{Q}$ and at which θ is integral. Infinitely many such primes exist by Chebotarev's theorem [22, Section 2.1, Theorem 1]. Excluding the finitely many ramified primes, denominators, and previously used primes does not affect existence. Choose a prime $\mathfrak{p}_\theta$ of K above ℓ_θ. Its residue field is $\mathbb{F}_{\ell_\theta}$.

Choose a nonzero nonsquare $u_\theta \in \mathbb{F}_{\ell_\theta}$ and impose the integer congruence

$$C \equiv \overline{\theta} - u_\theta \pmod{\ell_\theta}.$$

Every such C makes $\theta - C$ a unit at $\mathfrak{p}_\theta$ with nonsquare residue. It cannot be a square in K: a square which is a unit has a unit square root and square residue. The Chinese remainder theorem imposes all these congruences simultaneously and supplies the desired arithmetic progression.

The construction is effective given exact number-field data: enumerate rational primes, test complete splitting by finite factorization outside the discriminant, compute residues, and test squares in the finite residue fields. Chebotarev guarantees that each required search terminates. No conjectural effective prime bound is needed for this assertion. □

For the scalar formulas one may take Θ to consist of the five rational contacts together with those roots of $T^2 + T + 23$ which lie in K. Choose $C \geq 2402$ in the resulting progression. Then both r_β and the normalizing denominator are nonzero for every $X \in K$, and the conditional definition over $\mathbb{Q}$ remains valid by Proposition 16.1. The shift is allowed to depend on K, but remains an ordinary fixed integer coefficient, not an extra variable. The field-independent choice 2402 suffices when a real embedding exists.

No fixed integer shift can remove these contacts over every number field. For any fixed C, the algebraic integer $X = \sqrt{-C}$ in $\mathbb{Q}(\sqrt{-C})$ satisfies $X^2 + C = 0$. The field-dependent choice in Lemma 16.2 is therefore essential if all number fields are to be considered.

17 The compiled polynomials with forty-one and forty-three coordinates

We apply the guarded finite-algebra theorem to the direct Lucas-polynomial source of Section 15. All formal generators, monic relations, coefficient conversions, denominator guards, and finite index ranges are supplied in the corresponding polynomial specification.

Theorem 17.1. *Under the stated source arithmetic inputs and Sun's simultaneous prime-pattern theorem, the supplied polynomial P_{41} defines $\mathbb{Z}$ in $\mathbb{Q}$ using forty-one total coordinates, including the input. It has forty existential witnesses and satisfies*

$$\deg P_{41} \leq 2^{9566673}, \qquad H(P_{41}) \leq 2^{2^{86098382}}.$$

The semilocal-anchor alternative P_{43}^{loc} uses forty-three total coordinates, forty-two existential witnesses, and satisfies

$$\deg P_{43}^{\mathrm{loc}} \leq 2^{236}, \qquad H(P_{43}^{\mathrm{loc}}) \leq 2^{2^{240}}.$$

Both are explicit indexed integer polynomials. These are upper bounds for degree and height, not exact expansion statistics.

Proof. The source has forty retained rational coordinates and nineteen formal algebra generators, nine of degree four and ten of degree two. Its maximum finite rank is 2^{28}. The semilocal alternative adds two retained coordinates and one quadratic generator, giving maximum rank 2^{29}. Their rational input projections are $\mathbb{Z}$ by Proposition 15.2.

The source-to-integer conversion is the same exact procedure proved in Section 14. For every rational square equation $q^2 = N/D$, its denominator is guarded and the substitution $Z = Dq$ gives $Z^2 = ND$. Subsequent powers-of-two clearings preserve the triangular monic presentation. Every quadratic generator is split into its zero case and nonzero case, with the appropriate closed equation or derivative guard. The nine point quartics retain their derivative guards. Thus the required geometric simplicity holds at every guarded point, including exceptional parameter specializations.

Theorem 9.1 eliminates these formal generators with one additional coordinate. It gives the stated totals and preserves the rational projection of each source. The emitted specifications have respectively 6018 and 6039 integer coefficient-DAG nodes. The independent checker verifies the source hash, all affine and rational-square changes, guard coverage, monicity, triangularity, every finite index range, and the recursive degree and coefficient majorants. The bounds are exactly those displayed above. The fixed elliptic exclusion product is treated by the canonical clearing and height estimate already proved for that product. □

The packages are `research/phase3/finite_fibre/polynomial-41/` and `research/phase3/finite_fibre/polynomial-43-semilocal/`. Each contains its complete indexed specification, construction report, and passing independent verification receipt. The large multiplication matrices and outer products are not expanded. The semilocal variant has a materialized coefficient DAG; the forty-one-coordinate variant additionally contains the explicitly indexed elliptic exclusion product.

The same producer also emits the two versions with input shift $C = 2402$ from Proposition 16.1. Their retained counts, maximum ranks, and displayed degree and height bounds are the same. They have 6019 and 6040 coefficient-DAG nodes, respectively, and separate passing receipts. The additional coefficient node is a fixed integer, not a quantified coordinate. These variants remove the scalar-contact obstruction over fields with a real embedding; they do not resolve the other arithmetic obstructions over extensions.

18 The quadratic compilation with Lucas proxy coordinates

The finite-algebra construction minimizes the number of coordinates by allowing a very high degree. For a degree-four presentation, auxiliary coordinates serve a different purpose: they keep each arithmetic relation quadratic. We now apply this method to the shared-point arithmetic system, using the common height point, the fixed square scales, comparison order 52, scalar order 32, and the semilocal size condition. The scalar pole in this quartic retains its preparation coordinate. The separate normalization which removes that coordinate in the smallest-variable construction is not used here; its extra arithmetic would cost more quadratic intermediate coordinates.

18.1 Comparison order fifty-two with an explicit larger anchor

The high comparison order can be lowered from 64 to 52 while keeping beta order 16 and scalar-height order 48. This does not change the number of retained coordinates of the shared

construction. It shortens eight reciprocal logarithms, and can therefore reduce the number of variables in a quartic circuit presentation. Such a circuit count is a separate, implementation-dependent assertion.

Proposition 18.1. *Assume the arithmetic inputs of the shared-point construction and* $1 \le H < 2^{4783227}$*. Replace every order-*64 *comparison by order* 52*, and replace the corresponding logarithms* $zG_{15}(Df^2)$ *by* $zG_{12}(Df^2)$*. Retain the beta order* 16*, the scalar order* 48*, and all other equations and guards. Either of the following size conditions is sufficient:*

$$|a| > (21H)^{13/2},$$

or the semilocal anchor with fixed constant

$$C = 3^{2^{26}}, \qquad u^2 + 6v^2 - 2w^2 - 3(f_0^2 + f_0 f_1 - f_1^2)^2 + 2C^4 = 0.$$

Here $f_0 + f_1\sqrt{2} = f(S)$*. Alternatively the finite exclusion condition can use* $M_{52} = 2^{31091008}$*. Each version has the same integer input projection as the order-*64 *construction under the stated arithmetic inputs.*

Proof. We first check that the changed logarithm has the required properties. The formal identity

$$\log_E(z) = z \sum_{j \ge 0} c_j (Df^2)^j, \qquad c_1 = -2/5, \quad c_j = c_{j-1}(4j-5)/(4j+1),$$

is independent of the chosen truncation. At a place above $p > 76176$, normalized by $v(p) = 1$, a nonzero point in the formal group has $r = v(z) \ge 1$ and $v(Df^2) = 4r$. The denominator of c_j divides $(4j+1)!$. For $j \ge 13$ with $4j + 1 < p$, the jth term therefore has valuation at least $52r$. Otherwise $j \ge 19044$ and

$$v\big(c_j (Df^2)^j\big) \ge 4jr - \frac{4j+1}{p-1} \ge 4jr - j \ge 52r.$$

These lower bounds tend to infinity, so they control the whole tail. Consequently

$$v\big(\log_E(z) - zG_{12}(Df^2)\big) \ge 53r, \qquad \frac{zG_{12}(Df^2)}{\log_E(z)} \in 1 + p^{52}\mathcal{O}_{K,\mathfrak{p}}.$$

This proof applies to either a split or an inert good prime, since both local extensions are unramified. The source's global nonvanishing argument applies to every G_J with $J \ge 2$ and hence to G_{12}. Thus all logarithm denominators remain nonzero.

The two-anchor comparison and the detector proof use only the specified precision; they now give order 52 for the three Lucas labels and order 16 for beta. The transformed contact-height estimate is unchanged. Put $a_0 = |a|$ and $B = \max(|n|, |a|, |t|, |b|)$. If the integer Lucas residual Δ is nonzero, the resulting estimates are

$$B^2 - 6 < 7H|\Delta|^{24/52}, \qquad |\Delta| < 5B^4.$$

Since $B, H \ge 1$, they imply

$$B^{2/13} < \big(7\,5^{6/13} + 6\big)H < 21H, \qquad B < (21H)^{13/2}.$$

The strict constant inequality follows by raising $5^{6/13} < 15/7$ to the thirteenth power; equivalently, $5^6 7^{13} < 15^{13}$. In particular replacing the factor by $14H$ would not be justified: $7\,5^{6/13} > 14$. The size hypothesis contradicts $B \geq a_0$, so $\Delta = 0$ and the unchanged Lucas lemma gives $\log B > a_0 - 3$.

The scalar estimate remains $h(\beta) \leq H\sqrt{a_0}$. If $\beta \neq n$, the order-16 argument remains

$$\log B < 2\log(32H) + 3h(\beta).$$

The new size hypothesis implies $a_0 > (20H)^2$, because $21^{13} > 20^4$ and $H \geq 1$. Hence $h(\beta) < a_0/20$. Also $\log(32H) < 20H^2 < a_0/20$ for $H \geq 1$. It follows that $\log B < a_0/4 < a_0 - 3$, a contradiction. Thus $\beta = n$ and $X \in \mathbb{Z}$ as before.

For the explicit constants,

$$\log_2\big((21H)^{13/2}\big) < \frac{13}{2}(5 + 4783227) = 31091008.$$

The exact source subgroup calculation for an anchor $C = 3^{2^q}$ gives

$$0 \neq a \in 2\,3^{2^{q-1}-1}\mathbb{Z}.$$

For $q = 26$, using $3^2 > 2^3$,

$$\log_2 |a| \geq 1 + (2^{25} - 1)\log_2 3 > 1 + \frac{3}{2}(2^{25} - 1) = 50331647.5 > 31091008.$$

The semilocal anchor is therefore sufficient. The alternative indexed exclusion with $M_{52} = 2^{31091008}$ is sufficient by the same inequality. Its enormous number of factors is part of the explicit indexed specification, not an expanded coefficient table.

Completeness still chooses the anchor index in an arbitrarily divisible nonzero subgroup, or larger than a fixed finite exclusion. All the prime-pattern, congruence, local-square, and real-smallness arguments are unchanged: lowering a required comparison precision only weakens the necessary formal divisibilities. This proves the claimed equality of input projections. □

For the detector arithmetic, with $F = F_5(A\lambda)$ and $\delta = A\lambda - 1$, the three possible weights become

$$q_1 = \frac{A^5\delta^2}{F}, \qquad q_{16} = \frac{A^{155}\delta^{152}}{F^{31}}, \qquad q_{52} = \frac{A^{515}\delta^{512}}{F^{103}}.$$

These follow from $q_m = A^{10m-5}\delta^{10m-8}/F^{2m-1}$. There are eight order-52 logarithms and two order-16 logarithms. Each changed logarithm loses exactly three terms. The fixed coefficient in the anchor is $2C^4 = 2 \cdot 3^{2^{28}}$; using it as a literal coefficient adds no variable.

Corollary 18.2 (Scalar order thirty-two)**.** *Under the hypotheses and the enlarged size condition of Proposition 18.1, the two order-*48 *scalar-height tests can be replaced by order-*32 *tests. This preserves the integer input projection and does not change the point inventory.*

Proof. The scalar-height argument with order m gives $h(\beta) \leq H|a|^{24/m}$. At $m = 32$ it gives $h(\beta) \leq H|a|^{3/4}$. The size condition implies

$$|a| > (21H)^{13/2} > (20H)^4,$$

because $H \geq 1$ and $21^{13} > 20^8$. Thus $h(\beta) < |a|/20$, exactly the estimate used in the last contradiction of Proposition 18.1. The Lucas part uses only the order-52 comparisons and is unchanged. Completeness chooses S deeper than a fixed scalar-height point, and decreasing this order preserves that construction, also when the height point is shared as proved above. This proves both implications. □

18.2 The domains of the division equations

Put $k = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, and $D = d^2$.

Lemma 18.3. *On $E : y^2 = x^3 - Dx$, the equation*

$$(2y)m = 3x^2 - D$$

forces $y \neq 0$ at every affine point, and hence defines the usual doubling slope. Every finite point of $2E(k)$ has nonzero abscissa and nonzero ordinate.

Proof. If $y = 0$, the curve equation gives either $x = 0$ or $x^2 = D$. The numerator $3x^2 - D$ is then respectively $-D$ or $2D$, both nonzero. Thus the slope equation cannot be satisfied when $y = 0$. This argument uses the curve equation, and does not need a separate assertion that $3x^2 - D$ never vanishes elsewhere.

Let $Q = 2R$ be finite. The point R is affine with nonzero ordinate, and its doubling formula gives

$$x(Q) = \left(\frac{x(R)^2 + D}{2y(R)} \right)^2 .$$

Since $D = d^2$ and k embeds in $\mathbb{R}$, the numerator is nonzero. Thus $x(Q)$ is a nonzero square in k. A point with zero ordinate has abscissa $0, d$, or $-d$. The last two cannot be squares in k: their norm to $\mathbb{Q}$ is 7, whereas the norm of a square in k is a rational square. Consequently $y(Q) \neq 0$ as well. □

The lemma shows that a halfpoint on the curve followed by the direct doubling-slope equations defines a finite point of $2E(k)$ with the required nonzero coordinates. Both usual coordinates of the doubled point are then given by their standard polynomial expressions in the slope, so they lie on E. Repeating this operation is valid by the same argument. If the usual subtraction slope is defined by its existing abscissa-inequality guard, the resulting finite difference of two points in $2E(k)$ also belongs to $2E(k)$ and has the same nonvanishing properties.

Therefore the parameter $z(Q) = -x(Q)/y(Q)$ may be specified by a direct two-coordinate quotient for every such point Q. It is itself nonzero. For the reciprocal logarithms used in the source,

$$\ell_J(Q) = z(Q) G_J\big(D/x(Q)^2\big),$$

the separately proved global nonvanishing of G_J for $J \geq 2$ makes $\ell_J(Q) \neq 0$. All truncations used here have $J \in \{3, 12, 15\}$. Ratios with these logarithms as denominators may likewise use direct quotient equations. This last step depends on the source's stated nonvanishing lemma, not on numerical evaluation.

The abscissa inequality for the subtraction slope is retained: if two points have the same abscissa, an unguarded quotient equation could have both sides zero. None of the preceding arguments removes that guard.

18.3 The exact arithmetic conventions

Represent a rational intermediate value by an affine linear form in the coordinates already introduced. Addition, subtraction, and multiplication by a fixed rational coefficient then require no new coordinate. A product of two nonconstant affine forms a, b is represented by one coordinate v and the quadratic equation $v - ab = 0$. A rational inverse uses $av - 1 = 0$. More generally, an

arbitrary quadratic expression q can be assigned to one new coordinate with $v - q = 0$. Common expressions are reused.

An element of $K = \mathbb{Q}(\sqrt{2})$ is a pair of rational affine forms. Its product with another pair is assigned to two coordinates, using

$$v_0 - a_0 b_0 - 2a_1 b_1 = 0, \qquad v_1 - a_0 b_1 - a_1 b_0 = 0.$$

This costs two coordinates, although an arithmetic algorithm minimizing scalar multiplications might instead use three products and free additions. The objective here is the number of existential coordinates. An inverse uses two coordinates and

$$a_0 v_0 + 2a_1 v_1 = 1, \qquad a_0 v_1 + a_1 v_0 = 0.$$

These equations hold exactly when the represented element is nonzero and (v_0, v_1) is its inverse. They enforce the necessary domain without introducing a separate norm and a separate inverse of that norm.

When the domain lemmas above already prove that $b \neq 0$, a quotient $z = a/b$ uses just the two equations

$$b_0 z_0 + 2b_1 z_1 = a_0, \qquad b_0 z_1 + b_1 z_0 = a_1.$$

Their determinant is $b_0^2 - 2b_1^2 \neq 0$, so their solution is unique. This rule is used for doubling slopes, local parameters, logarithm ratios, and the scalar local-parameter ratio, with the domain proof supplied in the preceding subsection. The subtraction slope retains an explicit inverse of the difference of abscissas. These domain arguments are part of the proof of the quartic presentation; a coefficient-table check alone would not justify dropping a denominator condition.

18.4 Reusing polynomial evaluations

Two elementary reorganizations save many auxiliary coordinates. First, write a polynomial with rational coefficients in blocks:

$$G(W) = \sum_{r=0}^{s} W^{br} \left(\sum_{j=0}^{b-1} c_{br+j} W^j \right).$$

After computing $1, W, \ldots, W^b$, each parenthesized expression is an affine linear combination of their two rational components. Horner evaluation in W^b finishes the calculation. This is the Paterson–Stockmeyer principle [11]. Here $b = 3$ for G_{12} and $b = 4$ for G_3. The identity is obtained simply by grouping the coefficients according to their exponent modulo b; no approximation of the logarithm is introduced by this evaluation rule.

Second, the eighteen field tests D_j are the same at every coordinate detector. If $\delta = A\lambda - 1$, the corresponding factor is

$$1 + q_m D_j^2, \qquad q_m = R^{1-2m} \delta^{-2m-2}, \qquad m \in \{1, 16, 52\}.$$

For each order m, form once the polynomial

$$E_m(U) = \prod_{j : m_j = m} (1 + U D_j^2) \in K[U].$$

Its coefficients are the elementary symmetric polynomials in the listed D_j^2. There are respectively ten, two, and six factors. For each of the nine detectors, evaluate $E_1(q_1)E_{16}(q_{16})E_{52}(q_{52})$. This is exactly the product defining W, with the field coefficients reused across all nine detectors.

Finally, for $W = w_0 + w_1\sqrt{2}$,

$$\mathrm{Tr}(W^{-1}) = \frac{2w_0}{w_0^2 - 2w_1^2}, \qquad B = \frac{R(w_0^2 - 2w_1^2)^2}{4w_0^2}.$$

The inverse of w_0 enforces the trace condition. Since $w_0 \neq 0$, the norm of W is nonzero in the quadratic field. Together with $R \neq 0$, this also gives $B \neq 0$, so an additional inverse of B would be redundant.

18.5 Finite choices at degree four

The point occurrences use four rational halfpoint coordinates and two curve equations. The three-coordinate quartic chart is more economical for finite-algebra elimination, but is not needed in this degree-four construction. The nine point occurrences are the four root points and the five shared comparison points; the first comparison point is also the scalar-height point.

The fixed square scales proved above remove all zero-component case choices. Only the scalar correction remains. For each $p \in \{2, 3, 5, 7, 11, 13, 17\}$ introduce a Boolean coordinate ϵ_p, and introduce one sign coordinate ϵ_-. Impose $\epsilon^2 - \epsilon = 0$ for all eight coordinates and compute

$$c = (1 - 2\epsilon_-) \prod_p \big(1 + (p-1)\epsilon_p\big).$$

Over $\mathbb{Q}$, this realizes exactly the required 256 signed squarefree corrections. Unlike a finite product of alternative polynomials, this keeps the total degree four after quadratic compilation, at the cost of eight coordinates.

18.6 Removing redundant quadratic coordinates

The compiler next performs two reversible operations. Replacing a residual q_i by $q_i + cq_j$, while retaining q_j, does not change the common zero set. If a residual has the form

$$av + q = 0, \qquad a \in \mathbb{Q}^\times, \quad v \notin q,$$

we may substitute $v = -q/a$ in all the other residuals and remove this coordinate and equation. Every zero of the new system reconstructs one and only one value of v; the converse follows by substitution. The compiler accepts such a substitution only if every resulting residual still has degree at most two. In particular a quadratic replacement is permitted only when that coordinate occurs linearly elsewhere. Its exact row operations and substitutions are retained as a replayable certificate.

After these operations, clear the rational denominator of each residual and divide its integer coefficients by their common divisor. Neither operation changes its rational zero set. If the resulting residuals are $q_1, \ldots, q_s$, define

$$P = \sum_{i=1}^{s} q_i^2.$$

The rational zero set of P is exactly their common zero set, because each summand is nonnegative. All these equivalences concern rational solutions and do not require numerical sampling.

Theorem 18.4 (The supplied quartics)**.** *Assume the source arithmetic inputs and Sun's simultaneous prime-pattern theorem used in the shared-point construction. The supplied polynomial P_{1367} has 1367 total variables, 1366 existential witnesses, and total degree four, and defines $\mathbb{Z}$ in $\mathbb{Q}$. It is specified by a complete sparse coefficient table over $\mathbb{Z}[\kappa]$, where*

$$\kappa = 3^{2^{28}}$$

is a fixed integer, not a coordinate. There is also a fully expanded ordinary integer quartic P_{1391} with 1391 total variables and primitive coefficient height less than 2^{1020}, with the same defining property.

Proof. Apply the just-described rules to the shared-point source with the precision and size choices specified at the start of this section. Each arithmetic coordinate has its prescribed value, except where the explicit finite Boolean choices are intended; the paired quotient domain lemmas establish the same assertion for the divisions. The reversible quadratic operations preserve the rational input projection. The source arithmetic theorem therefore identifies the final projection with $\mathbb{Z}$.

The exact implementation introduces 1367 coordinates after sixty quadratic-preserving eliminations. Its 1322 primitive residuals expand to 29,419 nonzero encoded entries over $\mathbb{Z}[\kappa]$, giving 29,415 distinct coordinate monomials after collecting the fixed constant's powers. The independent checker replays every arithmetic relation, row operation, and elimination and independently convolves every pair of residual terms. This checks the entire sparse table, including mixed coefficients. Its maximum variable degree is exactly four.

For the ordinary integer alternative, start with the fixed integer 3^{16} and use twenty-four successive squarings to compute $3^{16 \cdot 2^{24}} = 3^{2^{28}}$. The associated coordinates are retained to avoid expanding this integer into the coefficients. Their defining equations have degree at most two and uniquely determine their values. The resulting 1391-coordinate system has 1346 residuals and 29,489 nonzero expanded integer terms. The independent expansion check gives coefficient bit length at most 1020. The coefficient content is one. These facts prove the degree, height, and count assertions. □

The supplied data are in `research/phase2/quartic/quartic-best/polynomial.json.gz` and `research/phase2/quartic/quartic-small-coefficients/polynomial.json.gz`. An entry $[\mathbf{m}, k, c]$ means $c\kappa^k \prod_{i \in \mathbf{m}} X_i$, with repetitions in $\mathbf{m}$ recording powers. The index zero is the input. In the ordinary integer table every k is zero. This convention, the complete tables, and the exact construction give ordinary finite integer polynomials. No gate or coefficient symbol is silently omitted from the variable count.

The program `build_shared_quartic.py` is a producer; it is not trusted by the verification program `verify_shared_quartic.py`, which does not import it. The separate mathematical review checks the source formulas and the domain lemmas. Neither computational check is claimed to verify the external arithmetic theorems. No global minimality claim is made for 1367 variables: the theorem supplies a concrete upper bound for the quartic problem, while the unconditional lower bound for this sum-of-squares architecture is five total variables.

19 Quartics obtained from the direct Lucas test

The direct test also improves the degree-four construction. For this purpose we retain the scalar block with its preparation coordinate, because its quadratic evaluation is less expensive than the

normalized scalar source used for finite-algebra compression. We retain the proved comparison order 52, scalar order 32, and fixed size constant $\kappa = 3^{2^{28}}$.

Replace the three proxy comparisons and the exact Lucas equation by the one field-valued test

$$F(L_2, L_3, L_4), \qquad F(U, V, W) = V^2 - U(U-2)VW + (U-2)^2W^2 - 1,$$

of order 52. Keep $L_1 - \beta$ of order 16 and all four two-anchor cross comparisons. The direct-polynomial congruence lemma applies with $k = 52$ just as with $k = 64$. The precision-52 height proof uses the same integer residual Δ and its bound $|\Delta| < 5B^4$, so its threshold and the chosen anchor are unchanged. The scalar-order-32 proof is unaffected. Completeness uses the same fixed-point neighborhoods: a polynomial in the original logarithm ratios remains bounded there, introduces no new denominator, and has the required prime-power precision.

The quadratic compiler evaluates the new field test by computing

$$A = U(U-2), \qquad B = VW, \qquad C = (U-2)W, \qquad F = V^2 - AB + C^2 - 1.$$

Each field multiplication uses its two rational component equations as before. The coordinate detectors now have ten tests of order one, two of order sixteen, and four of order fifty-two. Their symmetric-product coefficient lists are shared across all nine coordinate detectors. This removes more arithmetic coordinates than merely deleting the three proxies would do. All reciprocal-domain lemmas, Boolean scalar choices, and quadratic-preserving substitutions from the preceding quartic construction are retained.

Theorem 19.1. *Under the arithmetic inputs stated for the shared construction, the supplied polynomial P_{1315} defines $\mathbb{Z}$ in $\mathbb{Q}$, has 1315 total variables and degree exactly four. It is the sum of squares of 1272 primitive quadratic residuals and has 28,991 nonzero encoded entries in its complete table over $\mathbb{Z}[\kappa]$, with the specified fixed integer $\kappa = 3^{2^{28}}$. After collecting powers of this constant, there are 28,987 distinct monomials in the quantified coordinates.*

The alternative P_{1339} has 1339 total variables, degree exactly four, 1296 quadratic residuals, and 29,061 expanded ordinary integer terms. Its primitive coefficient height is less than 2^{1020}.

Proof. The preceding argument proves the arithmetic source replacement at precision 52. The exact quadratic compilation and the domain equivalences were proved in Section 18. For the present source the independent checker verifies 1285 arithmetic gates, eleven row additions, and sixty coordinate eliminations. It then independently squares and convolves every residual and compares the complete resulting coefficient table. The active coordinate count is 1315, the number of primitive residuals is 1272, and the number of nonzero table entries is 28,991. The encoding separates different powers of κ in a coefficient. Collecting these powers merges four pairs of entries: the constant coefficient is positive, of the form $c_0 + 1024\kappa^2$, and three further coefficients are $1 - 192\kappa \neq 0$. Thus there are 28,987 monomials in the actual coordinates, with no cancellation after substituting the specified integer. A nonzero degree-four coefficient occurs, so the degree is exactly four. Over $\mathbb{Q}$ a sum of squares vanishes exactly when all its residuals vanish, giving the asserted input projection.

For the ordinary integer alternative, retain the same twenty-four successive squaring coordinates computing κ from 3^{16}. The count becomes 1339, and the checker verifies 1309 arithmetic gates with the same eleven row additions and sixty eliminations. Its 1296 primitive residuals give exactly 29,061 nonzero integer terms. Their largest coefficient has bit length at most 1020, and their content is one. This proves the degree and height claims without leaving any coefficient expansion implicit. □

The two tables are in `research/phase3/quartic/quartic-direct/` and `research/phase3/quartic/quartic-direct-small-coefficients/`. Their producer is `build_direct_quartic.py`. The independently written checker `research/phase2/quartic/verify_shared_quartic.py` is reused without modification and does not import this producer. These algebraic checks do not certify the external arithmetic lemmas. The number-field section separately examines both literal quartics and faithful combinations of their residual systems.

20 Reducing the anchor coefficients and shortening its constant chain

The following refinements are exact transformations of the supplied quadratic residual systems. They require no additional arithmetic theorem. The original tables remain available; the transformed tables are separate artifacts with certificates referring to their frozen parents.

20.1 An affine change in the three private anchor coordinates

Put

$$\kappa = 3^{2^{28}}, \qquad s = 3^{2^{27}}, \qquad s^2 = \kappa.$$

In the 1315-coordinate table the anchor residual is

$$R = -3A^2 + 16u^2 + 96v^2 - 32w^2 + 32\kappa, \qquad A = X_{52} + X_{53} - X_{54},$$

where $(u, v, w) = (X_{55}, X_{56}, X_{57})$. These three coordinates occur in no other residual. The other three displayed coordinates are constrained arithmetic values, so they cannot be treated as independent anchor witnesses.

Make the rational affine substitution

$$u = s + \frac{U - 3V + 3W}{4}, \qquad v = \frac{s}{2} + \frac{V}{4}, \qquad w = \frac{3s}{2} + \frac{W}{4}. \tag{4}$$

Its linear determinant is $1/64$, and its inverse is

$$U = 4u + 12v - 12w + 8s, \qquad V = 4v - 2s, \qquad W = 4w - 6s.$$

It is therefore a bijection over every characteristic-zero field, fixing the input and every other coordinate. Direct expansion gives

$$\widetilde{R} = -3A^2 + U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2 + 8sU. \tag{5}$$

Indeed the constant terms sum to $16s^2 + 24s^2 - 72s^2 + 32s^2 = 0$, and the linear terms sum to $8sU$. All coefficients of the new residual are integers, with s regarded as the specified fixed integer, not a coordinate.

Proposition 20.1. *Replacing R by $\widetilde{R}$ in the frozen* 1315*-coordinate residual list gives a polynomial $\widetilde{P}_{1315}$ with exactly the same input projection as P_{1315} over every characteristic-zero field. Its degree is exactly four, its primitive integer coefficient height is*

$$H(\widetilde{P}_{1315}) = 64\kappa < 2^{429496736},$$

and its complete table has 29021 *nonzero ordinary coordinate monomials. It has* 1272 *quadratic residuals and* 1314 *existential witnesses. In particular, under the already stated arithmetic inputs, it defines $\mathbb{Z}$ in $\mathbb{Q}$.*

Proof. The polynomial identity

$$\widetilde{P}_{1315} = P_{1315} \circ \text{the substitution in (4)}$$

and the explicit inverse prove equality of input projections. This conclusion concerns the literal single polynomial as well as the common residual zero set; it does not use positivity of squares. No new coordinate or equation is introduced, and every residual remains quadratic.

For the height assertion write

$$B = -3A^2 + U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2.$$

Then $\widetilde{R}^2 = B^2 + 16sUB + 64s^2U^2$. Since U, V, W occur nowhere else, the coefficient of the monomial U^2 in the sum of squares is exactly $64s^2$. It is the only coefficient involving s^2. The coefficients involving s have absolute value at most $288s$. The independently expanded table records that every remaining integer coefficient has absolute value less than 2^{1020}, and no monomial receives two different powers of s. The value $s = 3^{2^{27}}$ makes $64s^2$ larger than both other bounds. The input coefficient of X^4 is exactly one. Thus the integer content is one, the height is exactly $64s^2 = 64\kappa$, and the degree is exactly four. Finally $3^5 < 2^8$ gives

$$\log_2(64\kappa) < 6 + \frac{8}{5}\,2^{28} < 429496736.$$

The displayed residual list and its exact convolution give the stated monomial count. □

20.2 Eliminating the last coordinate in the constant chain

In the ordinary-coefficient 1339-coordinate table, the coordinate X_{81} occurs in exactly two residuals:

$$X_{80}^2 - X_{81}, \qquad R_0 + 32X_{81},$$

where

$$R_0 = -3(X_{52} + X_{53} - X_{54})^2 + 16X_{55}^2 + 96X_{56}^2 - 32X_{57}^2.$$

Substitute $X_{81} = X_{80}^2$, delete its defining residual, and renumber all subsequent coordinates by subtracting one. The second residual becomes $R_0 + 32X_{80}^2$ and is still quadratic.

Proposition 20.2. *The resulting system has* 1338 *total coordinates and* 1295 *quadratic residuals. Its sum of squares P_{1338} has degree exactly four,* 29059 *nonzero ordinary integer monomials, primitive integer content one, and coefficient height less than 2^{1020}. Its rational input projection equals that of P_{1339}.*

Proof. On a common zero of the parent residuals, the first displayed equation uniquely determines $X_{81} = X_{80}^2$. Deleting this coordinate therefore gives a solution of the new system. Conversely, inserting that square in a new solution restores all parent equations. This is a bijection of common residual zero sets over every characteristic-zero field and fixes the input.

Over a formally real field, in particular over $\mathbb{Q}$, a sum of squares vanishes precisely when all its residuals vanish. Hence the same equivalence holds for the literal sum-of-squares input projections over those fields. No such converse for the literal polynomials over nonreal fields is inferred from this elimination.

The substitution keeps every residual quadratic. Exact convolution of the remaining residuals gives 29059 nonzero coefficients, all of absolute value below 2^{1020}. The unchanged coefficient of the input monomial X^4 is one, proving content one and degree four. The complete maximum coefficient is recorded in the construction report. The coordinate and residual counts each decrease by one. □

20.3 Inlining the three initial constants

The first three coordinates of the remaining constant chain are uniquely determined as

$$X_{58} = 3^{32}, \qquad X_{59} = 3^{64}, \qquad X_{60} = 3^{128}.$$

Their three defining residuals are

$$3^{32} - X_{58}, \qquad X_{58}^2 - X_{59}, \qquad X_{59}^2 - X_{60}.$$

They have no other occurrences except in the next equation of the chain. Substitute the three constants and delete these coordinates and residuals. The first remaining chain equation becomes $3^{256} - X_{61} = 0$ before renumbering, or $3^{256} - X_{58} = 0$ afterwards.

Theorem 20.3 (The preferred ordinary-coefficient quartic)**.** *The supplied polynomial P_{1335} has* 1335 *total coordinates,* 1334 *existential witnesses,* 1292 *quadratic residuals, degree exactly four, and* 29050 *nonzero ordinary integer monomials. Its primitive coefficient height is unchanged from P_{1338} and is less than 2^{1020}. Under the same arithmetic hypotheses, it defines $\mathbb{Z}$ in $\mathbb{Q}$.*

Proof. Each eliminated coordinate equals the displayed fixed integer at every common zero of the parent residuals. Substitution and reinsertion are inverse maps on the corresponding common zero sets, over every characteristic-zero field, and fix the input. All remaining residuals have degree at most two. Over $\mathbb{Q}$ their sum of squares therefore has the same input projection.

The first remaining chain residual contributes the constant 3^{512} on squaring, and

$$3^{512} < 2^{812} < 2^{1020}.$$

For an elementary check of the first inequality, $3^{41} < 2^{65}$ and $512 \cdot 65 < 812 \cdot 41$. The only changed constant coefficient is the old one minus 3^{64} plus 3^{512}; it has bit length at most 826. The new linear coefficient has absolute value $2 \cdot 3^{256} < 2^{407}$. Every other coefficient is inherited, up to coordinate renumbering, or removed. The previous coefficient of maximum height occurs in an unaffected monomial, so the exact height is unchanged. These assertions are also checked against the complete coefficient table.

Twenty constant-chain coordinates remain, representing $3^{2^8}, 3^{2^9}, \ldots, 3^{2^{27}}$. The final square is used directly in the anchor. Thus the ordinary-coefficient presentation adds twenty coordinates to the 1315-coordinate literal-constant presentation. Exact convolution gives 29050 nonzero terms. The unchanged coefficient of the input monomial X^4 is one, proving primitive content one and degree four. □

The preferred fresh packages are `research/phase3/quartic/quartic-direct-translated/` and `research/phase3/quartic/quartic-direct-small-coefficients-1335/`. The intermediate 1338-coordinate package is retained separately under `quartic-direct-small-coefficients-1338/`. Each contains the full residual list, complete coefficient table, parent hash, and exact transformation certificate. Their producer is `transform_direct_quartics.py`. It verifies both the residual substitutions and the full polynomial identities against the frozen parent tables. The separate program `verify_final_transforms_independent.py` imports no producer code and independently checks every transformed residual, every expanded coefficient, all coordinate maps, and the stated heights and supports. All three checks pass. The translated table uses the coefficient symbol S for the fixed integer s; an entry $[\mathbf{m}, k, c]$ means $cs^k \prod_{i \in \mathbf{m}} X_i$. This symbol contributes no variable and no degree in the quantified coordinates.

21 Exact quartics and their computational verification

The preceding anchor formula is applied to the actual retained residual system, not only to an informal count. The old private anchor uses two scaling coordinates a_i with $Ca_i = f_i$, and, for each $i = 0, 1$, the seven-witness equations

$$b_{i0}^2 - 2b_{i1}^2 - 3b_{i2}^2 + 6b_{i3}^2 = 1,$$
$$(a_i - 2b_{i0})^2 - 8c_{i1}^2 - 12c_{i2}^2 + 24c_{i3}^2 = 4.$$

The norm-one trace-sum theorem for the quaternion algebra $(2, 3)$ identifies this condition with $a_i \in R$; its ramification set is $\{2, 3\}$ and it splits at infinity [13, Proposition 6]. The second equation has scalar part $(a_i - 2b_{i0})/2$, so the trace convention agrees exactly. The retained constant is $C = 3^{2^{24}} \neq 0$.

Proposition 21.1. *There are exact integer quartics P_{12493} and P_{12465} with the same rational input projection as the released source. Their statistics are*

	total variables	*residuals*	*monomials*	$\log_2 H$ *bound*
P_{12493}	12493	11936	135667	< 80
P_{12465}	12465	11908	130427	$< 3247.$

The height of P_{12493} equals the height of the release. Both complete expanded integer coefficient tables are supplied.

Proof. For the first polynomial replace the six old anchor equations by the three equations of Corollary 5.3. Their sixteen private coordinates occur nowhere else. The five new coordinates retain quadratic residuals and have exactly the same projection at the interface (C, f_0, f_1). Eleven variables and three residuals are removed. The sum of the squares of all retained residuals is a quartic.

For the second polynomial first perform twenty-eight additional exact eliminations and thirteen invertible row operations. An elimination uses a row $cY + g = 0$, with $c \in \mathbb{Q}^\times$ and g independent of Y, substitutes $Y = -g/c$ into every occurrence, and removes that row and coordinate. Both rational solution maps are immediate. A row operation replaces f by $af + bg$ while retaining g, with $a \in \mathbb{Q}^\times$; again both solution sets agree. The certificate records every affected row, all denominator clearings, and the variable remapping. Every resulting residual is still quadratic. Apply the same eleven-variable anchor replacement afterward.

The independent checker replays the pinned baseline certificate, checks every extra operation using exact fractions without importing the producer, checks the literal six anchor rows and their full incidence, verifies the nonzero defining chain of C, and expands every square by ordered products. It compares every coefficient with the supplied table and verifies that each reported variable occurs. This gives the displayed finite statistics. The semantic anchor equivalence is proved above; the checker verifies its exact placement. Neither part uses an unproved arithmetic optimization of the source. □

The small-coefficient package provides a useful alternative when file size matters. Under the variable-first ordering, P_{12465} is preferable despite its coefficient increase. Its largest coefficient has 978 decimal digits; the exact integer is recorded in the verification receipt, rather than reproduced here.

21.1 Removing constants without expanding their decimal digits

There are still fifteen fixed coordinates in P_{12465}, beginning with $c_0 = 3^{1024}$ and continuing by $c_{k+1} = c_k^2$ for $0 \le k < 14$. Thus $c_{14} = C = 3^{2^{24}}$. The new anchor coordinate $D_0 = C^2$ is fixed as well. Substitute these sixteen constants into every remaining residual and delete their defining rows. This cannot increase degree. The only remaining large constant is

$$M = C^4 = 3^{2^{26}},$$

in the anchor residual

$$u^2 + 6v^2 - 2w^2 - 3r^2 + 2M.$$

Here M is a fixed integer, not a variable.

Proposition 21.2. *The constant substitution gives a quartic in exactly 12,449 total variables, with 11,892 residuals and 130,378 monomials, having the same rational input projection. Its primitive integer coefficient height is exactly*

$$H = 4 \cdot 3^{134217728} + H_0, \tag{6}$$

where

$$\begin{aligned} H_0 = {} & 373684652 \cdot 10^{144} \\ & + 3552493711110554154609653170407475896167777859 74 \cdot 10^{96} \\ & + 770831188591770167322175457505857070439628239726 \cdot 10^{48} \\ & + 443171972269170621942059432904883611120549682273. \end{aligned}$$

Its coefficients are completely specified by a table over $\mathbb{Z}[M]$ of degree at most two in the fixed constant M.

Proof. Each deleted coordinate is uniquely fixed by its defining equation. Direct incidence checking shows that, outside the deleted chain, these coordinates occur only in the displayed anchor places. Thus substitution and reconstruction give inverse maps on the retained coordinates. The exact expansion over $\mathbb{Z}[M]$ gives the stated counts. No nonconstant monomial disappears on specializing the fixed constant: its coefficient is $a + bM$ with $|a| < 2^{3247} < M$, so it is nonzero when $b \neq 0$; a nonzero constant coefficient remains nonzero as well. Only the coefficient of the constant monomial has an M^2 term. The constant coefficient is $4M^2 + H_0$. Every other coefficient has absolute value at most $2^{3247} + 24M$, which is strictly smaller. The coefficient of u^4 is one, so the integer content is one and the degree is four. Each remaining variable occurs in a residual, and a sum of squares of real polynomials cannot lose its dependence on such a variable: its leading coefficient in that variable is a nonzero sum of squares. This verifies the exact variable count. □

The integer in (6) is intentionally not converted to a decimal string. This is an exact polynomial specification by an ordinary integer power, not a numerical approximation. An elementary upper bound is $H < 2^{268435459}$. The smaller variable count is purchased with very large coefficients, in accordance with the specified priority.

21.2 Verification boundaries

The files separate producer output from independent verification. The constant table is checked as a polynomial over $\mathbb{Z}[M]$, so there is no floating-point arithmetic or gigantic decimal conversion in that check. For the preceding elimination suffix, seven deliberate mutations of pivots, solved numerators, affected-row lists, substitution patches, and row operations were rejected. These checks test the advertised finite transformations. They neither test all rational inputs nor certify the manuscript's restricted converse, height estimate, or prime-pattern argument.

22 A fully specified polynomial with 854 variables

This section gives a finite formula for an actual polynomial, not just an existence argument. Its degree is much higher than four. Arithmetic expressions in this section introduce no quantified coordinates unless explicitly listed. The accompanying JSON circuit makes every addition, multiplication, power, residual numerator, and nonzero condition literal.

22.1 The finite list of rational equations

Use the curve and constants of Section 4, and write σ for the automorphism of $K = \mathbb{Q}(\sqrt{2})$ sending $\sqrt{2}$ to $-\sqrt{2}$. For a nonidentity point $Q = (x, y)$ define

$$f(Q) = 1/x, \qquad z(Q) = -x/y.$$

All the denominators occurring here and below are required to be nonzero. A field inverse is written using its rational norm, so both rational coefficient formulas are specified.

There are six root points A, S, T, Z, J_0, J, and five private points $Q_{\ell,0}, \dots, Q_{\ell,4}$ for each $1 \le \ell \le 41$. Their total number is 211. For each point use the three rational coordinates (a_0, a_1, C_Q) and the common chart parameter $j \in \{0, \dots, 211\}$ of Proposition 7.1. Namely put $a = a_0 + a_1\sqrt{2}$, write $a^3 - Da = A_Q + B_Q\sqrt{2}$, reconstruct $b = b_0 + b_1\sqrt{2}$, impose

$$C_Q^2 V_Q - (B_Q/2 + jV_Q)^2 = 0, \qquad V_Q = \frac{C_Q^2 - A_Q - jB_Q}{j^2 - 2},$$

and use the double $Q = [2](a, b)$ in every subsequent expression. Require $C_Q \neq 0$, $N_{K/\mathbb{Q}}(b) \neq 0$, and nonzero norms for the abscissa and ordinate of the output point. The source-chart equivalence and its inverse maps are proved in Section 10.

Set $\beta = X^2 + 2$. Introduce rational proxies s, τ, z and impose

$$\tau^2 - s(s-2)\tau z + (s-2)^2 z^2 - 1 = 0. \tag{7}$$

Use one rational θ, with $\varepsilon = \theta^2 - 2$. The letter z without an argument in (7) is a proxy; $z(Q)$ is the specified elliptic rational function.

The finite reciprocal logarithms are

$$\mathcal{L}_L(Q) = z(Q) G_{L/4-1}(Df(Q)^2), \qquad L \in \{16, 64\},$$

where $G_J(W) = 1 + \sum_{r=1}^{J} a_r W^r$, $a_1 = -2/5$, and $a_r = a_{r-1}(4r-5)/(4r+1)$. Their nonvanishing on the required rational point charts is proved in the finite-fiber application.

We next specify the norm operation used in every contact. Given a nonzero rational r, an ordered list of tests t_i with positive orders L_i, and $L_1 = 1$, form the normalized leaves $1, t_1/r, t_2/r^{L_2}, \ldots$. First pair $1, t_1/r$ by $N_r(a,b) = a^2 + rb^2$. Group all remaining tests by increasing order, preserving their original order within each group; pair successive normalized leaves and carry an unpaired last leaf. At each higher level pair successive nodes by N_r and carry any last node until one root W_r remains. The input of the pole equations is $b = rW_r^2$. This prescription is finite and fixes the order of every operation.

For a rational input b, a pole occurrence has five coordinates t, e_p, p, q, h_p. Define

$$U = \frac{b(1-bt^2)}{1+bt^2}, \qquad B = 17b, \qquad H = e_p(B - U^2),$$

and impose

$$p^2 = B(1-e_p)(1+3e_p), \qquad Bq^2 = (h_p^2 - BH)(h_p^2 + 3BH). \tag{8}$$

The denominator $1+bt^2$ cannot vanish on the original pole equations; its explicit nonzero condition is retained here.

For the height occurrence take

$$r_0 = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1)$$

and the six rational coefficients of $f(J_0), f(S), z(S)/z(J_0)$, in that order. Their orders are $(1,1,1,1,48,48)$. Replace the resulting pole input by $cr_0W_{r_0}^2$, where the fixed finite choice is

$$c \in \mathcal{C} = \left\{ \pm \prod_{p \in I} p : I \subseteq \{2,3,5,7,11,13,17\} \right\}.$$

There are 256 such integers, covering the source's eight Boolean choices exactly.

Introduce a nonzero coordinate e, and let $a_1, \ldots, a_9$ be e followed by the eight rational coefficients of $ex(A), ex(S), ex(T), ex(Z)$. Introduce two contact coordinates u_c, v_c. With $W_0 = 25878772920$, the forty-one contacts are

$$\frac{v_c}{32W_0}, \quad u_c, \quad u_c - v_c, \quad u_c - 2401v_c, \quad u_c + \frac{343}{32}v_c,$$

and, for each $i = 1, \ldots, 9$, on putting $A_i = 47027a_i + 3$,

$$u_c - A_iv_c, \quad u_c - A_iv_c + v_c, \quad u_c - A_iv_c + 2401v_c, \quad u_c - A_iv_c - \frac{343}{32}v_c.$$

Write these contacts as $r_1, \ldots, r_{41}$ in the displayed order.

For contact ℓ, the five target points are $(4P_*, 4A, 4S, 4T, 4Z)$. Subtract them respectively from the five private points to obtain $Y_{\ell,i}$, using the affine subtraction formula and its nonzero abscissa-difference denominator. The ordered twenty field tests are the following:

1. $f(Q_{\ell,i})$ for $0 \le i \le 4$;
2. $f(J)^\sigma$;
3. $f(Y_{\ell,i})^\sigma$ for $0 \le i \le 4$;

4. $\mathcal{L}_{16}(Q_{\ell,1})/\mathcal{L}_{64}(Q_{\ell,0}) - \beta$, and the analogous three ratios with numerator precision 64 and proxies s, τ, z;

5. $z(Y_{\ell,i})^{\sigma}/z(J)^{\sigma}$ for $0 \le i \le 4$.

Replace each field test $A + B\sqrt{2}$ by the two rational tests $\varepsilon B, A + \theta B$, and prepend ε. The resulting forty-one orders are

$$(\underbrace{1, \dots, 1}_{23}, 16, 16, \underbrace{64, \dots, 64}_{8}, 16, 16, \underbrace{64, \dots, 64}_{6}).$$

Apply the norm prescription at r_ℓ and then one copy of (8). This makes 42 pole occurrences in total.

Finally, write $f(S) = f_0 + f_1\sqrt{2}$, set $F = f_0^2 + f_0 f_1 - f_1^2$, and let $C = 3^{2^{24}}$ be the fixed anchor constant. The three-witness anchor is

$$u_a^2 + 6v_a^2 - 2w_a^2 - 3F^2 + 2C^4 = 0.$$

We next absorb its private coordinate u_a and all denominator conditions into one coordinate, without increasing their combined count.

22.2 Eliminating the denominator witness

For fixed (j, c), clear the denominators of all rational equations other than the anchor, and call their polynomial numerators $E_1, \dots, E_{296}$. There are 211 point equations, 84 pole equations, and one Lucas equation. During the construction record every nonzero numerator required for a division, including rational norms for divisions in K, and the explicit point and coordinate guards. Let G be their product, with repeated factors omitted if desired. All the rational functions used are defined whenever $G \neq 0$.

Write

$$6v_a^2 - 2w_a^2 - 3F^2 + 2C^4 = N/B$$

with integer polynomial numerator N and denominator B. Choose $F = R_0/S_0$ in the rational-expression construction. Then

$$B = S_0^2, \qquad N = S_0^2(6v_a^2 - 2w_a^2 + 2C^4) - 3R_0^2.$$

The actual circuit's cleared anchor has exactly this form. All denominator factors are included among the recorded conditions. The three anchor coordinates occur nowhere in G or the other residuals, and B is independent of them.

The anchor always has a solution with $u_a \neq 0$ whenever it has any solution. Indeed its three witnesses cannot all vanish, since $3F^2 = 2C^4$ is impossible for rational F/C^2. If $u_a = 0$ and $w_a \neq 0$, replace (u_a, w_a) by $(4w_a, 3w_a)$; this preserves $u_a^2 - 2w_a^2$. If $u_a = w_a = 0$ and $v_a \neq 0$, replace (u_a, v_a) by $(-12v_a/7, -5v_a/7)$; this preserves $u_a^2 + 6v_a^2$. In either case u_a becomes nonzero without changing any outside equation.

Rescale $t_a = S_0 u_a$, so that the cleared anchor is $t_a^2 + N = 0$. Replace its private coordinate by U and put

$$\Psi = 1 + NG^2U^2.$$

Define the branch polynomial

$$F_{j,c} = \sum_{i=1}^{296} E_i^2 + \Psi^2, \tag{9}$$

and the required integer polynomial

$$P_{854} = \prod_{j=0}^{211} \prod_{c \in \mathcal{C}} F_{j,c}. \tag{10}$$

Proposition 22.1. *The polynomial* (10) *uses 854 total variables and has the same rational input projection as* $\mathcal{S}$*. A supplied integer arithmetic circuit specifies it completely. Its degree is at most* 63570781474816*, and its coefficient height is at most* $2^{757866260454400}$*.*

Proof. If $GU = 0$, then $\Psi = 1$, so (9) has no rational zero. Otherwise $G \neq 0$ also implies $S_0 \neq 0$, and

$$\Psi = (GU)^2\big((1/(GU))^2 + N\big).$$

A zero therefore gives an original anchor witness $u_a = 1/(GS_0U)$. Conversely, use the preceding transformations to choose $u_a \neq 0$, then put $U = 1/(GS_0u_a)$. This preserves the existential input projection. It need not preserve a fiber in which both of the other private anchor witnesses have been fixed, and no such claim is needed. The sum of squares imposes all the outside equations and all the divisions used before taking numerators.

The point substitutions and the pole substitutions have the inverse maps proved above and below. Among the 212 fixed chart choices there is a valid common choice for any list of 211 source halfpoints. The 256 fixed corrections cover the eight Boolean source choices. A product of branch polynomials permits reuse of the same witness coordinates, since only one branch must vanish. In particular, no denominator condition from an inactive branch is imposed.

The coordinate count is

$$633 + 210 + 3 + 3 + 1 + 1 + 2 + 1 = 854 :$$

point coordinates, pole coordinates, anchor coordinates (U, v_a, w_a), proxies, θ, e, two contact coordinates, and the input. The final U also enforces the denominators, so no inverse coordinate is added. All 854 coordinates occur in the supplied specification.

For the bounds, assign each integer constant its absolute coefficient sum, each variable degree one, and each fixed parameter its maximum absolute value over its stated finite list. At a sum, degrees take a maximum and coefficient sums add; at a product, degrees add and coefficient sums multiply; at a power both bounds scale accordingly. These exact integer recurrences bound every node. Finally multiply the branch bounds over $212 \cdot 256 = 54272$ branches. Replaying these recurrences on the supplied circuit gives the displayed bounds. They are upper bounds, not exact expansion statistics. □

This explicit circuit and the 558-variable indexed construction serve different purposes. The former can be regenerated directly from the finite formulas and inspected as a circuit; the latter uses further finite-algebra elimination whose indexed output is specified and checked, although its rank-sized matrices and full expansion are not materialized. Neither is presented as a practical solver for Diophantine equations.

23 An explicit reduction to 853 variables

Two further elementary steps combine the private anchor root with the Lucas root. Their proof concerns rational solutions. It is not a claim about geometric fibers over an algebraic closure.

23.1 The pole denominators are automatic over the rationals

Lemma 23.1. *After clearing the denominator $\delta = 1 + bt^2$ from (8), its two equations already imply $\delta \neq 0$ over $\mathbb{Q}$.*

Proof. Put $B_p = 17b$ and $A = B_p\delta^2 - b^2(1 - bt^2)^2$. The cleared second equation is

$$B_p q^2 \delta^4 = (h_p^2\delta^2 - B_p e_p A)(h_p^2\delta^2 + 3B_p e_p A).$$

If $\delta = 0$, then $bt^2 = -1$, so b, t, B_p are nonzero and $A = -4b^2$. The equation becomes $0 = -48B_p^2 b^4 e_p^2$, whence $e_p = 0$. The first equation now gives $p^2 = 17b = -17/t^2$, or $(pt)^2 = -17$, impossible in $\mathbb{Q}$.

This also applies to the reduced rational numerators used by the circuit. In each actual second conic, the denominator contains the numerator of δ to exactly the fourth power, and every other factor is among the retained nonzero conditions. The first-conic denominator uses only retained factors. Multiplying back yields the displayed polynomial identity even when $\delta = 0$. These exact denominator-factor statements are checked in the independent verification. □

More generally the same proof works with $B_p = cb$, over a field of characteristic different from two and three, whenever $-c$ is a nonsquare. It deliberately does not give the corresponding assertion over every number field.

Remove precisely the 42 pole-denominator conditions from the product G of the preceding section, and call the remaining product G_0. Its factors depend only on the input, point coordinates, coordinate denominator, and contact coordinates. In particular G_0 is independent of all three proxies, all pole coordinates, θ, and the private anchor coordinates. This follows from the displayed construction and was checked on the entire circuit. All other rational denominators are covered by G_0; the omitted ones are enforced by Lemma 23.1.

23.2 Combining two square roots and the nonzero conditions

Write the Lucas equation as

$$\tau^2 - a\tau + b_0 = 0, \quad a = s(s-2)z, \quad b_0 = (s-2)^2 z^2 - 1, \quad \Delta = a^2 - 4b_0.$$

Thus $v = 2\tau - a$ satisfies $v^2 = \Delta$. The scaled anchor witness satisfies $t_a^2 = -N$, where N is independent of τ. Choose its nonzero representative as in the preceding section. Its sign can be selected so that $W = t_a + v \neq 0$: the two choices $t_a + v$ and $-t_a + v$ cannot both vanish when $t_a \neq 0$. Consequently no new finite selector is necessary.

The two square-root equations have the exact single relation

$$h(W) = W^4 + 2(N - \Delta)W^2 + (N + \Delta)^2 = 0, \tag{11}$$

with reconstruction

$$v = \frac{W^2 + N + \Delta}{2W}, \qquad t_a = \frac{W^2 - N - \Delta}{2W}, \qquad \tau = \frac{a + v}{2}.$$

Indeed $4W^2(v^2-\Delta) = h(W) = 4W^2(t_a^2+N)$. Conversely, $W = t_a + v$ and the two square equations imply (11) and these reconstruction formulas.

Set $W = 1/(G_0U)$ and define

$$\Psi_* = 1 + 2(N-\Delta)G_0^2U^2 + (N+\Delta)^2G_0^4U^4, \qquad \tau = \frac{1 + 2aG_0U + (N+\Delta)G_0^2U^2}{4G_0U}.$$

The equation $\Psi_* = 0$ forces $G_0U \neq 0$ and is the reciprocal form of (11). Substitute the displayed rational formula for τ into all the other residuals. Choose a syntactic upper bound k for each residual's degree in τ, and multiply its substituted expression by $(4G_0U)^k$ to obtain a polynomial numerator. At addition gates take the maximum of the children's bounds; at multiplication gates add them; at power gates multiply by the exponent. This specifies the emitted circuit even if cancellation lowers the actual degree. The operation is reversible on $G_0U \neq 0$. Delete the old Lucas residual and the anchor residual; denote the remaining 295 polynomial numerators by E_i'. The branch polynomial and the full polynomial are

$$F_{j,c}^* = \sum_{i=1}^{295}(E_i')^2 + \Psi_*^2, \qquad P_{853} = \prod_{j=0}^{211}\prod_{c\in\mathcal{C}} F_{j,c}^*.$$

Proposition 23.2. *The integer polynomial P_{853} has 853 total variables and the same rational input projection as $\mathcal{S}$. The supplied finite arithmetic circuit has degree at most* 894090701148160 *and coefficient height at most* $2^{29154205340326912}$.

Proof. The forward construction may change only the three private anchor witnesses, making $t_a \neq 0$ and then choosing its sign to make $W \neq 0$. Set $U = 1/(G_0W)$; every transformed residual vanishes. Conversely, $\Psi_* = 0$ gives $G_0U \neq 0$. Reconstruct W, v, t_a and τ by the formulas above, then $u_a = t_a/S_0$. They satisfy the old anchor and Lucas equations. Undoing each homogenization gives the other original numerator equations. The retained guards and Lemma 23.1 make every original rational division valid. The source point-chart equivalences then apply as before. These operations prove equality of rational input projections.

Two coordinates, u_a and τ, have been replaced by the single coordinate U. The count is

$$633 + 210 + 2 + 2 + 1 + 1 + 2 + 1 + 1 = 853,$$

respectively point coordinates, pole coordinates, two remaining anchor coordinates, two remaining proxies, θ, the coordinate denominator, two contact coordinates, the input, and U. There are still 54272 finite branches. Exact circuit bounds are propagated as in Proposition 22.1; the degree and height statements are upper bounds. The maximum homogenizing degree in τ is 1280. □

The omission of the pole guards is justified here over $\mathbb{Q}$. The 558-variable geometric-fiber argument retains those guards and does not use this omission. This distinction prevents exceptional geometric fibers from being silently reintroduced into that argument.

24 Unconditional lower bounds

The lower bounds in this section do not assume the correctness of any proposed Diophantine definition. In fact, they hold over every number field. Let K be a number field and let $\mathcal{O}_K$ be its ring of integers. For a polynomial

$$F(T, Y_1, \ldots, Y_r) \in K[T, Y_1, \ldots, Y_r]$$

write

$$S_F = \{t \in K : \text{there exist } y_1, \dots, y_r \in K \text{ with } F(t, y_1, \dots, y_r) = 0\}.$$

The variable T is the input; the r variables Y_i are the witnesses. The degree is the total degree, including T.

Theorem 24.1. *If $S_F = \mathcal{O}_K$, then*

$$r \geq 2 \qquad \text{and} \qquad \deg F \geq 4.$$

In particular, a Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$ by one polynomial needs at least three total variables and has degree at least four. Thus a valid quartic construction has the smallest possible degree, regardless of its number of variables.

We prove the two assertions separately. The variable bound uses the integral form of Hilbert irreducibility. The cubic obstruction uses Kollár's theorem on unirational cubic hypersurfaces; the treatment of cones and reducible polynomials is included explicitly.

24.1 One witness and Hilbert irreducibility

We use the following standard form of Hilbert irreducibility. Given finitely many irreducible polynomials $f_i(T, Y) \in K(T)[Y]$, each of positive degree in Y, and finitely many forbidden values of T, there are infinitely many $a \in \mathcal{O}_K$ outside those values for which all the specializations are defined, preserve their Y-degrees, and remain irreducible in $K[Y]$. See [16, Chapter 4, Section 4]; the integral number-field form is also recalled in [17, Introduction]. The restriction to integral specializations is essential here: the Hilbertianity of K alone, without that strengthening, is not the assertion being applied.

Proposition 24.2. *Let $G \in K[T, Y]$. If S_G contains all but finitely many elements of $\mathcal{O}_K$, then S_G contains all but finitely many elements of K.*

Proof. The conclusion is immediate if $G = 0$. If $G \neq 0$ and has degree zero in Y, then S_G is finite, contrary to the hypothesis. Otherwise factor in $K(T)[Y]$:

$$G = c(T) \prod_{i=1}^{s} f_i(T, Y)^{e_i},$$

where $c(T) \neq 0$, the e_i are positive integers, and the f_i are monic irreducible polynomials of positive Y-degree. If some f_i is linear, it has a root $u(T) \in K(T)$. The identity $G(T, u(T)) = 0$ then specializes for all but finitely many $t \in K$, proving the conclusion.

If every f_i has Y-degree at least two, integral Hilbert irreducibility provides infinitely many $a \in \mathcal{O}_K$ at which $c(a) \neq 0$ and all $f_i(a, Y)$ remain irreducible of degree at least two. None has a root in K, so $G(a, Y)$ has no root in K. This contradicts the hypothesis. □

Since $K \setminus \mathcal{O}_K$ is infinite, the proposition rules out one witness. For example, $1/n \notin \mathcal{O}_K$ for every integer $n \geq 2$, because a rational algebraic integer is an integer. With no witness a nonzero polynomial in T has only finitely many roots, whereas the zero polynomial defines all of K. This proves the variable assertion of Theorem 24.1.

24.2 A topological obstruction to rational parametrizations

Let r_1 and r_2 denote the numbers of real embeddings and conjugate pairs of complex embeddings of K. Put

$$K_\infty = \mathbb{R}^{r_1} \times \mathbb{C}^{r_2},$$

viewed as a real vector space. The diagonal embedding $\iota : K \longrightarrow K_\infty$ has dense image, while $\iota(\mathcal{O}_K)$ is a discrete lattice. Indeed, an integral basis of K identifies these images with the images of $\mathbb{Q}^{[K:\mathbb{Q}]}$ and $\mathbb{Z}^{[K:\mathbb{Q}]}$ under an invertible real linear map.

Lemma 24.3. *Let $R \in K(U_1, \ldots, U_m)$ be nonconstant, and let U be a nonempty Zariski-open subset of $\mathbb{A}^m_K$ on which R is defined. Then $R(U(K))$ is not contained in $\mathcal{O}_K$.*

Proof. Assume to the contrary that all these values lie in $\mathcal{O}_K$. After shrinking U we may take U to be a principal open subset, defined by one nonzero polynomial. Apply every archimedean embedding of K to its coefficients and to those of R. This gives a continuous map, on a nonempty Euclidean open subset of K_∞^m,

$$R_\infty : \prod_{v|\infty} K_v^m \longrightarrow \prod_{v|\infty} K_v = K_\infty.$$

Choose a small connected open ball B in its domain. The diagonal image of K^m is dense in B. By hypothesis its image under R_∞ lies in the closed discrete set $\iota(\mathcal{O}_K)$; by continuity the same is true on all of B. A continuous map from a connected set to a discrete set is constant. Each archimedean component of R is therefore constant on an open set, so its numerator minus that constant times its denominator is the zero polynomial. This forces R to be constant, a contradiction. □

For $K = \mathbb{Q}$, this argument is simply the density of $\mathbb{Q}$ in $\mathbb{R}$ and the discreteness of $\mathbb{Z}$. For a general number field one must use all archimedean embeddings together; $\mathcal{O}_K$ need not be discrete in one selected embedding.

An integral variety X/K is called *unirational over K* if some projective space admits a dominant rational map to X, defined over K.

Corollary 24.4. *If X is unirational over K and $f \in K(X)$ is nonconstant, then the values of f at the K-points where it is defined cannot all belong to $\mathcal{O}_K$.*

Proof. Compose f with a dominant rational parametrization of X, and restrict the source to a suitable nonempty open subset of affine space. Dominance makes the resulting rational function nonconstant. Apply Lemma 24.3. □

24.3 Quadratic polynomials

Lemma 24.5. *If $\deg F \leq 2$ and $S_F \subseteq \mathcal{O}_K$, then S_F has at most two elements.*

Proof. Suppose that $a, b, c \in K^{r+1}$ are zeros of F with three distinct input coordinates. If these points are collinear, the restriction of F to their line has degree at most two and three distinct roots. Thus the entire line is contained in $F = 0$. The input is nonconstant on that line and takes all values in K, a contradiction.

Otherwise the three points span an affine plane L over K. The input coordinate is a nonconstant affine linear function on L. If $F|_L = 0$, a nonvertical line in L again gives a

contradiction. If $F|_L$ factors into affine linear factors over K, one of its at most two lines contains two of the selected points, hence has nonconstant input; this gives the same contradiction. A nonzero linear polynomial cannot vanish at three noncollinear points.

The remaining case is an irreducible quadratic polynomial on L. It defines an irreducible conic with a smooth K-point: a singular conic has at most one singular point, so the three distinct points cannot all be singular. Projection from a smooth K-point parametrizes the conic by rational functions over K. The input is nonconstant on this conic, for otherwise an irreducible quadratic curve would be contained in an affine line. Its composition with the parametrization is a nonconstant rational function taking only values in $\mathcal{O}_K$, contrary to Lemma 24.3. □

24.4 Cubic cones and the sharp degree obstruction

We use the following published theorem, in its characteristic-zero case.

Theorem 24.6 (Kollár)**.** *Let K be a perfect field and let $X \subseteq \mathbb{P}^{n+1}_K$ be a geometrically integral cubic hypersurface of dimension $n \geq 2$ which is not a cone. If $X(K) \neq \varnothing$, then X is unirational over K.*

This is the geometrically integral case of Theorem 2 of [15]; the theorem there is stated for irreducible cubics. We handle geometric reducibility separately below, so no convention about *irreducible* is needed. Its cone exclusion must not be omitted: a cone over a smooth plane cubic need not be unirational.

For a homogeneous cubic H on a finite-dimensional K-vector space E, define its vertex subspace by

$$W = \{v \in E : D_v H = 0 \text{ as a polynomial on } E\}, \qquad D_v H(x) = \frac{d}{ds} H(x + sv)\Big|_{s=0}.$$

The equations defining W are linear over K. In characteristic zero, $D_v H = 0$ is equivalent to $H(x + sv) = H(x)$ for all x, s. Choosing a basis of W and extending it to a basis of E therefore makes H independent of the coordinates along W. Consequently H descends to a homogeneous cubic $\overline{H}$ on E/W. By maximality of W, this quotient cubic has no nonzero vertex direction and is not a cone. These statements remain true after extending scalars: the kernel of a matrix over K commutes with field extension.

Proof of the degree assertion in Theorem 24.1. Suppose $S_F = \mathcal{O}_K$ and $\deg F \leq 3$. The polynomial F is neither zero nor a nonzero constant. If F is reducible, each of its irreducible factors has degree at most two. In that case, the projection is the union of the projections of finitely many linear or quadratic factors. Every such projection is contained in $\mathcal{O}_K$, hence finite by Lemma 24.5. This cannot give $\mathcal{O}_K$. The same lemma excludes $\deg F \leq 2$. We may therefore suppose that F is irreducible of degree three.

Homogenize F to an irreducible cubic

$$H(X_0, X_1, \ldots, X_{r+1}) = X_0^3 F(X_1/X_0, \ldots, X_{r+1}/X_0).$$

Here X_1/X_0 is the input function. We first exclude geometric reducibility. Since the characteristic is zero, H has no repeated geometric components. Irreducibility over K makes the Galois group act transitively on them. Their degrees are equal and sum to three; thus if H is geometrically reducible, it is a product of three conjugate linear forms. A K-rational zero belongs to every

one of these hyperplanes, because it is fixed by Galois. Consequently the entire line joining any two K-rational zeros lies in all three hyperplanes. Taking affine zeros with inputs 0 and 1 would produce every input in K, a contradiction. We may therefore assume H is geometrically integral.

Let W be its vertex subspace. For any $v \in W$, any affine rational zero x normalized by $x_0 = 1$, and any $s \in K$ with $1 + sv_0 \neq 0$, the point $x + sv$ is a zero of H. Its input is

$$\frac{x_1 + sv_1}{1 + sv_0}.$$

If $v_1 - x_1 v_0 \neq 0$, this is a nonconstant Möbius function of s; its image omits at most one element of K. It cannot be contained in $\mathcal{O}_K$, because $K \setminus \mathcal{O}_K$ is infinite. Hence $v_1 = x_1 v_0$ at every such point. Apply this to zeros with input 0 and input 1, whose existence follows from $S_F = \mathcal{O}_K$. It follows that

$$v_0 = v_1 = 0 \qquad (v \in W).$$

Thus both X_0 and X_1 descend to independent linear forms on E/W. Choose coordinates on the quotient beginning with these forms. The irreducible quotient cubic $\overline{H}$ is not a cone, and its affine chart $X_0 = 1$ has exactly the same input projection as H. Indeed, every rational point on the quotient lifts through a K-linear splitting of $E \to E/W$, preserving X_0 and X_1, and H is constant along the fibers.

If the projective quotient hypersurface has dimension at least two, Theorem 24.6 makes it unirational, since it has a K-point with $X_0 = 1$. Its input function X_1/X_0 is nonconstant, as its image contains all of the infinite set $\mathcal{O}_K$. Corollary 24.4 now gives a nonintegral input, a contradiction.

If the quotient has dimension one, its affine chart is defined by a polynomial in the input and just one further coordinate. Its projection cannot be $\mathcal{O}_K$ by Proposition 24.2. If the quotient has dimension zero, the affine chart has only finitely many points and likewise cannot project onto $\mathcal{O}_K$. These cases exhaust the possibilities, proving $\deg F \geq 4$. □

Remark 24.7 (Scope of the optimality statement)**.** The conclusion concerns one polynomial equation with all witnesses in the field, and total degree in all variables. It does not prohibit a system of quadratic equations: a sum of their squares over $\mathbb{Q}$ is quartic. It also does not determine the smallest possible number of witnesses at degree four, or the smallest degree among definitions attaining the unknown minimum number of witnesses. The unconditional numerical bounds established here are two witnesses and degree four; no example meeting both bounds is asserted.

24.5 A stronger one-witness obstruction

For completeness there is a useful strengthening which uses Siegel's theorem rather than just Hilbert irreducibility. In its rational-function form, Siegel's theorem says that if C/K is a smooth projective geometrically integral curve of positive genus and $f \in K(C)$ is nonconstant, then

$$\{P \in C(K) : f(P) \text{ is defined and belongs to } \mathcal{O}_K\}$$

is finite. One can derive this from the usual theorem on affine integral points by taking C minus the poles of f: the map to $\mathbb{A}^1$ defined by f is finite, and after inverting finitely many primes its coordinate functions are integral over $\mathcal{O}_{K,S}[f]$. This reduction is also used in [17, Section 2.1]; see [18, Chapter 8].

Proposition 24.8. *If $G \in K[T, Y]$ satisfies $S_G \subseteq \mathcal{O}_K$, then S_G is finite. Thus no infinite subset of $\mathcal{O}_K$ is the projection of the K-zeros of a single polynomial with just one witness.*

Proof. Factor G and treat its finitely many irreducible factors separately. Factors depending only on T give finite projections. For another factor, let C be its plane curve. If C is not geometrically irreducible, its K-points lie in the intersections of distinct geometric components, a finite set. Otherwise pass to its smooth projective normalization $\widetilde{C}$. Away from the finitely many singular points of C, rational points lift to $\widetilde{C}(K)$, and the input defines a nonconstant rational function t on $\widetilde{C}$.

If $\widetilde{C}$ has positive genus, Siegel's theorem gives only finitely many points for which t is integral. If its genus is zero and it has any rational point, then $\widetilde{C} \simeq \mathbb{P}^1_K$. The input then becomes a nonconstant rational function on $\mathbb{P}^1_K$, and on a nonempty open subset all its values would be integral, contradicting Lemma 24.3. If it has no rational point, only the finitely many previously excluded singular points remain. Each case gives a finite projection. □

Corollary 24.9. *Let $F \in K[T, Y_1, \ldots, Y_r]$ have total degree at most three. If $S_F \subseteq \mathcal{O}_K$, then S_F is finite.*

Proof. For an infinite projection the proof of Theorem 24.1 applies unchanged: choose any two distinct input values in the geometric-reducibility argument and in the vertex elimination. In the curve case use Proposition 24.8 instead of Proposition 24.2. The other cases already give a contradiction to infinitude. □

25 Stronger obstructions for small polynomial presentations

The unrestricted bounds established in the preceding lower-bound section are two existential witnesses and total degree four. The arguments below give stronger bounds for important classes of presentations. They do not prove that every definition needs more than two witnesses. In particular, the general case of a sign-changing quartic in three total variables remains unresolved here.

For a number field K, write $\mathcal{O}_K$ for its ring of integers. If $V \subseteq \mathbb{A}^n_K$ is an algebraic set with first coordinate T, put

$$\pi(V(K)) = \{t \in K : \text{some } (t, y_1, \ldots, y_{n-1}) \in V(K)\}.$$

Thus n counts all coordinates, and $n - 1$ counts existential witnesses. All assertions in this supplement are independent of the proposed arithmetic proof in the repository.

25.1 Curves and rational parametrizations

We use the following consequences of Siegel's theorem and the topology of a number field. They apply to open subsets, which is important when using birational maps.

Lemma 25.1. *Let V be an algebraic set over K of dimension at most one, and let $f : V \to \mathbb{A}^1_K$ be a regular function. If $f(V(K)) \subseteq \mathcal{O}_K$, then $f(V(K))$ is finite. The same conclusion holds for a rational function restricted to any Zariski-open subset on which it is defined.*

Proof. There are finitely many irreducible components. A zero-dimensional component contributes finitely many points, and a component on which f is constant contributes at most one value. On

an irreducible curve which is not geometrically irreducible, every K-point lies on the intersection of two distinct conjugate geometric components. There are only finitely many such points.

For a geometrically integral curve, normalization and projective completion discard or add only finitely many points. Let C be the resulting smooth projective curve and let $f \in K(C)$ be nonconstant. If C has positive genus, Siegel's theorem in its rational-function form gives only finitely many $P \in C(K)$ for which $f(P)$ is defined and integral. One obtains this form from the usual theorem by removing the poles of f and enlarging a finite set of primes so that the coordinate functions of that affine curve are integral over $\mathcal{O}_{K,S}[f]$; see [18, Chapter 8].

If C has genus zero and has a K-point, it is K-isomorphic to $\mathbb{P}^1$. A nonconstant rational function on a nonempty open subset of $\mathbb{P}^1(K)$ cannot take only integral values. Indeed, embed K diagonally in $K_\infty = \mathbb{R}^{r_1} \times \mathbb{C}^{r_2}$. The image of K is dense, while that of $\mathcal{O}_K$ is a discrete lattice. Continuity would force the rational function to be constant on a small connected open ball, and hence identically constant. If $C(K)$ is empty, only the previously discarded singular points remain. These cases prove the assertion. □

Lemma 25.2. *Let X/K be an integral unirational variety, let $U \subseteq X$ be nonempty and open, and let $f \in K(X)$ be nonconstant and defined on U. Then $f(U(K))$ is not contained in $\mathcal{O}_K$.*

Proof. Compose f with a dominant rational map from projective space to X. After restricting to an affine open set, the composition is a nonconstant rational function in finitely many variables. The density and discreteness argument in the preceding proof, now on K_∞^m, gives a contradiction if all its values are integral. □

Lemma 25.3. *Let C/K be a geometrically integral curve, let $m \geq 1$, and let X be birational over K to $C \times \mathbb{P}^m$. Suppose $U \subseteq X$ is nonempty and open and $U(K)$ is Zariski dense. A nonconstant $f \in K(X)$, defined on U, cannot satisfy $f(U(K)) \subseteq \mathcal{O}_K$.*

Proof. Restrict the birational correspondence to isomorphic nonempty opens and replace C by its smooth projective model. We may work on an open subset of $C \times \mathbb{A}^m$. If f is nonconstant in the affine coordinates over $K(C)$, then outside finitely many points of C its restriction to a fiber is nonconstant. Outside another finite set that fiber meets the chosen open subset in a nonempty open subset of $\mathbb{A}^m$. Density of $U(K)$ supplies a rational base point outside these exceptional sets. The restriction to this fiber violates Lemma 25.2.

Otherwise f belongs to $K(C)$ and is nonconstant there. For every $c \in C(K)$ outside a finite set, the open fiber has a K-point, so $f(c)$ is integral. Lemma 25.1 gives only finitely many such values, hence only finitely many such c, because a nonconstant map of projective curves has finite fibers. This contradicts the density of $U(K)$ in $C \times \mathbb{A}^m$. □

Remark 25.4. The preceding lemma concerns a variety birational over K to a product with projective space. Geometric ruledness after extending scalars is not sufficient for its proof. In particular, it must not be applied without further argument to an arbitrary conic bundle over $\mathbb{P}^1_K$ which has no rational section.

Lemma 25.5. *Let V/K be an algebraic set and $f : V \to \mathbb{A}^1_K$ a regular function. If $f(V(K))$ is infinite and contained in $\mathcal{O}_K$, the Zariski closure of $V(K)$ has a geometrically integral component X of dimension at least two such that $X(K)$ is Zariski dense and $f|_X$ is nonconstant.*

Proof. The closure has finitely many irreducible components, each with dense K-points on the open set obtained by deleting the other components. If such a component were geometrically

reducible, its rational points would lie in the proper closed intersections of its conjugate geometric components, a contradiction. Components on which f is constant contribute finitely many values, and components of dimension at most one contribute finitely many by Lemma 25.1. Thus at least one component has the required properties. □

25.2 Two quadrics in projective four-space

The next argument uses only the elementary degree bound $\deg X \geq 1 + \operatorname{codim}(X)$ for a nondegenerate projective integral variety, Bézout's theorem, and the following theorem of Kollár: a geometrically integral cubic hypersurface of dimension at least two over a perfect field is unirational if it has a rational point and is not a cone [15, Theorem 2]. The elementary degree bound is recalled in [19, Introduction]. We give the projection argument explicitly in order to include reducible and nonreduced intersections.

Proposition 25.6. *Let $Q_1, Q_2 \in K[X_0, \ldots, X_4]$ be coprime homogeneous quadratics, and let X be a geometrically integral surface component of $V(Q_1, Q_2)$. Suppose $X(K)$ is Zariski dense. Then X is either unirational over K or birational over K to $C \times \mathbb{P}^1$ for a geometrically integral curve C/K.*

Proof. The complete intersection $W = V(Q_1, Q_2) \subseteq \mathbb{P}^4$ has degree four. More precisely, if m_X is its multiplicity along X, then $m_X \deg X \leq 4$. This follows either from Bézout's theorem with multiplicities or from the complete-intersection Hilbert series $(1 - t^2)^2/(1 - t)^5$ and additivity of degree.

If $\deg X \leq 2$, the elementary degree bound shows that its linear span has dimension at most three. It is therefore a plane or an integral quadric surface in its span. Its linear span is defined over K, and density supplies a smooth K-point in the quadric case. Projection from that point makes the quadric rational; a plane is rational as well.

Suppose $\deg X \geq 3$. The degree inequality forces $m_X = 1$. Since the characteristic is zero, a nonempty open subset of X is smooth in the complete intersection W. Choose a K-point there, using density, and change coordinates so that it is $p = (1 : 0 : 0 : 0 : 0)$. The quadratics have the form

$$Q_1 = X_0 L_1 + q_1, \qquad Q_2 = X_0 L_2 + q_2,$$

where L_i are linear and q_i are quadratic in $X_1, \ldots, X_4$. Smoothness of W at p means that L_1, L_2 are independent. The cubic

$$H = L_1 q_2 - L_2 q_1$$

is not identically zero. Otherwise coprimality of L_1, L_2 gives $q_1 = L_1 L_3$ and $q_2 = L_2 L_3$ for a linear form L_3, contradicting coprimality of Q_1, Q_2.

The surface X cannot be contained in the plane $L_1 = L_2 = 0$, because its degree is at least three. Interchange the two quadratics if necessary so that L_1 does not vanish identically on X. Projection from p then has the rational inverse

$$X_0 = -q_1/L_1$$

on the corresponding open set. Thus X is birational to a geometrically integral surface component Y of the cubic $V(H) \subseteq \mathbb{P}^3$. The rational points of Y are dense.

If Y is a plane or a quadric, it is rational as above. If it is a cubic which is not a cone, Kollár's theorem makes it unirational. Finally, a geometrically integral cubic surface which is a cone has

a single vertex point defined over K. Indeed the vertex directions form the kernel of a linear map over K given by the directional derivatives of its cubic equation; more than one independent vertex direction would leave a binary cubic, which is geometrically reducible. The complement of the vertex is birational over K to the product of its plane cubic base with $\mathbb{P}^1$. This proves the result. □

Theorem 25.7. *Let $q_1, \ldots, q_s \in K[T, Y_1, \ldots, Y_{n-1}]$ have total degree at most two, where $n \leq 4$. If the rational projection*

$$S = \{t \in K : \exists \mathbf{y} \in K^{n-1}, \quad q_1(t, \mathbf{y}) = \cdots = q_s(t, \mathbf{y}) = 0\}$$

is contained in $\mathcal{O}_K$, then S is finite.

Proof. Adding unused coordinates reduces the assertion to $n = 4$. Zero residuals can be removed; if all residuals vanish identically, then $S = K$, contrary to the hypothesis. A nonzero constant residual gives the empty set. Let V be their common zero set.

Suppose S is infinite, and choose $X \subseteq V$ as in Lemma 25.5. Any common polynomial factor g of the residuals has degree at most two. The entire hypersurface $g = 0$ lies in V, and its input projection is therefore contained in $\mathcal{O}_K$. The quadratic obstruction proved in the first report says this projection has at most two elements. Consequently X is not contained in $g = 0$. Dividing the residuals by their common greatest divisor does not remove X and leaves quadratics with greatest common divisor one.

If their K-linear span contains a nonzero affine linear polynomial, restrict to its affine hyperplane, isomorphic to $\mathbb{A}^3_K$. Not all restricted residuals vanish identically, since that would give the original quadratics a common linear factor. The resulting surface X is a hypersurface component of a nonzero quadratic in three variables, hence is a plane or a quadric. Density gives a rational parametrization, contradicting Lemma 25.2.

Otherwise choose two coprime quadratics in the linear span of the residuals. To see that this is possible, first choose a nonzero quadratic q and then a linear combination not divisible by any irreducible factor of q; finitely many proper linear subspaces cannot cover a vector space over the infinite field K. Their affine common zero set has dimension two. Thus X is a surface component. Their degree-two homogenizations remain coprime, and the projective closure of X is a component of their intersection in $\mathbb{P}^4$.

Proposition 25.6 makes this component unirational or birational to a product of a curve and $\mathbb{P}^1$. The input is a nonconstant rational function and is integral on the rational points of its affine open subset. This contradicts Lemma 25.2 or Lemma 25.3. □

Corollary 25.8. *Suppose*

$$P(T, \mathbf{Y}) = \sum_{j=1}^{s} c_j q_j(T, \mathbf{Y})^2, \qquad c_j \in \mathbb{Q}_{>0}, \quad \deg q_j \leq 2.$$

If the rational existential projection of $P = 0$ is $\mathbb{Z}$, then P uses at least five total variables, hence at least four existential witnesses. The same lower bound holds over a number field for any quartic obtained by applying an anisotropic quadratic form to quadratic residuals and defining its ring of integers.

Proof. In the rational case, the sum is zero exactly when all q_j vanish. For an anisotropic combining form the same equivalence holds by definition of anisotropy. Apply Theorem 25.7. □

This improves the unrestricted three-total-variable bound for the sum-of-squares quartic architecture used by the released polynomial and all the expanded quartic variants constructed above. It does not rule out a sign-changing quartic in three or four total variables.

25.3 Nonnegative polynomials in three variables

Proposition 25.9. *Let $P \in \mathbb{Q}[T, Y, Z]$ be squarefree and nonnegative on $\mathbb{R}^3$. If its rational input projection is contained in $\mathbb{Z}$, that projection is finite.*

Proof. Every real zero of a differentiable nonnegative function is a local minimum, so all three partial derivatives of P vanish there. The singular locus of a squarefree hypersurface in characteristic zero has codimension at least one in that hypersurface. In this case it has dimension at most one. All rational zeros lie in this singular locus. Apply Lemma 25.1 to its input function. □

Theorem 25.10. *Let $P \in \mathbb{Q}[T, Y, Z]$ be nonnegative on $\mathbb{R}^3$. If its rational input projection is an infinite subset of $\mathbb{Z}$, then P has an irreducible factor of degree at least four and multiplicity at least two. In particular* $\deg P \geq 8$.

Proof. Choose a component X of the closure of its rational zeros as in Lemma 25.5. It has dimension two, so its prime ideal is generated by an irreducible factor g of P. Write $P = g^m h$ with $g \nmid h$. Density supplies a rational point on X which is smooth on $g = 0$ and does not lie on $h = 0$. If $m = 1$, then $\nabla P = h \nabla g$ is nonzero there, contrary to nonnegativity of P. Thus $m \geq 2$.

The projection of the whole hypersurface $g = 0$ is contained in the projection of $P = 0$, and is infinite because the input is nonconstant on X and $X(\mathbb{Q})$ is dense. The unconditional degree-three obstruction of the preceding lower-bound section gives $\deg g \geq 4$. Therefore $\deg P \geq m \deg g \geq 8$. □

Nonnegativity only on rational inputs gives the same result, because $\mathbb{Q}^3$ is dense in $\mathbb{R}^3$ and P is continuous. The degree-eight conclusion is restricted to nonnegative polynomials; it is not an unconditional degree-eight lower bound for two witnesses.

25.4 Further necessary conditions

Proposition 25.11. *If a polynomial $P(T, Y_1, \ldots, Y_r) \in K[T, \mathbf{Y}]$ defines $\mathcal{O}_K$ in K, its degree in each witness which occurs is at least two. Its degree in T is also at least two.*

Proof. Suppose $P = AY_r + B$, where $A \neq 0$ is independent of Y_r. For all but finitely many $t \in K$, the specialization $A(t, Y_1, \ldots, Y_{r-1})$ is a nonzero polynomial. Since K is infinite, choose the other witnesses so that it does not vanish and then solve uniquely for Y_r. The projection is cofinite in K, which is incompatible with $\mathcal{O}_K$.

If $P = AT + B$ with $A \neq 0$ independent of T, its projection includes the values of $-B/A$ on the nonempty open set $A \neq 0$. Integrality of all these values forces this rational function to be constant, by the density and discreteness argument. Thus $B = -cA$ for some $c \in K$, and $P = A(T - c)$. If A has a K-zero, every input is accepted; otherwise only c is accepted. Neither is $\mathcal{O}_K$. If P is independent of T, its projection is empty or all of K. □

Proposition 25.12. *Suppose a quartic $P \in \mathbb{Q}[T, Y, Z]$ defines $\mathbb{Z}$ in $\mathbb{Q}$. Then P is irreducible and geometrically irreducible, its rational points are Zariski dense on its affine surface, and P takes both positive and negative values on $\mathbb{R}^3$. The input function is nonconstant on that surface. Every*

curve birational over $\mathbb{Q}$ to $\mathbb{P}^1$ on which the input is defined is vertical (the input is constant), and the same holds for every geometrically integral genus-one curve on the surface with infinitely many rational points.

Proof. If P factors nontrivially, every factor has degree at most three. Each factor's rational projection is contained in $\mathbb{Z}$ and is finite by the earlier degree-three obstruction. Their finite union cannot be $\mathbb{Z}$. If the irreducible P is geometrically reducible, its rational points lie in intersections of distinct conjugate components, of dimension at most one; Lemma 25.1 again gives a finite projection. The same lemma shows that a proper Zariski closure of the rational points would give a finite projection. Thus the surface is geometrically integral and its rational points are dense. The input is nonconstant because its image is infinite.

If P had only one sign, apply Theorem 25.10 to P or $-P$. Finally, a nonvertical rational curve with a rational point would violate Lemma 25.2. On a genus-one curve with infinitely many rational points a nonconstant input function would take only finitely many integral values at rational points by Siegel's theorem, whereas every rational point where the input is defined would have to give such a value. This is impossible. □

25.5 Why the unrestricted two-witness case remains difficult

A smooth quartic surface in $\mathbb{P}^3$ has trivial canonical bundle and is a K3 surface. It is not covered by the unirational cubic theorem. Zariski density of rational points, even when known, does not itself imply density in the real topology or the existence of a nonintegral value of a specified rational function. Likewise, potential density after a finite field extension cannot be substituted for a statement about rational points over $\mathbb{Q}$. For examples and the distinction between density over the ground field and potential density, see [20].

The same limitation appears naturally in the improved bound for quadratic systems. A smooth complete intersection of three quadrics in $\mathbb{P}^5$ is a K3 surface: adjunction gives $K_X = \mathcal{O}_X(2+2+2-6) = \mathcal{O}_X$, and the complete-intersection cohomology gives $H^1(X, \mathcal{O}_X) = 0$. Thus five affine coordinates already permit a class of surfaces not handled by the two-quadric argument. This observation is a limit of the proof, not an existence claim for a five-variable definition.

Proposition 25.13 (Conditional obstruction for general-type surfaces)**.** *Assume the weak Bombieri–Lang conjecture for surfaces over $\mathbb{Q}$: the rational points of a smooth projective surface of general type are not Zariski dense. Then no polynomial in three total variables whose geometrically integral projective surface has a smooth projective model of general type can have an infinite rational input projection contained in $\mathbb{Z}$. In particular, under this conjecture a polynomial in three variables with smooth projective closure of degree at least five cannot define $\mathbb{Z}$.*

Proof. Lemma 25.1 forces Zariski density of rational points on a surface with such a projection. Passing to isomorphic dense opens of a smooth projective model preserves density, contradicting the conjecture. A smooth degree-d surface in $\mathbb{P}^3$ has canonical bundle $\mathcal{O}_X(d-4)$, which is ample when $d \geq 5$. □

Mazur's conjecture about finitely many real connected components of the closure of rational points would rule out every Diophantine definition of $\mathbb{Z}$ in $\mathbb{Q}$, in any number of variables. It therefore cannot supply a conditional numerical improvement compatible with simultaneously assuming the proposed arithmetic theorem. Proposition 25.13, by contrast, is a restriction on one class of possible two-witness surfaces.

26 A restriction on the degree in two witnesses

The total degree can conceal an important distinction: a polynomial may have arbitrarily high degree in the input, while remaining quadratic in its witnesses. With only two witnesses, such a polynomial cannot define the ring of integers. We prove a simultaneous version which also rules out a finite disjunction of such equations.

Theorem 26.1. *Let K be a number field and let $F_1, \dots, F_s \in K[T, Y, Z]$ have total degree at most two in the pair (Y, Z), with no restriction on their degree in T. Put*

$$S = \{t \in K : \text{for some } i \text{ and } y, z \in K,\ F_i(t, y, z) = 0\}.$$

Exactly one of the following alternatives holds:

1. *at least one equation has a solution $(Y, Z) \in K(T)^2$, and S is cofinite in K;*
2. *no equation has such a solution, and $K \setminus S$ contains infinitely many elements of every coset $a + \mathfrak{a}$, where $a \in \mathcal{O}_K$ and $\mathfrak{a}$ is a nonzero ideal of $\mathcal{O}_K$.*

In particular, no single polynomial quadratic in two witnesses, or finite disjunction of such polynomials, defines $\mathcal{O}_K$ in K.

The cofinite assertion in the first alternative follows by specializing the rational functions in a generic solution away from their finitely many poles. The substantive issue is to find integral specializations which simultaneously preserve the obstructions in the second alternative. We use Faddeev's residue sequence, Chebotarev's theorem and integral Hilbert irreducibility. The argument is a consequence of these classical tools; no claim of historical priority is made.

26.1 Quaternion obstructions at integral specializations

For a field F of characteristic different from two, (a, b) denotes the Brauer class of the quaternion algebra generated by i, j with $i^2 = a$, $j^2 = b$ and $ij = -ji$. A smooth conic is split, or equivalently has an F-point, precisely when its associated quaternion class is zero. For this correspondence and the residue sequence below see [21, Theorem 5.1 and Section 6.1].

We recall explicitly the residue convention. At the closed point of $\mathbb{A}^1_K$ given by a monic irreducible $f(T)$, write

$$a = f^r u, \qquad b = f^s v,$$

where u, v are units in $K[T]_{(f)}$. If θ is a root of f and $L = K(\theta)$, the residue of (a, b) is the square class of

$$(-1)^{rs} u(\theta)^s / v(\theta)^r \quad \text{in } L^\times / L^{\times 2}.$$

Faddeev's theorem implies that a class in $\operatorname{Br} K(T)$ with all finite residues zero comes from $\operatorname{Br} K$. Indeed the sum-of-corestrictions relation first makes the residue at infinity zero, since its residue field is K; exactness then gives a constant class. In characteristic zero the sequence applies to the whole Brauer group because $\operatorname{Br} \overline{K}(T) = 0$.

We shall also use the following elementary local formula. If F is a nonarchimedean local field with odd residue field κ, if ϖ is a uniformizer, and if u, v are units, the quaternion $(\varpi^r u, \varpi^s v)$ is split exactly when

$$(-1)^{rs} \overline{u}^{\,s} / \overline{v}^{\,r} \quad \text{is a square in } \kappa^i mes.$$

One obtains this by bilinearity of the Hilbert symbol: two unit parameters give a split algebra, (ϖ, u) is detected by the square class of $\overline{u}$, and $(\varpi, \varpi) = (\varpi, -1)$. These facts also follow from the norm criterion for the unramified quadratic extension; see [23, Proposition 5.5].

Lemma 26.2. *Let $\alpha \in \operatorname{Br} K(T)$ be a nonzero quaternion class. If α is constant, every specialization outside a finite set is nonzero. If α is not constant, then outside any prescribed finite set of primes of K there is a prime $\mathfrak{p}$ and a residue class $c \bmod \mathfrak{p}^2$ such that, outside a finite set of inputs, every $t \in \mathcal{O}_K$ in that residue class has*

$$\alpha(t) \neq 0 \quad \textit{in } \operatorname{Br} K_{\mathfrak{p}}.$$

Proof. The assertion for a constant class follows from specialization on an open set where its equality with the chosen quaternion presentation is defined. Otherwise choose a nonzero finite residue using Faddeev's theorem. With the notation just introduced, let

$$d = (-1)^{rs} u(\theta)^s / v(\theta)^r \in L^\times$$

be its nonsquare representative. Choose a finite Galois extension M/K containing $L(\sqrt{d})$, and an element $g \in \operatorname{Gal}(M/L)$ which sends $\sqrt{d}$ to $-\sqrt{d}$. Such an element exists because d is not a square in L.

Chebotarev's theorem gives infinitely many primes of K with Frobenius conjugate to g; see [22, Section 2.1, Theorem 1]. Choose one outside the prescribed finite set and all further primes of bad reduction for the displayed algebraic data, including primes above two. Choose a prime of M whose Frobenius is exactly g, and let $\mathfrak{q}$ be its restriction to L. Since g fixes L, this prime of L has residue degree one over $\mathfrak{p}$. Since g acts nontrivially on $\sqrt{d}$, the reduction of d at $\mathfrak{q}$ is a nonsquare. The excluded bad primes ensure that it is a nonzero unit.

The corresponding completion of L is $K_{\mathfrak{p}}$, so f has a simple root $\theta_{\mathfrak{p}}$ in $\mathcal{O}_{K_{\mathfrak{p}}}$. Choose a residue class $c \bmod \mathfrak{p}^2$ for which

$$v_{\mathfrak{p}}(t - \theta_{\mathfrak{p}}) = 1 \qquad (t \equiv c \bmod \mathfrak{p}^2).$$

There is such a class because the reduction map from $\mathcal{O}_K$ onto $\mathcal{O}_{K_{\mathfrak{p}}}/\mathfrak{p}^2$ is surjective. For all these inputs, $v_{\mathfrak{p}}(f(t)) = 1$, and $u(t), v(t)$ are units whose reductions equal those at θ and $\mathfrak{q}$. The local formula therefore makes the specialized quaternion nonsplit, since its residue square class is that of d. The finitely many inputs where a rational presentation is undefined may be discarded. This proves the assertion. □

Corollary 26.3. *Let $C_1, \dots, C_m$ be smooth conics over $K(T)$ with no $K(T)$-point. Given any coset $a + \mathfrak{a}$ as in Theorem 26.1, it contains a smaller nonempty coset of a nonzero ideal such that, outside finitely many inputs, none of the specialized conics has a K-point.*

Proof. Constant nonzero quaternion classes already exclude every good specialization. For each remaining conic, use the preceding lemma with a fresh prime, distinct from all previously chosen primes and coprime to $\mathfrak{a}$. The Chinese remainder theorem simultaneously imposes its bad residue class modulo the square of that prime and the original coset. Every resulting good specialization is locally insoluble at the selected prime for each corresponding conic. □

26.2 Degenerate conics and the simultaneous theorem

Proof of Theorem 26.1. Write $F = K(T)$. If one equation has a point in F^2, its rational functions specialize to a point at every input outside a finite set. This gives the first alternative. Suppose

henceforth that none does. We classify each equation over F, and discard finitely many inputs so that all rational coordinate changes, ranks and nonzero coefficients in this classification specialize correctly.

An equation of witness degree zero is a nonzero element of F, and has no point at any good input. Witness degree one always gives an affine F-point, so cannot occur in the case under consideration. For witness degree two, homogenize in Y, Z with a third coordinate W, giving a ternary quadratic form and its projective conic C. If the form has rank three and its quaternion class is zero, then $C \simeq \mathbb{P}^1_F$. The affine open $W \neq 0$ is nonempty and hence has an F-point, a contradiction. Thus every rank-three case has a nonzero quaternion class, and Corollary 26.3 simultaneously excludes all of them on a smaller coset of the given integral progression.

A rank-one ternary form is a scalar times the square of an F-linear form. That line cannot be the line at infinity, since the original polynomial has witness degree two. It therefore has an affine F-point, again a contradiction. A split rank-two form is the product of two distinct F-linear forms. At least one line is not the line at infinity, and supplies the same contradiction.

The remaining possibility is a nonsplit rank-two form. After a linear change over F it is a nonzero scalar times

$$U^2 - d(T)V^2, \qquad d(T) \notin F^{\times 2}.$$

Its only projective F-point is its radical point, defined by $U = V = 0$. If that point were affine, it would give an F-point of the original equation. Therefore its original W coordinate is zero. At every good specialization for which $d(t)$ remains a nonsquare, the same reasoning leaves only this radical point, still at infinity, and the specialized affine equation has no solution.

There are only finitely many nonsquare functions $d(T)$ to treat. The smaller coset already obtained contains a progression $t = c + bz$ with $c \in \mathcal{O}_K$, $0 \neq b \in \mathcal{O}_K$ and $z \in \mathcal{O}_K$: choose b in its defining nonzero ideal. The polynomials

$$Y^2 - d(c + bZ)$$

remain irreducible over $K(Z)$, because $T \mapsto c + bZ$ is an isomorphism of rational function fields. The integral Hilbert irreducibility theorem, simultaneously for this finite list, gives infinitely many integral z for which all these specializations remain irreducible and all previously excluded finite conditions are avoided. See the integral form recalled in the original lower-bound section and [17, Introduction]. If the list is empty, any infinitely many good inputs in the coset suffice.

The resulting inputs lie in the original coset $a + \mathfrak{a}$, and none admits a solution of any equation. This proves the second alternative and the theorem. □

Corollary 26.4. *If $P(T, Y, Z)$ defines $\mathbb{Z}$ in $\mathbb{Q}$, then its total degree in the two witnesses Y, Z is at least three. More generally, after factoring over $\mathbb{Q}(T)$, at least one factor has total witness degree at least three. This remains true after every invertible affine change of the witness coordinates over $\mathbb{Q}(T)$, provided the transformed equation and inverse change are defined at all but finitely many input values.*

Proof. The first assertion is the theorem with one equation. If all factors had witness degree at most two, clear their rational-function denominators. Outside finitely many inputs, the original zero condition is their finite disjunction, contradicting the theorem. An invertible affine change over $\mathbb{Q}(T)$ preserves the rational solution sets at every good specialization, so the same argument applies after it. □

Corollary 26.5. *Let $S \subset K$ contain $\mathcal{O}_K$, and suppose that $K \setminus S$ is infinite. Then S is not a finite union of projections of equations in two witnesses which are jointly quadratic in those witnesses, even if the degree in the input is unrestricted.*

Proof. The second alternative of Theorem 26.1 would omit infinitely many elements of $\mathcal{O}_K$, while the first would make $K \setminus S$ finite. Both contradict the stated assumptions. □

For example, take

$$S = \mathbb{Z}_{(2)} \cap \mathbb{Z}_{(3)} = \{a/b \in \mathbb{Q} : \gcd(a,b) = 1,\ b > 0,\ 2 \nmid b,\ 3 \nmid b\}.$$

This set contains $\mathbb{Z}$ and has infinite complement, for instance the distinct elements 2^{-n} for $n \geq 1$. Thus a quadratic three-witness definition of this semilocal ring cannot be replaced by a quadratic two-witness definition, or a finite union of such definitions. The obstruction concerns joint witness degree; it allows arbitrary input degree and does not assert a lower bound for equations of higher witness degree.

This is an unconditional restriction on a possible two-witness definition at every input degree. It does not exclude general quartic surfaces, whose generic fibers over the input can have positive genus. It also does not extend by this proof to quadratic equations in three witnesses: their fibers are quadric surfaces, and are not controlled by a single quaternion obstruction of the kind used above.

27 Lower bounds for systems of polynomial equations

Let K be a number field, and let $\mathcal{O}_K$ be its ring of integers. A polynomial system in n total coordinates has the form

$$F_1(T, Y_1, \dots, Y_{n-1}) = \cdots = F_s(T, Y_1, \dots, Y_{n-1}) = 0.$$

Its degree is $\max_i \deg F_i$, including the input T. Allowing several equations does not change the number of quantified coordinates. We also allow finitely many nonzero conditions when stating obstructions: these specify a Zariski open subset of the common zero set, and strengthen rather than weaken the conclusions for closed systems. A finite union of such presentations is treated componentwise.

The main new restriction is that *two quadratic equations cannot suffice in any number of coordinates.* The proof includes singular, reducible, and nonreduced intersections, rather than assuming a smooth complete intersection.

27.1 Integral-valued functions on simple varieties

Lemma 27.1. *Let X be an integral K-variety which is K-unirational, let $U \subset X$ be nonempty and open, and let $f : U \to \mathbb{A}^1_K$ be a regular function. If $f(U(K)) \subseteq \mathcal{O}_K$, then f is constant. In particular, its image is finite.*

Proof. A dominant rational parametrization reduces the assertion to a rational function on a nonempty open subset of affine space. If this function were nonconstant, one of its partial derivatives would be nonzero. Choose a K-point at which that derivative and every relevant denominator are nonzero. Such a point exists because K is infinite.

Put $K_\infty = \prod_{v|\infty} K_v$, with one factor for each real place and one complex factor for each pair of complex embeddings. The diagonal copy of K is dense in K_∞: a $\mathbb{Q}$-basis of K identifies this assertion with the density of $\mathbb{Q}^{[K:\mathbb{Q}]}$ in its real vector space. In contrast, $\mathcal{O}_K$ is a lattice, hence is closed and discrete there. The inverse-function theorem at the chosen point, at every archimedean place, gives a nonempty open subset of K_∞ in the local image of the rational function. Density of K makes its K-values dense in this open subset. They cannot all belong to the discrete set $\mathcal{O}_K$. This is the required contradiction. □

Lemma 27.2. *Let X be an algebraic set over K of dimension at most one, let $U \subseteq X$ be open, and let $f : U \to \mathbb{A}^1_K$ be regular. If $f(U(K)) \subseteq \mathcal{O}_K$, then its image is finite.*

Proof. Reduce to the finitely many irreducible components. Zero-dimensional components give finite sets. If an irreducible curve is not geometrically integral, all its K-points lie in the intersections of distinct geometric components, a finite set. Otherwise take its smooth projective normalization C. Except for finitely many points, rational points lift to $C(K)$, and f becomes a rational function on C. There is nothing to prove when this function is constant.

If C has genus zero and a K-point, it is isomorphic to $\mathbb{P}^1_K$, and Lemma 27.1 excludes a nonconstant integral-valued function on its permitted open subset. If it has no K-point, only the finitely many excluded points remain. If the genus is positive, Siegel's theorem makes the set of K-points where a fixed nonconstant rational function is integral finite. For this last formulation, remove the poles of the function: the resulting affine curve is finite over the affine line. After inverting a fixed finite set of primes, its coordinate ring is integral over the ring generated by that function. Integral function values therefore give points integral outside this fixed set, to which Siegel's theorem applies; see [18, Chapter 8]. □

Lemma 27.3. *Let $X \subseteq \mathbb{P}^N_K$ be a hypersurface defined by a nonzero homogeneous polynomial of degree at most three. For an open subset $U \subseteq X$ and a regular function $f : U \to \mathbb{A}^1_K$, the containment $f(U(K)) \subseteq \mathcal{O}_K$ implies that $f(U(K))$ is finite. The same conclusion holds for $X = \mathbb{P}^N_K$.*

Proof. The assertion for projective space follows from Lemma 27.1. Factor the defining polynomial over K and treat each irreducible factor separately; multiplicities do not change the rational point set. If such a factor of degree two or three is not geometrically integral, its geometric factors are conjugate linear forms, since these degrees are prime. Every K-point lies in their common intersection. That intersection is a K-linear subspace, so the preceding lemma for unirational varieties applies. Linear factors are handled in the same way.

A geometrically integral quadric with a smooth K-point is rational by projection from that point. If it has no smooth K-point, all its K-points lie in its linear vertex. Both cases give the result. For a geometrically integral cubic of dimension at most one, use Lemma 27.2. In dimension at least two, Kollár's theorem says that a cubic which is not a cone and has a K-point is K-unirational [15, Theorem 2]. A number field is perfect, as required in that theorem.

It remains to treat a cubic cone. Its maximal linear vertex is defined over K: in characteristic zero it is the projectivization of the vector subspace of directions along which its defining form is constant. Quotient by these directions and choose a complementary K-linear space. This gives a geometrically integral cubic base B which is not a cone. Its dimension is at least one; a geometrically integral zero-dimensional cubic would be impossible. Outside the vertex, projection to B has affine-space fibers, whose projective closures are the linear spaces joining a point of B to the vertex. The cone is birational to $B \times \mathbb{P}^{r+1}$ if the vertex has dimension r.

If $B(K)$ is empty, all rational points of the cone lie in the vertex. If $B(K)$ is finite, each permitted fiber is an open subset of a linear space; the function is constant on it by Lemma 27.1. Together with the vertex this gives a finite image. If $\dim B \geq 2$ and $B(K)$ is nonempty, Kollár's theorem makes B, and hence the cone, unirational.

The only remaining case is a cubic curve B with infinitely many K-points. Pass to its normalization C, discarding finitely many exceptional base points. On a dense open of the cone we may write f as a rational function on $C \times \mathbb{P}^{r+1}$. If it depends nontrivially on the projective-space coordinates, its specialization is nonconstant for all base points outside a finite subset of C. A rational base point outside that subset then contradicts Lemma 27.1 on the fiber. Thus $f \in K(C)$. For all but finitely many points of $C(K)$, the permitted fiber is a nonempty open subset of projective space and has a K-point. Consequently f is integral at all these base points. Lemma 27.2 forces f to be constant, because a nonconstant function on a curve has finite fibers and $C(K)$ is infinite. Being constant in the function field, it is constant wherever it is regular on the irreducible cone. This also treats all points omitted from the birational chart and completes the proof. □

27.2 Two quadrics in any dimension

Theorem 27.4. *Let $X \subseteq \mathbb{P}^N_K$ be the common zero set of at most two homogeneous quadratic forms, let $U \subseteq X$ be open, and let $f : U \to \mathbb{A}^1_K$ be regular. If $f(U(K)) \subseteq \mathcal{O}_K$, then $f(U(K))$ is finite. No smoothness, irreducibility, or reducedness assumption is made.*

Proof. We induct on N. The cases $N \leq 1$ follow from the curve lemma. A zero equation may be discarded. With at most one nonzero equation, use Lemma 27.3. If the two quadratic forms have a common factor of degree two, they are proportional. If their greatest common divisor is a linear form L, write them as LL_1, LL_2; their common zero set is the union of the linear spaces $L = 0$ and $L_1 = L_2 = 0$. The assertion again follows from the unirational lemma. We may therefore suppose that the two forms Q_1, Q_2 are coprime.

If $U(K)$ is empty there is nothing to prove. Choose $p \in U(K)$ and coordinates with $p = (1 : 0 : \cdots : 0)$. Since $Q_i(p) = 0$, we can write

$$Q_1 = X_0 L_1(X_1, \ldots, X_N) + q_1(X_1, \ldots, X_N), \qquad Q_2 = X_0 L_2(X_1, \ldots, X_N) + q_2(X_1, \ldots, X_N),$$

where the L_i are linear and the q_i are quadratic. We consider the rank of the span of L_1, L_2.

If the rank is two, projection from p is an isomorphism between

$$X \setminus V(L_1, L_2) \quad \text{and} \quad V(L_1 q_2 - L_2 q_1) \setminus V(L_1, L_2) \ \subseteq \mathbb{P}^{N-1}_K.$$

On the chart $L_i \neq 0$ the inverse is given by $X_0 = -q_i/L_i$; these formulas agree on the overlap. The cubic $L_1 q_2 - L_2 q_1$ is nonzero. Otherwise coprimality of the two linear forms would give $q_1 = L_1 M$, $q_2 = L_2 M$ for a linear form M, making $X_0 + M$ a common factor of Q_1, Q_2. The image of f on this open part is finite by Lemma 27.3. The exceptional part $X \cap V(L_1, L_2)$ is cut out by two quadrics in the smaller linear space $V(L_1, L_2) \simeq \mathbb{P}^{N-2}_K$. The induction hypothesis applies to the restriction of f and the open subset inherited from U.

If the rank is one, replace the two forms by an invertible linear combination so that $L_1 \neq 0$ and $L_2 = 0$. Away from $L_1 = 0$, projection from p is an isomorphism to the open subset

$$V(q_2) \setminus V(L_1) \subseteq \mathbb{P}^{N-1}_K,$$

with inverse $X_0 = -q_1/L_1$. The polynomial q_2 is nonzero, since the second equation was nonzero. The quadric lemma gives a finite image here. On the exceptional linear space $L_1 = 0$ we again have two quadrics in a smaller projective space, so induction gives a finite image there as well.

If the rank is zero, both equations are independent of X_0, and p is a vertex of the intersection. For every $q \in U(K)$, the line joining p to q lies in X. The restriction of f to the open part of this line has only integral values at its K-points, hence is constant by the unirational lemma. It is defined at both p and q, so $f(q) = f(p)$. The image is a singleton. The three cases prove the induction. □

Corollary 27.5. *A system of at most two equations of total degree at most two, in any number of rational coordinates over K, cannot define $\mathcal{O}_K$ in its input. More generally, whenever its projection, possibly restricted by finitely many polynomial nonzero conditions, lies in $\mathcal{O}_K$, that projection is finite. The same holds for a finite union of such presentations.*

Proof. Homogenize each equation to degree two and take the open chart where the homogenizing coordinate is nonzero. The input becomes a regular rational function on that chart. Apply the theorem; a finite union of finite images is finite. □

Linear equations can first be solved by Gaussian elimination. On the resulting affine subspace, the input is an affine linear function and the remaining residuals are still quadratic. Thus the same obstruction applies whenever those residuals span a vector space of dimension at most two. A quadratic system defining $\mathcal{O}_K$ must have at least three independent quadratic residuals after the linear equations have been removed. This assertion counts equations, not variables.

27.3 Restrictions on the number of coordinates

Lemma 27.6. *Let U be a locally closed subset of affine space over K and let $f : U \to \mathbb{A}^1_K$ be regular. If $f(U(K))$ is an infinite subset of $\mathcal{O}_K$, the Zariski closure of $U(K)$ has an irreducible component Z of dimension at least two, with Zariski-dense K-points and nonconstant input function. The component is geometrically integral.*

Proof. Take the finitely many irreducible components of the closure. The rational points are dense on each component after removing its intersections with the others. A component that is not geometrically integral has all its K-points in the intersections of distinct geometric components; this contradicts density in characteristic zero. If every component had dimension at most one or constant input, the curve lemma would make the union of their input images finite. At least one component has the stated properties. □

Theorem 27.7. *Suppose a finite polynomial system, possibly with nonzero guards, has an infinite rational input projection contained in $\mathcal{O}_K$.*

1. *It uses at least three total coordinates. This holds at every degree and for a finite union of presentations.*

2. *If it uses three total coordinates, every nonzero residual has degree at least four. In fact the residuals have a common irreducible factor of degree at least four with nonconstant input.*

3. *If it uses at most four total coordinates, its residuals cannot all be quadratic. More strongly, in four coordinates it cannot contain two coprime residuals of degree at most two, regardless of the degrees of its other residuals.*

Proof. For the first assertion, a proper algebraic subset of affine two-space has dimension at most one, so the curve lemma applies. If the closure is all of affine two-space, a locally closed subset with that closure contains a nonempty open subset; its input cannot be integral-valued there by the unirational lemma. The cases with fewer coordinates are included. A finite union does not help, since an infinite image would require an infinite image from one member.

In three coordinates, choose the component Z in Lemma 27.6. If there is a nonzero residual, its common zero set is proper, so $\dim Z = 2$. Its height-one prime ideal is generated by an irreducible polynomial g. Every residual vanishes on Z, hence is divisible by g. The permitted open subset of $V(g)$ is contained in the source system and has dense rational points and nonconstant input. Lemma 27.3 excludes $\deg g \leq 3$. This proves the second assertion. If all residuals are zero, the source is open in affine space and is excluded by the unirational lemma.

For the stronger statement in four coordinates, suppose two residuals Q_1, Q_2 of degree at most two are coprime. Their common zero set has dimension at most two. Thus the horizontal component Z again has dimension two and is an irreducible component of this common zero set. Homogenize to quadrics, restrict to the affine chart, and remove all other components. Intersect further with the source's nonzero guards. This gives an open subset of the intersection of two projective quadrics whose rational points lie in the source system. The image is finite by Theorem 27.4, whereas density of the points on Z and nonconstancy of the input make that image infinite. This is a contradiction.

Finally suppose all residuals have degree at most two. A nonzero constant makes the source empty. A nonzero linear equation reduces the ambient affine dimension to at most three and is excluded by the second assertion. If the quadratic residuals have a common factor, either they are proportional quadrics or they are a common linear factor times linear forms. Their common zero set is then a quadric or a union of linear spaces, and the cubic-function lemma applies on every permitted open part. Otherwise choose one residual Q_1 and a linear combination Q_2 of the others which avoids every irreducible factor of Q_1. Such a choice exists over the infinite field K: each forbidden divisibility condition is a proper linear subspace of the coefficient choices. The resulting pair is coprime, so the preceding paragraph applies. □

In particular, a quadratic system defining the integers over the rationals needs at least five total coordinates and at least three independent quadratic equations after removal of linear constraints. A finite system of cubic equations in three total coordinates cannot define the integers either. These restrictions leave open systems in the requested range of five to ten coordinates. They do not provide a construction in that range and do not exclude it.

28 What generic projection can and cannot compress

The finite-fiber method eliminates algebraic choices over a retained base. It cannot automatically replace freely varying base coordinates by a single primitive element. The distinction is geometric, and it must not be confused with a lower bound for every possible Diophantine definition.

Proposition 28.1. *Let X be an integral K-variety of dimension d. If a rational map from a nonempty open subset of X to $\mathbb{A}^b_K$ has finite geometric fibers, then $b \geq d$. In particular, rational changes of coordinates followed by finite-fiber projection cannot reduce the dimension of the retained base below d.*

Proof. Let Y be the closure of the image. A generically finite map makes the function-field extension $K(X)/K(Y)$ finite. Their transcendence degrees over K are therefore equal. Thus

$d = \dim Y \leq b$. Restricting to a nonempty open or making a birational change does not change the function field or its transcendence degree. □

Corollary 28.2. *Suppose a source is described by a positive-rank finite free algebra over $K[X_1, \ldots, X_b]$, followed only by nonzero conditions. Every nonempty irreducible component of its geometric open locus has dimension b. A compression which preserves such a component up to birational equivalence and finite-fiber projection must retain at least b base coordinates.*

Proof. A finite algebra is integral over its base, and every minimal prime of a finite flat algebra over a domain lies over the zero prime. To see the latter assertion, use going down for a flat extension: a nonzero contracted prime would admit a strictly smaller prime below the proposed minimal prime. The induced finite dominant map from each irreducible component to the base preserves dimension. A nonempty open in that component still has dimension b. Apply the preceding proposition. □

The unstratified 43-coordinate source before the direct Lucas-residual refinement has 42 retained base coordinates and a triangular monic finite algebra, followed only by guards. The new direct-residual source has 40 retained coordinates and the same property. Thus a generic projection does not, by itself, turn either source into a finite-fiber presentation over nine retained coordinates. Such an operation would have positive-dimensional geometric fibers. A proof that those fibers retain the required rational points would be new arithmetic information, not a consequence of the primitive-element theorem. The subsequent decomposition into zero and nonzero quadratic root cases can introduce closed conditions and smaller components; the dimension assertion just made concerns the original unstratified source, not each of those later cases separately.

Proposition 28.3. *Let a source variety with regular input T project to $\mathbb{A}^b_K$ with coordinates $(T, Z_2, \ldots, Z_b)$. If that projection has a rational section on a nonempty open subset of $\mathbb{A}^b_K$, then the source's input projection contains all but finitely many elements of K. Consequently a source whose rational input projection is $\mathcal{O}_K$ has no such section.*

Proof. Shrink the domain of the section to a nonempty principal open $G(T, \mathbf{Z}) \neq 0$. For all but finitely many $t \in K$, the specialization $G(t, \mathbf{Z})$ is a nonzero polynomial: at least one of its nonzero coefficient polynomials in T has only finitely many roots. Since K is infinite, one can choose $\mathbf{Z} \in K^{b-1}$ with $G(t, \mathbf{Z}) \neq 0$. The section then supplies a rational source point at input t. The ring $\mathcal{O}_K$ has infinite complement in K, so it cannot contain such a cofinite projection. □

The role of the base in this statement matters. A rational section over an already constrained subvariety is compatible with an arithmetic image. For example, in the presentation

$$Y^2 = T$$

of rational squares one may add arbitrarily many completely free coordinates. They can later be deleted with no change of rational projection, although the deleting map has positive-dimensional fibers. Its section is over the square-cover variety, not over the entire input affine line. Thus dimension alone is not a lower bound on the number of coordinates in every equivalent presentation.

There is a second reason for caution. The geometric dimension of one equation need not be the dimension of the closure of its rational solutions. The equation

$$U^2 + V^2 = 0$$

defines a one-dimensional hypersurface over an algebraic closure, but its rational solution set is just $(0, 0)$. Therefore a large geometric source can in principle have its rational points confined to a much smaller algebraic set. The dimension restriction must be applied to a component actually preserved by the proposed compression, not silently to all equivalent rational presentations.

Corollary 28.4. *For every polynomial-system definition of $\mathcal{O}_K$, the Zariski closure of its rational witness points has a horizontal component of dimension at least two. On that component the input map has positive-dimensional generic geometric fiber.*

Proof. Use Lemma 27.6. Its nonconstant input function is dominant onto the affine line, so the generic fiber has dimension $\dim X - 1 \geq 1$. □

This last restriction is intrinsic to the rational-point locus, unlike the preceding bound for a particular finite-flat presentation. It rules out compressing all witnesses to finitely many algebraic choices over the input alone. It does not rule out a two-dimensional arithmetic witness variety, nor a definition with three to ten total coordinates. No such new construction is furnished by these projection arguments.

29 Geometric limitations of compressing the existing witnesses

The bound of 41 total coordinates is not a lower bound for definitions of $\mathbb{Z}$. This section proves a more limited statement: certain proposed ways to compress the existing elliptic point modules cannot work. A new arithmetic construction can avoid these limitations by restricting or replacing those modules while preserving only the input projection.

29.1 Finite algebra requires geometric finiteness

Proposition 29.1. *Let V be a geometrically irreducible variety of dimension d over a field F, and let $S \subseteq V(F)$ be Zariski dense. Suppose there are finitely many varieties Y_j, morphisms $\pi_j : Y_j \longrightarrow \mathbb{A}^m_F$ with finite geometric fibres, and rational maps $g_j : Y_j \dashrightarrow V$, defined where they are used, such that*

$$S \subseteq \bigcup_j g_j(Y_j(F)).$$

Then $m \geq d$.

Proof. The dimension theorem gives $\dim Y_j \leq m$ because each geometric fibre of π_j is finite. Every irreducible component of the closure of $g_j(Y_j)$ therefore has dimension at most m. If $m < d$, the union of these finitely many closures is a proper closed subset of the irreducible variety V. It cannot contain the Zariski dense set S. □

In particular, a finite triangular monic algebra over m rational parameters, followed by rational decoding formulas, cannot cover a Zariski dense d-dimensional family when $m < d$. A finite list of exceptional charts does not change this conclusion.

This observation also distinguishes finite rational fibres from finite geometric fibres. Pasten proves that, for a positive-rank elliptic curve over a number field, there is an affine open set U and a morphism $U \times U \to \mathbb{A}^1$ which is injective on rational points, although those rational points are Zariski dense in the surface [31, Corollary 1.2]. Its rational fibres have at most one point. Its generic geometric fibre has dimension one. Proposition 29.1 shows that this injection does not supply a finite-algebra decoder for the pair from one rational coordinate.

29.2 The rank-one point group is dense in a surface

We now establish the obstruction for the exact elliptic curve used in the construction. The proof needs no height estimate, prime-pattern theorem, or restricted converse for elliptic curves.

Lemma 29.2. *Let F be a number field, let K/F be quadratic with involution σ, and let $d \in K^*$. Put $N = d\sigma(d) \in F^*$ and suppose $\sqrt{N} \notin K(i)$. On*

$$E_d : y^2 = x^3 - d^2x$$

let $P \in E_d(K)$ be nontorsion. The cyclic group generated by P, viewed in the Weil restriction $A = \mathrm{Res}_{K/F} E_d$, is geometrically Zariski dense in the two-dimensional variety A.

Proof. After extension to an algebraic closure, $A \simeq E_d \times E_{\sigma(d)}$. Put $u = d/\sqrt{N}$. The change of variables

$$E_{\sigma(d)} \longrightarrow E_d, \qquad (x, y) \longmapsto (u^2x, u^3y)$$

is an isomorphism because $u^4\sigma(d)^2 = d^2$. Write R for the image of $\sigma(P)$. There is an automorphism τ fixing $K(i)$ and changing the sign of $\sqrt{N}$. Consequently

$$\tau(P) = P, \qquad \tau(R) = -R.$$

The geometric endomorphism ring of E_d is $\mathbb{Z}[i]$, and all its endomorphisms are defined over $K(i)$. One can see this from complex uniformization: the order-four automorphism $(x, y) \mapsto (-x, iy)$ makes the period lattice a rank-one $\mathbb{Z}[i]$-module; since $\mathbb{Z}[i]$ is a principal ideal domain, its multiplier ring is exactly $\mathbb{Z}[i]$. In particular τ fixes every such endomorphism.

Suppose $\alpha(P) + \beta(R) = 0$ for endomorphisms $\alpha, \beta \in \mathbb{Z}[i]$. Applying τ gives $\alpha(P) - \beta(R) = 0$. Thus $2\alpha(P) = 2\beta(R) = 0$. A nonzero endomorphism of an elliptic curve is an isogeny and has finite kernel. Since P and R are nontorsion, it follows that $\alpha = \beta = 0$.

For completeness, this absence of relations implies the claimed density. The Zariski closure of a subgroup of an algebraic group is an algebraic subgroup. If the closure of $\mathbb{Z}(P, R)$ were proper in E_d^2, its identity component would have dimension zero or one. Dimension zero would make (P, R) torsion. In dimension one, let C be that identity component. Both projections $C \to E_d$ are nonzero isogenies: otherwise a nonzero multiple of P or R would vanish. If these maps are ϕ, ψ, a dual isogeny to ϕ gives a nonzero relation

$$[\deg \phi]y - \alpha x = 0 \quad \text{on } C$$

for an endomorphism α of E_d. A nonzero multiple of (P, R) lies in C, so this is a nontrivial endomorphism relation between P and R, contradicting the previous paragraph. The standard isogeny facts used here may be found in [33, Sections 8 and 12]. □

Corollary 29.3 (The point modules used here)**.** *For $K = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$, and the explicit point*

$$P = (57/8 - 7\sqrt{2}/2, -165/8 + 417\sqrt{2}/32),$$

the set $\mathbb{Z}P$ is Zariski dense in $\mathrm{Res}_{K/\mathbb{Q}} E_d$. Every nonzero finite-index subgroup has the same property. Nine independent copies of the point module consequently form a Zariski dense subset of a variety of dimension eighteen.

A construction which covers all those independent point choices using finitely many rational decoding charts over a finite algebra needs at least eighteen retained rational parameters. In particular, replacing each point module separately by one rational parameter and finitely many algebraic choices is impossible.

Proof. Here $N(d) = 7$ and the quadratic subfields of $\mathbb{Q}(\sqrt{2}, i)$ are $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(i)$, and $\mathbb{Q}(\sqrt{-2})$. Thus $\sqrt{7} \notin K(i)$.

The stated point is nontorsion by an elementary local calculation. The curve has good reduction at three, the reduction of P is the nonzero two-torsion point $(\sqrt{2}, 0)$, and

$$z(2P) = \frac{3396356159088 - 4135204317756\sqrt{2}}{3287811625151}, \qquad z = -x/y,$$

has valuation one at the unramified quadratic place above three. For a point of positive integral depth there, multiplication by a three-adic unit preserves depth, while multiplication by three increases depth by exactly one. Indeed the integral formal group has odd multiplication series, whose first term is nz; the higher terms in $[3]z$ have valuation at least $3v(z)$, strictly greater than $1 + v(z)$. No nonzero integer can therefore annihilate $2P$. The displayed exact calculation is reproduced by `check_elliptic_density_data.py`.

Apply Lemma 29.2. Replacing P by a nonzero multiple preserves its nontorsion property and the same argument. A Cartesian product of Zariski dense sets is dense in the product variety, which has dimension eighteen. The final assertion follows from Proposition 29.1. □

Scope of this obstruction. The nine point choices are not independent after every arithmetic condition of the final definition has been imposed. The corollary therefore gives a limitation on separate replacement of the point modules and on witness-preserving finite-algebra parametrization. It is not a lower bound of eighteen witnesses for $\mathbb{Z}$, and it does not exclude a new definition with at most ten total coordinates. Such a reduction must exploit additional arithmetic relations among the points, replace the point construction, or use a different kind of representation.

30 Limits of finite tests on denominator powers

The local obstruction behind the pole construction excludes odd negative valuations. It is natural to try to replace its many occurrences by one product, or to impose finitely many such conditions directly on the input. The following results rule out two precise versions of those strategies. They do not identify the image of every pole equation with the sets defined below; that identification would require an additional converse theorem.

Definition 30.1. For an integer $m \geq 2$, let

$$\mathcal{D}_m = \{q \in \mathbb{Q}_{>0} : \text{the positive reduced denominator of } q \text{ is an } m\text{th power}\}.$$

Equivalently, $q \in \mathcal{D}_m$ if and only if $q > 0$ and $m \mid v_p(q)$ whenever $v_p(q) < 0$. In particular, $\mathcal{D}_2$ consists of the positive rationals with square denominator.

30.1 Multiplying local obstructions loses information

Proposition 30.2. *Let $c_1, \dots, c_r \in \mathcal{D}_2$, let $A = (a_{ji}) \in \mathrm{Mat}_{r\times n}(\mathbb{Z})$, and put*

$$M_j(b_1, \dots, b_n) = c_j \prod_{i=1}^{n} b_i^{a_{ji}}.$$

If A has rank less than n over $\mathbb{F}_2$, there are positive rationals $b_1, \dots, b_n$, some outside $\mathcal{D}_2$, such that every $M_j(\mathbf{b})$ belongs to $\mathcal{D}_2$. Consequently an equivalence

$$\big(b_i \in \mathcal{D}_2 \text{ for every } i\big) \quad \Longleftrightarrow \quad \big(M_j(\mathbf{b}) \in \mathcal{D}_2 \text{ for every } j\big)$$

on all positive rational inputs requires $r \geq n$ and full column rank of A modulo two. These necessary conditions are not asserted to be sufficient.

Proof. Choose a nonzero vector $\epsilon \in \ker(A : \mathbb{F}_2^n \to \mathbb{F}_2^r)$ and a prime p outside the numerator and denominator supports of all c_j. Set $b_i = p^{-\epsilon_i}$. At least one b_i has reduced denominator p and is not in $\mathcal{D}_2$. But

$$v_p(M_j) = -\sum_i a_{ji}\epsilon_i$$

is even. Every other valuation of M_j is the corresponding valuation of c_j, so every negative valuation is even. Positivity is automatic. □

For example, the product of two inputs with denominator p has square denominator. The proposition shows that choosing several fixed monomial weights does not repair this cancellation unless the parity conditions retain at least as much independent information as the original conditions.

30.2 Fixed rational pullbacks cannot characterize integrality

A positive Boolean combination means a finite expression formed using only conjunction and disjunction. Rational functions are evaluated only where their denominators are nonzero. The result below permits arbitrary fixed rational functions and different denominator exponents.

Theorem 30.3. *Let $S \subseteq \mathbb{Q}$ be defined by a finite positive Boolean combination of predicates of the following forms:*

$$f(X) \in \mathcal{D}_m \quad (f \in \mathbb{Q}(T),\ m \geq 2), \qquad g(X) = 0, \qquad g(X) \neq 0 \quad (g \in \mathbb{Q}[T]).$$

If S contains all but finitely many rational integers, then there is an integer $n \in S$ and infinitely many nonintegral elements of S converging to n in the real topology. In particular no such finite combination defines $\mathbb{Z}$ in $\mathbb{Q}$.

We first record the elementary prime-divisor fact needed in the proof.

Lemma 30.4. *Let $h \in \mathbb{Z}[T]$ be nonconstant and squarefree over $\mathbb{Q}$. Outside any prescribed finite set of primes, there is a prime p and an integer residue class $a \pmod{p^2}$ such that*

$$v_p(h(n)) = 1 \quad \text{for every } n \equiv a \pmod{p^2}.$$

One may simultaneously require finitely many polynomials coprime to h over $\mathbb{Q}$ to be nonzero modulo p on this class.

Proof. A nonconstant integer polynomial has infinitely many prime divisors of its nonzero integer values. Here is a proof. Translate its argument so its constant term c is nonzero. If all such prime divisors belonged to a finite set Π, choose M divisible by every prime in Π. The nonconstant integer polynomial

$$H(T) = h(cMT)/c$$

has constant term one and satisfies $H(t) \equiv 1 \pmod{M}$. Every prime divisor of a nonzero value of H would belong to Π, but none could divide that value. Thus every nonzero value would be 1 or -1, a contradiction.

Now exclude the prescribed primes, the prime divisors of the leading coefficient and discriminant of h, and those of the resultants with the other polynomials. A remaining prime divisor of a value supplies a simple root a_0 modulo p. Among its p lifts modulo p^2, exactly one is a root modulo p^2. Any other lift has valuation one. The resultant conditions give the last assertion. $\square$

Proof of Theorem 30.3. Write the defining expression in disjunctive normal form, as a finite union of conjunctions, called branches in this proof. A branch containing a nonzero polynomial equality can contain only finitely many integers; branches containing an identically false predicate are empty. Discard these branches and enlarge the finite exceptional set of integers accordingly. All remaining polynomial conditions are identities or nonvanishing conditions.

Write each nonzero rational function in reduced form P/Q over $\mathbb{Q}[T]$. Call a remaining branch bad if one of its predicates $P(X)/Q(X) \in \mathcal{D}_m$ has an irreducible factor of Q whose multiplicity e is not divisible by m. For each bad branch choose one such predicate and factor. Exclude the prime supports of the rational scalar factors and of all coefficient denominators in these factorizations. By Lemma 30.4, choose a distinct prime p and a residue class modulo p^2 on which this factor has valuation one, while P and every other denominator factor are units. Then

$$v_p(P(n)/Q(n)) = -e$$

on that class, so the chosen predicate, and hence that branch, fails. The Chinese remainder theorem combines these finitely many classes into an infinite arithmetic progression on which all bad branches fail.

Choose an integer n in this progression outside the finite exceptional set. Since $n \in S$, it belongs to a remaining branch which is not bad. Every denominator factor in each predicate of this branch has multiplicity divisible by its exponent m_j. After absorbing a nonzero rational constant into the numerator, write all its denominator predicates as

$$f_j(X) = P_j(X)/B_j(X)^{m_j} \in \mathcal{D}_{m_j}, \qquad \gcd(P_j, B_j) = 1.$$

These rational functions are defined and positive at n.

Let Π be a finite set of primes containing all coefficient denominators, leading-coefficient supports, nonzero resultant supports, and the numerator and denominator supports of the nonzero rational numbers $f_j(n)$. Enlarge it if needed so that outside Π all the displayed polynomials are integral with unit leading coefficient and each pair (P_j, B_j) has unit resultant. For each $q \in \Pi$, continuity in $\mathbb{Q}_q$ gives an integer $e_q \geq 1$ such that

$$x - n \in q^{e_q}\mathbb{Z}_q \quad \Longrightarrow \quad v_q(f_j(x)) = v_q(f_j(n)) \quad \text{for every } j.$$

Choose a positive integer M divisible by every q^{e_q}, and put $L = \operatorname{lcm}(m_1, \ldots, m_s)$. If there are no denominator predicates, set $L = 2$. For any sufficiently large prime $p \notin \Pi$ not dividing M, define

$$x_p = n + M/p^L.$$

These are distinct nonintegers and converge to n in $\mathbb{R}$.

We check every denominator condition of the chosen branch. At $q \in \Pi$ the valuation agrees with its value at n, which satisfies that condition. At p we have $v_p(x_p) = -L$; the unit leading coefficients give

$$v_p(f_j(x_p)) = -L \deg P_j + m_j L \deg B_j,$$

a multiple of m_j. At a prime $q \notin \Pi \cup \{p\}$, the number x_p is integral. If $B_j(x_p)$ is a unit, then $v_q(f_j(x_p)) \geq 0$. Otherwise the unit resultant makes $P_j(x_p)$ a unit and

$$v_q(f_j(x_p)) = -m_j v_q(B_j(x_p)),$$

again a permitted valuation. Real continuity makes all $f_j(x_p)$ positive for sufficiently large p, and preserves the finitely many polynomial nonvanishing conditions. Thus all sufficiently large choices of p give $x_p \in S \setminus \mathbb{Z}$. □

Corollary 30.5. *The conclusion of Theorem 30.3 also holds for arbitrary finite Boolean combinations of the listed predicates, allowing negation.*

Proof. Use disjunctive normal form with both positive and negative denominator literals. Negated polynomial equalities and inequalities are already among the allowed predicates. Discard finite branches as before, and use the same Chinese remainder construction to eliminate every branch with a bad positive denominator literal. Choose the resulting admitted integer n also away from the finitely many zeros and poles of all nonzero rational functions in the formula.

Consider a negative literal $f(n) \notin \mathcal{D}_m$ in the chosen branch. An identically zero f makes it automatically true. Otherwise either $f(n) < 0$, which persists in a real neighborhood, or $f(n) > 0$ and there is a prime q with $v_q(f(n)) < 0$ not divisible by m. The latter failure persists in a sufficiently small q-adic neighborhood of n. Add one such prime for each relevant negative literal to Π, and choose M sufficiently divisible to preserve all those valuations as well. The same numbers $x_p = n + M/p^L$ then satisfy every positive and negative literal for all sufficiently large permitted primes p. They are nonintegral and converge to n. □

The theorem is deliberately about predicates on fixed rational functions of the input, with no extra existential coordinates. It does not rule out auxiliary elliptic points, variable rational functions, or other arithmetic predicates. It also does not replace the independent proof needed to identify the exact image of a given pole system. Its conclusion is that even arbitrarily complicated fixed rational pullbacks of perfect-denominator conditions cannot alone remove every nonintegral rational number.

31 What rational pairing and existential rank do, and do not, supply

The finite-algebra method removes algebraic choices over retained parameters. A proposed further compression must justify the decoding of the remaining rational parameters. Set-theoretic countability and injectivity of a coding function do not give this justification.

31.1 Integer pairing is not rational pairing

The familiar polynomial

$$C(u, v) = \frac{(u + v)(u + v + 1)}{2} + v$$

pairs nonnegative integers, but has infinitely many rational points in every rational fibre. Indeed, for any $c, s \in \mathbb{Q}$, set

$$v = c - \frac{s(s + 1)}{2}, \qquad u = s - v.$$

Then $C(u, v) = c$. Thus using this pairing on rational witnesses introduces spurious decodings. Restricting its arguments to integers would require precisely the integrality predicates whose cost is being optimized.

Poonen proves that a suitable non-density consequence of the Bombieri–Lang conjecture implies a polynomial injection $k^2 \to k$ over a number field [30, Theorem 1.1 and Remark 1.2]. Even such an injection would not supply rational functions recovering the two inputs, nor a finite geometric fibre for primitive-element compression. Every nonconstant morphism $\mathbb{A}^2 \to \mathbb{A}^1$ has a one-dimensional generic geometric fibre. The elliptic-surface example of Pasten in Section 29 makes the distinction between rational injectivity and geometric finiteness unconditional.

31.2 A recent explicit result has a restricted domain

Hirvensalo's Theorem 2 gives an explicit polynomial injection on

$$\Lambda_\Pi = \mathbb{Z}[1/p : p \in \Pi] \cap [0, 1],$$

where Π is a fixed finite nonempty set of odd primes [32]. With $M = \operatorname{lcm}_{p\in\Pi}(p - 1)$ and $N \geq 2$ coprime to every $p(p - 1)$, its construction is a positive constant multiple of

$$(x^M + y^M)^N + y^M.$$

The degree is MN; for $\Pi = \{5\}$ one can take $M = 4, N = 3$, giving degree twelve. The theorem does not assert an injection on all rational pairs. The variable denominators in the present elliptic construction are not restricted to such a fixed set of primes, and decoding would still need an independent existential definition. It therefore gives no unconditional reduction of the present count.

31.3 Existential rank gives the appropriate distinction

For a Diophantine set $D \subseteq K^r$, its existential rank is the least number of quantified field variables in an existential definition of D. The rank can be smaller than the dimension of a particular witness variety, since a different definition need only preserve the input projection. Daans–Dittmann–Fehm's essential fibre dimension formalizes this distinction, and their characteristic-zero bound is [6, Corollary 4.21]

$$\mathrm{rk}_K^{\exists,+}(D) \leq \mathrm{efd}_K(D) + 1.$$

The explicit finite-algebra construction used here is an effective special case of that general principle, with all exceptional algebraic fibres handled in the specification. It does not show that positive dimensional arithmetic fibres have finite algebraic decoders.

The same paper's Remark 8.8 excludes a one-witness existential definition of the ring of integers of any number field. It does not provide an upper bound of nine witnesses for $\mathbb{Z}$ in $\mathbb{Q}$. Its MRDP-based bounds are conditional on an existing integrality definition and multiply its witness cost; they do not remove that cost [6, Remark 8.23].

31.4 What remains necessary for a count of at most ten

The 41-coordinate definition has one input, eighteen retained coordinates for nine elliptic points, one common slope, twenty coordinates for ten pole pairs, and one final algebraic coordinate. Replacing the finite algebra by another finite primitive-element construction does not change

these forty retained coordinates. Proposition 29.1 rules out separately encoding the full nine-point family over fewer than eighteen parameters. Proposition 30.2 rules out one proposed direct aggregation of the local denominator conditions. Corollary 30.5 further rules out every finite Boolean combination of fixed rational pullbacks of perfect-denominator predicates as a direct definition of integrality.

Neither statement is a lower bound for every definition of $\mathbb{Z}$. A substantially smaller construction would need new arithmetic relations that reduce the independent point choices, a different method of enforcing integrality, or a proved simultaneous construction of pole witnesses. Merely requiring the ten independent pole conics to use a common parameter is not justified by the separate existence theorems used in completeness. No argument in this work establishes a definition with two through ten total variables.

32 Quantitative consequences for rational Diophantine problems

There is a useful refinement of the reduction in Section 2. The literature gives a bound of $12N$ witnesses for computably enumerable subsets of $\mathbb{Q}$ when $\mathbb{Z}$ has a definition with N witnesses [6, Remark 8.23]. Using a small positivity condition together with the constructive intersection bound gives the following alternative.

Lemma 32.1. *For an integer t,*

$$t \geq 0 \quad \Longleftrightarrow \quad \exists u, v, w \in \mathbb{Q} : \ t = u^2 + v^2 + w^2 + w.$$

Proof. If $t \geq 0$, the three-square theorem applies to $4t+1$. Its three squares have one odd and two even entries, since $4t + 1 \equiv 1 \pmod 4$. Write them as $(2w+1)^2, (2u)^2, (2v)^2$ with integers u, v, w and divide by four. Conversely, a rational solution gives $t + 1/4 = u^2 + v^2 + (w + 1/2)^2 \geq 0$. The assumption $t \in \mathbb{Z}$ then gives $t \geq 0$. This is the integer positivity identity used in [12, Equation (1.1)], with its witnesses now allowed to be rational. The integer hypothesis must be retained. □

Proposition 32.2. *Suppose $\mathbb{Z}$ is defined in $\mathbb{Q}$ with $N \geq 1$ existential witnesses. Every computably enumerable subset of $\mathbb{Q}$ then has a one-polynomial definition with at most $11N + 1$ existential witnesses. Furthermore, polynomial solvability over $\mathbb{Q}$ restricted to at most $9N + 2$ variables is undecidable.*

Proof. We use the published integer representation theorem [12, Theorem 1.1(i)]: for every computably enumerable $B \subseteq \mathbb{Z}_{\geq 0}$ there is an integer polynomial g_B such that, for $c \geq 0$,

$$c \in B \quad \Longleftrightarrow \quad \exists y_1, \ldots, y_9 \in \mathbb{Z} : \ y_9 \geq 0, \quad g_B(c, y_1, \ldots, y_9) = 0.$$

For a computably enumerable $A \subseteq \mathbb{Q}$, encode integer pairs by the injective polynomial

$$q(a) = 2a^2 - a, \qquad f(a, b) = (q(a) + q(b))(q(a) + q(b) + 1) + 2q(b).$$

The map $q : \mathbb{Z} \to \mathbb{Z}_{\geq 0}$ is injective, and the second formula is twice the usual pairing polynomial on nonnegative integers. For a fixed sum of the two arguments its values are distinct, and successive sums have disjoint ranges, proving injectivity. The set

$$B = \{f(a, b) : a, b \in \mathbb{Z}, \ b \neq 0, \ a/b \in A\}$$

is computably enumerable and contained in $\mathbb{Z}_{\geq 0}$. Thus $x \in A$ is equivalent to the existence of $y_0, \ldots, y_9 \in \mathbb{Q}$ satisfying

$$xy_0, y_0, y_1, \ldots, y_9 \in \mathbb{Z}, \quad y_0 \neq 0, \quad y_9 \geq 0, \quad g_B(f(xy_0, y_0), y_1, \ldots, y_9) = 0.$$

There are eleven integrality predicates, each using N private witnesses. Use Lemma 32.1 for $y_9 \geq 0$, writing $y_9 = u^2 + v^2 + w^2 + w$, and use $y_0\eta - 1 = 0$ for $y_0 \neq 0$.

Regard $g_B(f(xy_0, y_0), y_1, \ldots, y_8, Y)$ as a polynomial in Y. On the closed stratum where all its coefficients vanish, impose $y_9 = 0$; on the complement retain its root equation. This preserves the final projection to x. Indeed, on the former stratum the only other restrictions on y_9 are integrality and nonnegativity, both satisfied by zero. Its private integrality witnesses can be chosen afresh and we may set $u = v = w = 0$. We do not claim preservation of the projection to every larger tuple of private coordinates.

Normalize each integrality polynomial in its last private witness as in the proof of Proposition 8.3. Retain $y_0, \ldots, y_8$, the first $N - 1$ witnesses of each integrality predicate, and u, v. Besides x, this retains

$$9 + 11(N - 1) + 2 = 11N$$

coordinates. Over every geometric retained tuple, y_9 has finitely many choices; then each last integrality witness has finitely many choices, w satisfies the monic quadratic $w^2+w+u^2+v^2-y_9 = 0$, and η is unique whenever it exists. The resulting constructible system has finite geometric fibers. Theorem 8.1 replaces all these remaining choices by one witness, giving $11N + 1$ in total.

For the last assertion fix a nonrecursive computably enumerable $B \subseteq \mathbb{Z}_{\geq 0}$ and use g_B directly, with its input c supplied as a nonnegative integer parameter. Only the nine integrality predicates on $y_1, \ldots, y_9$ and the three-witness positivity predicate are needed. Apply the same normalization to $g_B(c, y_1, \ldots, y_8, Y)$. Retaining $y_1, \ldots, y_8$, the first $N - 1$ private coordinates in each predicate, and u, v leaves $8 + 9(N - 1) + 2 = 9N + 1$ coordinates. One final finite-choice witness gives $9N + 2$. An algorithm deciding rational solvability in this many variables would therefore decide $c \in B$, a contradiction. This step concerns an integer input parameter; it makes no assertion that the same formula defines B on arbitrary rational inputs without an integrality test on that parameter. □

With $N = 40$, the conditional 41-variable construction therefore gives 441 existential witnesses for each computably enumerable subset of $\mathbb{Q}$, and undecidability of rational polynomial solvability in at most 362 variables. These conclusions have the same arithmetic assumptions as that construction. Using only the source-preserving 558-variable construction instead, $N = 557$ gives the bounds 6128 and 5015. These are quantitative consequences of the defining-polynomial method, not claims of optimality among all undecidability reductions. In particular a specialized undecidability theorem and a definition of the ordinary subset $\mathbb{Z} \subset \mathbb{Q}$ are different objectives. The first bound improves $12N$ when $N > 1$. No polynomial achieving these MRDP-derived counts has been expanded in this article.

33 Number fields: what transfers, and what does not

Let K be a number field, that is, a finite extension of $\mathbb{Q}$, and let $\mathcal{O}_K$ be its ring of algebraic integers. Coefficients in K are allowed in a Diophantine definition over K. This section distinguishes the existence of a definition for each K from the much stronger assertion that one fixed equation works after every extension of scalars.

33.1 An exact counterexample to extending the released polynomial

Write $P(X_0, \dots, X_{12503})$ for the released quartic, with the variable indices used in its monomial table, and put

$$C = 5937913971887799417367.$$

The following calculation is independent of the arithmetic argument that P defines $\mathbb{Z}$ over $\mathbb{Q}$.

Proposition 33.1. *Over $\mathbb{Q}(i)$, the existential projection of the released polynomial P onto its input coordinate is all of $\mathbb{Q}(i)$. In particular, this polynomial does not define $\mathbb{Z}[i]$ in $\mathbb{Q}(i)$.*

Proof. Set every witness variable other than $X_{2242} = u$ and $X_{2243} = v$ equal to zero. The exact resulting polynomial is

$$P(t, 0, \dots, u, \dots, v, \dots, 0) = C + 4t^2 + t^4 + u^2 + v^2. \tag{12}$$

For any t in a field containing $\mathbb{Q}(i)$, put

$$R = C + 4t^2 + t^4, \qquad u = \frac{1 - R}{2}, \qquad v = i\frac{1 + R}{2}.$$

Then

$$R + u^2 + v^2 = R + \frac{(1 - R)^2 - (1 + R)^2}{4} = 0.$$

Thus every $t \in \mathbb{Q}(i)$ is accepted. For example, $t = 1/2$ is not in $\mathbb{Z}[i]$ and has the witnesses

$$X_{2242} = -\frac{9500662355020479067873}{32}, \qquad X_{2243} = i\frac{9500662355020479067905}{32},$$

with all other witness coordinates zero.

For reproducibility, `check_base_change.py` checks (12) in two independent ways: by retaining the relevant monomials of the expanded table, and by specializing and squaring the quadratic residuals. It also substitutes the displayed generic witnesses directly in both representations, using exact arithmetic in $\mathbb{Q}(i)[t]$. The source table has SHA-256

`7f5aa3117af82a0b323f0a4e74a18599204f5540408106d3ce99612b5ed830ca.`

□

Remark 33.2. Proposition 33.1 is a counterexample to direct extension of the particular equation. It is not a counterexample to the existence of a different Diophantine definition of $\mathbb{Z}[i]$ in $\mathbb{Q}(i)$. The obstruction is already present in the final sum-of-squares operation: $u^2 + v^2 = 0$ no longer forces $u = v = 0$ over $\mathbb{Q}(i)$. Even replacing this operation by a suitable norm form would not, by itself, establish that the original arithmetic system characterizes integrality over a new field.

33.2 A general obstruction to uniform scalar extension

Proposition 33.3. *Suppose that $P \in \mathbb{Q}[T, W_1, \dots, W_m]$ has total degree D and defines $\mathbb{Z}$ in $\mathbb{Q}$. There is a finite set $E \subset \mathbb{Q}$ with $|E| \le D - 1$ such that, for every $q \in \mathbb{Q} \setminus (\mathbb{Z} \cup E)$, there exist a number field K_q and $\mathbf{w} \in K_q^m$ satisfying*

$$2 \le [K_q : \mathbb{Q}] \le D, \qquad P(q, \mathbf{w}) = 0, \qquad q \notin \mathcal{O}_{K_q}.$$

Consequently no fixed polynomial defining $\mathbb{Z}$ in $\mathbb{Q}$ defines $\mathcal{O}_K$ in every number field K by scalar extension.

Proof. The polynomial must depend on at least one W_j: otherwise its rational zero set is finite or all of $\mathbb{Q}$. Expand it as

$$P(T, \mathbf{W}) = \sum_{\alpha} c_\alpha(T)\mathbf{W}^\alpha$$

and select $\alpha \neq 0$ with $c_\alpha \neq 0$. Let E be the rational zero set of c_α. Since $\deg c_\alpha \leq D - |\alpha|$, we have $|E| \leq D - 1$. If $q \notin E$, the polynomial $P(q, \mathbf{W})$ is nonconstant.

Choose a variable on which this polynomial depends. Its leading coefficient in that variable is a nonzero polynomial in the remaining variables. A nonzero polynomial over an infinite field cannot vanish at every rational point; by induction on the number of variables, we can therefore specialize all the remaining variables to rational numbers so that the resulting polynomial $h(U) \in \mathbb{Q}[U]$ is nonconstant. Its degree is at most D. Choose an irreducible factor of h, let θ be a root, and set $K_q = \mathbb{Q}(\theta)$. The specialized coordinates, with $U = \theta$, give the required zero of P. The degree of K_q is at most D. It cannot be one because $q \notin \mathbb{Z}$ and P defines $\mathbb{Z}$ over $\mathbb{Q}$.

Finally, $\mathcal{O}_{K_q} \cap \mathbb{Q} = \mathbb{Z}$: a rational algebraic integer has denominator one, by the rational-root theorem applied to a monic integral equation. Hence q is not integral. □

Remark 33.4. If one chosen witness has degree at most d in P and genuinely occurs after specialization, the same proof gives $[K_q : \mathbb{Q}] \leq d$. For a sum of squares whose residuals are affine linear in that witness, a nonconstant quadratic specialization with no rational zero has strictly negative discriminant. Indeed it is a sum of squares of real affine linear polynomials, so its discriminant is nonpositive, and discriminant zero would give a rational root. In this case an imaginary quadratic extension already supplies a false positive.

33.3 Combining equations over an arbitrary number field

The use of one polynomial rather than a finite system causes no definability obstruction over a number field, although it can affect the degree severely.

Lemma 33.5. *Let K be a number field, let $s \geq 1$, and let $f_1, \dots, f_s \in K[\mathbf{X}]$ have degree at most d. There is a polynomial $F \in K[\mathbf{X}]$, with no additional variables and degree at most sd, such that*

$$F(\mathbf{x}) = 0 \quad \Longleftrightarrow \quad f_1(\mathbf{x}) = \cdots = f_s(\mathbf{x}) = 0 \qquad (\mathbf{x} \in K^n).$$

Proof. Choose a field extension L/K of degree s and a K-basis $\beta_1, \dots, \beta_s$ of L. Such an extension exists: choose a discrete valuation of K with a uniformizer $\pi \in K$; the polynomial $U^s - \pi$ is Eisenstein. The norm form

$$N(Z_1, \dots, Z_s) = N_{L/K}\left(\sum_{j=1}^{s} \beta_j Z_j\right)$$

is homogeneous of degree s over K, and it vanishes on a vector in K^s precisely when that vector is zero. Take $F = N(f_1, \dots, f_s)$. □

For the degree objective, the preceding elementary norm bound can be substantially improved. The following construction uses only a single nonarchimedean place.

Lemma 33.6. *Every number field K admits an anisotropic quadratic form in four variables with coefficients in $\mathcal{O}_K$. Consequently a system of $s \geq 2$ polynomial equations of degree at most d over K can be combined, without additional variables, into one equation of degree at most*

$$2^{\lceil \log_4 s \rceil} d.$$

If K admits a real embedding, degree at most $2d$ suffices for every s.

Proof. Choose a prime ideal $\mathfrak{p}$ of $\mathcal{O}_K$ above an odd rational prime, and let v be its normalized discrete valuation. Choose $a \in \mathcal{O}_K$ whose image in $\mathcal{O}_K/\mathfrak{p}$ is a nonsquare, and choose $\pi \in \mathfrak{p} \setminus \mathfrak{p}^2$. For any $(x, y) \neq (0, 0)$ in K^2, reduction modulo $\mathfrak{p}$ after division by $\pi^{\min\{v(x),v(y)\}}$ gives

$$v(x^2 - ay^2) = 2 \min\{v(x), v(y)\}.$$

Indeed the two leading residues cannot cancel because a is a nonsquare in the residue field. It follows that

$$q(X_1, X_2, X_3, X_4) = X_1^2 - aX_2^2 - \pi X_3^2 + a\pi X_4^2$$

is anisotropic: equality of its two binary parts would equate an even valuation with an odd one, unless both parts vanish, in which case all four coordinates are zero.

Place the s residuals at the leaves of a complete tree of branching degree four and height $h = \lceil \log_4 s \rceil$, padding the unused leaves by zero. At every internal vertex apply q to its four child polynomials. Anisotropy shows inductively that the resulting polynomial vanishes exactly when every original residual vanishes. Each level at most doubles the degree, giving the stated bound. If K has a real embedding, the sum of the squares of all the residuals works directly, since a sum of real squares is zero only when every summand is zero. □

33.4 Precise transfer principles

Lemma 33.7 (Transitivity)**.** *Let $A \subset B \subset C$ be rings of characteristic zero whose fraction fields are not algebraically closed. If A is Diophantine in B and B is Diophantine in C, then A is Diophantine in C. Here the coefficients used to define A are regarded as constants in C.*

Proof. Take a defining existential system for A over B. Regard its variables as ranging over C, and append a defining system for membership in B to every variable, including the free variable. The resulting finite existential system defines exactly A in C. It can be combined into one equation by an anisotropic norm form, as in Lemma 33.5 for the number-field cases used here. For the stated more general rings, the same construction uses the homogenization of an irreducible polynomial over their fraction fields and clears denominators. □

Proposition 33.8. *If $\mathbb{Z}$ is Diophantine in a number field K of degree n, then $\mathcal{O}_K$ is Diophantine in K.*

Proof. Choose an integral basis $\omega_1, \ldots, \omega_n$ of $\mathcal{O}_K$. Then

$$x \in \mathcal{O}_K \quad \Longleftrightarrow \quad \exists z_1, \ldots, z_n \in \mathbb{Z} : \quad x = \sum_{j=1}^{n} \omega_j z_j.$$

Replace each condition $z_j \in \mathbb{Z}$ by its Diophantine definition over K, and combine the finitely many equations by Lemma 33.5. □

Proposition 33.9 (Quantitative integral-basis construction)**.** *Suppose a polynomial $G(T, \mathbf{W})$ of degree d defines $\mathbb{Z}$ over K, using m witnesses. If $[K : \mathbb{Q}] = n$, then $\mathcal{O}_K$ has a definition using at most*

$$n(m+1) - 1$$

witnesses. For $n \geq 2$ the total degree can be at most $2^{\lceil \log_4 n \rceil} d$, and at most $2d$ when K admits a real embedding. For $n = 1$, the original polynomial suffices.

Proof. In the integral-basis construction, eliminate the last coefficient by

$$z_n = \frac{x - \sum_{j<n} \omega_j z_j}{\omega_n}.$$

Apply G to each of $z_1, \dots, z_n$, using separate witness tuples. The remaining coefficient variables number $n-1$ and the witnesses inside the n copies of G number nm. Substitution of the displayed linear expression for z_n does not increase degree, and multiplication by a nonzero element of K clears any coefficient denominator. There are now n equations of degree at most d; apply Lemma 33.6. □

Proposition 33.10. *Let L/K be an extension of number fields. If $\mathcal{O}_L$ is Diophantine in L, then $\mathcal{O}_K$ is Diophantine in K.*

Proof. Choose a K-basis $\beta_1 = 1, \beta_2, \dots, \beta_e$ of L. In a defining polynomial for $\mathcal{O}_L$, replace each quantified L-variable by $\sum_{j=1}^{e} \beta_j y_j$, with the y_j quantified over K. Expand every coefficient in the same basis and equate the e coordinates of the resulting expression to zero. For a free variable $x \in K$, this system is equivalent to $x \in \mathcal{O}_L$, which is equivalent to $x \in \mathcal{O}_K$. Combine the equations using Lemma 33.5. □

33.5 The remaining arithmetic question

The theorem of Koymans–Pagano and, independently, Alpöge–Bhargava–Ho–Shnidman proves that

$$\mathbb{Z} \text{ is Diophantine in } \mathcal{O}_K \quad \text{for every number field } K. \tag{13}$$

This is Theorem 1.2 of [24] and Corollary 1.2 of [25]. These statements concern witnesses in $\mathcal{O}_K$; they do not supply unrestricted witnesses in K.

Theorem 33.11 (Conditional equivalence of the extension questions)**.** *Assume that $\mathbb{Z}$ is Diophantine in $\mathbb{Q}$. For a number field K, the following are equivalent:*

1. *$\mathcal{O}_K$ is Diophantine in K;*
2. *$\mathbb{Z}$ is Diophantine in K;*
3. *$\mathbb{Q}$ is Diophantine in K.*

The equivalence of the first two assertions uses the established theorem (13) *but does not require the assumption about $\mathbb{Q}$.*

Proof. The first assertion implies the second by (13) and transitivity. The second implies the first by Proposition 33.8. For the implication from the second to the third, use

$$x \in \mathbb{Q} \quad \Longleftrightarrow \quad \exists a, b \in \mathbb{Z}\ \exists c \in K : \quad bx = a, \quad bc = 1.$$

Finally, if $\mathbb{Q}$ is Diophantine in K, transitivity transports the assumed definition of $\mathbb{Z}$ over $\mathbb{Q}$ to K. □

These transfer statements do not construct a Diophantine definition of $\mathbb{Q}$ in an arbitrary number field. The rank-stability theorems behind (13) establish definitions over rings of integers, not that missing field-level assertion. The results proved here give no counterexample to the existence of a field-dependent definition of $\mathcal{O}_K$, and do not settle that general question. They do show that an extension argument must contain new arithmetic input: neither the reuse of the released equation nor an unrestricted change of the witness field suffices.

34 What the newer constructions do over number fields

Three different questions must be separated. A fixed polynomial over $\mathbb{Q}$ has an existential projection over every extension field. An equation-combining procedure may preserve the projection of a specified source system over that extension. Neither fact identifies the resulting set with the ring of integers of the extension. The following results locate explicit failures of the first and third assertions, and give an effective repair of the second in Section 35. The counterexamples and algebraic encoding results here do not depend on accepting the source manuscript's arithmetic theorem.

34.1 The four literal quartics accept every input over $\mathbb{Q}(i)$

Let P_N denote the recorded sum-of-squares quartic with N total coordinates, indexed from zero, where X_0 is the input. Set all witnesses to zero except the following two coordinates:

N	quadratic residuals	u coordinate	v coordinate
1315	1272	72	73
1339	1296	96	97
1367	1322	62	78
1391	1346	86	102

The exact restriction in each case is

$$P_N(T, 0, \dots, u, \dots, v, \dots, 0) = C_N + T^4 + u^2 + v^2. \tag{14}$$

Here C_N is the constant obtained by the same restriction at $T = u = v = 0$. For $N = 1315, 1367$ it includes the term $1024\kappa^2$, with the fixed integer $\kappa = 3^{2^{28}}$. The full integer constants, coordinate names, and hashes of the four complete coefficient tables are recorded in `quartic-base-change-verification_20261106_v1.json`. Defining C_N by this specialization avoids any ambiguity from rounding or decimal transcription.

Proposition 34.1. *Each of these four literal quartics has existential input projection equal to K over every characteristic-zero field K containing i. In particular none defines $\mathcal{O}_{\mathbb{Q}(i)}$ in $\mathbb{Q}(i)$.*

Proof. For any $T \in K$ put $R = C_N + T^4$ and take

$$u = \frac{1 - R}{2}, \qquad v = i\frac{1 + R}{2}.$$

Then $u^2 + v^2 = -R$, so (14) vanishes. Taking $T = 1/2$ gives a nonintegral input with witnesses already in $\mathbb{Q}(i)$. This identity is exact for the fixed integer κ as well as before its substitution. □

The independent script `check_quartic_base_change_20261106_v1.py` obtains the restriction twice: from the expanded sparse polynomial and by squaring the independently restricted primitive residuals. It compares the two results and substitutes the displayed generic witnesses using exact arithmetic in $\mathbb{Q}(i)[T,\kappa]$. The conclusion is therefore stronger than the general observation that sums of squares can cancel in a nonreal field.

34.2 An exact integral input rejected by the unshifted sources

The unshifted source uses $\beta = X^2 + 2$ and guards the scalar contact expression

$$r_\beta = \beta(\beta-1)(\beta-2401)(32\beta+343)(2\beta-1) \tag{15}$$

away from zero. Its later nonzero rational-square rescaling does not remove these zeros. The individual factors are present as nonvanishing guards in the indexed coefficient specifications. Consequently $X = \sqrt{2399}$, i, and $\sqrt{-2}$ are rejected by the unshifted guarded source, even though each is an algebraic integer in the corresponding quadratic field.

For the indexed polynomials the failure is particularly exact. Recall the notation of the finite-algebra specification: G is its product of guards, n is its algebra rank,

$$H(V) = \det(I - VM_{G\lambda}) = \sum_{k=0}^{n} a_k V^k, \qquad R_E(V) = \operatorname{Tr}\big(M_E \operatorname{adj}(I - VM_{G\lambda})\big).$$

Each branch has an index $1 \le d \le n$ and uses $g = a_d \operatorname{Res}(H_d, H_d')$, $L = a_d^2 g$,

$$\Psi = 1 + \sum_{j=1}^{d} a_{d-j} a_d^{2j-1} g^j U^j, \qquad \widetilde{R}_E = \sum_{k=0}^{n-1} [V^k] R_E(V) (LU)^{n-1-k}.$$

Its final residuals are Ψ, the high coefficients $a_{d+1}, \dots, a_n$, and the $\widetilde{R}_E$.

Proposition 34.2. *For each unshifted indexed polynomial with respectively 41, 43, 43, and 45 total coordinates from the direct and proxy constructions, the polynomial equals* 1 *identically in all remaining coordinates after imposing any of*

$$X^2 = 2399, \qquad X^2 = -1, \qquad X^2 = -2.$$

The assertion holds over every characteristic-zero field.

Proof. The indicated contact factor is zero as a polynomial in the retained base coordinates, so $G = 0$ in every specialized finite algebra, for every value of every finite parameter. Thus $M_{G\lambda} = 0$, $H = 1$, and all a_k with $k \ge 1$ vanish. Hence $g = L = 0$ and $\Psi = 1$. Moreover $R_E(V) = \operatorname{Tr}(M_E)$ is constant. Every branch retains nine quartic point generators, so $n \ge 4^9 > 1$. Therefore its reversed trace is zero: the sole possibly nonzero coefficient has the factor $(LU)^{n-1} = 0$. Every branch sum of squares is consequently 1, and so is their finite product. This calculation does not use anisotropy of a sum of squares. □

The contact checker verifies a literal multiplicative path from the zero factor to the materialized guard, evaluates that factor in $\mathbb{Q}[X]/(X^2 - a)$, and checks the branch-rank assertion. Its separate checks of the four quadratic systems follow exact input-dependent gate equations until an inverse equation becomes $-1 = 0$. The recorded reversible row operations transport this contradiction

to each final quadratic residual system. Thus a faithful norm combination of the unshifted residuals still rejects these inputs. Over $\mathbb{Q}(\sqrt{2399})$ the literal sum of squares is faithful as well, and therefore rejects that integral input. Over $\mathbb{Q}(i)$ the literal quartic instead accepts every input by Proposition 34.1; this is precisely the distinction between a system and its sum-of-squares encoding.

These finite contact obstructions admit the field-dependent shift repair of Section 16. The assertion above is about the unshifted artifacts and does not apply to the separately emitted $\beta = X^2 + 2402$ artifacts. Removing a finite list of excluded inputs does not restore the elliptic arithmetic argument considered next.

34.3 A totally real quartic obstruction to the finite elliptic anchor

Put $F = \mathbb{Q}(\sqrt{2})$, $d = 3 - \sqrt{2}$ and $D = d^2 = 11 - 6\sqrt{2}$. Let

$$E:\ y^2 = x^3 - Dx,$$

and let $P \in E(F)$ be the fixed point used in the finite anchor. Its particular coordinates are immaterial below; only $P \in E(F)$ is used. The anchor excludes equality with the finitely many abscissas $x([n]P)$, $1 \le n \le M$, by taking the product of their nonzero rational norms.

Proposition 34.3. *There is a totally real quartic number field K, with $\sqrt{2} \notin K$, and a point $Q \in 2E(K(\sqrt{2}))$ of fixed height such that*

$$x(Q) \in F, \qquad x(Q) \neq x([n]P) \quad \text{for every } n \in \mathbb{Z}$$

whenever the abscissa on the right is defined. The same fixed point therefore satisfies every finite anchor exclusion, for every positive bound M.

Proof. Let s be a root of

$$s^4 - 1696s^2 + 18432 = 0$$

and put $K = \mathbb{Q}(s)$. Set $F' = \mathbb{Q}(\sqrt{2737})$, where $2737 = 7 \cdot 17 \cdot 23$. The two values of s^2 are $848 \pm 16\sqrt{2737}$, both strictly positive. They have norm $18432 = 2 \cdot 96^2$, which is not a rational square. Thus neither is a square in F', and $[K : \mathbb{Q}] = 4$. All four conjugates of s are real, so K is totally real.

If K contained $\sqrt{2}$, it would equal $F'(\sqrt{2})$. Writing $s = a + b\sqrt{2}$ with $a, b \in F'$ and using $s^2 \in F'$ gives $ab = 0$. In either case the norm of s^2 from F' to $\mathbb{Q}$ would be a rational square: it would be $N(a)^2$ or $4N(b)^2$. This contradicts the computed norm. Hence $\sqrt{2} \notin K$.

In $K(\sqrt{2})$ set

$$b = \frac{s}{2} + \frac{48}{s}\sqrt{2}, \qquad R = (8, b).$$

The defining equation for s gives $b^2 = 424 + 48\sqrt{2} = 8^3 - 8D$, so $R \in E(K(\sqrt{2}))$. It has nonzero ordinate. The doubling formula gives

$$\begin{aligned} x(Q) = x([2]R) = \frac{(8^2 + D)^2}{4b^2} &= \frac{5697 - 900\sqrt{2}}{1696 + 192\sqrt{2}} \\ &= \frac{312741}{87584} - \frac{40941}{43792}\sqrt{2} \in F. \end{aligned}$$

This value is not $0, d$, or $-d$, as is already seen from its coefficient of $\sqrt{2}$. Thus $y(Q) \neq 0$.

On the other hand $b \notin F$. If it belonged to F, the norm of its square would be a square in $\mathbb{Q}$; but

$$N_{F/\mathbb{Q}}(b^2) = 424^2 - 2 \cdot 48^2 = 175168 = 2^6 \cdot 7 \cdot 17 \cdot 23$$

is not a square. The doubling slope is $(3 \cdot 8^2 - D)/(2b)$. Since $b^2 \in F$, the resulting ordinate $y(Q)$ is a scalar in F times b. That scalar is nonzero because $y(Q) \neq 0$, so $y(Q) \notin F$.

Every multiple $[n]P$ has coordinates in F. Equality of its abscissa with $x(Q)$ would imply $Q = [n]P$ or $Q = -[n]P$, and would put $y(Q)$ in F, a contradiction. Hence each defined difference $x(Q) - x([n]P)$ is a nonzero element of F and has nonzero rational norm. Every finite product is nonzero. □

The halfpoint is represented in the literal two-coordinate model by $a_0 = 8$, $a_1 = 0$, $b_0 = s/2$, $b_1 = 48/s$ in K. Thus the example applies to the extended point chart itself; it is not an artifact of a split quadratic algebra. The script `check_real_anchor_20261106_v1.py` checks all curve and doubling identities by exact arithmetic in the degree-eight compositum.

The conclusion disproves the extension of the inference "finite exclusion forces a large integer elliptic label." Over the original field that inference depends on a cyclic group description. Enlarging the field introduces points outside that fixed cyclic group. We have not used this single point to construct witnesses for every other source equation at a nonintegral input; the proposition is a counterexample to an essential arithmetic inference, not a claim of a complete false positive for the entire source.

34.4 A domain assertion that does extend to fields with a real place

One useful component survives more generally. Let

$$G_J(W) = 1 + \sum_{j=1}^{J} c_j W^j, \qquad c_1 = -\frac{2}{5}, \qquad \frac{c_j}{c_{j-1}} = \frac{4j-5}{4j+1} \quad (j \geq 2).$$

All c_j are negative, and induction gives the exact identity

$$G_J(1) = -\frac{4J-1}{2} c_J > 0.$$

Consequently $G_J(w) \geq G_J(1) > 0$ for $0 \leq w \leq 1$. At a real embedding of the curve above, $D > 0$. For an affine double $Q = [2](a, b)$ with the required denominators and $y(Q)$ nonzero,

$$x(Q) = \left(\frac{a^2 + D}{2b}\right)^2 > 0.$$

The curve equation then implies $x(Q) > \sqrt{D}$, and hence $0 < D/x(Q)^2 < 1$. Thus

$$-\frac{x(Q)}{y(Q)} G_J\left(\frac{D}{x(Q)^2}\right) \neq 0.$$

An embedding is injective, so this proves the nonvanishing in any number field with a real embedding. If the formal quadratic algebra splits, apply the argument to each component; the source denominator guards ensure both are defined.

The analogous assertion is false over unrestricted number fields. Choose an algebraic root w of G_J. It is neither zero nor one. Choose $x = d/\sqrt{w}$ and a nonzero square root y of $x^3 - Dx$.

This gives an algebraic point Q where the displayed logarithm truncation vanishes. Adjoin a halfpoint of Q, which exists over a finite extension because the multiplication-by-two morphism is surjective, and, if desired, adjoin i. The resulting number field supplies the asserted failure even on $2E$. Therefore a final norm combination repairs conjunctions, but does not automatically justify all quotient-domain simplifications originally proved over $\mathbb{Q}$.

34.5 No one fixed existential formula works over every number field

Theorem 34.4. *Let k_0 be a number field. Fix an existential formula over k_0 whose quantifier-free part is a finite union of systems of polynomial equations and nonvanishing conditions. Suppose its input projection over k_0 contains $\mathbb{Z}$. There is a finite extension K/k_0 over which its input projection contains a rational noninteger. In particular no one such fixed formula defines $\mathcal{O}_K$ in every finite extension K/k_0.*

Proof. Over an algebraic closure $\overline{k_0}$ the formula's witness locus is constructible. Its image in the affine input line is constructible by Chevalley's theorem [36, Tag 00FE]. A constructible subset of an affine line over an algebraically closed field is finite or cofinite. This image contains $\mathbb{Z}$, so it is cofinite. Choose $q \in \mathbb{Q} \setminus \mathbb{Z}$ in it. The finitely many algebraic witness coordinates belong to some finite extension K/k_0. The formula therefore holds at q over K. But a rational algebraic integer is an integer: if $q = a/b$ in lowest terms satisfies a monic integer polynomial, clearing denominators shows $b \mid a^m$ and hence $b = 1$. Thus $q \notin \mathcal{O}_K$. □

This includes any finite number of simultaneous equations, any finite alternatives, and any finite list of denominator guards. Increasing these counts cannot overcome the uniformity obstruction. It does not preclude choosing a different formula for each number field.

There is a useful real refinement when the coefficient field is $\mathbb{Q}$, or more generally when k_0 has a chosen real embedding. The real input projection is semialgebraic by quantifier elimination and contains all positive and negative integers. A semialgebraic subset of the line is a finite union of points and intervals; it therefore contains both sufficiently distant tails. Choose a nonintegral rational input in one tail. The transfer principle for real closed fields supplies real algebraic witnesses. They, together with k_0, generate a number field K with a real embedding where that input is accepted. Quantifier elimination and this transfer principle are stated, for example, in [37, Theorem 2.1] and [38, Corollary 4.5 and Example 4.6]. The resulting field need not be totally real; only a real embedding is asserted.

34.6 What remains open after these checks

None of these counterexamples proves that some $\mathcal{O}_K$ is not Diophantine in K. They prove that the literal formulas, fixed cyclic arithmetic argument, or fixed field-independent coefficient choice cannot supply the desired universal conclusion. A field-adapted norm form faithfully combines an already correct finite system; it does not produce the missing arithmetic definition. Any extension of the rational construction must additionally control the new field-valued point coordinates and arithmetic witnesses, or supply an independent existential definition of the embedded rational field. Under the claimed rational definition, such rational isolation would allow an integral basis construction of $\mathcal{O}_K$, but that additional existential assertion has not been established here for every number field.

35 Combining equations over number fields by division norms

This section concerns the faithful encoding of polynomial systems. It does not assert that extending the arithmetic source to a number field defines that field's ring of integers. The distinction is essential: a correct equation-combining operation preserves whatever solutions the extended source has, including any arithmetic defects.

35.1 A rational norm form adapted to a specified number field

The reduced norm of a central division algebra of degree d is a homogeneous form of degree d in d^2 coordinates, vanishing only at the origin. We give an explicit cyclic construction and a local proof of division, so that the effectivity and the coefficients are specified. Standard background on cyclic algebras is available in [34, Chapter IV]; the prime-selection step uses Chebotarev's theorem, for example [35, Section 2.4].

Theorem 35.1. *Let K be any specified number field and let $r \geq 1$. Put $d = \lceil \sqrt{r} \rceil$. There is an effectively specified homogeneous polynomial*

$$N \in \mathbb{Z}[Z_1, \ldots, Z_{d^2}]$$

of degree d such that, for $z_j \in K$,

$$N(z_1, \ldots, z_{d^2}) = 0 \quad \Longleftrightarrow \quad z_1 = \cdots = z_{d^2} = 0.$$

Consequently a system of r equations over K, of degrees at most D, can be combined into one equation of degree at most dD, with no additional variables. The combining form can have rational integer coefficients even though its choice depends on K.

Proof. For $r = 1$ use the equation itself. Assume $d \geq 2$. Let K' be the normal closure of $K/\mathbb{Q}$. Choose a rational prime $q \equiv 1 \pmod{d}$ unramified in K', and let L be the unique degree-d subfield of $\mathbb{Q}(\zeta_q)$. Such a prime exists by Dirichlet's theorem. Every nontrivial subfield of L is ramified at q, whereas K' is unramified there. Thus $L \cap K' = \mathbb{Q}$. Both extensions are Galois, so their compositum has the product Galois group.

Chebotarev's theorem supplies a rational prime $p \neq q$ which splits completely in K' and whose Frobenius generates $\mathrm{Gal}(L/\mathbb{Q})$. Hence p is inert in L. Fix a generator σ of this cyclic Galois group and form

$$A = (L/\mathbb{Q}, \sigma, p) = \bigoplus_{j=0}^{d-1} L u^j, \qquad u^d = p, \qquad ua = \sigma(a)u.$$

These relations give the usual associative cyclic algebra.

At a place of K above p, the completion is $\mathbb{Q}_p$, and the corresponding completion of L is the unramified degree-d extension $L_p/\mathbb{Q}_p$. On the local cyclic algebra define

$$w\left(\sum_{j=0}^{d-1} a_j u^j\right) = \min_{a_j \neq 0} \left(v_{L_p}(a_j) + \frac{j}{d} \right).$$

The minimum is attained at exactly one index, since the fractional parts j/d are distinct. Multiplying the unique lowest terms shows that $w(ab) = w(a) + w(b)$: the automorphism

preserves the valuation, and reducing u^d to p preserves the weight. No other product term has this lowest weight. The local algebra therefore has no zero divisors and is a division algebra, because it is finite dimensional over a field. It follows that $A \otimes_{\mathbb{Q}} K$ is also a division algebra: a zero divisor would remain one after extension to that completion. The same argument proves that A is division.

Here is a concrete basis of L consisting of algebraic integers for writing its norm form. Let g generate $(\mathbb{Z}/q\mathbb{Z})^*$, let $H = \langle g^d \rangle$, and put

$$\eta_i = \sum_{h \in H} \zeta_q^{g^i h}, \qquad 0 \le i < d.$$

The periods are linearly independent over $\mathbb{Q}$. Indeed a relation among them gives a polynomial in ζ_q of degree at most $q-1$ and constant coefficient zero. It must be a multiple of $1+T+\cdots+T^{q-1}$, whose constant coefficient is one, and therefore must be zero. Since the periods are fixed by H, they form a basis of L. Their sum is -1, so $1, \eta_1, \ldots, \eta_{d-1}$ is also a basis. Choose σ so that $\sigma(\eta_i) = \eta_{i+1}$, with indices modulo d.

Write $a_j = Z_{j,0} + \sum_{i=1}^{d-1} \eta_i Z_{j,i}$. Let $D(a) = \operatorname{diag}(a, \sigma(a), \ldots, \sigma^{d-1}(a))$ and define the matrix U by

$$Ue_j = e_{j-1} \ (1 \le j < d), \qquad Ue_0 = pe_{d-1}.$$

Then $U^d = pI$ and $UD(a) = D(\sigma(a))U$. The norm form is

$$N(Z) = \det \left(\sum_{j=0}^{d-1} D(a_j) U^j \right). \tag{16}$$

Applying σ to this matrix conjugates it by U, so its determinant has rational coefficients. The periods are algebraic integers and p is an integer, so those rational coefficients are integers. The form is homogeneous of degree d. A nonzero coordinate vector over K represents a nonzero, hence invertible, element of $A \otimes K$; its matrix is invertible and its norm is nonzero. This proves anisotropy.

The construction is effective. One can search for q, compute its cyclotomic periods, and search primes p testing complete splitting in K' and the indicated Frobenius class in L. The cited existence theorems ensure termination. Finally substitute the r residual polynomials for the first r coordinates of N, and zeros for all remaining coordinates. Anisotropy proves equivalence, and homogeneity gives the degree bound without introducing any quantified coordinate. □

Remark 35.2 (Coefficient bound)**.** In every complex embedding, $|\eta_i| \le (q-1)/d$. Every entry of the matrix in (16) is a linear form whose coefficient absolute sum is at most $(q-1)p$. Expanding the determinant and applying the triangle inequality therefore gives

$$\|N\|_1 \le d!\big((q-1)p\big)^d.$$

This includes all determinant permutations and all monomials. The form is primitive: assigning $Z_{0,0} = 1$ and all other coordinates zero represents the identity and gives $N = 1$. If all input residuals have coefficient norm at most M, their combined polynomial has coefficient norm at most $\|N\|_1 M^d$.

35.2 The degree-36 form and the untransformed direct systems

The untransformed direct-Lucas systems have respectively 1315 and 1339 total coordinates, with 1272 and 1296 quadratic residuals. Both residual counts are at most $36^2 = 1296$. For these systems take the completely explicit choices

$$d = 36, \qquad q = 37, \qquad p = 2.$$

The element 2 generates $(\mathbb{Z}/37\mathbb{Z})^*$: indeed $2^{18} \equiv -1$ and $2^{12} \equiv 26 \not\equiv 1 \pmod{37}$, which exclude the maximal proper divisors of 36. Thus $L = \mathbb{Q}(\zeta_{37})$ has degree 36 and two is inert. The preceding local valuation proof shows that

$$A = (L/\mathbb{Q}, \sigma, 2), \qquad \sigma(\zeta_{37}) = \zeta_{37}^2,$$

is division. Formula (16) therefore gives an integer norm form of degree 36 in 1296 coordinates. Here the period basis is simply $1, \zeta_{37}^2, \zeta_{37}^{2^2}, \ldots, \zeta_{37}^{2^{35}}$, with exponents reduced modulo 37.

Lemma 35.3. *Let A be a central division algebra of degree d over a field k. If K/k is finite and $\gcd(d, [K : k]) = 1$, then $A \otimes_k K$ is division.*

Proof. Write $A \otimes K \simeq M_s(B)$, where B is central division over K of degree d/s. A simple left module has K-dimension $s(d/s)^2 = d^2/s$. By restricting scalars it is a vector space over the division ring A, so its k-dimension is a multiple of d^2. Hence $s \mid [K : k]$. Since also $s \mid d$, coprimality gives $s = 1$. □

This proves that the fixed degree-36 form remains anisotropic whenever $\gcd(36, [K : \mathbb{Q}]) = 1$. The assertion must not be justified by choosing one completion of coprime local degree: that inference is false for composite d, whereas the module argument applies directly.

The same form also remains anisotropic over $\mathbb{Q}(i)$. Here is a local proof, independent of the coprime-degree lemma. At 37 put $\pi = \zeta_{37} - 1$. The residue field of L_{37} is $\mathbb{F}_{37}$ and $\sigma(\pi)/\pi \equiv 2 \pmod{\pi}$. The algebra $\mathcal{O}_{L_{37}}[u]$, $u^{36} = 2$, has residue algebra

$$\mathbb{F}_{37}[u]/(u^{36} - 2),$$

which is a field. Indeed a root α satisfies $\alpha^{37} = 2\alpha$, so its Frobenius orbit has length exactly 36 because 2 has multiplicative order 36. For $x = \sum a_j u^j$ define $w(x) = \min_j v_\pi(a_j)$. After factoring out its lowest power of π, a nonzero element has nonzero reduction in this residue field. Commuting π^s through powers of u changes reduction by the automorphism $u \mapsto 2^s u$; consequently the lowest reductions of a product are still a product of nonzero elements. Thus $w(xy) = w(x) + w(y)$ and the local algebra is division. Finally 37 splits in $\mathbb{Q}(i)$ because $6^2 \equiv -1 \pmod{37}$, so a completion after this base change is still $\mathbb{Q}_{37}$. Division at that completion proves division over $\mathbb{Q}(i)$.

Combining the two preferred quadratic systems gives polynomials with respectively 1315 and 1339 total coordinates and *degree exactly 72*. The exact residual inventories put the input monomial X_0^2 with coefficient one in the first residual; no other residual contains X_0. The first norm coordinate is the coefficient of the algebra identity, so the coefficient of X_0^{72} in the resulting polynomial is one. In particular the top degree cannot cancel, and each resulting integer polynomial is primitive.

The coefficient estimate becomes

$$\|N\|_1 \le 36!\, 72^{36} < 2^{361}.$$

The exact residual tables have coefficient norm at most 2^{511}. The 1339-coordinate version, with ordinary integer coefficients already materialized in those residuals, therefore has coefficient height at most

$$2^{18757}.$$

For the 1315-coordinate version substitute the fixed integer $\kappa = 3^{2^{28}}$ into residuals affine in κ. Using $\log_2 \kappa < 429496730$ gives coefficient height at most

$$2^{15461901037}.$$

The artifact names `norm_repair_1315_degree72_20261106_v1.json.gz` and `norm_repair_1339_degree72_20261106_v1.json.gz` specify the complete residual lists, cyclotomic basis, matrix entries, and determinant. They are exact finite polynomial specifications, not expanded coefficient tables of degree 72. The symbol ζ_{37} is eliminated by reduction modulo its cyclotomic polynomial and extraction of the constant coefficient; it is not a quantified variable.

For an arbitrary specified number field the fixed form may cease to be anisotropic. Theorem 35.1 provides a field-dependent choice of primes q, p with the same degree 36 and rational integer coefficients. Thus degree 72 and the coordinate counts persist, while the coefficient bound becomes the explicit prime-dependent bound in that theorem.

35.3 The preceding degree-37 form and its comparison bounds

The choices

$$d = 37, \qquad q = 149, \qquad p = 5$$

are explicit. The degree-37 subfield of $\mathbb{Q}(\zeta_{149})$ is unramified at five, and five is inert there: the Galois group has prime order 37 and $5^4 \equiv 29 \not\equiv 1 \pmod{149}$. The local valuation argument above proves division.

This same rational norm form remains anisotropic over every number field K for which $37 \nmid [K : \mathbb{Q}]$. To see this directly, use $\sum_{v|5} e_v f_v = [K : \mathbb{Q}]$ to choose a place with $37 \nmid e_v f_v$. The unramified degree-37 extension remains a field after this local base change. Normalizing the valuation of K_v gives $v(5) = e_v$, and the weights

$$v(a_j) + \frac{e_v j}{37}$$

again have distinct fractional parts. The same lowest-term argument proves that the local cyclic algebra is division. For fields with degree divisible by 37, the field-dependent prime choices in the theorem still supply degree 37 and rational integer coefficients.

For comparison, there are 1322 primitive quadratic residuals in the 1367-coordinate system and 1346 in the 1391-coordinate system. Both numbers are at most $37^2 = 1369$. Thus the norm construction combines either system into one polynomial of degree exactly 74, with the same respective total coordinate counts. Exactness of the degree follows without expansion: the highest homogeneous quadratic parts of the residuals are not all zero. At some rational specialization this tuple is nonzero, and anisotropy shows that its norm is nonzero. It is exactly the homogeneous degree-74 part of the combined polynomial.

For the explicit $(37, 149, 5)$ choice,

$$\|N\|_1 \le 37!\, 740^{37} < 2^{496}.$$

The exact residual tables give maximum coefficient norm at most 2^{511}. Hence the ordinary integer version with 1391 total coordinates has coefficient height at most

$$2^{19403}.$$

For the 1367-coordinate version the residuals are affine in the fixed integer $\kappa = 3^{2^{28}}$. Since $\log_2 \kappa < (8/5)2^{28} < 429496730$, a valid integer coefficient-height bound is

$$2^{15891398413}.$$

These bounds concern the norm-compiled polynomials. Their degree-74 determinant specifications are separate from the frozen degree-four sum-of-squares tables.

35.4 The finite-algebra compiler over every number field

All algebraic steps in the guarded finite-algebra theorem work over an arbitrary characteristic-zero field: the finite separating-parameter bound, simple rational eigenline, trace-adjugate identities, and reciprocal guard are unchanged. Only the final conjunction by a sum of squares requires an ordering. Over a number field with a real embedding it remains valid as written. Over an arbitrary number field replace each branch's sum of squares by the norm form of Theorem 35.1.

If n bounds the algebra rank and c bounds the number of closed residuals, a branch contains at most $n + c$ residuals: one reciprocal residual, at most $n - 1$ high-coefficient conditions, and at most c reversed traces. Use one norm form of degree

$$d = \lceil \sqrt{n + c} \rceil$$

and pad shorter lists with zero. This introduces no new coordinate. Relative to the recorded sum-of-squares degree bound, multiply each branch bound by $d/2$ before multiplying over its finite indices. This operation preserves the rational projection over the specified number field of the guarded finite source itself.

The exact recorded integer bounds give the following convenient rounded estimates. All counts include the input.

source	total coordinates	d	degree bound over K
direct finite anchor	41	16385	$2^{9566686}$
direct semilocal	43	23171	2^{250}
proxy finite anchor	43	23171	$2^{9566693}$
proxy semilocal	45	32769	2^{257}

The combining form has integer coefficients chosen using K; the number of coordinates and the displayed degree bound do not depend on $[K : \mathbb{Q}]$. These are field-adapted encodings of the corresponding source systems. They do not establish that those systems define $\mathcal{O}_K$.

36 Field behavior of the final quartic transformations

The last transformations improve the quartic tables without new arithmetic input. They give the preferred translated 1315-coordinate quartic, with coefficient height 64κ for $\kappa = 3^{2^{28}}$, and the preferred ordinary integer-coefficient quartic with 1335 coordinates and height less than 2^{1020}. Their field behavior follows precisely from the type of transformation involved. The four preceding coefficient tables and their certificates remain unchanged.

36.1 The affine translation works over every field of characteristic zero

Put $S = 3^{2^{27}}$, so $S^2 = \kappa$. The private anchor coordinates $u = X_{55}$, $v = X_{56}$, $w = X_{57}$ in the 1315-coordinate table occur only in the row

$$-3A^2 + 16u^2 + 96v^2 - 32w^2 + 32S^2, \qquad A = X_{52} + X_{53} - X_{54}.$$

Make the rational affine substitution

$$u = S + \frac{U - 3V + 3W}{4}, \qquad v = \frac{S}{2} + \frac{V}{4}, \qquad w = \frac{3S}{2} + \frac{W}{4}. \tag{17}$$

The new row is exactly

$$-3A^2 + U^2 - 6UV + 6UW + 15V^2 - 18VW + 7W^2 + 8SU.$$

The inverse map is

$$U = 4u + 12v - 12w + 8S, \qquad V = 4v - 2S, \qquad W = 4w - 6S.$$

Both maps are defined over $\mathbb{Q}$, with S an ordinary fixed integer. Therefore the substitution is a bijection of witness tuples over every characteristic-zero field. It preserves the literal polynomial's input projection, as well as that of the residual system. No use of an ordering is required for this assertion.

For the generic $\mathbb{Q}(i)$ section of the preceding 1315-coordinate quartic, the old private anchor coordinates are all zero. Their inverse images are

$$(X_{55}, X_{56}, X_{57}) = (8S, -2S, -6S).$$

Keep the Gaussian coordinates at indices 72 and 73, and set all remaining witnesses to zero. The restriction is the same $R(T) + a^2 + b^2$ as before, with $\kappa = S^2$. Choosing $a = (1 - R)/2$ and $b = i(1 + R)/2$ proves that the translated literal quartic also accepts every input over every characteristic-zero field containing i.

36.2 Eliminating a private squaring coordinate preserves the system

In the preceding 1339-coordinate table, the coordinate $z = X_{81}$ occurs only in the two rows

$$X_{80}^2 - z, \qquad R_0 + 32z,$$

where R_0 is independent of z. Delete the first row and z, and replace the second row by $R_0 + 32X_{80}^2$. This gives 1338 coordinates and 1295 quadratic residuals. Over every field, solutions of the new residual system lift uniquely by $z = X_{80}^2$, and every old system solution is obtained this way.

The literal sums of squares need a separate statement. Every zero of the new sum lifts to a zero of the old sum over every field, because the deleted row then vanishes and the remaining rows agree. The converse follows over fields with a real embedding: a zero sum of squares forces each row to vanish, and hence forces $z = X_{80}^2$. Over a general nonreal field, an old zero might instead arise by cancellation among nonzero rows. System elimination alone does not prove that such a zero descends to the new sum. We make no unrestricted converse assertion for literal sums.

Nevertheless the old generic $\mathbb{Q}(i)$ section has $X_{80} = X_{81} = 0$, so it does descend. Its Gaussian coordinate indices 96 and 97 become 95 and 96 after deletion. Thus the new 1338-coordinate quartic, too, has input projection equal to the whole field whenever i belongs to that field. The script `check_quartic_base_change_20261106_v2.py` verifies these new generic sections from the full sparse tables and, independently, by squaring the restricted residuals. It writes a separate receipt, leaving the preceding four certificates unchanged.

36.3 Inlining three fixed constants

The final ordinary-coefficient refinement substitutes the fixed integers 3^{32}, 3^{64}, and 3^{128} for coordinates 58, 59, and 60 of the 1338-coordinate system, and deletes their three defining rows. The remaining system has 1335 coordinates and 1292 quadratic residuals. These are private polynomial graph equations, so substitution and reinsertion give inverse maps on system solutions over every field. As in the preceding elimination, the new literal sum of squares always lifts to the old one; its converse follows over fields with a real embedding, and is not asserted over all nonreal fields merely from the system equivalence.

The largest newly introduced coefficient in a residual is 3^{256}, whose bit length is 406. The associated new quartic coefficient 3^{512} is less than 2^{812}. The full unchanged bound $H < 2^{1020}$ therefore persists, as also verified from the complete expanded coefficient table. The generic Gaussian section now uses coordinates 92 and 93, with all other witnesses zero. Direct restriction of the new table again gives $R(T) + a^2 + b^2$ for an explicitly recorded constant-plus-T^4 polynomial R; the same Gaussian witness formulas prove the whole-field projection over any characteristic-zero field containing i. This section is checked directly, rather than assuming that an old cancellation solution survives fixed-constant substitution.

36.4 The preferred degree-72 norm specifications

All three transformed residual systems fit the same degree-36 cyclic division norm: their residual counts are 1272, 1295, and 1292, respectively, each at most $36^2 = 1296$. The resulting norm polynomials have 1315, 1338, and 1335 total coordinates, respectively. Their degree is exactly 72 and their coefficient of X_0^{72} is one, since the first residual still contains $+X_0^2$ and all other residuals are independent of the input coordinate.

The new translated residuals have only one coefficient involving S, namely $8S$ at U in the anchor row. The coefficient norm of every constant part is less than $2^{511} < S$. Hence every residual has coefficient norm strictly less than

$$9S = 3^{2^{27}+2}.$$

Since $3^5 < 2^8$,

$$\log_2(9S) < \frac{8}{5}(2^{27} + 2) = 214748368.$$

Combining this with $36!72^{36} < 2^{361}$ gives the explicit bound

$$H(P_{1315}^{\mathrm{norm,tr}}) < 36!72^{36}\, 3^{36(2^{27}+2)} < 2^{7730941609}.$$

The ordinary 1338- and 1335-coordinate residuals still have coefficient norm at most 2^{511}, giving

$$H(P_{1338}^{\mathrm{norm}}), \quad H(P_{1335}^{\mathrm{norm}}) \le 2^{18757}.$$

The complete specifications are `norm_repair_1315_translated_degree72_20261106_v2.json.gz` `norm_repair_1338_degree72_20261106_v2.json.gz`, and `norm_repair_1335_degree72_20261106_v2.json.gz`. Each supplies the full residual list, cyclic matrix, and cyclotomic coefficient-extraction rule. Neither expands the full degree-72 coefficient table or introduces a quantified cyclotomic coordinate.

The fixed norm is faithful over $\mathbb{Q}(i)$ and over extensions of degree coprime to 36, by the preceding local and module arguments. For any other specified number field, the field-dependent

prime construction supplies a faithful degree-36 norm with the same coordinate counts and degree 72, with its stated prime-dependent coefficient bound. Because these transformations preserve the residual systems over all fields, these norm polynomials preserve their input projections wherever the combining norms are faithful.

In particular the unshifted scalar-contact contradictions still hold for the transformed residual systems and their faithful norm combinations. Over a field with a real embedding they also hold for the literal transformed sums of squares. These facts concern the unshifted artifacts; they do not extend the contact counterexample to the separately checked shift-2402 constructions. Neither transformation changes the unresolved arithmetic question of defining $\mathcal{O}_K$ over a general number field.

37 The remaining optimization problem

The preferred construction in this article uses forty-one total coordinates, including the input. Its degree is bounded by $2^{9566673}$, and its arithmetic interpretation remains conditional on the named source inputs and Sun's simultaneous prime-pattern theorem. The forty-three-coordinate semilocal alternative has the substantially smaller degree bound 2^{236}. For degree four, the supplied counts are 1315 and 1335, the latter with coefficient height less than 2^{1020}. These are constructions and upper bounds; no optimality claim follows from the comparisons.

37.1 What a reduction to ten coordinates would have to accomplish

The forty retained coordinates of the preferred source consist of the input, eighteen components for nine elliptic abscissas, one common slope, and twenty parameters in ten pole predicates. The finite-algebra compiler adds one final witness. It has already removed the ordinate roots, pole-conic roots, and the arithmetic evaluation coordinates. The direct Lucas test removes the remaining two retained proxy coordinates and their third, algebraic proxy. Replacing this compiler by another primitive-element argument would leave the same retained base and would not, by itself, decrease the count.

The geometric results above explain why each independent elliptic point cannot simply be encoded by one rational parameter and a finite algebraic choice. The specific cyclic point group is dense in its two-dimensional Weil restriction. The denominator results also exclude compressing independent parity conditions by a rank-deficient list of monomials, or replacing the entire integrality condition by finitely many fixed rational pullbacks of perfect-denominator predicates. These are restrictions on particular methods, not general lower bounds of eighteen or forty coordinates.

The most promising remaining arithmetic question is whether several point choices or pole witnesses can be tied together while preserving simultaneous rational solvability. A proposed sharing must prove this simultaneous assertion. Separate existence for each pole conic, or local solvability at every prime for each one independently, does not provide a common rational parameter. Another possibility is an integer relation with a different height argument requiring fewer independently chosen elliptic labels. The direct-polynomial congruence lemma supplies a precise interface for testing such relations, including the condition $k > 12 \deg F$ for its displayed height estimate.

No construction with two through ten total coordinates has been proved here. Two total coordinates are impossible, even for arbitrary finite systems. The unrestricted cases of three through ten coordinates remain unresolved by this work. For quadratic systems the proved

restrictions are stronger: at least five total coordinates and at least three independent quadratic residuals are necessary. A successful quadratic system in five through ten coordinates would immediately give a quartic in exactly the same coordinates over $\mathbb{Q}$.

37.2 Degree and coefficients at a fixed coordinate count

The fixed elliptic exclusion product dominates the enormous degree of the forty-one-coordinate construction. A better explicit height constant, or a different size condition costing no new coordinates, would be more consequential than a small improvement to its finite algebra. The semilocal alternative illustrates this trade explicitly. The remaining finite ranks are 2^{28} and 2^{29}; their determinant and separating-parameter bounds are conservative, and sparse formulas could improve degree and coefficient bounds without saving coordinates.

For quartics, the preferred 1315-coordinate version also uses an invertible affine change of its three private anchor witnesses, giving exact primitive coefficient height $64 \cdot 3^{2^{28}}$. Removing the last constant-chain square and inlining its first three constants gives the 1335-coordinate alternative at height less than 2^{1020}. The direct field-valued Lucas test has reduced both the number of tests and the lengths of their shared symmetric-product computations. Further useful computation should operate on the actual quadratic residual system and preserve an exact reconstruction certificate. Changes of residual basis and invertible coordinate changes may also reduce coefficient height. All comparisons must use primitive integer coefficients after clearing numerical denominators. Computing a huge fixed integer with extra squaring coordinates remains a distinct trade, which is why both quartic tables are supplied.

37.3 The arithmetic hypotheses still requiring independent review

The most consequential review remains the source manuscript's restricted 2-converse, its fixed-field coefficient transfer, and the complete five-contact height argument with its explicit constant. The shared-point and direct-Lucas constructions additionally use Sun's Theorem 4.1. The stated theorem and its order-compatibility consequences were checked; the complete analytic proof of its prime-pattern estimates has not been independently reproduced here. The distinction between the unconditional Artin-symbol theorem of Frei–Tinková and their conditional primitive-root theorem matters [9]. Sun's further large-prime sieve and its use of polynomial-prime transference [10] remain external inputs, not conclusions of the polynomial checkers.

This extra input has a specific role. A common comparison index must satisfy simultaneous congruences at different selected primes. Their compatibility requires

$$\gcd(d_{p,-}, d_{q,+}) = 4 \qquad \text{for all } p, q.$$

An odd common divisor would make the residues zero and four incompatible. One cannot justify the common index by reusing only the original private-witness existence statements. The conditional proofs in this article retain the stronger simultaneous hypothesis explicitly.

37.4 Number fields and the limits of a uniform formula

The explicit $\mathbb{Q}(i)$ sections and the rejected integral contact inputs show that the literal newer polynomials do not extend unchanged to rings of integers. Translating the input square removes the finite contact obstruction; a field-adapted division norm repairs conjunction without adding variables. These are useful repairs, but neither restores the fixed cyclic elliptic-group argument.

The totally real quartic example gives a concrete point outside that group satisfying every finite anchor exclusion.

The general constructibility theorem further shows that no one fixed finite existential formula can define the rings of integers in all number-field extensions. The field-dependent existence question remains open here. Under the claimed rational definition, an existential definition of the embedded rational field would permit the integral basis construction in Theorem 33.11. Undecidability or MRDP over the ring of integers does not supply that rational-field isolation. The next arithmetic work should address this missing assertion or develop a different source adapted to each field.

38 A conditional geometric obstruction

The numerical lower bounds and the conic obstruction above are unconditional. There is also a useful conjectural obstruction of a different kind. Mazur's finite-component conjecture states that the closure of the rational points of a variety over $\mathbb{Q}$ in its real points has finitely many connected components [26, Conjecture 3]. It would rule out a Diophantine definition of $\mathbb{Z}$ with any degree or number of variables. It must not be used as an established theorem against the proposed source.

Proposition 38.1. *If a polynomial defines $\mathbb{Z}$ in $\mathbb{Q}$, the real closure of its rational zero set has infinitely many connected components.*

Proof. Let V be its affine zero set and let C be the closure of $V(\mathbb{Q})$ inside $V(\mathbb{R})$. The input projection is continuous. Its value on every convergent sequence of rational points is a limit of integers, and hence an integer; therefore its image on C is contained in $\mathbb{Z}$. Every integer occurs already on $V(\mathbb{Q})$. A connected component of C has connected image in the discrete set $\mathbb{Z}$, hence has only one input value. Infinitely many components are necessary. □

Mazur's formulation uses reduced schemes of finite type over $\mathbb{Q}$, so it applies directly to the reduced affine zero set used here [26, Section 1 and Conjecture 3]. This topological obstruction clarifies the strength of the proposed theorem. It does not add a premise to our exact algebraic transformations.

A Permissible precision reductions

The following elementary argument isolates exactly what the growth comparison needs. It does not establish the arithmetic estimates from which it starts.

Proposition A.1. *Let $a, B, n, \Delta \in \mathbb{Z}$, with $B \geq a \geq 1$, let $\beta \in \mathbb{Q}$, and let $H \geq 1$ be real. Write $h(\beta) = \log \max\{|b|, c\}$ when $\beta = b/c$ is in lowest terms with $c > 0$. If $\beta \neq n$, let N be the absolute value of the numerator of $\beta - n$ in lowest terms. Suppose an arithmetic certificate supplies the following estimates for some $\kappa > 48$, $\lambda > 12$, and $\mu > 24$:*

1. *if $\Delta \neq 0$, then $B^2 - 6 < 7H|\Delta|^{24/\kappa}$ and $|\Delta| < 5B^4$;*
2. *if $\Delta = 0$, then $\log B > a - 3$;*
3. *if $\beta \neq n$, then $B^2 < 7HN^{24/\lambda} + 6$ and $N \leq 2e^{h(\beta)}B$;*
4. *$h(\beta) \leq Ha^{24/\mu}$.*

Put

$$C_\kappa = 7\,5^{24/\kappa} + 6, \qquad C_\lambda = 7\,2^{24/\lambda} + 6, \qquad A_\lambda = \frac{\lambda}{2\lambda - 24}, \qquad B_\lambda = \frac{12}{\lambda - 12}.$$

If

$$a > \max\left\{6, (C_\kappa H)^{\kappa/(2\kappa-96)}, 4A_\lambda \log(C_\lambda H), (4B_\lambda H)^{\mu/(\mu-24)}\right\},$$

then $\Delta = 0$ *and* $\beta = n$.

Proof. For $\Delta \neq 0$, absorb the constant six using $B, H \geq 1$ to obtain $B^{2-96/\kappa} < C_\kappa H$, contradicting $B \geq a$. Thus $\Delta = 0$. If $\beta \neq n$, put $r = 24/\lambda < 2$. The third hypothesis and $B, H \geq 1$ imply

$$B^{2-r} < C_\lambda H e^{rh(\beta)}.$$

Consequently

$$\log B < A_\lambda \log(C_\lambda H) + B_\lambda h(\beta) < a/2 < a - 3,$$

contrary to the second hypothesis. □

The thresholds $48, 12, 24$ are thresholds of this particular comparison of exponents. This proof does not show that the corresponding boundary values are impossible by a different argument.

Corollary A.2. *The choices*

$$(\kappa, \lambda, \mu) = (52, 13, 32), \qquad a > (21H)^{13/2}$$

satisfy the conclusion of Proposition A.1. If $\log_2 H < 4783227$*, the existing type of anchor with twenty-six successive squarings, giving* $a \geq 3^{2^{25}-1}$*, is sufficient.*

Proof. Here $C_\kappa < 21$, $C_\lambda < 32$, $A_\lambda = 13/2$, $B_\lambda = 12$, and $24/\mu = 3/4$. The assumed lower bound for a implies $a > (48H)^4$ because $21^{13/2} > 48^4$ and $H \geq 1$; hence $h(\beta) < a/48$. It also implies $a > 26\log(32H)$: use $\log(32H) < 32H$ and $21^{13/2} > 832$. Thus the last contradiction reads $\log B < (13/2)\log(32H) + 12h(\beta) < a/2 < a - 3$. For the numerical anchor,

$$\log_2\left((21H)^{13/2}\right) < \tfrac{13}{2}(5 + 4783227) = 31091008,$$

whereas

$$\log_2\left(3^{2^{25}-1}\right) > \tfrac{3}{2}(2^{25} - 1) = 50331646.5.$$

□

For the source system's scalar batch the order 48 is used solely to obtain $M_5(\beta)^{48} \mid s$, hence $h(\beta) \leq H\sqrt{|s|}$. Replacing that order by 32 gives instead $h(\beta) \leq H|s|^{3/4}$, which Corollary A.2 shows is sufficient. Completeness still allows all the required divisibilities: the index s is chosen in an arbitrarily divisible nonzero subgroup. The batch's recursive power construction uses eight multiplications for the required powers r^2, r^{96}, and six for r^2, r^{64}. Thus this change removes two source power definitions. Its exact effect on the compiled polynomial must be measured after the full pipeline.

For comparison, orders 49 and 52 require the same thirteen terms in the logarithm truncation used by the source, but the source's shared power recursion is less economical for $(49, 13)$: it uses twelve multiplications for r^2, r^{26}, r^{98} and nine for r^2, r^{26}, r^{104}. A smaller numerical precision does not automatically mean fewer variables. Joint addition-chain optimization is the correct comparison.

The source-preserving released-count packages retain the original precisions. The $(52, 13)$ experiment on the original source did not improve its coordinate count, because its power chains were less economical. The new shared-point quartics of Section 18 use precision 52 and scalar order 32, together with stronger size conditions and a different evaluation scheme. Their certified counts are 1367 and 1391, as proved there.

B Audit of the arithmetic source and the remaining dependencies

B.1 Scope and conclusion

The source examined here is `quartic_12504.tex` at commit `28e87b1b62af0effd954605af7f70b2699dd5672`, with SHA–256

`38d458ba4178083766c5ae56a894da595fe27170c1ef07f6b49750fc5edf7c37.`

The line numbers below refer to this fixed file. Labels are also given because subsequent edits can change line numbers. We examined the arithmetic interpretation of the released reciprocal comparison source, its principal local arguments, its final height comparisons, and selected substantial inputs to completeness. We did not independently verify the entire new restricted converse argument or every application of the arithmetic-geometric height theorems.

No rational counterexample to the source system, and no invalid step in the elementary local and numerical deductions listed below, was found in this audit. This statement is substantially weaker than an independent proof of the asserted Diophantine definition. The exact polynomial transformations elsewhere in this paper preserve the projected rational zero set. Identifying that set with $\mathbb{Z}$ still requires the arithmetic dependencies isolated here.

Two dependencies deserve particular emphasis. The pole predicate's one-sided obstruction has an elementary valuation proof, but its completeness uses a new restricted 2-converse. The asserted five-contact height theorem has a lengthy geometric and modularity proof; a calculation of its displayed constant is not a proof of that theorem. Neither dependency is discharged by checking the polynomial compiler.

B.2 The local obstruction and norm argument

For $b \in \mathbb{Q}$, put $B = 17b$ and let $\Psi(b)$ assert the existence of $t, U, e, p, q, s, h \in \mathbb{Q}$ satisfying

$$U(1{+}bt^2) = b(1{-}bt^2), \qquad p^2 = B(1{-}e)(1{+}3e), \qquad h = e(B{-}U^2), \qquad Bq^2 = (s^2{-}Bh)(s^2{+}3Bh).$$

The relevant source statements are `hyp:o8-pole` and `lem:original-seven-parity-soundness` (lines 7532–7597).

Checked one-sided statement. If $b \neq 0$ and $\Psi(b)$, then $v_\ell(b)$ is not negative and odd for any prime $\ell \notin \{2, 3, 17\}$. Indeed, suppose $k = v_\ell(b) < 0$ is odd. The first equation rules out $1 + bt^2 = 0$. The valuations of $1 - bt^2$ and $1 + bt^2$ agree, so $v_\ell(U) = k$; this also holds when $t = 0$. Consequently $v_\ell(B - U^2) = 2k$. The second equation rules out $e = 0$ and forces $v_\ell(e) = 0$: a nonzero valuation would make its right side have odd valuation. Thus $h \neq 0$ and $v_\ell(Bh) = 3k$ is odd. For any s, the right side of the final equation is nonzero and has even valuation; the two terms in each factor have unequal valuation when $s \neq 0$, and the assertion is immediate when $s = 0$. The left side is either zero or has odd valuation, a contradiction. This proof includes $e = 1$, $e = -1/3$, and $p = 0$. No omitted inverse condition on a witness is needed.

Checked norm propagation. The binary operation used by the compiler is $N_r(a,b) = a^2 + rb^2$. If $R = v_\ell(r)$ is odd, terms with both inputs nonzero have distinct valuation parity. Hence

$$v_\ell N_r(a,b) = \min\{2v_\ell(a), R + 2v_\ell(b)\}.$$

In particular a tree containing the constant leaf 1 cannot have zero root at such a prime. If the constant is on the all-left path and $R < 0$, the root W has $v_\ell(W) \le 0$; then rW^2 has negative odd valuation, contradicting the pole obstruction. Thus $R > 0$. The same obstruction then implies $R + 2v_\ell(W) > 0$. Descending the preceding minimum formula shows that a leaf X/r^e has valuation greater than $-R$. Therefore

$$v_\ell(X) > (e-1)R \ge e-1, \qquad \text{and hence} \qquad v_\ell(X) \ge e.$$

This verifies the strict inequality and integer rounding used in the mixed-precision batches; it also explains why a global condition $W \neq 0$ is unnecessary for soundness. See lines 7789–7804 and 8080–8120 of the source.

The finite squareclass correction in lines 7755–7804 is separate from this soundness argument. Its extra factors have support in $\{2, 3, 5, 7, 11, 13, 17\}$, already excluded from the relevant contacts. Completeness chooses the root to be a unit at primes dividing the raw integer or the correction. This condition is needed: multiplying an arbitrary rational square by an integer does not by itself guarantee a square reduced denominator.

B.3 The reciprocal logarithm, common point, and final estimates

Checked formal approximation. For $E : y^2 = x^3 - Dx$, $D = 11 - 6\sqrt{2}$, set $f = 1/x$, $z = -x/y$, $a_0 = 1$, $a_1 = -2/5$, and $a_j = (4j-5)a_{j-1}/(4j+1)$. The relations

$$f = z^2(1 - Df^2), \qquad 4w(1-w)G'(w) + (1+w)G(w) = 1 - w, \quad G(w) = \sum_{j \ge 0} a_j w^j,$$

give $\log_E(z) = zG(Df^2)$. At a good split prime $p > 76176$, with $r = v_p(z) \ge 1$, one has $v_p(f) = 2r$. For $e \in \{16, 64\}$, $m = e/4$, and $L_e = z\sum_{j=0}^{m-1} a_j (Df^2)^j$, the source bounds the entire tail, including indices whose denominators are divisible by p:

$$v_p(\log_E(z) - L_e) \ge (4m+1)r, \qquad L_e/\log_E(z) \in 1 + p^e\mathbb{Z}_p.$$

The factorial denominator bound used for the tail is sufficient. For $4j + 1 \ge p$ one has $j \ge 19044$ and $v_p(a_j) \ge -(4j+1)/(p-1) \ge -j$, which gives the required bound and convergence. Truncating before the first denominator divisible by p alone would not justify the infinite-tail estimate; the source does not make that shortcut. See `lem:quartic-reciprocal-log`, lines 8753–8816.

The separate global nonvanishing argument is also valid for the truncations used here. In $\mathbb{Q}(\sqrt{2})$ the valuation extending v_2 has value group $\frac{1}{2}\mathbb{Z}$, $v(D) = 0$, and $v(a_j) = 1$ for $j \ge 1$. If $v(f) \ge 0$, the constant term of $G_J(Df^2)$ is uniquely least; if $v(f) < 0$ and $J \ge 2$, its last term is uniquely least. Thus the logarithm inverses introduce no exceptional rational point. See `lem:quartic-reciprocal-nonzero`, lines 8818–8842.

Checked compatibility of the common auxiliary point. In the completeness construction the private congruences are

$$h_p \equiv 0 \pmod{d_{p,1}}, \quad h_p \equiv 4 \pmod{d_{p,2}}, \quad h_p \equiv 4 \pmod{p^{64+v_p(j)}}, \qquad j = \operatorname{lcm}_p d_{p,2}.$$

The conditions $\gcd(d_{p,1}, d_{p,2}) = 4$ and $p \nmid d_{p,1}d_{p,2}$ suffice for these congruences at each individual p. The construction does not need pairwise coprimality between group orders belonging to different primes. A selected prime may divide j because of another group order; the term $v_p(j)$ already accounts for this possibility. Avoiding the finitely many forbidden values of h_p preserves the point-chart guards. The same verification does not authorize replacing all h_p by a single shared h: that would require compatibility between congruences with moduli $d_{p,1}$ and $d_{q,2}$ for different primes p, q, and the source provides no such cross-prime hypothesis. Sharing this witness is therefore a blocked optimization unless an additional prime-pattern theorem or a different construction is supplied. This checks the stated use of the ordinary-prime pattern, rather than independently reproving the full prime-pattern theorem; see `subsec:o8-common-J-proof`, lines 8394–8468, and lines 8964–8984.

Checked final numerical implication. Assume the stated height and radical-divisibility inputs, and put $a = |s|$, $B = \max(|n|, |s|, |t|, |z|)$, $H = H_*$. The anchor gives $a > (20H)^2$. If the integer Lucas residual

$$\Delta = t^2 - s(s-2)tz + (s-2)^2z^2 - 1$$

is nonzero, the source obtains $B^2 - 6 < 7H|\Delta|^{3/8} < 14HB^{3/2}$. Dividing by $B^{3/2}$, with $B \geq 1$, yields $\sqrt{B} < 14H + 6 \leq 20H$, a contradiction. Thus $\Delta = 0$. The elementary Lucas-unit argument then gives $\log B > a - 3$.

If $\beta \neq n$, let N be the absolute numerator of $\beta - n$ in lowest terms. The precision-16 argument gives

$$B^2 < 7HN^{3/2} + 6, \qquad N \leq 2e^{h(\beta)}B.$$

Consequently $\log B < 2\log(32H) + 3h(\beta)$. The scalar estimate $h(\beta) \leq H\sqrt{a} < a/20$ and $\log(32H) < a/20$ make the right side less than $a/4 < a - 3$. There is no loss of the exponent when the beta precision changes from 64 to 16. Finally $\beta = n$ and $X^2 = \beta - 2 \in \mathbb{Z}$ imply $X \in \mathbb{Z}$ by coprime numerator and denominator. See `thm:quartic-source-arithmetic`, lines 8867–8985. This is a conditional verification of the final implication, not a verification of its height hypotheses.

B.4 The two substantial arithmetic dependencies

Restricted converse and pole completeness. The exact converse needed is

$$\operatorname{corank}_{\mathbb{Z}_2} \operatorname{Sel}_{2^\infty}(E_\ell/\mathbb{Q}) = 1 \quad \Longrightarrow \quad \operatorname{ord}_{s=1} L(E_\ell, s) = 1, \qquad E_\ell : y^2 = x(x-\ell)(x+3\ell),$$

for positive squarefree $\ell \equiv 11 \pmod{12}$. It is `hyp:RC` at lines 1016–1028. The manuscript supplies its own long argument in Sections `sec:rc`, `sec:arithmetic`, and `sec:limits` (lines 1771–5755), with appendices at lines 9589–13699. It is not an immediate consequence of ordinary two-descent or of the published forward analytic-rank-zero-and-one theorems.

The distinction is essential. A two-Selmer group of dimension three, in the presence of full rational two-torsion, implies $r + \dim_{\mathbb{F}_2} Ш[2] = 1$. The Cassels–Tate pairing permits one to infer full 2^∞-Selmer corank one, as in `lem:corank`; it does not by itself remove a divisible $\mathbb{Q}_2/\mathbb{Z}_2$ summand of Sha or prove analytic rank one. The new restricted converse is used precisely at this remaining step.

The algebraic shape of the simultaneous realization and its reduction to the converse have been checked, but a complete independent certification of the local descent tables and the converse is not supplied. The most important further audit within the latter is the passage from finite Heegner

classes and local conditions to the uniform valuation bound and the fixed-field coefficient transfer: `ex:limits-uniform-lower-bound` (lines 4920–5217), `ex:limits-fixed-field-clearing` (lines 5498–5657), and `ex:limits-coefficient-transfer` (lines 5658–5754). These statements assert uniformity over arbitrarily large active supports and are part of the new arithmetic argument. The present report neither proves nor disproves them.

Five-contact height theorem and its exact constant. The required statement is `hyp:o8-height` (lines 6666–7520):

$$h(a) \le H_* M_5(a)^{24}, \qquad \log_2 H_* < 4783227,$$

where the contacts are $\infty, 0, 1, 2401, -343/32$ and the product $M_5(a)$ omits primes at most 76176. The displayed bound on H_* is needed for the released anchor; an unspecified larger constant cannot simply be substituted. The source derives it using the discriminant-210 Shimura curve, a degree-at-most-576 normal fiber field, a comparison with stable Faltings height, and separate CM and non-CM estimates.

Two potential objections are resolved by the cited primary sources. Yuan's Theorem 1.1 and Section 2.1 of *Explicit Kodaira–Spencer map over Shimura Curves* [28, Theorem 1.1 and Section 2.1] do provide the stated maximal-stack regularity and semistability and the isometry whose image is $d_B\omega^{\otimes 2}$. Pan's Theorem 1.0.4 of *The Fontaine–Mazur conjecture in the residually reducible case* [27, Theorem 1.0.4] has no extra residual-image restriction at $p = 5$, provided the representation is irreducible, odd, finitely ramified, potentially semistable, and regular. Thus residual reducibility alone does not invalidate the source's use of Pan.

What remains to be checked is the complete chain of hypotheses in the source's application: the canonical normalizer action and constant fiber torsor, the integral height comparison with its normalization, the scalar cocycle splitting and ramification bounds, and the modular-Jacobian transfer preserving the displayed constant. The principal non-CM passage is `sec:h5-level-repair`, lines 6904–7345; the CM estimate follows at lines 7349–7500. These are substantive mathematical arguments, not consequences of the finite numerical certificate. The validity of these applications depends on verifying their hypotheses, not on administrative review annotations in the sources.

B.5 Consequences for the optimization claims

The unconditional conclusions established by an exact transformation certificate have the form

$$\{X : \exists \mathbf{y}\ F(X, \mathbf{y}) = 0\} = \{X : \exists \mathbf{z}\ F_{\mathrm{new}}(X, \mathbf{z}) = 0\}.$$

They remain meaningful independently of the source's main arithmetic claim. Until the dependencies above have been independently verified, the further conclusion that either set equals $\mathbb{Z}$ must be stated as conditional on the arithmetic theorem of the source. A counterexample to an individual intermediate lemma would require tracing its use and possible repair before concluding that the released polynomial fails; no such counterexample is supplied by this audit.

C Reproducibility and the scope of the computations

The accompanying repository contains the publication manuscript in `paper/diophantine_optimization_20261106_v3.tex` and preserves all original research inputs under `research/`. The archived research files are not overwritten by editorial changes to the paper. The manuscript

is self-contained: it uses no external TeX input files or BibTeX database. Program outputs, certificates, and full polynomial representations are organized by construction.

C.1 Exact objects and independent checks

The directories `quartic-anchor-packed/` and `quartic-anchor-packed-large/` contain the 12,493- and 12,465-variable quartics, respectively. Each supplies the complete integer coefficient table, quadratic residuals, variable map, transformation certificate, and an independent checker. The 12,449-variable package `quartic-12449/` stores its full coefficient table over $\mathbb{Z}[M]$ with the fixed specialization $M = 3^{2^{26}}$. Its checker performs all substitutions and all sum-of-squares expansions exactly, and proves that the monomial count and coefficient height survive the specialization.

The directories `polynomial-853/` and `polynomial-854/` contain the complete polynomial circuits, finite parameter sets, construction reports, and verification receipts. The predecessor `polynomial-855/` is included as a checked dependency. Every operation in these circuits is integer addition, multiplication, or a nonnegative integer power. The two finite parameters are replaced by each of their specified integer values and the resulting branch polynomials are multiplied. There is no rational division or unquantified variable in the final polynomial specification.

The 855-variable structural checker verifies the complete circuit syntax, denominator coverage, residual and branch inventories, and degree and coefficient-norm bounds. Its six reproducible finite-field comparisons of all 297 rational residuals are explicitly *sampled checks*, not proofs of identities. The source correspondence is justified by the algebraic proofs in this paper, the independent review `compact-audit-855.md`, and inspection of the pinned templates. The separate square-root checker verifies twelve identities by exact integer polynomial arithmetic. The 854- and 853-variable transformation checkers import no producer code; they prove their anchor and primitive- element identities exactly. In particular the latter checks every one of the 295 substitutions by expression-tree induction on $H \mapsto Q^k H(P/Q)$ and reruns the exact pole-denominator check.

The number-field section is independently verified in $\mathbb{Q}(i)[X]$ against both the complete released coefficient table and all released residuals. The finite-fiber inventory checker verifies the 557 retained coordinates and the finite-choice count. The proof of geometric finiteness is mathematical. The explicit finite-algebra compiler and its independent checker, described in Section 9, now supply the indexed 558-variable polynomial, including all exceptional cases. The enormous final matrices and products are not expanded.

All primary verification scripts use the Python standard library. From the root of the pinned repository one may run

```
python3 -E -B research/optimization/verify_all.py
```

The script records the commands and their outcomes in `verification/all-checks.json`; detailed logs and individual receipts remain separate. The checkers are reproducible finite calculations, not formal verification of the source arithmetic theorem.

C.2 Identifying the principal polynomial files

The following SHA-256 digests identify the exact objects used. The 12,449-variable digest is for the uncompressed JSON coefficient table; the other quartic digests are for the uncompressed text tables.

Released 12,504-variable quartic:
`7f5aa3117af82a0b323f0a4e74a18599204f5540408106d3ce99612b5ed830ca`

12,493-variable quartic:
`06c2eeb37871341588f232d72eee72a49b8a76a2c904dc8465518d280f3378a9`

12,465-variable quartic:
`de8e8080587ef83e19653e3545a1c15805526ac847547c66dc6cc21dd1bc466c`

12,449-variable quartic:
`f8813916debae6bd20559e1130ae9786eebf48503582e37da32f1c4b018985d0`

853-variable circuit:
`67992338a1d904d623d9b1b076f28b9da6b84e44736d34e90784f75f305e5d63`

854-variable circuit:
`aef6888809817e37e372471fd780f83d0241e58f04ddeafd03fd6b46cfa39d62`

The repository includes an SHA-256 manifest of the frozen source artifacts, independent check scripts, and construction reports. The publication manuscript is maintained separately from the frozen sources; typesetting can be reproduced with two runs of `pdflatex`. A successful typesetting run does not establish any arithmetic claim.

D Reproducing the constructions with Lucas proxies

The preferred new constructions are stored under `research/phase2/`. The complete paper is the standalone file `research/optimization/diophantine_optimization.tex`. All variable counts include the distinguished input. The principal specifications and their independent verification receipts are:

Total variables	Package relative to `research/phase2/`
43	`finite_fibre/polynomial-43-conic/`
45	`finite_fibre/polynomial-45-conic-semilocal/`
558	`finite_fibre/polynomial-558/`
1367	`quartic/quartic-best/`
1391	`quartic/quartic-small-coefficients/`

The first three contain a `polynomial-specification.json`; the two quartics contain a `polynomial.json.gz`. Every package includes its construction report and a `verification.json` describing the exact scope of its independent check. The complete SHA-256 manifest is generated with the review archive. File names and hashes identify mathematical objects; they are not additional mathematical hypotheses.

From the repository root, the single command

```
python3 -E -B research/phase2/verify_phase2.py
```

runs eleven checks and records their outputs in `research/phase2/verification/all-checks.json` and its `logs/` directory. It checks the frozen source DAGs, the scalar normalization constants, the conic parametrization and duplication identities, the precision inequalities, the fixed-point arithmetic, the three preferred finite-algebra specifications, and both complete quartic tables.

The checkers are separate programs from the producers. Exact integer and rational arithmetic is used for the new algebraic checks.

The finite-algebra interpreter also executes every determinant, resultant, coefficient reversal, and outer finite case on eight small examples, including nilpotents, excluded components, collisions, rank drops, irrational residue fields, and zero quadratic roots. It checks 177 factors and supplies a fully expanded degree-sixteen example. This is a check of the construction on the specified examples. The general theorem, and not extrapolation from those computations, justifies the large instances.

To rebuild the quartics, use the producer with `--share-height--precision52--scalar-order32`. The optional `--small-coefficients` selects the 1391-variable alternative. The package README files give the complete commands for rebuilding each source and finite-algebra specification, including all finite-chart and scalar-normalization options. The original-source comparison checks remain independently runnable using `research/optimization/verify_all.py`.

These receipts establish the stated finite identities, inventories, and bounds. They do not claim to verify the external prime-pattern theorem, the source's restricted pole converse, or its full explicit height theorem. Those arithmetic dependencies are stated in Section 1 and in the source-audit appendix. The huge indexed products and multiplication matrices are specified exactly but are not materialized. Only the quartics have complete expanded coefficient tables. No global minimum is claimed.

The source assembly program checks balanced environments and braces, unique labels, cross-references, and bibliography keys. The publication copy can be compiled with a standard `pdflatex` installation (twice to resolve cross-references); typesetting does not establish any of the arithmetic assertions.

E Reproducing the direct constructions and the field counterexamples

All files in this section are in the new `research/phase3/` directory. The previous complete paper, the preceding source files, and the released quartic remain unchanged. The standalone paper is `diophantine_optimization_20261106_v3.tex`. It contains its own bibliography and does not require external LaTeX inputs. Static checks verify balanced environments and braces, unique labels and bibliography keys, and defined references and citations. No rendered-PDF verification is claimed.

E.1 Indexed formulas and quartic tables

The four complete coefficient sources are `compression/source-41-direct.json.gz`, `compression/source-43-direct-semilocal.json.gz`, `compression/source-41-shift2402.json.gz`, and `compression/source-43-semilocal-shift2402.json.gz`. They are produced by `compression/build_direct_source.py`. The options `-anchor finite` and `-anchor semilocal` select the size condition; `-beta-shift 2402` selects the translated input.

The generic compiler and checker from the preceding construction are reused unchanged. From the repository root, one example is

```
python3 -E -B research/phase2/finite_fibre/compile_shared_source.py \
  research/phase3/compression/source-41-direct.json.gz \
  --output research/phase3/finite_fibre/polynomial-41
```

```
python3 -E -B research/phase2/finite_fibre/verify_shared_specification.py \
  research/phase3/compression/source-41-direct.json.gz \
  research/phase3/finite_fibre/polynomial-41/polynomial-specification.json
```

The other three sources have separate matching specification packages and passing receipts. The verification concerns integer coefficient conversion, triangular monicity, guards, all exceptional quadratic cases, the finite index sets, and degree and coefficient majorants. It is not a verification of the external arithmetic hypotheses.

The direct quartic producer is `quartic/build_direct_quartic.py`. Reproduction uses

```
python3 -E -B research/phase3/quartic/build_direct_quartic.py \
  --share-height --precision 52 --scalar-order 32 \
  --output research/phase3/quartic/quartic-direct
python3 -E -B research/phase2/quartic/verify_shared_quartic.py \
  research/phase3/quartic/quartic-direct/polynomial.json.gz
```

Add `-small-coefficients` and use the corresponding small-coefficient output directory for the 1339-variable polynomial. The checker imports no producer. It checks every gate, the reversible row operations and eliminations, and the complete polynomial obtained by independent exact squaring and convolution of the final residuals. No floating-point or randomized comparison is used.

The preferred tables additionally use the certified transformations in `quartic/transform_direct_quartics.py`. The corresponding independent checker is `quartic/verify_final_transforms_independent.py`. It checks every affine and constant substitution, the inverse coordinate maps or unique lifts, and both the complete substituted polynomial and the independently squared residual list. It imports no producer. The fresh transformed packages preserve their frozen parents for comparison. The distinction between encoded coefficient entries and coordinate monomials after specializing a fixed integer is checked separately by `compression/check_fixed_coefficient_support.py`.

E.2 Independent checks for the field assertions

The following scripts use exact arithmetic and have separate receipts:

Quartic base change. `number_fields/check_quartic_base_change_20261106_v1.py` checks the generic $\mathbb{Q}(i)$ section against both the complete tables and the independently squared restricted residuals, for all four quartics.

Rejected contacts. `number_fields/check_contact_exclusions_20261106_v1.py` checks the three exact input relations in the four unshifted indexed sources and four quadratic systems. The shifted sources are expressly outside this counterexample's scope.

The totally real extension. `number_fields/check_real_anchor_20261106_v1.py` checks the field, point, and doubling calculations in the displayed degree-eight compositum.

Division norms. `number_fields/build_division_norm_specs_20261106_v1.py` produces all four determinant specifications. The independently written `number_fields/verify_division_norm_specs_independent_20261106_v1.py` imports neither this producer nor an earlier checker. It compares every embedded residual with pinned source bytes, reconstructs each matrix-entry index, verifies formal Galois conjugacy, checks the explicit

inert primes, and proves the stated exact degrees from the unit coefficient of the highest pure input power. It also recomputes the coefficient bounds and the unrounded finite-algebra degree majorants before their displayed powers-of-two rounding.

The accompanying independent proof review explains the local division arguments, including the degree-36 norm over $\mathbb{Q}(i)$, and the coprime-extension module argument. The norm determinants are exact finite specifications, not expanded tables. The counterexamples and norm equivalences are unconditional algebraic results. They do not certify that the changed source defines any extension field's ring of integers.

The final affine and constant-chain transformations have separate number-field certificates and degree-72 specifications with dated suffix `20261106_v2`. These are checked independently in addition to the preceding specifications. In particular, elimination of a defining residual preserves common zero sets over every field, but does not by itself preserve every cancellation among nonzero squares over a nonreal field. The explicit Gaussian sections are therefore verified again for each new literal quartic.

E.3 Rebuild and integrity records

`assemble_phase3.py` builds this paper from the preceding complete report and the new mathematical supplements, and checks that the preceding report's bytes are unchanged. The verification index records the passing receipts and hashes of the checked artifacts. `make_checkpoint.py` packages the complete research tree, the pinned source manuscript, and the unchanged released baseline. It hashes the same bytes it writes and verifies the resulting ZIP archive. Its manifest makes the checkpoint independently inspectable.

F Verification of the new source specification

The computational checks in this section concern the explicitly recorded source equations and their translation. They do not certify the external arithmetic hypotheses used to prove the integer input projection. That distinction is essential: an exact compiler can correctly translate an arithmetic system whose completeness or soundness has not been independently established.

The program `check_direct_source.py` compares the abstract syntax trees of the frozen producer with proxies and the direct-Lucas producer. All eight auxiliary methods are identical. After removing the two retained proxies, their finite quadratic root, and their three comparison tests, the remaining construction statements are identical apart from the direct polynomial test, its corresponding counts and metadata, and the optional shift in β. The previously preferred sharing, scalar normalization, conic parametrization and fixed scale options are now the defaults; their mathematical formulas are unchanged.

The new direct test is also checked by a separate sparse polynomial implementation over $\mathbb{Q}$. Substitute

$$u = u_0 + u_1\sqrt{2}, \quad v = v_0 + v_1\sqrt{2}, \quad w = w_0 + w_1\sqrt{2}.$$

Evaluation of the producer's actual expression agrees exactly with the independently expanded rational and $\sqrt{2}$ components of

$$v^2 - u(u-2)vw + (u-2)^2w^2 - 1.$$

These components have respectively 23 and 19 nonzero monomials and total degree four. The complete coefficient dictionaries are in `direct-source-check.json`.

The normalization is verified by exact Laurent-polynomial identities. For each precision $m \in \{1, 16, 64\}$ and $R = A^{-5}F\delta^{-6}$, the coefficient multiplying the square of the field test is

$$\frac{R}{\delta^{2m+2}R^{2m}} = A^{10m-5}\delta^{10m-8}F^{1-2m}.$$

For $W = w_0 + w_1\sqrt{2}$, the guarded reciprocal trace is $2w_0/(w_0^2 - 2w_1^2)$, so its pole input is exactly

$$\frac{R}{(\mathrm{Tr}_{K/\mathbb{Q}}(W^{-1}))^2} = \frac{F(w_0^2 - 2w_1^2)^2}{4A^5\delta^6 w_0^2}.$$

The checker reads these expressions from the producer and compares their expanded dictionaries, rather than relying only on sample numerical evaluations.

The four archived source files reproduce byte for byte. Their integer coefficient DAGs are acyclic, every monic relation depends only on retained coordinates and earlier generators, and all counts and finite parameter lists agree with the displayed source. There are exactly sixteen comparison tests and no remaining Lucas proxy variable, equation or exceptional-root case. The verified counts are

anchor	C	retained coordinates	finite relation degrees	coefficient DAG nodes
finite	2	40	$4^9\,2^{10}$	16039
semilocal	2	42	$4^9\,2^{11}$	16085
finite	2402	40	$4^9\,2^{10}$	16040
semilocal	2402	42	$4^9\,2^{11}$	16086

Here C means $\beta = X^2 + C$. The products in the fourth column describe the nine quartic and ten or eleven quadratic finite relations; they are not degrees of the final polynomial. The independent finite-algebra compiler supplies the final degree and height bounds.

The source check is reproducible with the command

```
python3research/phase3/compression/check_direct_source.py
```

The separate program `check_elliptic_density_data.py` verifies the exact curve and point calculations used in Corollary 29.3. Its scope is those fixed algebraic calculations; the geometric density assertion is proved in the text.

References

[1] *A polynomial definition of the integers in the rationals*, repository `elliotglazer/diophantine-z-q`, revision `28e87b1b62af0effd954605af7f70b2699dd5672`, accessed 6 November 2026. Manuscript `quartic_12504.tex`; package `quartics/quartic-12504`. `https://github.com/elliotglazer/diophantine-z-q`.

[2] Yu. V. Matiyasevich, *The Diophantineness of enumerable sets*, Doklady Akademii Nauk SSSR **191** (1970), no. 2, 279–282; English translation, Soviet Mathematics Doklady **11** (1970), 354–357. `https://www.mathnet.ru/eng/dan35274`.

[3] Yu. Matiyasevich and J. Robinson, *Reduction of an arbitrary Diophantine equation to one in 13 unknowns*, Acta Arithmetica **27** (1975), 521–553. `https://doi.org/10.4064/aa-27-1-521-553`.

[4] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), no. 3, 549–571. https://doi.org/10.2307/2273588.

[5] N. Daans, *Universally defining $\mathbb{Z}$ in $\mathbb{Q}$ with 10 quantifiers*, Journal of the London Mathematical Society **109** (2024), no. 2, e12864. https://doi.org/10.1112/jlms.12864. Primary preprint: https://arxiv.org/abs/2301.02107.

[6] N. Daans, P. Dittmann, and A. Fehm, *Existential rank and essential dimension of diophantine sets*, arXiv:2102.06941v5 (2025), in particular Theorem 1.4, Corollary 4.21, and Remark 8.23. https://arxiv.org/abs/2102.06941v5.

[7] G.-R. Zhang and Z.-W. Sun, *$\mathbb{Q} \setminus \mathbb{Z}$ is Diophantine over $\mathbb{Q}$ with 32 unknowns*, Bulletin of the Polish Academy of Sciences, Mathematics **70** (2022), no. 2, 93–106. https://arxiv.org/abs/2104.02520.

[8] Z.-W. Sun, *86 unknowns for Hilbert's tenth problem over the rationals*, arXiv:2610.20691v2 (2026). https://arxiv.org/html/2610.20691v2. The new shared-point construction additionally assumes Theorem 4.1, not merely the algebraic devices of Sections 10.3–10.6. Its use is explicitly separated from source-preserving algebraic transformations.

[9] C. Frei and M. Tinková, with an appendix by C. Frei, J. König, and M. Tinková, *Linear constellations in primes with arithmetic restrictions*, arXiv:2408.17441v3 (2025), Theorem 1.4. https://arxiv.org/html/2408.17441v3. The Artin-symbol theorem cited here is unconditional; the separate primitive-root theorem has a Riemann-hypothesis assumption.

[10] T. Tao and T. Ziegler, *Polynomial patterns in the primes*, Forum of Mathematics, Pi **6** (2018), e1. https://arxiv.org/abs/1603.07817.

[11] M. S. Paterson and L. J. Stockmeyer, *On the number of nonscalar multiplications necessary to evaluate polynomials*, SIAM Journal on Computing **2** (1973), no. 1, 60–66. https://doi.org/10.1137/0202007.

[12] Z.-W. Sun, *Further results on Hilbert's Tenth Problem*, Science China Mathematics **64** (2021), no. 2, 281–306; in particular Theorem 1.1(i) and Equation (1.1). https://arxiv.org/abs/1704.03504v7.

[13] J. Koenigsmann, *Defining $\mathbb{Z}$ in $\mathbb{Q}$*, Annals of Mathematics **183** (2016), no. 1, 73–93. https://doi.org/10.4007/annals.2016.183.1.2. Primary preprint: https://arxiv.org/abs/1011.3424.

[14] J. Voight, *Quaternion algebras*, Graduate Texts in Mathematics **288**, Springer, 2021; in particular Proposition 14.6.7. https://doi.org/10.1007/978-3-030-56694-4.

[15] J. Kollár, *Unirationality of cubic hypersurfaces*, Journal of the Institute of Mathematics of Jussieu **1** (2002), no. 3, 467–476, Theorem 2. https://doi.org/10.1017/S1474748002000117. https://arxiv.org/abs/math/0005146.

[16] A. Schinzel, *Polynomials with special regard to reducibility*, Encyclopedia of Mathematics and its Applications **77**, Cambridge University Press, 2000, Chapter 4, Section 4. https://doi.org/10.1017/CBO9780511542916.

[17] P. Müller, *Finiteness results for Hilbert's irreducibility theorem*, arXiv:math/0109071, 2001; introduction and Section 2.1. https://arxiv.org/abs/math/0109071.

[18] S. Lang, *Fundamentals of Diophantine geometry*, Springer, 1983, Chapter 8, pp. 188–224. https://doi.org/10.1007/978-1-4757-1810-2.

[19] D. Eisenbud, M. Green, K. Hulek, and S. Popescu, *Small schemes and varieties of minimal degree*, American Journal of Mathematics **128** (2006), no. 6, 1363–1389. https://doi.org/10.1353/ajm.2006.0043. https://eisenbud.github.io/papers/pdfs/2006-002.pdf.

[20] J. Harris and Y. Tschinkel, *Rational points on quartics*, Duke Mathematical Journal **104** (2000), no. 3, 477–500. https://arxiv.org/abs/math/9809015. https://cims.nyu.edu/~tschinke/papers/yuri/00quart/quart.pdf.

[21] J. Berg, C. Pagano, B. Poonen, M. Stoll, N. Triantafillou, B. Viray, and I. Vogt, *Brauer–Manin obstructions requiring arbitrarily many Brauer classes*, Bulletin of the London Mathematical Society **56** (2024), no. 5, 1587–1604; Theorem 5.1 and Section 6.1. https://doi.org/10.1112/blms.12999. https://math.mit.edu/~poonen/papers/many_brauer.pdf.

[22] J.-P. Serre, *Quelques applications du théorème de densité de Chebotarev*, Publications Mathématiques de l'IHÉS **54** (1981), 123–201; Section 2.1, Theorem 1. https://numdam.org/item/PMIHES_1981__54__123_0.pdf.

[23] J. Voight, *Identifying the matrix ring: algorithms for quaternion algebras and quadratic forms*, author's revised text, 21 May 2019, Proposition 5.5; arXiv:1004.0994. https://jvoight.github.io/articles/quatalgs-051919.pdf.

[24] P. Koymans and C. Pagano, *Hilbert's tenth problem via additive combinatorics*, Journal of the American Mathematical Society, published electronically 10 August 2026. https://doi.org/10.1090/jams/1081. https://arxiv.org/html/2412.01768v3.

[25] L. Alpöge, M. Bhargava, W. Ho, and A. Shnidman, *Rank stability in quadratic extensions and Hilbert's tenth problem for the ring of integers of a number field*, Inventiones Mathematicae **243** (2026), 1129–1139. https://doi.org/10.1007/s00222-025-01392-3. The cited Corollary 1.2 is that of the published version; the arXiv version 1 states a weaker Diophantine-model conclusion.

[26] B. Mazur, *Speculations about the topology of rational points: an up-date*, Astérisque **228** (1995), 165–181; Conjecture 3, p. 165. https://numdam.org/item/AST_1995__228__165_0/.

[27] L. Pan, *The Fontaine–Mazur conjecture in the residually reducible case*, Journal of the American Mathematical Society **35** (2022), 1031–1169. https://doi.org/10.1090/jams/991. https://arxiv.org/abs/1901.07166.

[28] X. Yuan, *Explicit Kodaira–Spencer map over Shimura curves*, Acta Mathematica Sinica, Chinese Series **67** (2024), no. 2, 227–249. https://doi.org/10.12386/A20220154. https://arxiv.org/abs/2205.11334.

[29] A. Merkurjev and R. Parimala, *Quadratic Forms Beyond Arithmetic*, Notices of the American Mathematical Society **72** (2025), no. 7, 711–718, doi:10.1090/noti3192. Author manuscript: https://www.math.ucla.edu/~merkurev/papers/quadratic8.pdf, Section 5.1.

[30] B. Poonen, *Multivariable polynomial injections on rational numbers*, Acta Arithmetica **145** (2010), no. 2, 123–127. doi:10.4064/aa145-2-2. https://math.mit.edu/~poonen/papers/QQ.pdf.

[31] H. Pasten, *Bivariate polynomial injections and elliptic curves*, Selecta Mathematica, New Series **26** (2020), article 22. doi:10.1007/s00029-020-0548-x. https://arxiv.org/abs/1811.02416.

[32] M. Hirvensalo, *On injective multivariate polynomials over rational numbers*, Theoretical Computer Science **1082** (2026), 116158. doi:10.1016/j.tcs.2026.116158. Theorem 2 has the restricted domain specified in the text.

[33] J. S. Milne, *Abelian Varieties*, in G. Cornell and J. H. Silverman (eds.), Arithmetic Geometry, Springer, New York, 1986, 103–150; corrected author version, 2 January 2022, Sections 8 and 12. https://www.jmilne.org/math/xnotes/AVs.pdf.

[34] J. S. Milne, *Class Field Theory*, version 4.03, 6 August 2020, Chapter IV, Sections 1–4. https://www.jmilne.org/math/CourseNotes/CFT.pdf.

[35] K. S. Kedlaya, *Notes on Class Field Theory*, Section 2.4 (Chebotarev's density theorem) and Section 7.6 (Brauer groups, especially Theorem 7.6.12), accessed 6 November 2026. https://kskedlaya.org/cft/.

[36] The Stacks Project Authors, *The Stacks Project*, Tag 00FE, Theorem 10.29.10 (Chevalley's theorem), 2026. https://stacks.math.columbia.edu/tag/00FE.

[37] S. Basu, *Algorithms in Real Algebraic Geometry: A Survey*, arXiv:1409.1534, 2014, Section 1.1 and Theorem 2.1. https://arxiv.org/pdf/1409.1534.

[38] D. Plaumann, *Real Algebraic Geometry*, IHP tutorial lectures, 12 October 2023, Corollary 4.5 and Example 4.6. https://wwwold.mathematik.tu-dortmund.de/~dplauman/notes/rag-ihp.pdf.